HRMA

Rocket Motor Preliminary Design and Analysis Software

Hybrid, solid and liquid rocket motors

User Guide

Getting started, worked examples and reference

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Scope. HRMA is software for the preliminary design and analysis of rocket motors, built on closed-form and one-dimensional engineering relations. It is not a flight-qualification tool. Results must be confirmed by physical testing before flight (Section 19).

Basis of results. Every value on screen comes from user input, from data with a stated source or from the model's solution. If a required input is missing, the calculation stops and names the missing field. When a default value is used, it is reported with its name and source. Outside the model's validity limits no verdict is issued and the reason is shown.

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Part I Getting Started and Worked Examples

1. General Information

1.1 Purpose and scope

HRMA is a desktop application for the preliminary design and analysis of hybrid, solid and liquid rocket motors. From the requirement and design parameters supplied by the user it computes the motor geometry, its performance and a list of warnings; it runs thermal, structural, flow, stability, feed-system, safety and uncertainty analyses on the same motor; and it exports the result as a report (PDF, XLSX), CAD (STEP, STL, DXF), engineering drawings and a flight-simulation file (`.eng`).

The intended users are engineers and teams who design motors: rocketry teams, research groups, companies developing motors. The guide assumes familiarity with the basic concepts of rocket propulsion (thrust, specific impulse, oxidizer-to-fuel ratio, nozzle expansion); definitions are in Appendix B, symbols in Appendix C.

1.2 Basis of results and validity checks

When HRMA cannot compute a quantity it shows no value for it. This behaviour is called fail-closed and appears on screen in four forms.

Table 1.1: Forms of fail-closed behaviour on screen

Form	On screen	Example (from this guide's examples)
Rejection by name	The request is not sent, or the server rejects it with 422; the missing or out-of-range field is named by its on-screen label.	In the Nozzle Flow panel an empty Chamber Pressure is rejected with "incomplete_flow_input" naming the field.
Declared default	The value used for a field left empty is listed with its name and reason.	In the hybrid example, the list VALUES THIS RUN ASSUMED FOR FIELDS YOU LEFT EMPTY: Combustion (c*) Efficiency = 89%.
Withheld verdict	No verdict is issued; "NOT EVALUATED" ("NOT_EVALUATED" in the dock panels), "VERDICT WITHHELD" or "NOT MODELLED" is shown with the reason.	In the hybrid example the wall was sized by HRMA, so the structural verdict is NOT EVALUATED and VERDICT WITHHELD.
Provenance line	The source of a published number is written next to it: user input, solver result or sourced default.	c* efficiency $\eta_{c^*} = 0.89$ (sourced default: median of 26 measured firings, range 0.77-0.99).

Warnings are not a sign of malfunction; they report a model limit or a design approaching a limit. Numbers are read together with their warnings (Section 17).

1.3 Structure and conventions of this guide

The guide has two parts.

- **Part I, Getting started and worked examples.** Installation, a 15-minute quick start, and a step-by-step study of the example shipped for each motor type. Each step states what to look at on screen and the physical meaning of the number.

- **Part II, Reference.** Purpose, inputs, outputs, verdicts, warnings and limits of every page, in tables. The input tables are generated from the application's own screen: label, identifier, unit, form limit, default and the "?" note next to the field.

Table 1.2: Conventions

Item	Convention
Interface names	Labels, buttons and tabs are written in this typeface exactly as on screen.
Field identifier	The name used in the project file (`.hrma`) and in server messages is written `in this typeface` (example: `chamber_pressure`). Identifier to page: Appendix D.
Units	SI. Pressures are absolute and in bar unless stated otherwise.
Numbers	The English text uses a decimal point (209.7 s). Example numbers were read from the screen of the version this guide describes (3.0.6).
Procedures	Numbered steps; each step gives the action and the expected result.

1.4 Local operation and data privacy

Design data never leaves the computer. The Python runtime and all libraries are inside the installer. On its own the application goes online only for the new-version check at startup (GitHub); if the request fails the application works normally, and the check can be switched off in Settings. The following accesses are started by the user:

1. Update now: the installer is downloaded to the Downloads folder (Section 2.4).
2. Report a bug and the GitHub links of the window menu: the GitHub page opens; submitting the issue is the user's decision.
3. The online enrichment on the Launch Site page: satellite tiles (NASA GIBS) and elevation and surface weather data (Open-Meteo). Both are off by default; when enabled, the request sends the coordinates of the viewed region to the provider.

1.5 Licence

HRMA 3.0.6 is distributed under the terms of the GNU General Public License, version 3 or (at your option) any later version (`GPL-3.0-or-later`). The licence text is the `LICENSE` file at the root of the source repository; the same text is included in the application folder of the installation package (`app/LICENSE`). The package includes the RocketCEA library (GPLv3) for the NASA CEA comparison; the licence notices of third-party components are in the package folder `app/licenses/` .

2. Installation and Start-up

2.1 System requirements

Table 2.1: Supported platforms and installers

Platform	Package	Note
Windows 10/11 (64-bit)	`HRMA-Setup-3.0.6.exe`	Per-user installation, no administrator rights needed. The package is unsigned; a SmartScreen warning may appear.
macOS 11 or later, Apple Silicon	`HRMA-Setup-3.0.6-macOS.dmg`	Does not run on Intel Macs. The package is unsigned; Gatekeeper asks for approval at first launch.

The Python 3.12 runtime and all dependencies (thermochemistry, CAD kernel, plotting libraries) are inside the package; nothing else needs to be installed.

2.2 Installation procedure

Windows

1. Run `HRMA-Setup-3.0.6.exe`. Press Next on the welcome page, Install on the folder page and Finish on the last page. If Launch HRMA now is ticked on the last page, the application opens.
2. The installation folder must be empty or contain a previous HRMA installation; if an unrelated folder is chosen, setup does not proceed and asks for another folder. The default folder is `%LOCALAPPDATA%\HRMA`.
3. If SmartScreen warns, press More info, then Run anyway.

macOS

1. Open `HRMA-Setup-3.0.6-macOS.dmg`.
2. Drag the `HRMA` icon to the `Applications` folder.
3. At first launch right-click the application (or Control-click it), choose Open and press Open again in the dialog. If the dialog has no Open button, press Open Anyway at the bottom of System Settings → Privacy & Security. This approval is requested only once.

2.3 First launch and project directory

The application shows a splash screen while the calculation engines load, then opens the home page in its own window. If the application is launched again while it is running, no second calculation server is started: the launcher finds the running application and connects the new window to it. Both windows use the same projects and the same calculation engine. When the first window is closed, the application closes.

Projects are stored as `.hrma` files in the user's `Documents/HRMA/projects` directory. At first launch three example projects are copied there (Table 2.2). They are copied once; a deleted example does not come back, and a copy the user has modified is never overwritten. An unmodified copy of an older example is refreshed with the current example in a new version.

Table 2.2: Example projects shipped with the application (3.0.6)

Project name	Design	Main result on screen
`Example Hybrid N2O-HTPB 3kN`	N_2O/HTPB hybrid, 3000 N, 10 s, 30 bar, O/F 6.0, showerhead injector	Isp 209.7 s (design point); delivered I_t 26 839 N · s (tank blowdown solution)
`Example Solid KNDX BATES 75mm`	KNDX solid, 3 BATES grains 75/32/360 mm (cartridge-loaded), 40 bar design pressure, graphite nozzle	Isp 125.3 s (sea level); peak thrust 1498 N; I_t 2957 N · s
`Example Liquid LOX-RP1 25kN`	LOX/RP-1 liquid, 25 kN, 50 bar, O/F 2.0, gas-generator cycle, regenerative cooling, 1.0 mm CuCrZr liner and closeout shell, 30 s burn (example input)	Engine Isp 254.8 s (sea level), 299.3 s (vacuum); I_t 750 000 N · s

Output files (reports, CAD, drawings, `.eng`, packages) are delivered as downloads. In the application window every download opens a save dialog; the dialog starts in the Downloads folder with the suggested file name, and the location and name are chosen there. The Open Output Folder item of the application menu opens the `Documents/HRMA` folder; it holds the project files (`projects`).

Downloaded files are not written to this folder automatically. The application log is `hrma_log.txt` in the same folder on Windows and `~/Library/Logs/HRMA.log` on macOS; attach it when reporting a problem.

2.4 Updates

When a new version is found at startup, a window offers three options.

Table 2.3: Update window options

Option	Effect
Update now	The installer is downloaded (with a percentage indicator), the application closes, installs and reopens. On macOS the installation takes a few minutes (release note: about 4 min); do not open the application manually during that time.
Later	The window closes; the application asks again at the next launch.
Skip this version	This version is not offered again. The decision is undone with Reset skipped version in Settings.

3. Getting Started: the First 15 Minutes

This section uses the hybrid example shipped with the application to open a motor, calculate it, read the results, change one input, and obtain the report and the CAD file. Section 4 explains each screen in detail.

3.1 Procedure

1. **Open the hybrid designer.** On the home page press LAUNCH DESIGNER on the Hybrid Motors card. The hybrid page opens with an empty form (Figure 3.1).
2. **Load the example.** In the project bar press Open and select `Example Hybrid N2O-HTPB 3kN`. The form fills with the example's inputs. The same example also opens from the RECENT PROJECTS list on the home page.
3. **Calculate.** Press Calculate below the form. The result sections open below the form. Calculation time depends on the computer; in the measurement for this guide the hybrid example finished in 17 s.
4. **Read the headline numbers.** The PERFORMANCE SUMMARY cards: ISP (DESIGN POINT, AT DESIGN-ALTITUDE AMBIENT) 209.7 s, THRUST 3000 N, CHAMBER PRESSURE 30.0 bar, MASS FLOW RATE 1.459 kg/s. Thrust and chamber pressure are inputs; Isp and mass flow are solved. Check: $\dot{m} = F/(I_{sp}\, g_0) = 3000/(209.7 \times 9.80665) = 1.459$ kg/s.
5. **Read the strip at the top of the results.** The strip says 6 physics items are NOT modelled in this hybrid result; expanding it lists what the model does not compute for this motor. The fifth card of the PERFORMANCE SUMMARY, STRUCTURAL MIN SF, reads NOT EVALUATED and WALL: SIZED BY HRMA; the verdict line under the wall block is VERDICT WITHHELD. No wall thickness was entered, so HRMA sized the wall and does not accept the wall it sized itself (Section 4.5).
6. **Read the warnings.** The WARNINGS section of this example carries six WARNING lines and five INFO lines, and no CRITICAL line. Each line names a quantity, its limit and the change that can be made.

7. **Change one input.** In the MIXTURE RATIO section raise O/F Ratio from 6 to 7 and press Calculate again. Isp rises from 209.7 to 210.0 s; a CRITICAL line is added to WARNINGS: The steady-state equilibrium temperature of the throat wall is 3313 K, above the 3300 K allowable temperature of Graphite (isostatic, nozzle insert grade) …. As the mixture ratio approaches stoichiometric the flame temperature rises; the Isp curve is flat in this region but the thermal load on the nozzle throat increases. Set the value back to 6 and calculate again.

8. **Get the report.** In EXPORT & CAD OPTIONS at the bottom of the page press Summary PDF or Technical PDF in the Analysis Reports group. Export Results at the top of the results produces the same file as Summary PDF.

9. **Get the CAD file.** In the 3D CAD Design group of the same section press Generate 3D Model (STEP): the page scrolls to the 3D view and the parametric solid model of the motor (chamber, nozzle contour, injector plate and assembly) is downloaded as a STEP package. The drawing PDF and DXF are the Technical Drawings (PDF) and DXF Drawing buttons of the same group. To get STEP, DXF, the drawing PDF and STL files in one archive use Generate All Files in the Complete Package group (Section 16).

10. **Save the project.** To keep the modified design under your own name press Save As in the project bar.

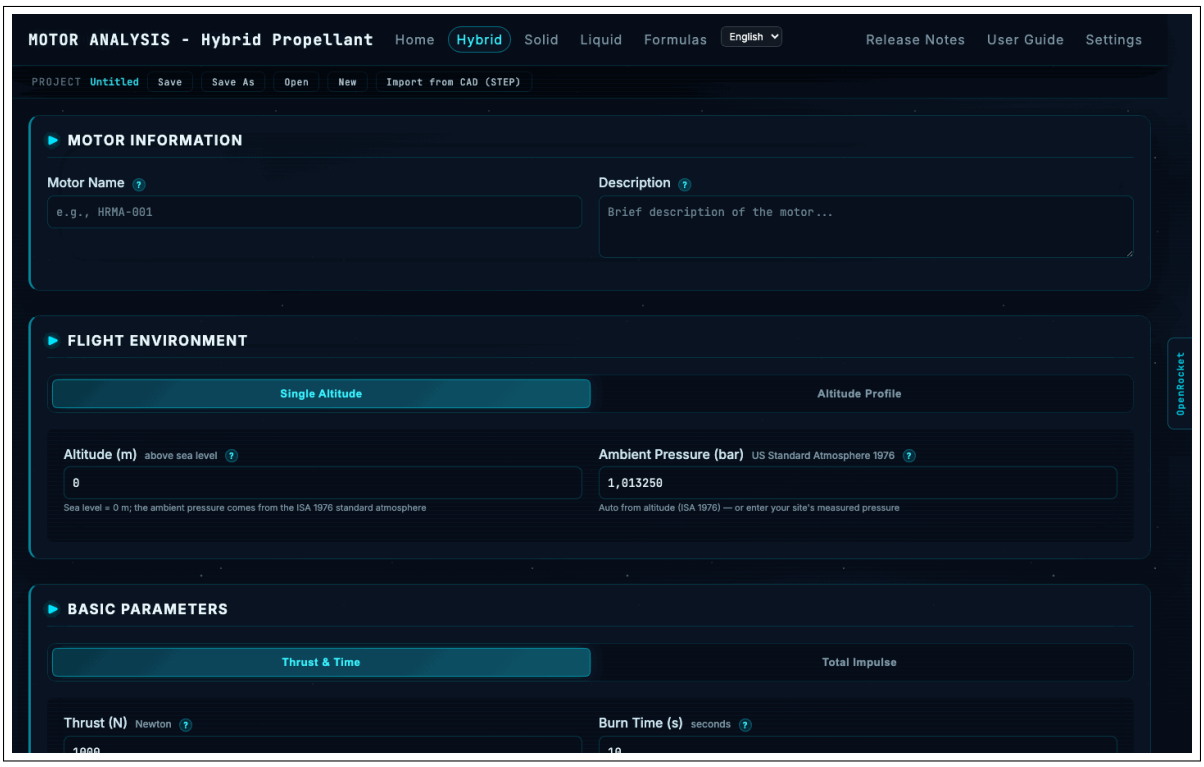


Figure 3.1: Hybrid page on opening: top bar, project bar (Save, Save As, Open, New, Import from CAD (STEP)) and the first form sections.

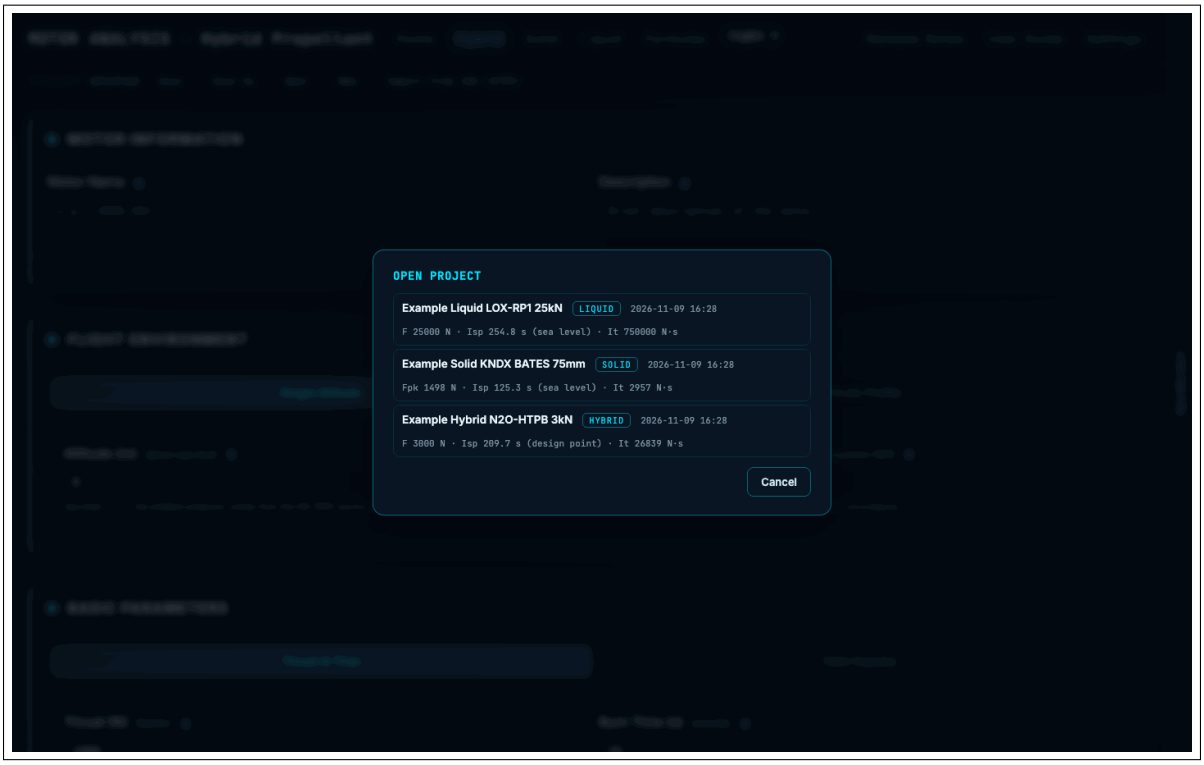


Figure 3.2: The Open window of the project bar: the three example projects, their motor types and the summary of the last calculation. Note which version computed each summary.

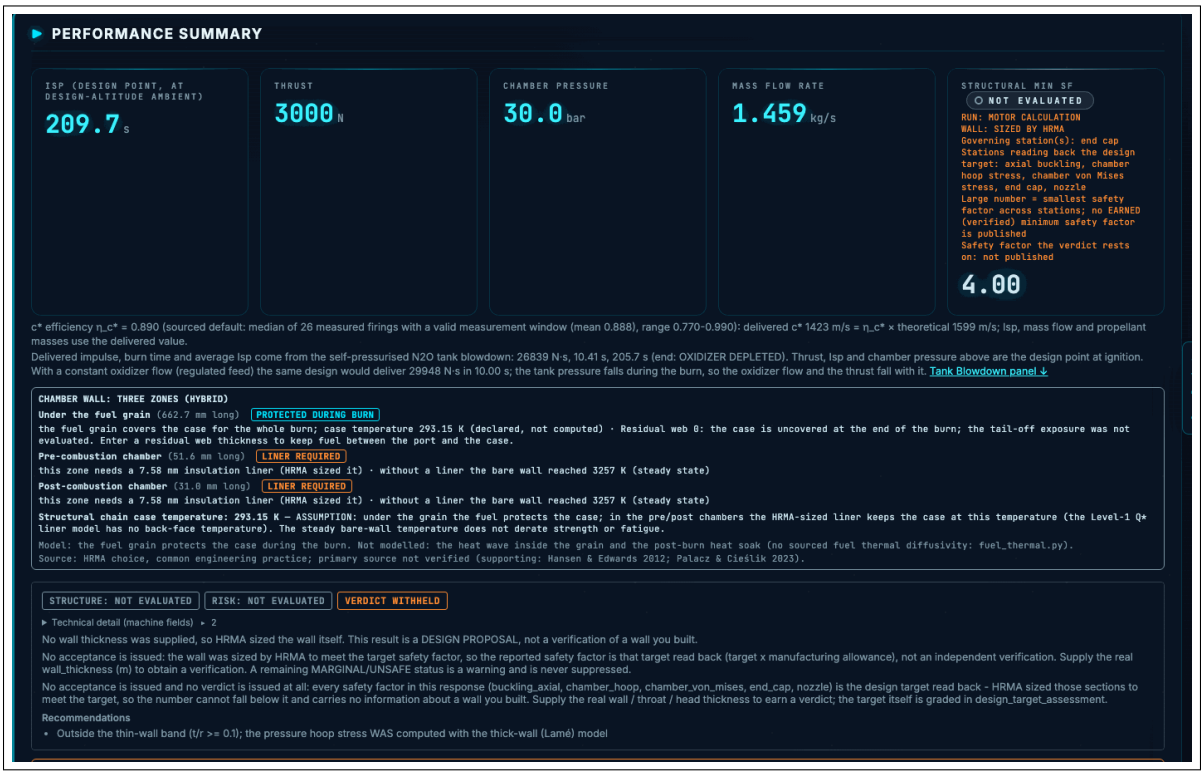


Figure 3.3: PERFORMANCE SUMMARY of the hybrid example: four headline cards, the STRUCTURAL MIN SF card and the c^* efficiency provenance. Note which cards are inputs (thrust, chamber pressure) and which are solved (Isp, mass flow).

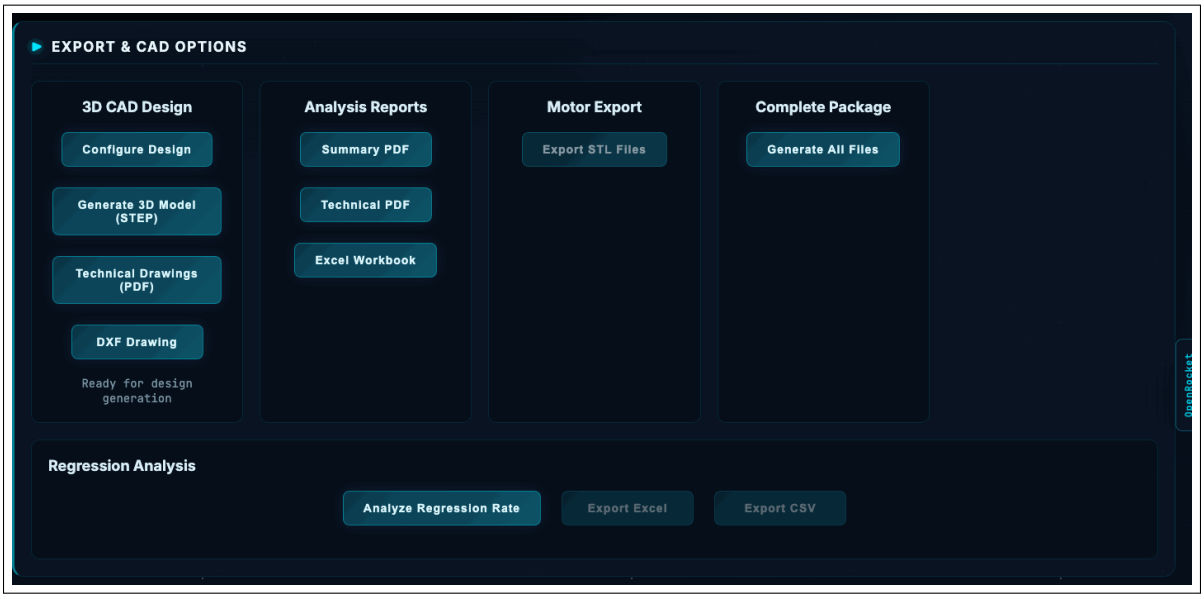


Figure 3.4: The EXPORT & CAD OPTIONS section: 3D CAD Design, Analysis Reports, Motor Export, Complete Package and Regression Analysis groups.

3.2 Next steps

- The full result screen of the hybrid example: Section 4.
- Solid and liquid examples: Sections 5 and 6.
- Starting a new motor from a requirement: Section 7.
- Finding what a field does: the "?" next to the field, or Appendix D (identifier-to-table index).

4. Example 1: Hybrid Motor (N_2O/HTPB, 3 kN)

4.1 Purpose

Read every section of the result screen on the hybrid example shipped with the application; see how one input changes both performance and a material limit; run an analysis panel. The full input list is in Tables 9.1–9.11.

Table 4.1: Main inputs of the hybrid example (`Example Hybrid N2O-HTPB 3kN`)

Input	**Value**	**Note**
Thrust (N), Burn Time (s)	3000 N, 10 s	Design mode Thrust & Time.
Chamber Pressure (bar)	30 bar	Absolute.
O/F Ratio	6.0	Design-point mixture ratio.
Fuel Type	HTPB, 920 kg/m^3	Regression $r = a\,G^n$: $a = 6.399 \times 10^{-5}$, $n = 0.554$. The example description gives the source: a log-log fit to the Carmicino et al. pure HTPB/N_2O firings in HRMA's validation records (the tests injected gaseous N_2O, the example injects liquid). With the fields empty the catalogue pair of the fuel is used ($a = 3.68 \times 10^{-5}$, $n = 0.555$).
Mass Flux (kg/m^2·s)	250	Initial oxidizer flux; inside the measured range of the fit. When empty, the motor's design flux (350) is used (Section 4.7).
Oxidizer Type, Phase, Temperature (K)	N_2O, liquid, 293.15 K	Tank Pressure Source: Self-pressurised: P_sat(T_tank); the tank pressure is the N_2O saturation pressure (50.53 bar).
L* Value (m)	1.0 m	Requested characteristic length.
Ae/At Ratio	0	0 = the model solves the expansion ratio.
Nozzle Type, Combustion Type	Conical, Infinite Area Combustion	
Injector Type	Showerhead	Target Velocity (m/s) 30, Plate Thickness (mm) 3.
Chamber Material, Nozzle Material	Stainless steel 304 (annealed); Graphite (isostatic, nozzle insert grade)	No wall thickness entered: HRMA sizes the wall.

4.2 The example

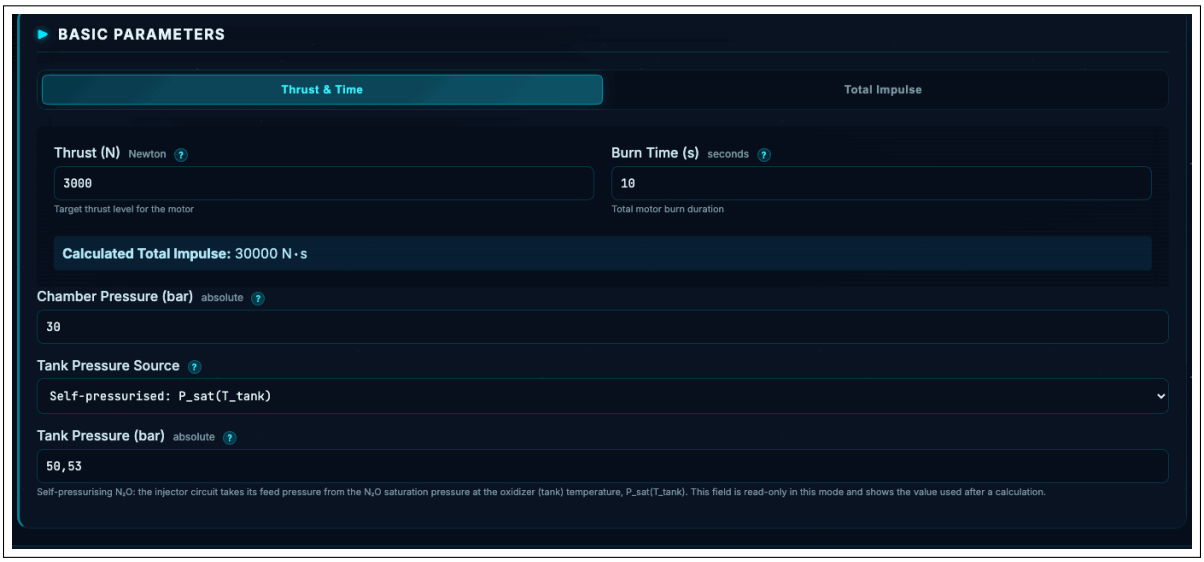


Figure 4.1: BASIC PARAMETERS after loading the example hybrid: thrust, burn time, chamber pressure and tank pressure source.

4.3 Performance: why Isp is 209.7 s

After calculation the PERFORMANCE SUMMARY shows four headline cards and the STRUCTURAL MIN SF card (Figure 3.3). Isp is the product of the thrust coefficient and the delivered characteristic velocity:

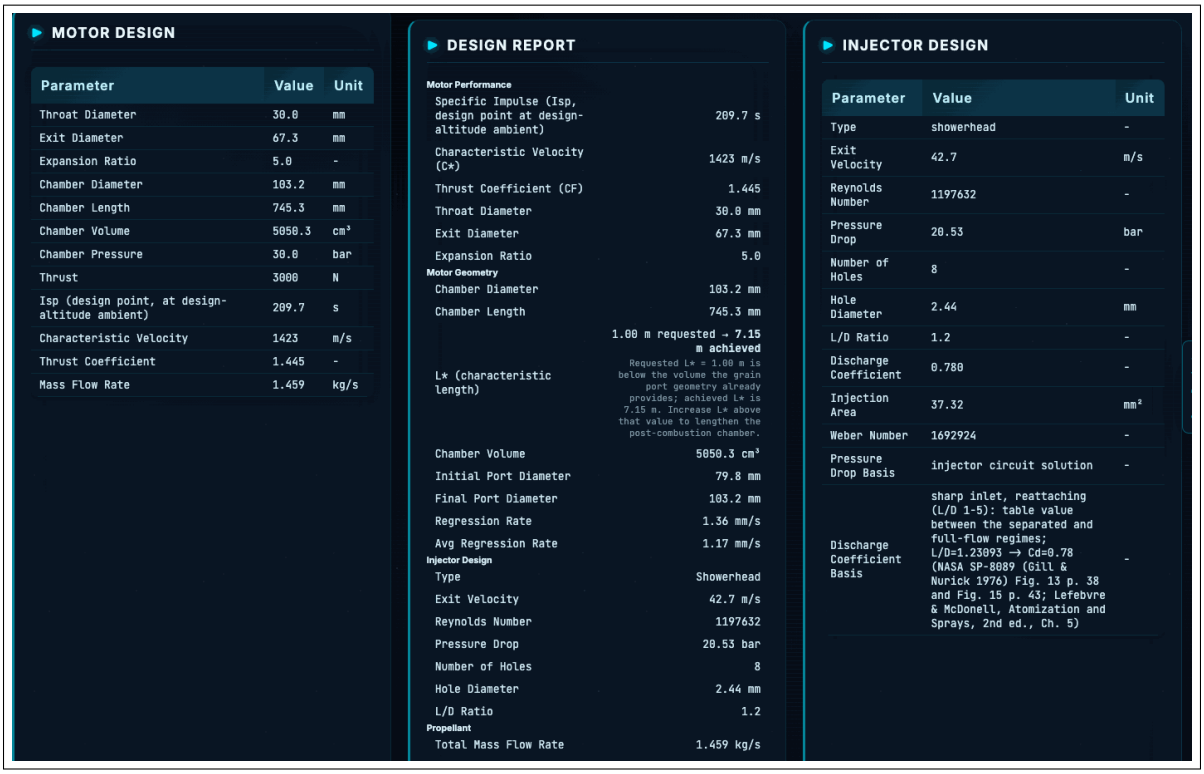
$$I_{sp} = \frac{c^* \, C_F}{g_0} = \frac{1423 \times 1.445}{9.80665} = 209.7 \text{ s}$$

The provenance line below the cards gives the source of c^*: delivered c* 1423 m/s = η_{c^*} × theoretical 1599 m/s. The theoretical value comes from equilibrium thermochemistry (Cantera) at the chamber pressure and O/F 6.0. In a real hybrid chamber mixing is incomplete and combustion does not finish; the c^* efficiency carries this loss. $\eta_{c^*} = 0.890$ is the median of 26 measured N_2O/HTPB firings with a valid measurement window in HRMA's validation records (mean 0.888, range 0.770–0.990). The field was left empty, so this value was used and declared in the VALUES THIS RUN ASSUMED FOR FIELDS YOU LEFT EMPTY list. If you have your own measurement, enter it in Combustion (c*) Efficiency (%); Isp, mass flow and propellant masses are recomputed with it.

The thrust coefficient $C_F = 1.445$ follows from the expansion ratio $\varepsilon = 5.0$ and the ambient pressure at the design altitude. Because Ae/At Ratio was entered as 0 the model solved the ratio that makes the exit pressure equal to ambient (5.0364); the warning list says so with Input field 'Ae/At Ratio' (0) is solved by the model, not read from the form.

Table 4.2: Hybrid example numbers in MOTOR DESIGN and DESIGN REPORT

Quantity	**Value**	**Meaning**
Throat / exit diameter	30.0 / 67.3 mm	Throat from mass flow and chamber pressure; exit from ε.
Chamber diameter / length	103.2 / 745.3 mm	The chamber diameter is the outer diameter of the fuel grain (end-of-burn port).
L*	1.00 m requested, 7.15 m achieved	The volume of the grain port already exceeds the requested L*; the note below the row says so.
Initial / final port diameter	79.8 / 103.2 mm	The port grows during the burn; the final port reaches the wall.
Regression rate	1.36 mm/s, average 1.17 mm/s	The flux G falls as the port grows, so the rate falls.
Mass flows	1.459 / 1.250 / 0.208 kg/s	Total / oxidizer / fuel; O/F = 1.250/0.208 = 6.0.
Total propellant (loaded)	15.00 kg	Oxidizer 12.95 kg (the mass in the tank at ignition: liquid and vapour), burned fuel 2.05 kg. At the end of the blowdown 1.62 kg of oxidizer remains in the tank (Residual (unused) oxidizer mass).
Fuel grain	662.7 mm long, L/D 6.42	Single port; below the L/D threshold of 10 (Section 4.7).
Injector	8 holes, 2.44 mm	Pressure drop 20.53 bar, exit velocity 42.7 m/s, $C_d = 0.78$.
Total impulse, burn time	26 839 N · s, 10.41 s	Delivered values: the integral of the thrust curve of the self-pressurised tank blowdown solution (Section 4.4).



MOTOR DESIGN

Parameter	Value	Unit
Throat Diameter	30.0	mm
Exit Diameter	67.3	mm
Expansion Ratio	5.0	-
Chamber Diameter	103.2	mm
Chamber Length	745.3	mm
Chamber Volume	5050.3	cm³
Chamber Pressure	30.0	bar
Thrust	3000	N
Isp (design point, at design-altitude ambient)	209.7	s
Characteristic Velocity	1423	m/s
Thrust Coefficient	1.445	-
Mass Flow Rate	1.459	kg/s

DESIGN REPORT

Motor Performance	
Specific Impulse (Isp, design point at design-altitude ambient)	209.7 s
Characteristic Velocity (C*)	1423 m/s
Thrust Coefficient (CF)	1.445
Throat Diameter	30.0 mm
Exit Diameter	67.3 mm
Expansion Ratio	5.0
Motor Geometry	
Chamber Diameter	103.2 mm
Chamber Length	745.3 mm
L* (characteristic length)	1.00 m requested → 7.15 m achieved Requested L* = 1.00 m is below the volume the grain port geometry already provides; achieved L* is 7.15 m. Increase L* above that value to lengthen the post-combustion chamber.
Chamber Volume	5050.3 cm³
Initial Port Diameter	79.8 mm
Final Port Diameter	103.2 mm
Regression Rate	1.36 mm/s
Avg Regression Rate	1.17 mm/s
Injector Design	
Type	Showerhead
Exit Velocity	42.7 m/s
Reynolds Number	1197632
Pressure Drop	28.53 bar
Number of Holes	8
Hole Diameter	2.44 mm
L/D Ratio	1.2
Propellant	
Total Mass Flow Rate	1.459 kg/s

INJECTOR DESIGN

Parameter	Value	Unit
Type	showerhead	-
Exit Velocity	42.7	m/s
Reynolds Number	1197632	-
Pressure Drop	28.53	bar
Number of Holes	8	-
Hole Diameter	2.44	mm
L/D Ratio	1.2	-
Discharge Coefficient	0.780	-
Injection Area	37.32	mm²
Weber Number	1692924	-
Pressure Drop Basis	injector circuit solution	-
Discharge Coefficient Basis	sharp inlet, reattaching (L/D 1-5): table value between the separated and full-flow regimes; L/D=1.23093 → Cd=0.78 (NASA SP-8089 (Gill & Nurick 1976) Fig. 13 p. 38 and Fig. 15 p. 43; Lefebvre & McDonell, Atomization and Sprays, 2nd ed., Ch. 5)	-

Figure 4.2: The MOTOR DESIGN, DESIGN REPORT and INJECTOR DESIGN tables. Note the requested and achieved values on the L* row.

4.4 Design point and tank blowdown

Thrust, Isp, chamber pressure and mass flow in the PERFORMANCE SUMMARY belong to the design point at ignition. The example uses a self-pressurised N_2O tank; as it empties, tank pressure, oxidizer flow and thrust fall. Delivered impulse, burn time and average Isp come from this blowdown solution, and the line below the cards says so: 26839 N · s, 10.41 s, 205.7 s (end: OXIDIZER DEPLETED). The same line states that with a constant oxidizer flow (regulated feed) the same design would deliver 29 948 N · s in 10.00 s; the difference is the falling tank pressure. Total Impulse and Burn Time in the Design Summary, the thrust curve, the PDF report, Trajectory and 6-DOF use the same delivered numbers. The link on the line leads to the Tank Blowdown panel, which plots tank pressure, flow and thrust against time and gives the same impulse and burn time.



Figure 4.3: The Tank Blowdown panel: pressure, flow and thrust decay of the self-pressurised N_2O tank. Note the end reason and the total impulse.

4.5 Chamber wall: three zones

The CHAMBER WALL: THREE ZONES (HYBRID) block evaluates the wall in three zones.

Table 4.3: Wall zones in the hybrid example

Zone	Verdict	Reason
Under the fuel grain (662.7 mm)	PROTECTED DURING BURN	The fuel covers the case; case temperature 293.15 K (declared, not computed). With residual web 0 the case is uncovered at the end of the burn; this tail-off exposure was not evaluated.
Pre-combustion chamber (51.6 mm)	LINER REQUIRED	Without a liner the bare wall reaches 3257 K in steady state; HRMA sized a 7.58 mm insulation liner (recession allowance plus conduction allowance, with the solved burn time).
Post-combustion chamber (31.0 mm)	LINER REQUIRED	Same reason and same liner thickness.

The same block gives the structural verdict as STRUCTURE: NOT EVALUATED and VERDICT WITHHELD. Reason: no wall thickness was entered, so HRMA sized the wall itself to meet the target safety factor. The reported safety factor is then the target read back, not an independent verification; the screen states This result is a DESIGN PROPOSAL. To obtain a structural verdict, enter the real thickness of the wall to be manufactured; HRMA then tests that wall.

4.6 Reading the warnings

The warning list carries lines of four classes (Table 17.1). The example has no CRITICAL line; it has six WARNING and five INFO lines. The engineer asks the same question on each line: under

which assumption did the number come out, and does that assumption hold for this design.

Table 4.4: Warnings of the hybrid example and how to read them

Warning (summary)	**Meaning and possible action**
Injector pressure drop is 0.68 of chamber pressure.	In a self-pressurised N_2O tank the drop is not chosen; it is the vapour pressure minus chamber pressure. To lower it, lower the tank temperature or raise the chamber pressure.
You overrode the built-in HTPB regression coefficients: a differs from the catalogue value by 73.9 %, n by −0.2 % (INFO).	The example carries its own regression pair (Table 4.1); the line keeps the source of the regression rate visible, nothing is corrected. The same motor with the catalogue pair is in Section 4.7.
Pre- and post-combustion chamber bare wall 3257 K; 7.58 mm liner needed.	Table 4.3.
Grain protection, residual web 0, case temperature 293.15 K for the structural results (three INFO lines).	The assumptions the model uses for the case; Table 4.3.
'Ae/At Ratio' and 'Chamber Diameter (mm)' fields (0) are not read from the form.	0 means “let the model solve it”; the analysis uses its solved values 5.0364 and 103.229.
Saturated N_2O inlet, $K_c = 0$ (INFO).	In a self-pressurised tank saturated liquid enters the injector; the flow is flashing two-phase and is solved in the injector circuit with the Dyer NHNE model.
Injector L/D 1.2 outside the 3–5 band (HRMA heuristic threshold).	Plate thickness and hole diameter together set L/D; the hydraulic-flip verdict is sensitive to L/D.
Target injection velocity 30.0 m/s is not set by hole count; achieved 42.7 m/s.	The velocity follows from flow and the pressure drop solved by the circuit; the ΔP needed for the target (5.81 bar) is stated on the line.



Figure 4.4: WARNINGS section of the hybrid example. Note the class of each line (WARNING, INFO) and the change proposed at the end of the line.

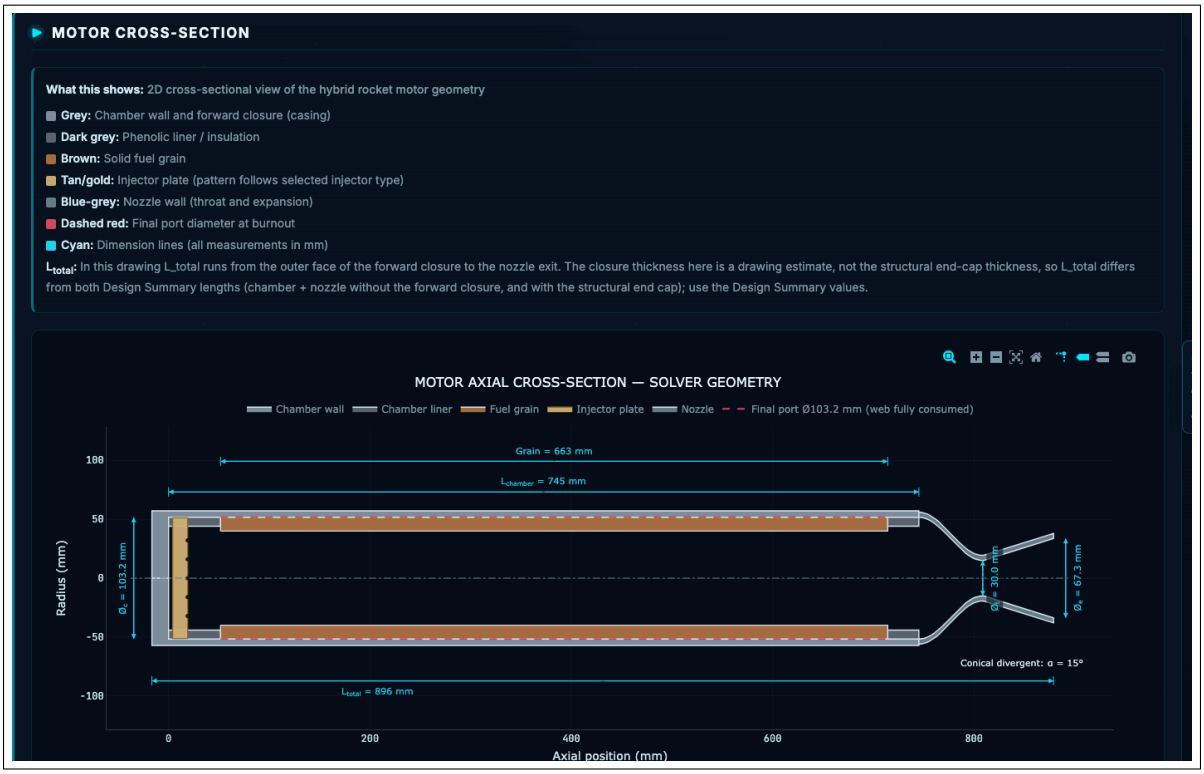


Figure 4.5: Motor cross-section: chamber, fuel grain, injector plate and nozzle dimensions. Note the ratio of grain length to chamber diameter (L/D).

4.7 Fuel grain slenderness: single-port L/D

In a single-port hybrid the fuel flow comes from the port surface: $\dot{m}_f = \rho_f \, \pi \, D_p \, L \, r$. If the regression rate is small, the required fuel flow is reached only with a long port. When the ratio of grain length to outer diameter exceeds 10 a line is added to the warning list: Grain L/D = … and chamber L/D = … exceed the single-port design-practice threshold of 10: the motor is an unusually long, thin tube. The threshold of 10 is an engineering-practice threshold used by HRMA; it is not a single published constant, and the warning states it as a number. The same relation is the reason for moving to a multi-port grain (Sutton and Biblarz, Ch. 16).

In the example the grain is 662.7 mm long and 103.2 mm in diameter (L/D 6.42) and there is no warning. Two inputs set this result: the example's regression pair and the initial oxidizer flux (Table 4.1). Table 4.5 gives the same motor with these inputs changed; in each row only the named fields were changed. Number of Ports is in the ADVANCED PARAMETERS section.

With the catalogue pair the average regression rate is 0.72–0.84 mm/s, with the example's pair 1.17–1.34 mm/s. For the same fuel flow the grain becomes almost twice as long and the L/D threshold is exceeded. In runs with the catalogue pair the warning list also states that this pair is biased against HRMA's validation records (median bias −39.5 % over 43 own records); a designer with own firing data fills in a and n. The effect of the initial flux is small: with a lower flux the port is larger, regression is slower, and the two effects largely cancel in length while the outer diameter grows. Increasing the port count shortens the grain, but the outer diameter and the dry mass grow. With a multi-port grain the result carries an INFO line: the ports are modelled as equivalent circular channels; interaction between channels, web thickness and channel merging are not modelled. The CAD generators draw a single circular port; for a multi-port grain no CAD file is produced and the reason is stated. Isp stays within 0.2 s in every case; the decision rests on length, diameter and mass.

Table 4.5: Effect of the regression pair, initial flux and port count on the fuel grain (hybrid example, from the screen)

Change	Grain length mm	Outer dia. mm	L/D –	Total length mm	Dry mass kg	Isp s	Warning
Example (flux 250)	662.7	103.2	6.42	879.1	12.72	209.7	no
Mass Flux 350	650.9	94.3	6.90	852.5	10.38	209.6	no
Regression fields empty (catalogue pair)	1145.8	94.1	12.18	1347.0	16.48	209.8	yes
Catalogue pair, Mass Flux 350	1125.0	84.1	13.37	1309.6	12.89	209.8	yes
Catalogue pair, Mass Flux 350, Number of Ports 2	795.8	127.4	6.25	1052.5	22.96	209.6	no

4.8 Changing one input: O/F 6.0 → 7.0

1. In the MIXTURE RATIO section type 7 in O/F Ratio.
2. Press Calculate.
3. Compare the results (Table 4.6).

Table 4.6: Effect of the O/F change (hybrid example, from the screen)

Quantity	O/F 6.0	O/F 7.0
Isp (design point)	209.7 s	210.0 s
Mass flow	1.459 kg/s	1.457 kg/s
Delivered / theoretical c^*	1423 / 1599 m/s	1421 / 1597 m/s
Fuel grain length	662.7 mm	573.6 mm
CRITICAL lines	none	1

Isp barely changes: the N_2O/HTPB Isp curve is flat in this region. The flame temperature, however, rises and so does the thermal load on the nozzle throat. The new CRITICAL line: The steady-state equilibrium temperature of the throat wall is 3313 K, above the 3300 K allowable temperature of Graphite (isostatic, nozzle insert grade) with Natural (heat sink) cooling. The line also states what the number means: the burn time is not used; this is the temperature the wall approaches when the burn lasts long enough, a conservative upper bound of the transient wall temperature. The remedies it names are a nozzle material with a higher allowable temperature, cooling, and a check of the transient wall temperature over the burn in the Thermal FEA panel. A small performance gain has produced a design that exceeds a material limit; this is why the example runs at O/F 6.0. Set the value back to 6 and calculate again.

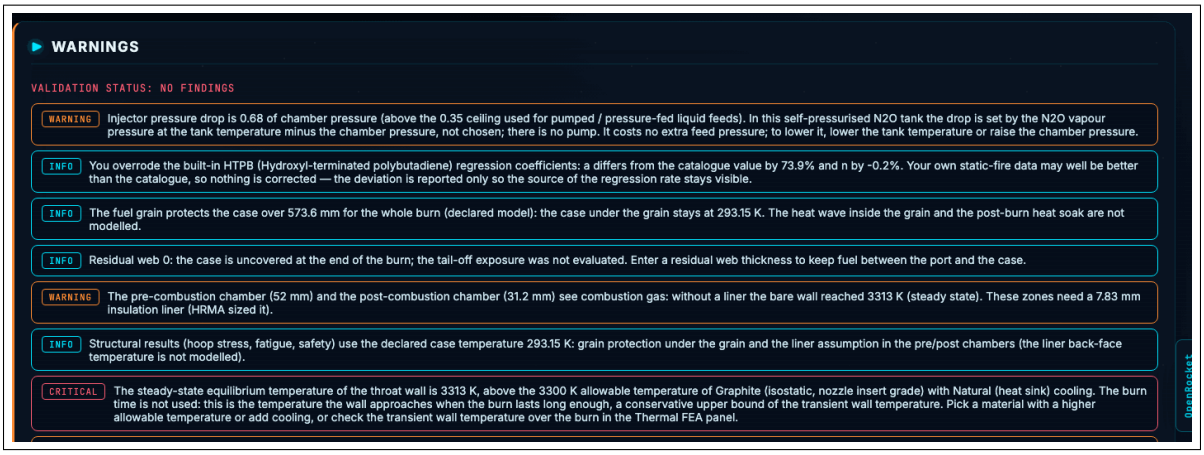


Figure 4.6: WARNINGS of the hybrid example calculated with O/F 7.0: the CRITICAL line (throat equilibrium temperature 3313 K, allowable 3300 K). Note in the same list that the grain length drops from 662.7 to 573.6 mm: a higher O/F means a lower fuel flow at the same thrust, hence a shorter grain.

4.9 Analysis Dock: the Nozzle Flow panel

1. In the ANALYSIS DOCK below the results open the FLOW tab.
2. The fields of the Nozzle Flow panel are filled from the last calculation; the badge next to each field reads COMPUTED or ASSUMED (Table 13.1).
3. Press Run Analysis.
4. Read the regime label first: PERFECT EXPANSION. The exit pressure equals ambient; this is expected, because ε was solved against the ambient pressure.
5. Compare CF EFFECTIVE with the main result: the panel gives 1.445, the same as the thrust coefficient of the main result. If the two differed, the two calculations would be solving different operating points.
6. Do not mistake the IDEAL cards for the motor's numbers. These cards are the quasi-1D solution of the chamber state with a c^* efficiency and a throat discharge coefficient of 1 (IDEAL THRUST 3150 N, IDEAL MASS FLOW 1.323 kg/s, IDEAL C* 1602 m/s). The numbers the motor delivers are printed unchanged in the Delivered by the motor - the numbers of the main result block below: THRUST (MOTOR) 3000 N, MASS FLOW (MOTOR) 1.459 kg/s, C* DELIVERED (MOTOR) 1423 m/s, C_F (MOTOR) 1.445. The difference is the c^* efficiency (0.89; Section 4).

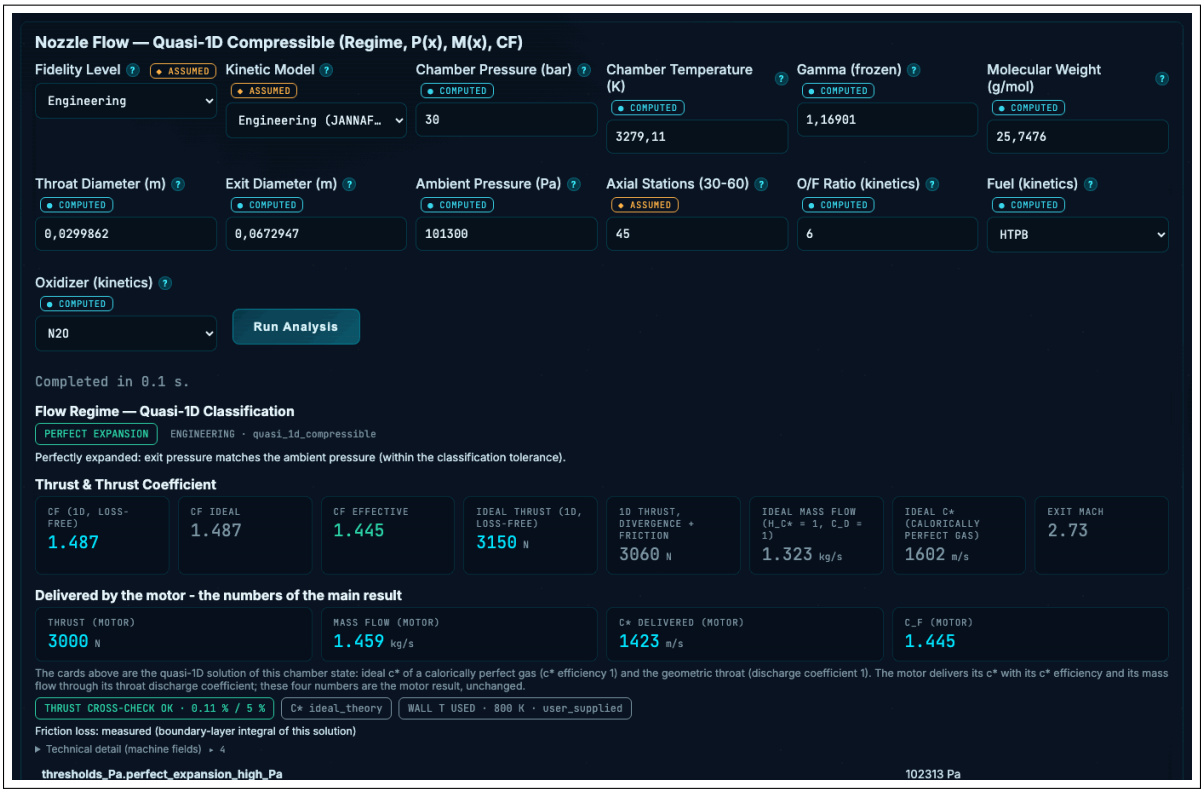


Figure 4.7: Nozzle Flow panel of the hybrid example after the run: regime label, the IDEAL cards and, below them, the numbers the motor delivers. Note that CF EFFECTIVE and C_F (MOTOR) are the same.

The panel inputs are in Table 13.11; the panel verdicts are in Section 13.

4.10 Report and CAD

The report and the CAD files are obtained with steps 8 and 9 of Section 3. Exports use the geometry published by the main calculation; if the geometry is incomplete or a gate is closed, no file is produced and the reason is given (Section 16).

5. Example 2: Solid Motor (KNDX, 3 × BATES 75 mm)

5.1 Purpose

Read the link between burning area and pressure (Kn), the shape of the thrust curve, and the two-phase and throat-erosion declarations on the solid example; change the design pressure and see how burn time and thrust change. The full input list is in Tables 10.1–10.12.

5.2 The example

5.3 Performance and Kn

The PERFORMANCE SUMMARY shows ten cards: SEA LEVEL ISP 125.3 s, VACUUM ISP 141.5 s, AVERAGE THRUST 1299 N, PEAK THRUST 1498 N, PEAK CHAMBER PRESSURE 40.0 bar, BURN TIME 2.28 s, TOTAL IMPULSE 2957 N · s, C* EFFICIENCY 95.0 %, THRUST COEFFICIENT 1.41, PROPELLANT MASS 2.41 kg.

The Isp of KNDX is low because its flame temperature is low and its products have a high molar

Table 5.1: Main inputs of the solid example (`Example Solid KNDX BATES 75mm`)

Input	Value	Note
Propellant (catalogue record)	KNDX (KNO_3/dextrose 65/35)	Thermochemistry source: catalogue; ρ 1850 kg/m^3.
Grain Type, count	BATES, 3	Outer surface inhibited; end faces open.
Grain Support	Cartridge-loaded (free-standing)	The grain is cast outside the case and inserted; cure shrinkage is not restrained. Changes only the grain structural results (bore strain, strain safety factor), not the ballistics.
Outer diameter / core / length	75 / 32 / 360 mm	The length is the total length of the grain stack: 3 × 120 mm (the cross-section also shows 2 × 2 mm gaps).
Chamber Pressure (bar)	40 bar	Design pressure; the throat is sized for it.
Burn Rate Preset	KNDX — KNO3/Dextrose 65/35 (Nakka fit)	Piecewise law: 5 pressure regimes.
Nozzle Material	Graphite (isostatic, nozzle insert grade)	Convergent 45°, divergent 15°.
Case Material	Carbon steel (generic, A36-class)	Yield Strength σ_y (MPa): the screen states that the material value 250 was applied because the field was not supplied.

mass; c^* scales approximately with $\sqrt{T_c/\mathcal{M}}$. The solver takes the thermochemistry from NASA CEA at the motor's chamber pressure and writes the value it used under each box: at 40 bar T_c 1696.0 K, $\mathcal{M}$ 42.145 g/mol, γ 1.1206, c^* 914.5 m/s. The values in the catalogue row (915.9 m/s, 1713.1 K) are the record's single-pressure values and are not sent. With 95 % combustion efficiency the delivered c^* is 869 m/s.

In a solid motor the burning area sets the chamber pressure. With the ratio of burning surface to throat area $K_n = A_b/A_t$, the equilibrium pressure is $P_c = (K_n\, a\, \rho_p\, c^*)^{1/(1-n)}$. In the example Kn (initial) is 222 and Kn (final) is 210; in a BATES grain the bore surface grows while the end faces shrink, and the curve becomes progressive-regressive. The line under the thrust-curve chart names the shape. The throat is given with two diameters: machined 18.40 mm and effective (with discharge coefficient) 18.22 mm.



Figure 5.1: PERFORMANCE SUMMARY of the solid example and the two-phase status band: the Isp carries no verified two-phase correction.

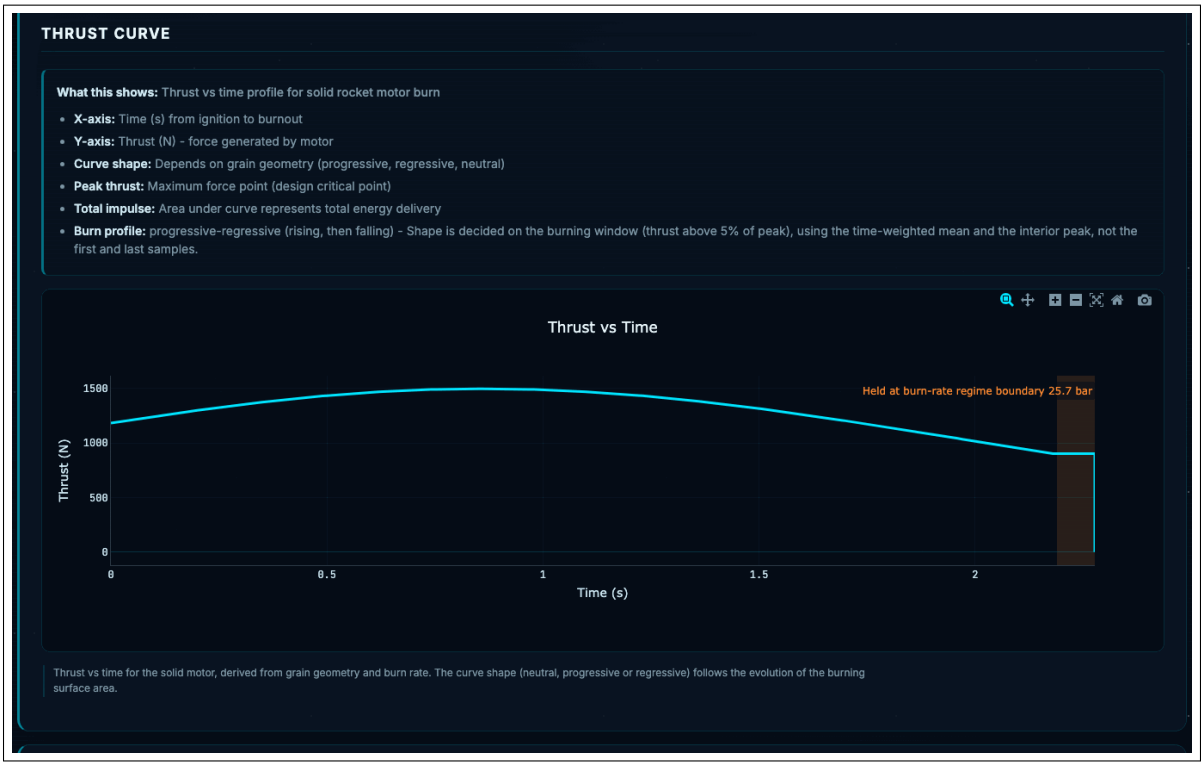


Figure 5.2: Thrust curve of the solid example: progressive-regressive shape and the regime-boundary note.

5.4 Declarations: two-phase flow, throat erosion, burn-rate regime

Table 5.2: CHECK lines of the solid example and how to read them

Line (summary)	Meaning
The published Isp carries no verified two-phase correction.	KNDX products contain a condensed phase (K_2CO_3); the particle-lag loss is not verified for this state and the applied two-phase efficiency is 1. Do not read this Isp as verified delivered performance.
Throat erosion is not modelled in this run: the area would grow by about 9.1 % (threshold 5 %).	The graphite throat wears during the burn; wear lowers chamber pressure and thrust. The number is an estimate.
The throat erosion law is extrapolated at this pressure (the law was measured for 69–450 bar).	The estimate above carries a large uncertainty.
Secondary coefficients (erosive burning, grain mechanics) are inherited from the KNSU (potassium nitrate/sucrose 65/35) record.	A family estimate, not KNDX-specific data.
Piecewise burn-rate law active with 5 pressure regimes.	A regime table replaces a single a–n pair. A chart note marks the moment the curve touches a regime boundary (25.7 bar in this example).

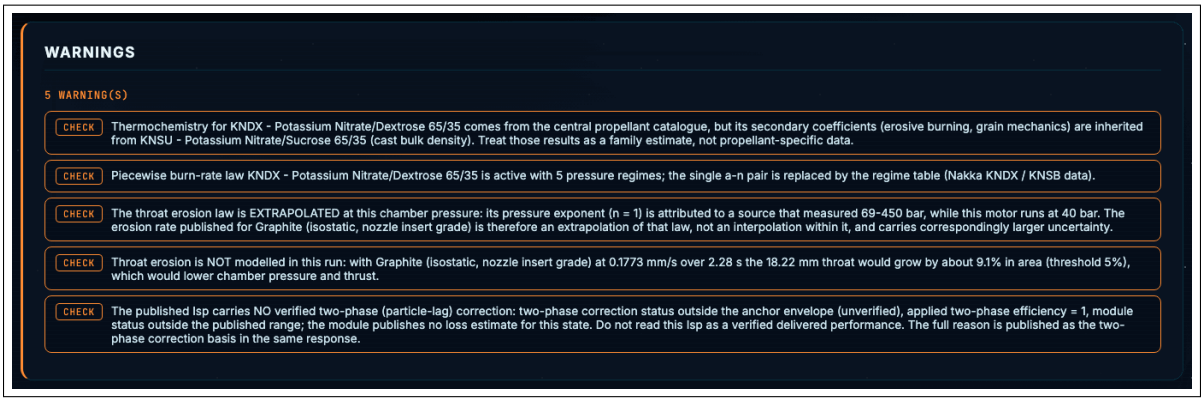


Figure 5.3: WARNINGS of the solid example: five CHECK lines (Table 5.2).

5.5 Changing one input: design pressure 40 → 50 bar

1. In the 4. NOZZLE section type 50 in Chamber Pressure (bar).
2. Press Calculate Motor Performance.

Table 5.3: Effect of the design pressure (solid example, from the screen)

Quantity	**40 bar**	**50 bar**
Sea-level / vacuum Isp	125.3 / 141.5 s	128.5 / 144.3 s
Average / peak thrust	1299 / 1498 N	1554 / 1795 N
Burn time	2.28 s	1.95 s
Thrust coefficient	1.41	1.45
Propellant mass	2.41 kg	2.41 kg
Total impulse (card)	2957 N · s	3032 N · s
CHECK lines	5	6

A higher design pressure gives a smaller throat and a higher K_n. Because the burn rate $r = a\,P^n$ rises with pressure, the web burns out sooner: the burn time shortens and thrust rises. The grain is unchanged, so the propellant mass is unchanged; total impulse is propellant mass times Isp and rises in the same proportion as Isp (2.5 %). The Isp gain comes from the higher pressure ratio (P_c/P_a). The new CHECK line concerns erosive burning: the initial channel mass flux of 1425 kg/m^{2}s is above the erosive-burning threshold of 1400 kg/m^{2}s; the head-end thrust spike and the shortened burn carry a large uncertainty. The cost of the higher pressure is the load on the case and closure; the wall and closure rows are resized accordingly.



Figure 5.4: PERFORMANCE SUMMARY of the solid example at a design pressure of 50 bar: burn time 1.95 s, average thrust 1554 N. Note that the propellant mass (2.41 kg) is unchanged.

5.6 Analysis Dock: the Structural Safety panel

1. In the ANALYSIS DOCK open the STRUCTURAL tab.
2. The Wall Thickness (m) field of the Structural Safety panel is filled with the wall sized by the main calculation (0.002 m in the example, COMPUTED badge). Press Run Analysis.
3. Read the verdict line: the panel states STRUCTURE: NOT_EVALUATED, VERDICT WITHHELD and WALL: SIZED BY HRMA. The reason under Why no acceptance was issued is The safety factor could not be earned: the analysis compared the wall against a thickness it had sized itself: HRMA does not approve a wall it has sized itself.
4. For a verification run, enter the thickness of the wall to be manufactured in the Wall Thickness (m) field and run the panel again. The badge becomes WALL: AS SUPPLIED and the entered wall is checked against the design pressure. If the wall misses the safety target a verdict is issued (VERDICT ISSUED, STRUCTURE: UNSAFE). If it meets the target, the response as a whole is still not accepted, because the closure and nozzle stations are not verified in this panel: the verdict stays VERDICT WITHHELD and the stations not verified are listed under Why no acceptance was issued.

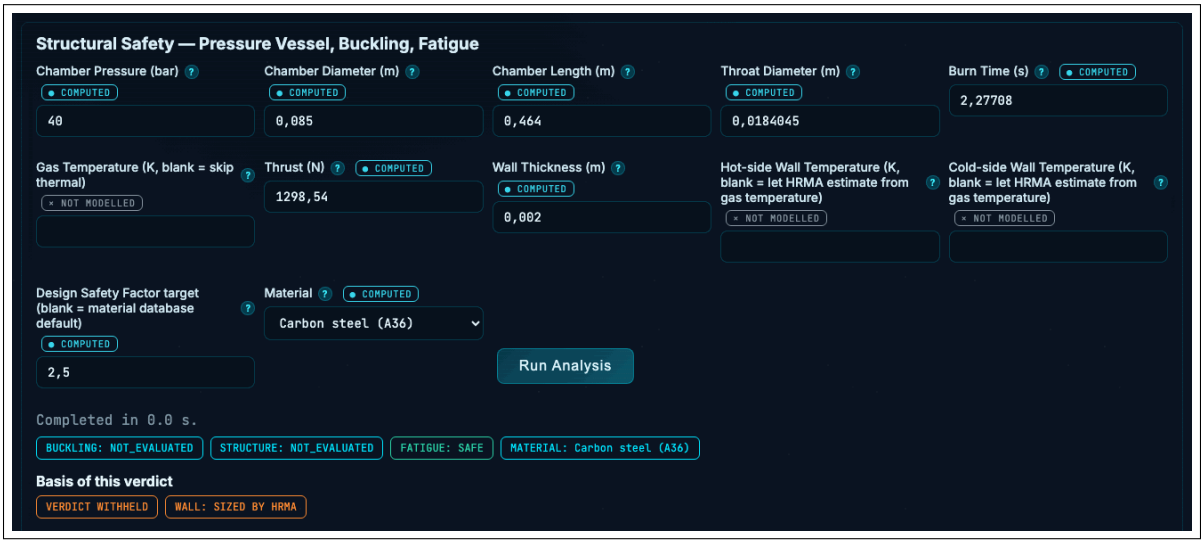


Figure 5.5: Structural Safety panel of the solid example after the run: inputs, value-source badges and the verdict line (VERDICT WITHHELD, WALL: SIZED BY HRMA).

The panel inputs are in Table 13.6. The bore-strain check specific to solid motors is in the Grain Section FEA panel (Section 13).

5.7 Grain support and bore strain

The example's Grain Support field is Cartridge-loaded (free-standing): the grain is cast outside the case and inserted, and cure shrinkage is not restrained (NASA SP-8073). The Structural Analysis tab of the Motor Specifications card then gives a bore strain of 0.376 % and a strain safety factor of 5.319 (margin of safety 4.319); the Crack Propagation Risk row reads Strain capability not exceeded and the Thermal Expansion Compatibility row Thermal shrinkage within the strain capability. Because NASA SP-8073 gives no class boundary above a zero margin of safety, the rows name no low/medium/high class; the verdict is whether the capability is exceeded. If the same grain is set to Case-bonded, the case restrains cure and cool-down shrinkage and the bore carries that thermal strain on top of the pressure strain. Measured result: bore strain 2.209 %, strain safety factor 0.906 (margin of

safety −0.094), Strain capability exceeded; the thermal shrinkage alone uses 91.6 % of the capability, the total load exceeds it. A CHECK line for bore cracking risk is added to the WARNINGS. The strain capability (2 %) is a literature-band input, not measured on the batch; measure the propellant before accepting the grain. The field changes only the grain structural results; the ballistic numbers stay the same.

6. Example 3: Liquid Engine (LOX/RP-1, 25 kN)

6.1 Purpose

Read the cycle card, the wall temperatures of regenerative cooling, the RP-1 carbon-deposit declarations and the NASA CEA comparison on the liquid example; see the effect of O/F on the wall limit. The full input list is in Tables 11.1–11.9.

6.2 The example

Table 6.1: Main inputs of the liquid example (`Example Liquid LOX-RP1 25kN`)

Input	Value	Note
Fuel / oxidizer	RP-1 / LOX	RP-1 density empty: the server uses its own propellant-table value and declares it.
Target Mixture Ratio (O/F), combustion efficiency	2.0; 97 %	
Engine Cycle Type	Gas generator	Turbopump efficiency 75 %, gas-generator temperature 900 K, turbine pressure ratio 4.
Thrust / chamber pressure	25 000 N / 50 bar	
Injector Type	Unlike doublet (F-O impinging), 100 elements	ΔP_{inj} 20 bar, C_d 0.7.
L* / contraction ratio	1.2 m / 4	
Chamber Material, wall	CuCrZr (C18150, aged), 1.0 mm	Two-layer: liner and a closeout shell sized by HRMA (closeout fields empty).
Cooling / nozzle	Regenerative; Bell Nozzle (80% length), ε 12	Coolant flow 100 %, film 0 %, inlet 300 K.
Burn time	30 s	Example input (HRMA choice); propellant mass and total impulse follow from it.

6.3 Performance and cycle

In an open cycle the gas generator consumes propellant that does not enter the main chamber. The Performance Summary therefore gives two Isp pairs. ENGINE ISP — PER TOTAL PROPELLANT CONSUMPTION is 254.8 s at sea level and 299.3 s in vacuum: the engine thrust (main chamber and turbine exhaust) divided by the total propellant flow. MAIN THRUST CHAMBER + NOZZLE ISP — PER MAIN-CHAMBER FLOW is 257.1 / 301.1 s: the thrust and flow of the main chamber alone. SPACE OPTIMIZED ISP 314.8 s (same chamber, vacuum expansion ratio 30), VACUUM THRUST 29.4 kN. Total mass flow 10.006 kg/s (oxidizer 6.546, fuel 3.460 kg/s), throat diameter 66.52 mm, exit diameter 230.45 mm, delivered c^* 1738.3 m/s, chamber temperature 3328 K. Of the 25 kN at sea level, 24.704 kN comes from the main chamber and 0.296 kN from the turbine exhaust (note under the altitude chart). The Engine Specifications card gives the optimum O/F as 2.568; the example deliberately runs at 2.0 (Section 6.6).

The difference between sea-level and vacuum Isp comes from the exit pressure: at $\varepsilon = 12$ the exit pressure is 0.474 bar, below ambient; the nozzle is overexpanded at sea level and loses $(P_e - P_a)A_e$ of thrust.

The Engine Cycle, Throttle & Pressurization card gives the power balance of the gas-generator cycle (Converged), the pump discharge pressures (oxidizer 73, fuel 77 bar) and the open-cycle Isp loss (2.3 s). The Feed pressure feasibility row reads Not evaluated: the requirement is published, but no independent feed pressure limit is modelled for this cycle, so no feasibility verdict is given. The Throttle map lists five points from 40 % to 100 % against two criteria: the ratio rule ($\Delta P/P_c \geq 0.20$) and the chug loop model. At 40 % the rule is marginal; at 40, 55 and 70 % the loop model reads UNSTABLE even where the rule is met, and these points are not counted as stable. The loop model also uses the sensitive time lag and the chamber time constant; the rule looks only at $\Delta P/P_c$. The throttle note also states that below 85.5 % of thrust the nozzle would separate at sea level and that the sea-level Isp of those points is not published.



Figure 6.1: Engine Cycle, Throttle & Pressurization: power balance, pump discharge pressures and the two chug columns of the throttle map (ratio rule and loop model).

6.4 Cooling and carbon deposits

RP-1 leaves a carbon deposit on the hot wall. The deposit is a thermal resistance: it insulates the wall and lowers the hot-face temperature. If it spalls, the wall underneath sees the deposit-free heat flux. HRMA computes both cases:

Table 6.2: Wall temperatures of the liquid example (from the warning lines)

Quantity	Value	Comparison
Peak hot wall, with deposit	615 K	Below the CuCrZr allowable.
Hot wall where the deposit spalls	1345 K	CuCrZr melts at 1349 K; the margin is narrow.
Coolant-side wall (RP-1 face)	605 K	Above the 561 K coking warning level, below the 728 K NASA SP-8087 design criterion.

Staying below 728 K does not show that the wall stays deposit-free; the screen says so at the end of the line. The deposit resistance is modelled for LOX/RP-1 inside the NASA TP-2862 envelope (O/F 1.77–3.32, P_c 4.1–13.8 MPa); outside it the result is declared out of envelope.



Figure 6.2: WARNINGS of the liquid example: carbon deposit, coking risk and INPUT lines.



Figure 6.3: The Regenerative Cooling panel: wall and coolant temperatures station by station, coking levels and the carbon-deposit band.

6.5 NASA CEA comparison

The NASA CEA COMPARISON block places the RocketCEA (NASA CEA) result at this run's operating point (P_c = 50 bar, O/F 2.0, ε 12) next to HRMA's delivered values.

Table 6.3: CEA comparison in the liquid example (from the screen)

Quantity	HRMA	CEA	Δ %	Explained by
c^* (m/s)	1738.3	1792.1	−3.000	Combustion efficiency 97 %.
Main chamber + nozzle Isp, vacuum (s)	301.10	320.86	−6.157	Applied delivered efficiency 93.843 %.
Main chamber + nozzle Isp, sea level (s)	257.11	276.43	−6.987	Not a single published factor; see the efficiency breakdown.
Chamber temperature (K)	3328.4	3328.4	0.000	Same number: HRMA publishes the CEA value.
Molar mass (kg/kmol)	20.799	20.799	0.000	Same number.
γ	1.2134	1.1713	+3.597	Different definition: HRMA's γ is the equivalent exponent that reproduces the CEA thrust coefficient.

The comparison is a code verification, not a validation: it gives the same input to two codes. HRMA's delivered numbers come after its loss chain and are lower than CEA's ideal values; the efficiency that explains each difference is stated on the row. The comparison changes no design number.

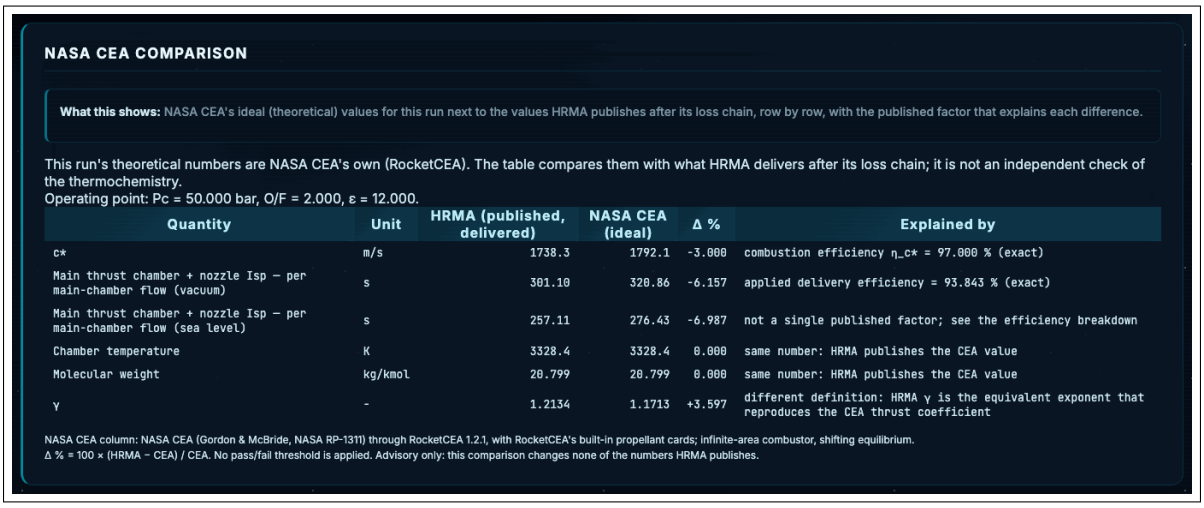


Figure 6.4: NASA CEA Comparison in the liquid example: CEA ideal and HRMA delivered values, with the efficiency column that explains the difference.

6.6 Changing one input: O/F 2.0 → 2.1

1. In the 2. O/F OPTIMIZATION DATA section type 2.1 in Target Mixture Ratio (O/F).
2. Press Calculate Engine Performance.

Engine Isp rises from 254.8 to 257.6 s at sea level and from 299.3 to 302.3 s in vacuum (main chamber: 257.1 → 260.0 and 301.1 → 304.2 s): O/F has moved towards the optimum (2.568). A CRITICAL line is added to the warning list, stating that where the carbon deposit spalls the hot wall reaches 1385 K, above the 1349 K melting point of CuCrZr (C18150, aged), which means local liner damage. Higher O/F means hotter gas and higher heat flux; for a 1.0 mm CuCrZr liner the valid

point is O/F 2.0. Moving the design to O/F 2.1 requires revisiting the cooling (channel geometry, coolant flow) or the liner. Set the value back to 2.0.



Figure 6.5: WARNINGS of the liquid example calculated with O/F 2.1: the CARBON DEPOSIT line (hot wall with the deposit and where it spalls) and the CRITICAL line (1385 K, CuCrZr melting point 1349 K).

6.7 Analysis Dock: the Thermal Safety panel

1. In the ANALYSIS DOCK open the THERMAL tab.
2. In the Thermal Safety panel press Run Analysis.
3. The wall-verdict line gives 614.6 / 605.0 K for the supplied wall: the panel uses the hot- and cold-face temperatures solved by the main calculation; these numbers are the same as in Table 6.2.

The panel inputs are in Table 13.3; the station-by-station solution of regenerative cooling is in the Regenerative Cooling panel (Table 13.5).

7. Workflow: from Requirement to Detailed Analysis

7.1 Flow

A new motor is developed in the following order. The section used in each step is given in parentheses. The basis of the first decisions (mixture ratio, chamber pressure, expansion ratio, characteristic length, injector pressure drop) is given in Section 7.2.

1. **Requirement.** Derive thrust, burn time or total impulse, the envelope (diameter, length) and the mass limit from the mission. In total impulse mode give either thrust or burn time; the other is solved from $I_t = F\,t_b$ (Table 9.3).
2. **Motor type and propellant pair.** Choose the page; enter the propellant pair and mixture ratio. A propellant not in the catalogue is rejected by name.
3. **Operating point.** Enter chamber pressure and expansion ratio (or design altitude); ratios left empty are solved by the model and reported in the warnings.

4. **Calculation.** Calculate. Read the strip at the top of the results (items not modelled), then the headline numbers, then the warnings (Section 17).
5. **Analysis.** The Analysis Dock and Analysis Center panels are filled from the last calculation; run the relevant panels (Section 13).
6. **Revision.** Resolve a CRITICAL line with the change it proposes; follow the effect in the dock's What changed block or in the Comparative Analysis panel. Each revision can be kept as a separate project with Save As.
7. **Verification inputs.** Enter known values such as the wall thickness to be manufactured, a measured c^* efficiency, regression or burn-rate coefficients; HRMA then runs a verification instead of a proposal.
8. **CAD.** Obtain STEP, STL, DXF and drawings (Section 16). Drawing dimensions are the geometry of the main calculation.
9. **Transfer to detailed analysis.** Take the STEP model to CAD and finite-element software and the `.eng` file to flight simulation. The PDF report carries the source of every number and the items not modelled.
10. **Test.** Before flight, carry out physical testing (static firing); compare the measured thrust curve with the HRMA prediction in the User Data Validation panel.

7.2 Choosing the design parameters

The first decisions of a preliminary design are the mixture ratio, the chamber pressure, the expansion ratio, the characteristic length and the injector pressure drop. For each of them the subsections below give three things: the relation that governs the choice and its source, what HRMA does when the field is left empty, and the effect measured on the shipped examples. A range is written only when it rests on a published source; the sources are listed in Table 7.2.

7.2.1 Mixture ratio (O/F)

The mixture ratio that maximises Isp lies below the stoichiometric ratio, on the fuel-rich side: the flame temperature drops slightly, the mean molar mass of the products drops more, and $c^* \propto \sqrt{T_c/\mathcal{M}}$ rises (Sutton and Biblarz, Ch. 5). HRMA scans this ratio from thermochemistry. On the liquid page the Engine Specifications card states the optimum O/F (2.568 in the liquid example); on the hybrid page the Find Optimum button finds the ratio that maximises Isp for the selected propellant pair.

The optimum O/F need not be the design point. As the ratio approaches stoichiometric, flame temperature and wall heat flux rise; the material limit of the wall or of the nozzle throat also governs the choice, and HRMA states that limit with its number in the warning list. Effect measured on the examples:

- Liquid example, O/F 2.0 → 2.1: engine Isp (sea level) 254.8 → 257.6 s; hot wall where the carbon deposit spalls 1345 → 1385 K, CuCrZr melting point 1349 K; CRITICAL (Section 6.6).
- Hybrid example, O/F 6.0 → 7.0: Isp 209.7 → 210.0 s; throat equilibrium temperature 3313 K, allowable temperature of graphite 3300 K; CRITICAL (Section 4.8).

In both examples the Isp gain is small (1.1 % in the liquid example, 0.1 % in the hybrid) and a material limit is exceeded; this is why the examples run below the optimum. In a solid motor O/F is not a design variable: the ratio is the composition of the propellant and comes from the catalogue record.

7.2.2 Chamber pressure

Chamber pressure changes several quantities together. The relations HRMA uses:

- Throat area $A_t = \dot{m}\, c^*/P_c$: at the same flow a higher pressure gives a smaller throat and a smaller motor.
- At the same ambient pressure a larger P_c/P_a allows a larger matched expansion ratio and a larger C_F (Sutton and Biblarz, Ch. 3).
- The gas-side heat transfer coefficient rises with $(P_c/c^*)^{0.8}$ (Bartz 1957); the cooling and liner load grow with pressure.
- In a thin wall the hoop stress $\sigma_\theta = P_c\, D/(2t)$ rises linearly with pressure; for the same safety factor the wall thickness and mass grow.

In a solid motor the chamber pressure is set by the burning area (K_n, Section 5); the burn rate depends on pressure through $r = a\, P^n$. In the solid example raising the design pressure from 40 to 50 bar shortens the burn time from 2.28 to 1.95 s and raises the average thrust from 1299 to 1554 N and the sea-level Isp from 125.3 to 128.5 s (Table 5.3). In a hybrid fed by self-pressurised N_2O the vapour pressure of the tank caps the chamber pressure: the injector drop is the tank pressure minus the chamber pressure (Table 4.4).

7.2.3 Expansion ratio

At a given ambient pressure thrust is largest when the nozzle exit pressure equals the ambient pressure ($P_e = P_a$) (Sutton and Biblarz, Ch. 3). If Ae/At Ratio is left at 0 on the hybrid page, HRMA solves this ratio from the isentropic relation for the ambient pressure of the design altitude (example: 5.0364) and states it in the warning list. At zero ambient pressure (vacuum) the matched ratio is unbounded; HRMA computes no ratio in this case and asks for the field by name. A vacuum nozzle is sized by mass and envelope.

A large-ratio nozzle operating at sea level is over-expanded and loses $(P_e - P_a)A_e$ of thrust (liquid example: $\varepsilon = 12$, $P_e = 0.474$ bar). When the wall pressure falls below about 0.4 times the ambient pressure the flow separates from the wall (Summerfield criterion; HRMA factor 0.40, literature band 0.35–0.40). For this reason the throttle note of the liquid example publishes no sea-level Isp for points below 85.5 % thrust (Section 6).

7.2.4 Characteristic length (L*)

$L^* = V_c/A_t$ sets the residence time of the combustion gas in the chamber. A short L* leaves combustion incomplete; a long L* adds mass and thermal load. The liquid page takes L* per propellant pair from the table of Huzel and Huang (NASA SP-125, Ch. IV, Table 4-1, p. 87; inch values converted to metres).

Table 7.1: L* bands of the liquid page per propellant pair (NASA SP-125, Table 4-1)

Propellant pair	L* (m)	SP-125 (in)
LOX / RP-1	1.02–1.27	40–50
LOX / LH_2 (liquid hydrogen injection)	0.76–1.02	30–40
LOX / NH_3	0.76–1.02	30–40
N_2O_4 / MMH, UDMH, hydrazine, A-50	0.76–0.89	30–35
RFNA, IRFNA / hydrazine, MMH, UDMH	0.76–0.89	30–35
H_2O_2 / RP-1 (with catalyst bed)	1.52–1.78	60–70
LF_2 / LH_2	0.56–0.76	22–30

If the field is empty HRMA uses the midpoint of the band and states the band with its source. For a pair not in the table the general range of the SP-125 text (0.38–3.05 m; residence time 0.002–0.040 s) is declared; it is not a pair-specific value. The liquid example uses an L* of 1.2 m, inside the LOX/RP-1 band.

In a hybrid motor most of the chamber volume is the port of the fuel grain. In the example the requested L* is 1.0 m and the reached one 7.15 m (Table 4.2): the grain length follows from the regression rate and the O/F, and L* is not the quantity that sizes this design.

7.2.5 Injector pressure drop ($\Delta P/P_c$)

The injector pressure drop decouples the feed system from pressure oscillations in the chamber. The classical design band is 15–25 % of chamber pressure (Sutton and Biblarz, Ch. 8; NASA SP-194, Ch. 5–6): below 0.15 the feed system can couple with the chamber in a low-frequency instability (chug), 0.20 is the recommended value, and a drop above 0.25 is paid for in feed pressure and pump work. The band is a design rule, not a sharp physical boundary.

If Injector Pressure Drop is left empty on the liquid page, the drop is the chamber pressure times the ratio of the element type (unlike impinging 0.22, coaxial 0.18, showerhead 0.28, pintle 0.15, other types 0.20), and the result states this with its source label. In the liquid example the drop is 20 bar at a chamber pressure of 50 bar: $\Delta P/P_c = 0.40$, above the band; the injector schematic shows it as dP/Pc 40%. In a self-pressurised hybrid the drop is not chosen; it follows from the tank and chamber pressures (0.68 in the hybrid example).

Table 7.2: Sources cited in this section

Short form	Work
Sutton and Biblarz	G. P. Sutton, O. Biblarz, *Rocket Propulsion Elements*, 9th ed., Wiley. Ch. 3 nozzles, Ch. 5 mixture ratio, Ch. 8 injectors.
NASA SP-125	D. K. Huzel, D. H. Huang, *Design of Liquid Propellant Rocket Engines*, NASA SP-125, 2nd ed., 1971. Ch. IV, Table 4-1.
NASA SP-194	D. T. Harrje, F. H. Reardon (eds.), *Liquid Propellant Rocket Combustion Instability*, NASA SP-194, 1972. Ch. 5–6.
Bartz 1957	D. R. Bartz, “A Simple Equation for Rapid Estimation of Rocket Nozzle Convective Heat Transfer Coefficients”, *Jet Propulsion* 27, 1957.
Summerfield criterion	M. Summerfield, C. R. Foster, W. C. Swan, “Flow Separation in Overexpanded Supersonic Exhaust Nozzles”, *Jet Propulsion* 24, 1954.

Part II Reference

8. Interface Layout

8.1 Page map

8.2 Home page

The home page carries three motor-type cards (Hybrid Motors, Solid Motors, Liquid Engines), each with a LAUNCH DESIGNER button. The RECENT PROJECTS list shows the projects in the project directory with motor type, main result (thrust, Isp and its basis, total impulse) and save time; a row opens the project on its own page.

Table 8.1: Pages of the application

Page	Address	Purpose	Reference
Home page	`/`	Motor type selection, recent projects, language selector.	8.2
Hybrid designer	`/hybrid`	Liquid (or gaseous) oxidizer and solid fuel motor.	9
Solid designer	`/solid`	Solid propellant motor.	10
Liquid designer	`/liquid`	Bipropellant liquid engine.	11
Analysis Dock and Analysis Center	inside the designer pages	Detailed analyses on the last calculation.	13
Launch Site	`/launch-site`	Site selection and the solved flight path.	14
Formulas	`/formulas`	Relations used by the solvers and their sources.	15

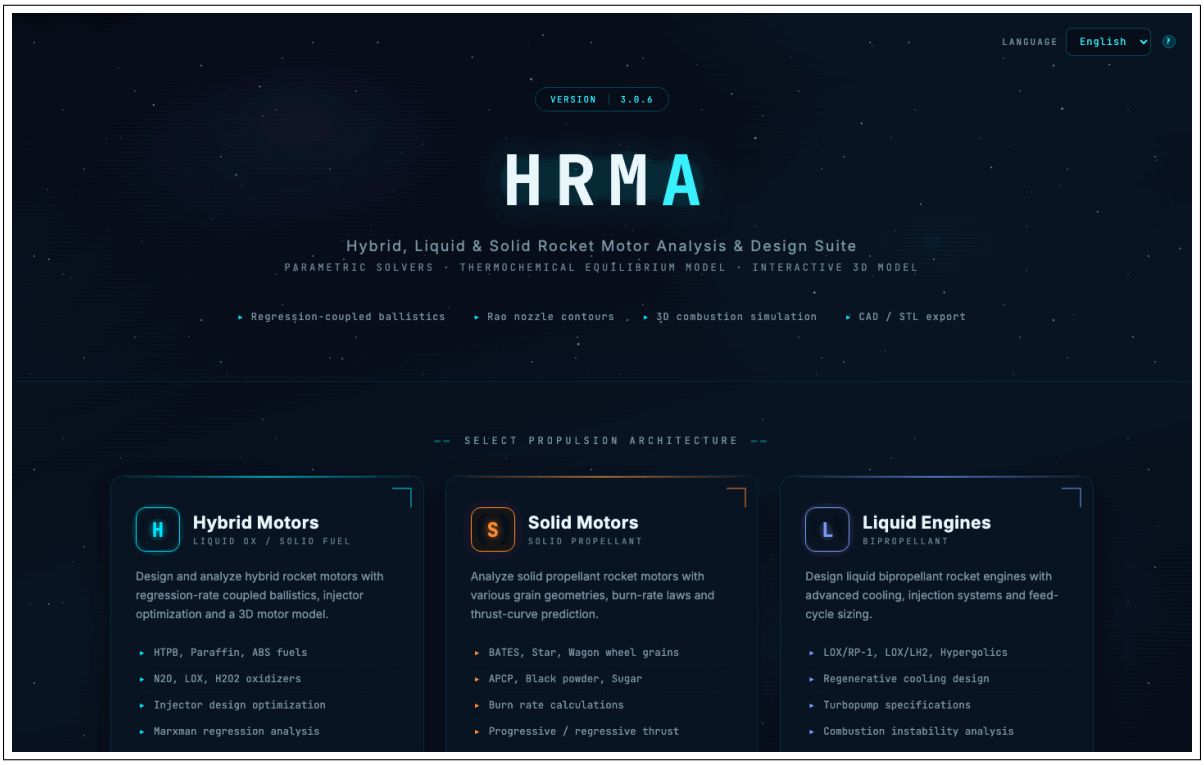


Figure 8.1: Home page: motor-type cards, the RECENT PROJECTS list and the language selector in the top bar.

Table 8.2: Top-bar items

Item	Function
Language selector	Changes the interface language (English, Türkçe) without reloading the page. The choice stays in sync with Settings.
Release Notes	Opens the change list of the installed version inside the application; no internet needed.
User Guide	Opens this document as a PDF in the system viewer, in the current interface language.
Settings	Window with four sections: Language; Updates (Check for updates at startup, Check now, Reset skipped version); Report a bug; application information (installed version, the source the version was read from, GitHub link).

8.3 Top bar and Settings

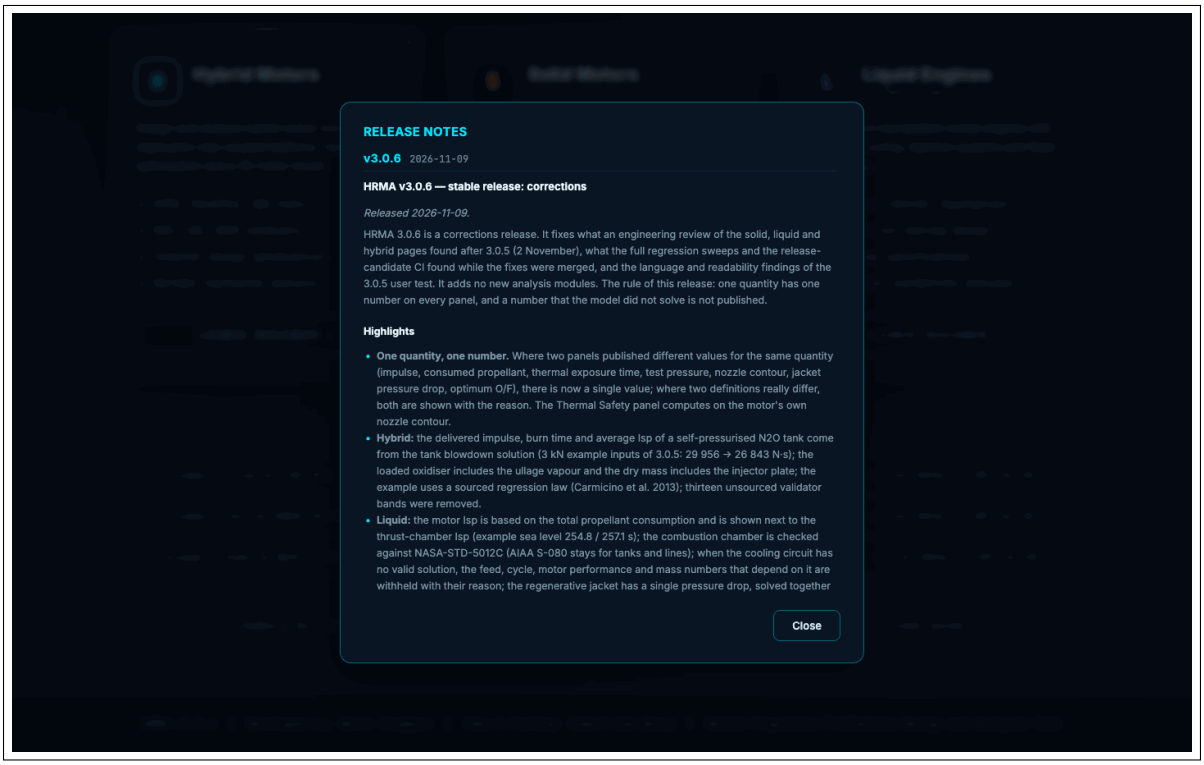


Figure 8.2: The Release Notes window: change list of the installed version; no internet required.

The Report a bug section opens a problem as a pre-filled issue on GitHub. If Attach diagnostic information is ticked, the numeric inputs of the current page and the console errors captured after it loaded are added to the text. File paths, user names and e-mail addresses are removed; the launch-site coordinates are not added. The text is shown before sending; the user presses the submit button on the GitHub page.

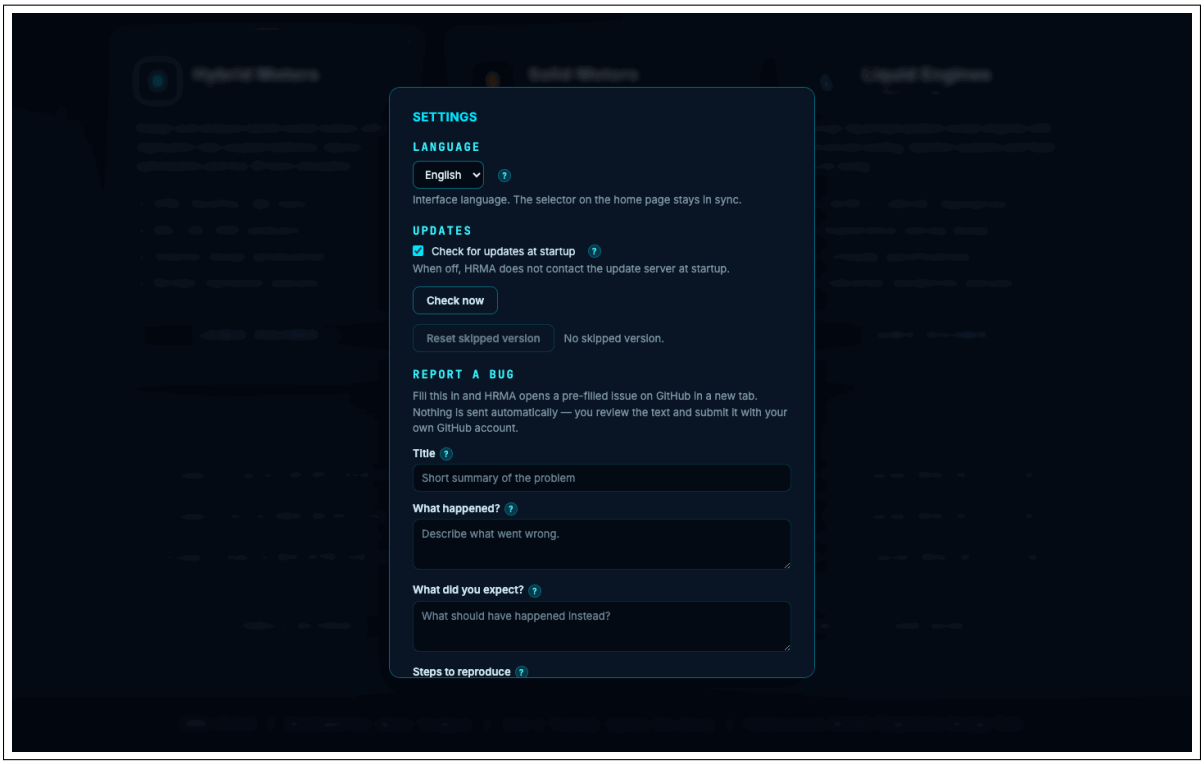


Figure 8.3: The Settings window. With Check for updates at startup off, the application does not contact the update server at startup.

8.4 Project bar

The project bar at the top of the three designer pages shows the project name, its save state and the buttons.

Table 8.3: Project-bar buttons

Button	Function
Save	Writes all form fields, the tab state and the summary of the last calculation (`results_summary`) to the `.hrma` file. A name is asked on the first save.
Save As	Saves the project under a new name. The dialog asks for the name; when the page has no description field it also asks for the description, filled with the existing one.
Open	Lists the projects in the project directory; the list states which version computed each summary. A project of another motor type opens on its own page.
New	Asks for the name of the new project and reopens the page with an empty form (the page defaults). With unsaved changes a confirmation opens first: Discard changes or Cancel. Open asks the same confirmation.
Import from CAD (STEP)	Reads the meridian section of a STEP file; shows the section drawing, the Detected surfaces list and a mapping proposal for the form fields with confidence badges. Apply to form writes the approved values into the form; a field not found stays empty and is marked not found.

Project names may contain Turkish letters; a rejected name states the offending character. 3.0.0 and 3.0.5 projects open in 3.0.6; the project file format has not changed. If the file was saved with another version, a load warning names both versions. 3.0.6 also writes the setup of the Flight Dynamics (6-DOF), Transient Analysis and Trajectory Analysis panels to the project file; in an

older file without this setup the panels open with the page defaults and say so in their own notes. The only declared migration concerns solid projects: in a solid project older than 3.0.1, if Chamber Diameter (mm) is not larger than the grain diameter (in older versions the field meant the grain diameter) the field is cleared on load, the server derives the case bore, and the migration is reported by a named load warning; the file on disk is not changed.

8.5 Common layout of a designer page

The three designer pages follow the same order.

1. **Form.** Numbered or named sections; a "?" note next to every field (opened with the mouse, with Tab and Enter or Space from the keyboard, or by touch; Esc closes it).
2. **Calculation button.** Calculate on the hybrid page, Calculate Motor Performance on the solid page, Calculate Engine Performance on the liquid page. The hybrid and liquid pages also have analysis-type tabs: Single Point Analysis, Parametric Analysis, Trajectory Analysis; on the liquid page the tabs appear after the first calculation.
3. **Design with AI.** The panel below the calculation button (Section 8.6).
4. **Results.** Performance summary, warnings, design tables, cross-section and charts, side panels, NASA CEA comparison.
5. **Analysis Dock and Analysis Center** (Section 13).
6. **Exports** (Section 16).

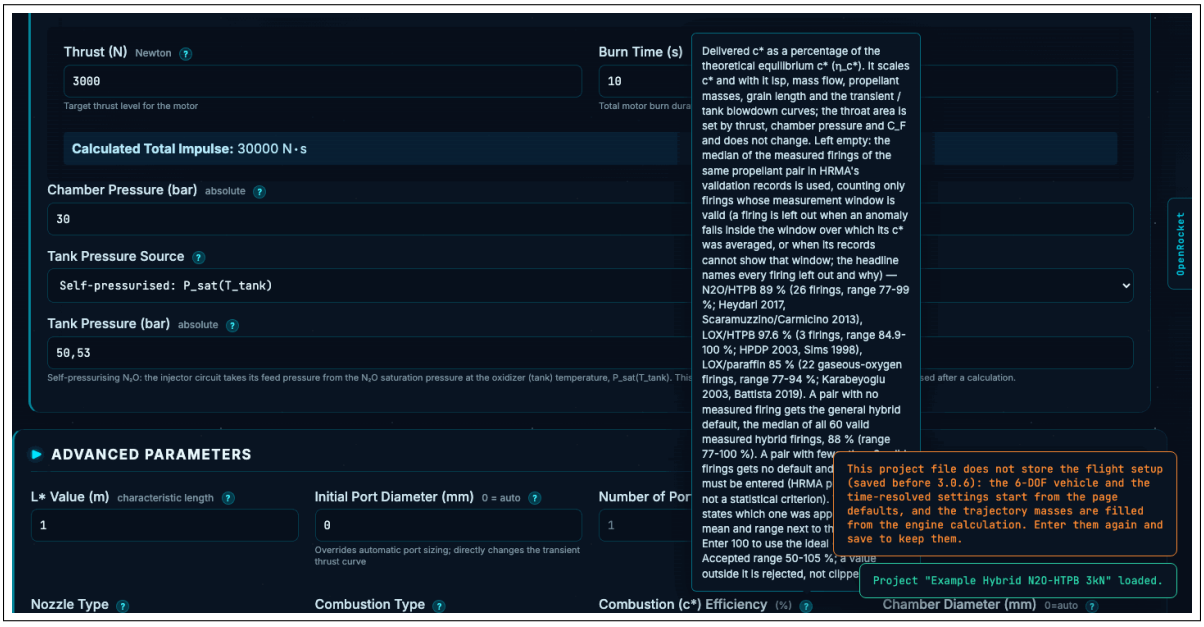


Figure 8.4: A field's "?" note open (Combustion (c*) Efficiency): what it is, its unit, its effect on the calculation, what happens if blank, and the accepted range.

Each chart carries a "what this shows" note and a reading note. Charts are interactive: hovering a point shows its value with units.

8.6 Design with AI

HRMA calls no AI service. The panel is for a manual text exchange with an assistant chosen by the user and also works before a calculation.

1. Type the design goal in the panel's text area.
2. Press Copy context. The text carries the entire form (label, value, unit, accepted range, source, the solver's "not used" declarations), the headline numbers if any, the warnings, the withheld verdicts and the CEA block. Show text opens the same text in the panel.
3. Paste the text into the assistant; paste the field values it proposes back into the panel.
4. Press Check suggestion. HRMA accepts only the page's own fields, as finite numbers or valid options; it rejects unit mismatches and out-of-range values by name and shows a before/after table with Apply and Apply and calculate.
5. Copy continuation package tells the assistant what changed and the new result.

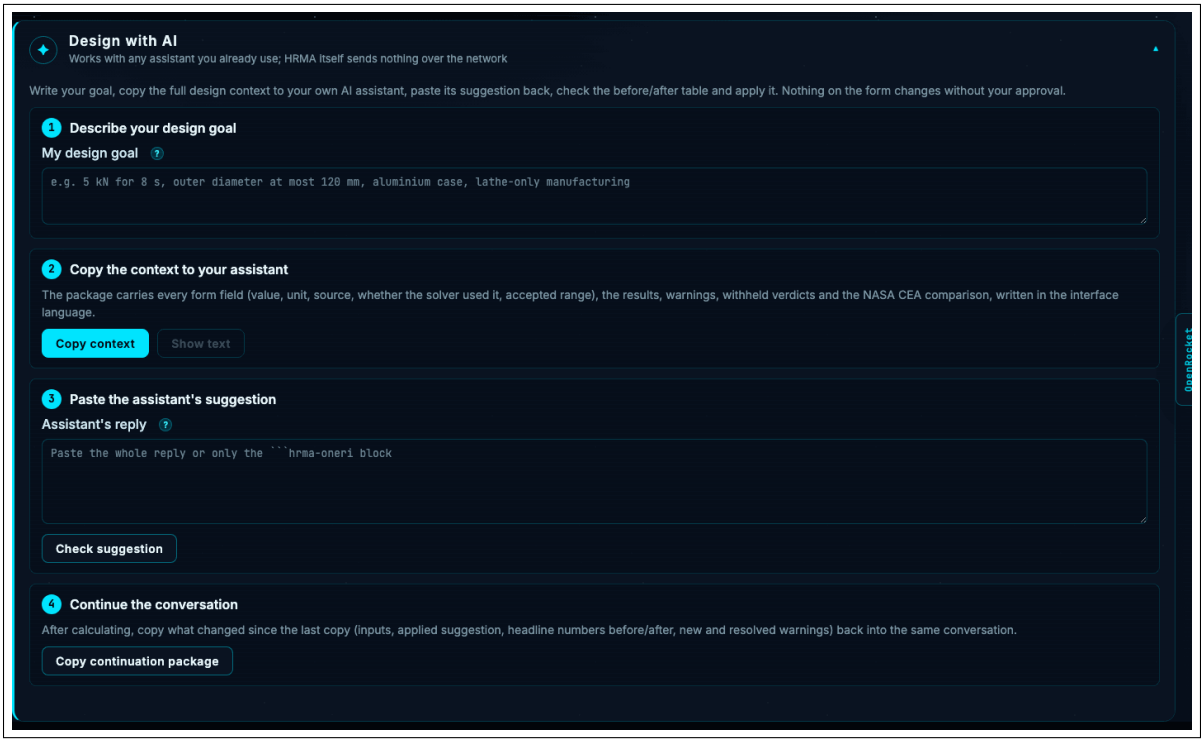


Figure 8.5: The Design with AI panel: design goal, Copy context and Check suggestion.

The assistant's suggestion is not a result; the calculation, the warnings and the rejections decide.

9. Hybrid Motor Page

9.1 Purpose

Sizes the design point, fuel grain, injector, nozzle and wall of a motor with a liquid (or gaseous) oxidizer and a solid fuel (hybrid); solves thrust and chamber pressure against time, port growth and O/F shift.

9.2 Screen

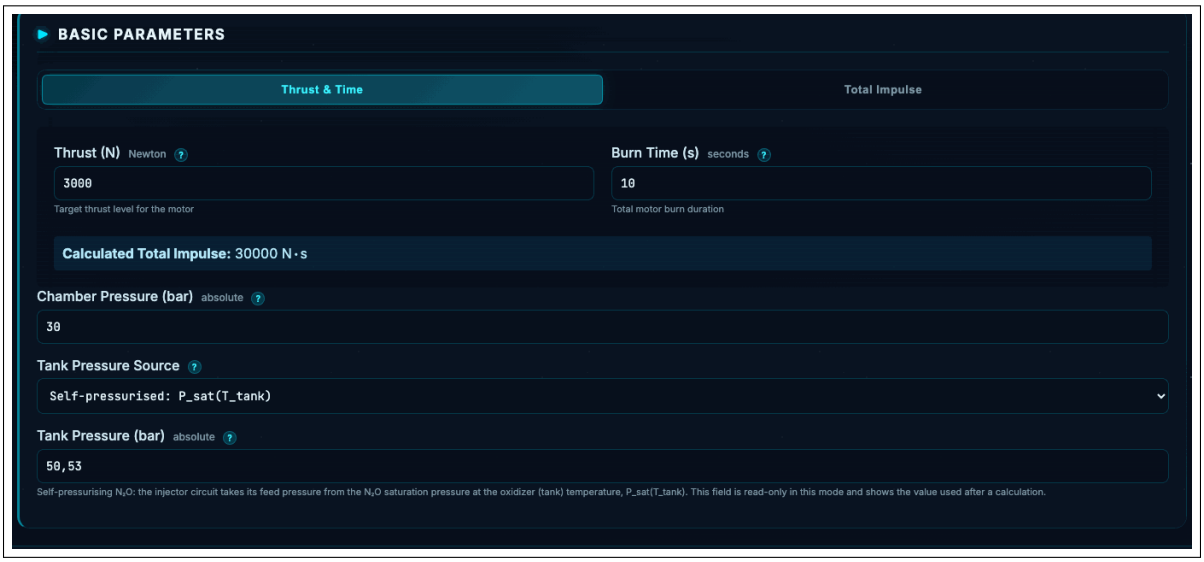


Figure 9.1: Hybrid page: the BASIC PARAMETERS section. Note the Thrust & Time and Total Impulse design-mode tabs and the Tank Pressure Source selector.

9.3 Inputs

Tables 9.1–9.11 give every field of the hybrid form in screen order. The Form limit column is the browser check; the server's canonical bands are in Appendix E. Form default is the value of an empty page; the example projects carry their own values.

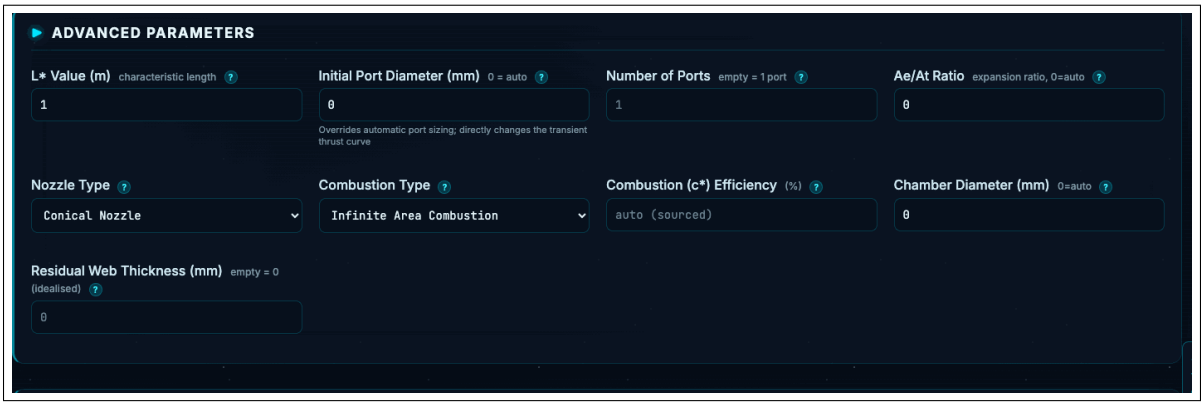


Figure 9.2: ADVANCED PARAMETERS: L*, port, expansion ratio, combustion type, c^* efficiency and residual web.

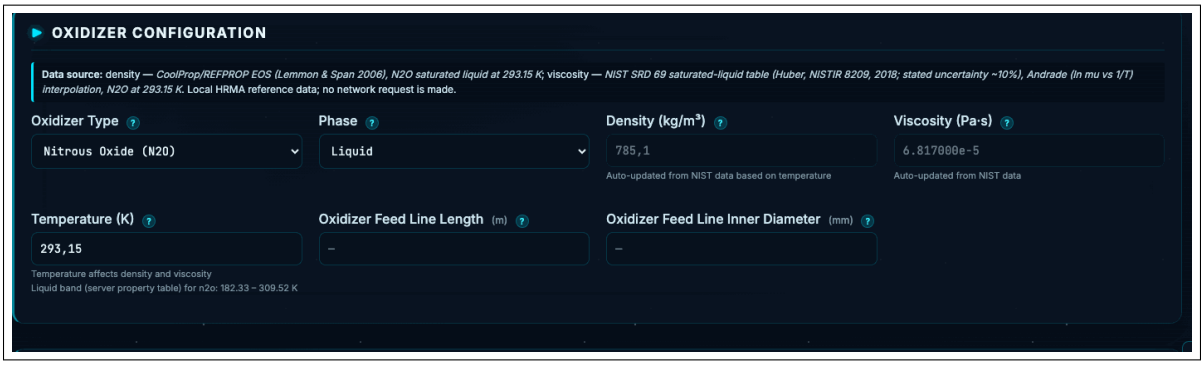


Figure 9.3: OXIDIZER CONFIGURATION and the tank pressure source.

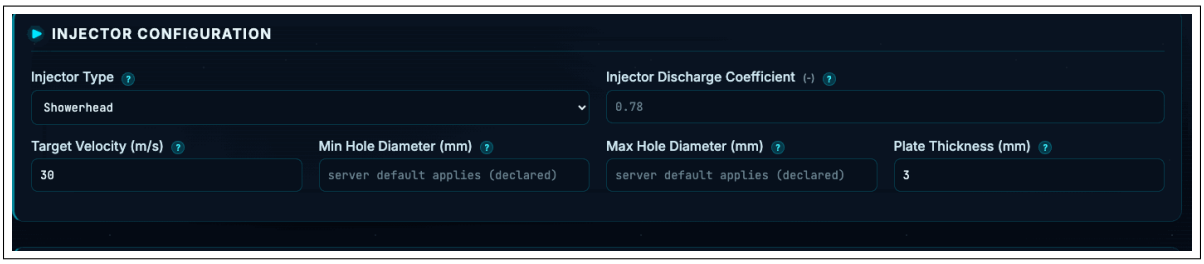


Figure 9.4: INJECTOR CONFIGURATION: injector type and type-specific fields.



Figure 9.5: MATERIALS before the calculation: chamber, nozzle and injector material and the properties the calculation reads.

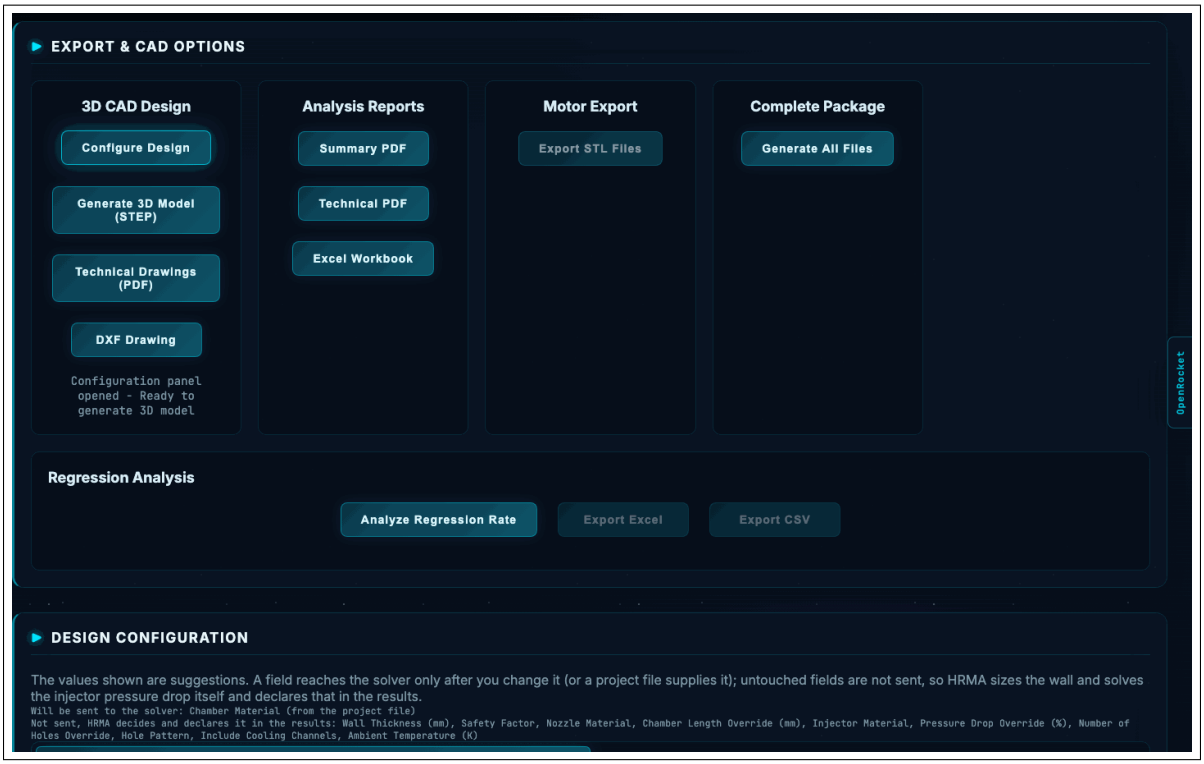


Figure 9.6: The Design Configuration window: motor and injector design parameters.

Table 9.1: Hybrid page: Motor Information

Input	Identifier	Unit	Form limit	Form default
Motor Name	`motor_name`			(blank)
Effect on the calculation: Unique identifier for your motor design. Use a systematic naming convention including your organization/project code, motor type, and version number for proper documentation and tracking.				
Description	`motor_description`			(blank)
Effect on the calculation: Brief technical description of the motor design including purpose, key features, and intended application. This helps document design objectives and requirements for future reference.				

Table 9.2: Hybrid page: Flight Environment

Input	Identifier	Unit	Form limit	Form default
Altitude (above sea level), h	`single_altitude`	m	0 – 50000	0
Effect on the calculation: Altitude of the design point above sea level (m). It fills the ambient pressure field below with the ISA 1976 pressure of that altitude; that pressure is the nozzle back pressure, so it changes thrust, C_F and Isp (a lower ambient pressure gives a higher Isp). Form range 0-50000 m.				
Ambient Pressure (US Standard Atmosphere 1976), P_a	`single_pressure`	bar	0 – 1.2	1.013250
Effect on the calculation: Computed from altitude using the US Standard Atmosphere 1976 model (ISA). You can also type your launch site's measured barometric pressure directly – the field then switches to site-specific mode until the altitude is changed again.				

continued on next page

Input	Identifier	Unit	Form limit	Form default
Initial Altitude	`alt_start`	m	0 – 50000	0
Effect on the calculation: Start altitude of the altitude profile (m). The calculation runs at this altitude: its ISA 1976 ambient pressure is the nozzle back pressure, as in the single-altitude mode. Form range 0-50000 m. *If blank:* Empty: 0 m.				
Final Altitude	`alt_end`	m	0 – 50000	10000
Effect on the calculation: End altitude of the altitude profile (m). Only the Preview Profile chart uses it (ISA 1976 ambient pressure from the start to the end altitude); the calculation does not read it. Form range 0-50000 m.				
Number of Points	`alt_points`		3 – 100	10
Effect on the calculation: Number of points of the Preview Profile curve between the start and end altitude. The calculation does not read it. Form range 3-100.				

Table 9.3: Hybrid page: Basic Parameters

Input	Identifier	Unit	Form limit	Form default
Thrust, F	`thrust`	N	10 – 1000000	1000
Effect on the calculation: Design-point thrust of the motor, in N. With the burn time it fixes the total impulse (total impulse = thrust × burn time) and, through Isp, the mass flow and propellant masses. Accepted range 10-1000000 N. *If blank:* Required: if it is left empty the run is rejected by name; no thrust is assumed.				
Burn Time, t_b	`burn_time`	s	0.1 – 300	10
Effect on the calculation: Design burn time, in s. With the thrust it fixes the total impulse (total impulse = thrust × burn time) and so the propellant masses and the fuel grain. Accepted range 0.1-300 s. *If blank:* Required: if it is left empty the run is rejected by name; no burn time is assumed.				
Total Impulse (Newton·seconds), I_t	`total_impulse`	N·s	1000 – 1000000	10000
Effect on the calculation: Total impulse the motor must deliver, in N·s (thrust × burn time at the design point). It fixes the propellant masses through Isp. With thrust given, the burn time is solved; with burn time given, the thrust is solved. Required in this mode. Form range 1000-1000000 N·s; a solved thrust or burn time outside its own band is rejected by name.				
Thrust, F	`thrust_impulse`	N	10 – 1000000	(blank)
Effect on the calculation: Design-point thrust, in N. In total-impulse mode give this OR the burn time: the other is solved from total impulse = thrust × burn time. Accepted range 10-1000000 N. *If blank:* If both are empty the run is rejected by name; HRMA does not pick an 'optimal' thrust.				
Burn Time, t_b	`burn_time_impulse`	s	1 – 300	(blank)
Effect on the calculation: Design burn time, in s. In total-impulse mode give this OR the thrust: the other is solved from total impulse = thrust × burn time. Form range 1-300 s. *If blank:* If both are empty the run is rejected by name.				
Chamber Pressure, P_c	`chamber_pressure`	bar	1 – 500	20
Effect on the calculation: Design chamber pressure, in bar. It sets the nozzle expansion and C_F, the throat area for the requested thrust, the equilibrium combustion (c*, flame temperature) and the wall and closure sizing. Accepted range 1-500 bar. *If blank:* Required: if it is left empty the calculation is not sent and the field is named.				
Tank Pressure Source	`tank_pressure_source`			Self-pressurised: P_sat(T_tank)
Effect on the calculation: Where the injector feed pressure comes from. Oxidizer default: self-pressurising N_2O uses P_sat(T_tank); LOX/H_2O_2 use the design pressure-drop ratio. User-supplied tank pressure: for a supercharged (He/N_2) N_2O tank or a pressure-fed LOX/H_2O_2 tank; the tank pressure you enter becomes the injector inlet pressure. In this mode the self-pressurised tank blowdown history is not published (pressurant expansion and dissolution are not modelled); only the design point is computed. Options: Self-pressurised: P_sat(T_tank); User-supplied tank pressure.				

continued on next page

Input	Identifier	Unit	Form limit	Form default
Tank Pressure (absolute), P_{tank}	`tank_pressure`	bar	≥ 1	(blank)

Effect on the calculation: Oxidizer tank pressure (absolute). How it is used depends on the Tank Pressure Source: with the oxidizer default the field is read-only and shows, after a calculation, the feed pressure the injector circuit used (self-pressurising N_2O: the saturation pressure at the oxidizer temperature, P_sat(T_tank); LOX/H_2O_2: P_c·(1 + ΔP/P_c) from the design pressure-drop ratio). With User-supplied tank pressure the value you enter feeds the injector circuit (P_1 = P_tank, feed line loss not included) and must exceed the chamber pressure; for N_2O it must also be at least P_sat(T_tank).

Table 9.4: Hybrid page: Advanced Parameters

Input	Identifier	Unit	Form limit	Form default
L* Value, L^*	`l_star`	m	≥ 0.1	(blank)

Effect on the calculation: Characteristic length L* = V_c / A_t, in m. The chamber volume is V_c = L*·A_t, so it sets the chamber length (and with it the chamber and case mass) unless you give a chamber length override. Zero or a negative value is rejected by name. *If blank:* Empty: 1.0 m is used and declared in the results as a default (HRMA placeholder, not a fuel-specific value).

Initial Port Diameter, $D_{p,0}$	`initial_port_diameter`	mm	0 – 500	0

Effect on the calculation: Starting fuel grain port diameter. Drives initial oxidizer mass flux, regression rate and the whole thrust-time curve. Leave 0 to size the port automatically from the design mass flux (350 kg/m2.s, or the chamber Gox you enter under finite combustion). *If blank:* 0 = auto

Number of Ports empty = 1 port, N_p	`port_count`		≥ 1	(blank)

Effect on the calculation: Number of equivalent circular combustion ports in the fuel grain. It scales the burning area, the port flow area and the acoustic mode count, so it changes thrust, chamber pressure and the stability verdict. Above one port the solver states that port interaction, web thickness and burn-through are not modelled. *If blank:* Leave empty for a single port.

Ae/At Ratio expansion ratio, 0=auto, ε	`expansion_ratio`		≥ 0	(blank)

Effect on the calculation: Nozzle expansion ratio ε = A_e/A_t (exit area over throat area). It sets the exit pressure and so C_F, Isp and thrust at the ambient pressure of the design point, and the nozzle exit diameter and length. *If blank:* Empty or 0: the solver computes the pressure-matched ratio (exit pressure = ambient pressure of the design point) and the result reports the ratio used; at zero ambient pressure no such ratio exists and the run is rejected by name, so give a value for a vacuum design.

Nozzle Type	`nozzle_type`			Conical Nozzle

Effect on the calculation: Nozzle contour: conical, bell or parabolic. It sets the divergence efficiency λ on the momentum term of C_F (and so Isp and the propellant masses), the drawn contour and the nozzle length. Conical: $\lambda = (1 + \cos\alpha)/2$ with α the divergent half-angle (0.983 at 15°; Sutton and Biblarz 9th ed. Eq. 3-34). Bell and parabolic: λ comes from the exit angle of the solved contour and depends on the expansion ratio. Default: conical. Options: Conical Nozzle; Bell Nozzle; Parabolic Nozzle.

Combustion Type	`combustion_type`			Infinite Area Combustion

Effect on the calculation: Chamber model of the equilibrium solution. Infinite area: the chamber cross-section is treated as infinite, so the gas enters the nozzle at rest and the injector-face pressure is the nozzle stagnation pressure. Finite area: with a finite contraction ratio the gas is accelerated in the chamber by combustion and loses stagnation pressure (Rayleigh flow), so the nozzle inlet stagnation pressure is lower; it needs a contraction ratio or chamber mass flux. Neither option models mixing. Options: Infinite Area Combustion; Finite Area Combustion.

continued on next page

Input	Identifier	Unit	Form limit	Form default
Combustion (c*) Efficiency, η_{c^*}	`combustion_efficiency`	%	50 – 105	(blank)

Effect on the calculation: Delivered c* as a percentage of the theoretical equilibrium c* (η_c*). It scales c* and with it Isp, mass flow, propellant masses, grain length and the transient / tank blowdown curves; the throat area is set by thrust, chamber pressure and C_F and does not change. A pair with no measured firing gets the general hybrid default, the median of all 60 valid measured hybrid firings, 88 % (range 77-100 %). A pair with fewer than 2 valid firings gets no default and the value must be entered (HRMA practical gate, not a statistical criterion). The headline states which one was applied, with the mean and range next to the median. Enter 100 to use the ideal (theoretical) c*. Accepted range 50-105 %; a value outside it is rejected, not clipped. *If blank:* Left empty: the median of the measured firings of the same propellant pair in HRMA's validation records is used, counting only firings whose measurement window is valid (a firing is left out when an anomaly falls inside the window over which its c* was averaged, or when its records cannot show that window; the headline names every firing left out and why) – N2O/HTPB 89 % (26 firings, range 77-99 %; Heydari 2017, Scaramuzzino/Carmicino 2013), LOX/HTPB 97.6 % (3 firings, range 84.9-100 %; HPDP 2003, Sims 1998), LOX/paraffin 85 % (22 gaseous-oxygen firings, range 77-94 %; Karabeyoglu 2003, Battista 2019).

Chamber Diameter, D_c	`chamber_diameter_input`	mm	≥ 0	(blank)

Effect on the calculation: Inner diameter of the combustion chamber, equal to the fuel grain outer diameter (liner/insulation is not modelled). Left at 0, HRMA sizes the grain to the end-of-burn ports: one port: final port diameter + 2 x residual web; N ports: single ring of final ports touching each other and the wall, final port x (1 + 1/sin(180°/N)) + 2 x residual web. A diameter you enter is kept; if the final ports do not fit in it the request is rejected with the minimum diameter, and the unburned fuel left outside the ports is reported. *If blank:* 0=auto

Residual Web Thickness, w_r	`residual_web_thickness`	mm	≥ 0	(blank)

Effect on the calculation: Fuel web kept between the end-of-burn port(s) and the grain outer surface. It adds 2 x this value to the grain outer diameter. *If blank:* Empty means 0: zero nominal residual fuel, an idealised sizing default; thermal protection (liner/insulation) and regression uncertainty are not included, and the port is modelled as axially uniform, so local burn-through near the aft end is not captured.

Contraction Ratio Ac/At, ε_c	`contraction_ratio`		0 – 10	(blank)

Effect on the calculation: Contraction ratio A_c/A_t (chamber cross-section over throat area). It sets the convergent cone length of the nozzle and, with the finite-area combustion type, the chamber Mach number and the injector-face to nozzle stagnation pressure ratio (isentropic area relation, Sutton and Biblarz Ch. 3). Below 2 a warning is raised. Form range 0-10; a value above 0 here clears and disables the mass flux. *If blank:* A value above 1 is used as given; empty or 0 (any value up to 1): it is derived from the chamber geometry, (D_ch/d_t)^2.

Mass Flux, G	`mass_flux_chamber`	kg/m^2·s	0 – 1000	0

Effect on the calculation: Initial oxidizer mass flux in the fuel port, G_ox = $\dot{m}$_ox / A_port, in kg/m^2·s. When no initial port diameter is given it sizes the port: A_port = $\dot{m}$_ox / G_ox (total over all ports); through the regression law $\dot{r} = a \cdot G_ox^{\wedge} n$ (the solver uses the oxidizer flux) it also sets the grain length. If you give an initial port diameter this field is not used and the flux is derived from the port. Above 600 kg/m^2·s a port flooding warning is raised (HRMA threshold). Form range 0-1000; a value above 0 here clears and disables the contraction ratio. *If blank:* Empty or 0: 350 kg/m^2·s is used (HRMA choice, not a measured value) and the result reports the design flux.

Table 9.5: Hybrid page: Fuel Configuration

Input	Identifier	Unit	Form limit	Form default
Fuel Type	`fuel_type`			HTPB (Hydroxyl-terminated polybutadiene)
Effect on the calculation: Solid fuel of the grain. The choice sets the fuel's catalogue density and regression coefficients a and n (filled into the fields below, which you can change) and the fuel side of the combustion equilibrium (c*, flame temperature, optimum O/F). Greyed-out entries are not modelled. Default: HTPB. Options: HTPB (Hydroxyl-terminated polybutadiene); Polyethylene; PMMA (Polymethyl methacrylate); Paraffin Wax; ABS Plastic; PLA Plastic; Carbon (Graphite); Aluminum Powder; Custom Mixture; Custom Fuel Composition.				
Fuel Density, ρ_f	`fuel_density`	kg/m^3	≥ 100	920
Effect on the calculation: Density of the solid fuel, in kg/m^3. It converts the regressed volume into fuel mass flow ($\dot{m}$_f = ρ_f · $\dot{r}$ · burning port surface), so it sets the grain length and the fuel and total masses. Form minimum 100 kg/m^3. *If blank:* When you pick a fuel it is filled with the HRMA catalogue density of that fuel; empty: the catalogue value is used.				
Regression Rate Coefficient (a), a	`regression_a`	m/s/(kg/m^2·s)^n	0.000001 – 0.01	0.0000368
Effect on the calculation: Coefficient a of the regression law r = a·G_ox^n (SI: r in m/s, G_ox in kg/m^2·s). HRMA's hybrid solver uses the OXIDIZER mass flux G_ox = $\dot{m}$_ox/A_port, the same basis the HRMA catalogue pairs are fitted on, so enter a pair fitted against G_ox. It sets the fuel regression rate and through it the fuel flow, grain length and port growth. Field range 0.000001-0.01. *If blank:* Empty: the selected fuel's catalogue value is used.				
Regression Rate Exponent (n), n	`regression_n`	-	0.1 – 1.0	0.555
Effect on the calculation: Power law exponent in regression rate equation r = a × G^n. Determines sensitivity to oxidizer mass flux. Lower values (0.3-0.5) provide more stable combustion, higher values (0.6-0.8) are more mass flux sensitive.				

Table 9.6: Hybrid page: Oxidizer Configuration

Input	Identifier	Unit	Form limit	Form default
Oxidizer Type	`oxidizer_type`			Nitrous Oxide (N2O)
Effect on the calculation: Oxidizer of the motor. The choice sets the oxidizer side of the combustion equilibrium (c*, flame temperature, optimum O/F), the reference density and viscosity filled below and the feed model: N2O is self-pressurising (stored as a saturated liquid; the injector flow is solved two-phase with the Dyer NHNE model and the tank blowdown history is computed); LOX is cryogenic. Greyed-out entries are not modelled. Default: N2O. Options: Nitrous Oxide (N2O); Liquid Oxygen (LOX); Hydrogen Peroxide (H2O2); Compressed Air; Custom Oxidizer.				
Phase	`oxidizer_phase`			Liquid
Effect on the calculation: Physical state of the oxidizer at the injector (N2O). It sets the reactant enthalpy the combustion equilibrium is solved with: liquid = saturated liquid at the oxidizer temperature, gas = gaseous N2O at the 298.15 K standard state. With a user-supplied tank pressure the check follows the phase: a liquid feed must be at or above the N2O saturation pressure at the oxidizer temperature, a gaseous feed at or below it (2 % tolerance); otherwise the request is rejected by name. The injector circuit and oxidizer tank are liquid-feed models, so with Gas the orifice plan, tank volume, slosh, tank vessel and blowdown history are not reported (a compressible orifice model is not available for the hybrid). Liquid is the default when left unchanged. Options: Liquid; Gas.				
Density, ρ_{ox}	`oxidizer_density`	kg/m^3		786.3
Effect on the calculation: Density of the oxidizer at the injector inlet, kg/m^3. Filled automatically for the selected oxidizer and temperature from HRMA's reference source (CoolProp/REFPROP equation of state for N2O, HRMA's propellant table otherwise); N2O is stored as a saturated liquid, so the value follows the tank temperature. When the injector circuit solves its own inlet state (the default), that state's density is used and a typed value is listed as not used; when you set the injector pressure drop yourself, the injector plate uses this value.				

continued on next page

Input	Identifier	Unit	Form limit	Form default
Viscosity, μ_{ox}	`oxidizer_viscosity`	Pa·s		6.834150e-5
Effect on the calculation: Dynamic viscosity of the liquid oxidizer, in Pa·s; used for the injector orifice Reynolds number and discharge losses. It is filled from HRMA's local reference data for the selected oxidizer at the tank temperature (the field shows the resolved value). If you leave it unedited the solver resolves it itself; a value you type is used as your input.				
Temperature, T_{ox}	`oxidizer_temp`	K	182.33 – 309.52	293
Effect on the calculation: Oxidizer temperature at storage/injection conditions. Affects density, viscosity, and vapor pressure. Room temperature: 293K (20°C). Critical for accurate property calculations and system design.				
Oxidizer Feed Line Length	`feed_line_length_m`	m	0.05 – 50	(blank)
Effect on the calculation: Oxidizer feed-line length from the tank outlet to the injector manifold. Together with the inner diameter it sets the feed-line inertance, which switches the chug (low-frequency) stability loop to its second-order form. *If blank:* Leave empty if unknown - the chug analysis then runs without line inertia and declares that assumption instead of guessing a layout.				
Oxidizer Feed Line Inner Diameter	`feed_line_inner_diameter_mm`	mm	0.5 – 500	(blank)
Effect on the calculation: Inner diameter of the oxidizer feed line. The flow area for the feed-line inertance is derived from it as a circular duct. *If blank:* Leave empty if unknown - the chug analysis then runs without line inertia and declares that assumption.				

Table 9.7: Hybrid page: Mixture Ratio

Input	Identifier	Unit	Form limit	Form default
O/F Ratio oxidizer/fuel, O/F	`of_ratio`		0.5 – 20.0	7.5
Effect on the calculation: Oxidizer-to-fuel mass ratio at the design point. Through the equilibrium solution it sets chamber temperature, c* and Isp; with the regression law it sizes the fuel grain. The maximum-Isp O/F depends on the propellant pair and chamber pressure: 'Find Optimum' solves it with the equilibrium solver for your chamber pressure. HRMA's in-repository reference table (a quoted design value, not a computed optimum) lists 7.5 for N2O/HTPB, 8.0 for N2O/paraffin and 2.3 for LOX/HTPB. Accepted range 0.5-20.				

Table 9.8: Hybrid page: Injector Configuration

Input	Identifier	Unit	Form limit	Form default
Injector Type	`injector_type`			Showerhead
Effect on the calculation: How the oxidizer is injected. A hybrid has ONE liquid stream - the oxidizer; the fuel is the solid grain. Showerhead: parallel axial jets from a drilled plate. Impingement: like-on-like doublets - pairs of oxidizer jets that hit each other. Pintle: a central post with tip holes and an annular sheet. Swirl: pressure-swirl elements fed by tangential slots (hollow spray cone). Coaxial: one element with an inner jet and an outer annulus, both oxidizer. Showerhead, impingement, pintle and swirl are sized by the injector circuit solver; coaxial has no circuit model and is sized by the plate model alone. Options: Showerhead; Impingement; Pintle; Swirl; Coaxial.				
Injector Discharge Coefficient, C_d	`discharge_coefficient`	-	0.20 – 1.0	(blank)
Effect on the calculation: Discharge coefficient C_d of the injector orifices. The orifice flow area is sized from it (mdot proportional to C_d), so it changes hole count/diameter and the Nurick flip verdict. Ignored for swirl injectors, where C_d is solved from geometry. *If blank:* Leave empty to use the solver default of 0.78, which is then declared in the results instead of being silently assumed.				

continued on next page

Input	Identifier	Unit	Form limit	Form default
Target Velocity, v_{inj}	`target_velocity`	m/s	≥ 0	(blank)
Effect on the calculation: Mean oxidizer velocity through the holes that you would like to reach. It does not size the holes: the hole area follows from the oxidizer flow and the pressure drop solved by the injector circuit, so the results report the achieved velocity and the pressure drop your target would need. *If blank:* Empty: 30 m/s is used for that comparison (server default, declared).				
Min Hole Diameter	`hole_diameter_min`	mm	≥ 0	(blank)
Effect on the calculation: Lower edge of the hole-diameter band the injector circuit plans in. The circuit uses the fewest holes whose diameter falls inside [min, max]; a hole count fixed in Injector Design overrides that search. *If blank:* Empty: 0.5 mm (design band 0.5-2.5 mm, Sutton & Biblarz 9th ed. p. 337; declared).				
Max Hole Diameter	`hole_diameter_max`	mm	≥ 0	(blank)
Effect on the calculation: Upper edge of the hole-diameter band the injector circuit plans in; it must be larger than the minimum. *If blank:* Empty: 2.5 mm (design band 0.5-2.5 mm, Sutton & Biblarz 9th ed. p. 337; declared).				
Plate Thickness, t_{pl}	`plate_thickness`	mm	≥ 0	(blank)
Effect on the calculation: Thickness of the injector plate, i.e. the length of each orifice. It sets the orifice L/D, which picks the discharge coefficient C_d and the hydraulic-flip verdict; the plate structural check also uses it. *If blank:* Empty: the circuit keeps its declared L/D = 4 for C_d and publishes no L/D or flip verdict.				

Table 9.9: Hybrid page: Materials

Input	Identifier	Unit	Form limit	Form default
Chamber Material	`chamber_material`			Stainless steel 304 (annealed)
Effect on the calculation: Material of the combustion chamber case. The structural chain sizes or verifies the wall against hoop stress with the yield and ultimate strength of this material, derated at the computed wall temperature, and takes the chamber mass from its density. The heat-transfer chain uses its thermal conductivity for the wall temperature and its allowable temperature for the thermal verdict. The structural, safety and thermal panels of the analysis deck show the same material. Options: AISI 4130 low-alloy steel (normalized); Carbon steel (generic, A36-class); Stainless steel 304 (annealed); Stainless steel 316 (annealed); Aluminum 6061-T6; Titanium Ti-6Al-4V; Inconel 718 (aged); CuCrZr (C18150, aged); Aluminum 7075-T6; Aluminum 2024-T3; Stainless 17-4PH (H900); AISI 4340 low-alloy steel (normalized); Inconel 625 (annealed); Titanium CP Grade 2; Beryllium copper C17200 (aged, AT); Magnesium AZ31B-H24.				
Nozzle Material	`nozzle_material`			Graphite (isostatic, nozzle insert grade)
Effect on the calculation: Material of the nozzle throat (insert). The throat thermal check solves the axial Bartz profile at the throat with the thermal conductivity and emissivity of this material, under the cooling assumption the motor assigns to it (shown below the selection), and compares the throat wall temperature with its allowable temperature. Throat erosion is computed only where a published erosion coefficient band exists (also shown below); otherwise a rigid throat is assumed and stated. The steady performance solution always assumes a rigid throat. Options: Graphite (isostatic, nozzle insert grade); Tungsten (wrought, stress-relieved); Carbon-carbon (2D/3D C-C); Molybdenum TZM (stress-relieved); Niobium C-103 (silicide coated); Copper (OFHC, annealed); CuCrZr (C18150, aged).				
Injector Material	`injector_material`			Not selected (plate check not run)
Effect on the calculation: Material of the injector plate. Used only in the injector plate structural check: a clamped circular plate (Roark Table 11.2, case 10b) with ASME PG-52 ligament efficiency gives the bending stress, the safety factor and the required thickness from the room-temperature yield strength, and the plate mass from the density. With no material selected the plate check is not run. Options: Not selected (plate check not run); AISI 4130 low-alloy steel (normalized); Carbon steel (generic, A36-class); Stainless steel 304 (annealed); Stainless steel 316 (annealed); Aluminum 6061-T6; Titanium Ti-6Al-4V; Inconel 718 (aged); CuCrZr (C18150, aged); Aluminum 7075-T6; Aluminum 2024-T3; Stainless 17-4PH (H900); AISI 4340 low-alloy steel (normalized); Inconel 625 (annealed); Titanium CP Grade 2; Beryllium copper C17200 (aged, AT); Brass C36000 (free-cutting, H02); Magnesium AZ31B-H24.				

Table 9.10: Hybrid page: Analysis Type (parametric sweep and trajectory)

Input	Identifier	Unit	Form limit	Form default
Parameter Type	`param_type`			O/F Ratio
Effect on the calculation: Parameter to vary during sweep analysis. O/F Ratio: optimize mixture ratio for performance. Chamber Pressure: study pressure effects. Expansion Ratio: optimize nozzle design. Altitude: analyze altitude performance. Options: O/F Ratio; Chamber Pressure; Expansion Ratio.				
Start Value	`param_start`			1.5
Effect on the calculation: First value of the sweep (in the unit of the swept parameter). The sweep runs from this value to the end value and includes both ends; the two ends are computed first, so a sweep cut short by the time budget still spans the whole range. For an O/F sweep, an untouched range is set to 50-150 % of the design O/F (HRMA choice, so the design point and the propellant's Isp peak lie inside).				
End Value	`param_end`			4.0
Effect on the calculation: Last value of the sweep (in the unit of the swept parameter); it is always computed. For an O/F sweep, an untouched range is set to 50-150 % of the design O/F (HRMA choice).				
Number of Steps	`param_steps`		3 – 50	10
Effect on the calculation: Number of sweep points between the start and end values; each point is a full motor calculation, so the run time grows with it and the sweep stops when its time budget is used up (the result says how many points were computed). Accepted by the server: 2-100; form range 3-50. *If blank:* Empty: 20 points (server default, declared).				
calculate_trajectory	`calculate_trajectory`			checked
Effect on the calculation: Whether Calculate also flies the trajectory. Checked: after the motor is solved, the flight is integrated with the motor result and the vehicle and launch inputs on this tab. Unchecked: the result carries no trajectory; it is reported as a skipped step with no flight numbers. Default: checked (the server default is also on). A value other than on or off (for example from an edited project file) is refused by name.				
Initial Altitude	`traj_alt_start`	m	0 – 50000	0
Effect on the calculation: Starting altitude of the trajectory (m); the flight starts there and the value is accepted within the range of the ISA atmosphere table. *If blank:* Empty or not a number: the trajectory is not computed and the field is named (no altitude is assumed).				
Final Altitude	`traj_alt_end`	m	0 – 100000	10000
Effect on the calculation: Upper altitude of the analysis (m). Informational: the trajectory calculation does not read it (measured: the result is the same with or without it, and the result lists it among the inputs not read). The flight is integrated through apogee to landing. *If blank:* Empty: not sent.				
Analysis Points	`traj_points`		5 – 100	20
Effect on the calculation: Number of output points along the trajectory. Informational: the trajectory calculation does not read it (measured: the result is the same with or without it, and the result lists it among the inputs not read); the solver uses its own time step. *If blank:* Empty: not sent.				
Initial Mass	`initial_mass`	kg	1 – 10000	50
Effect on the calculation: Total vehicle mass at ignition including propellant, structure, and payload.				
Final Mass	`final_mass`	kg	1 – 10000	25
Effect on the calculation: Vehicle mass after propellant consumption (dry mass + payload). Avionics, wiring and the recovery system belong in this mass: the engine dry (flying inert) mass does not contain them.				
Drag Coefficient	`drag_coefficient`		0.1 – 2.0	0.5
Effect on the calculation: Drag coefficient C_D of the vehicle (dimensionless), referred to the reference area below; the trajectory drag is $D = \frac{1}{2}\cdot\rho\cdot v^2\cdot C_D\cdot A_{ref}$. Must be greater than 0. Form range 0.1-2.0. *If blank:* Empty: Calculate Trajectory refuses and names the field; Calculate does not send it and the trajectory solver keeps its own coefficient.				

continued on next page

Input	Identifier	Unit	Form limit	Form default
Reference Area	`reference_area`	mm^2	100 – 10000000	17671

Effect on the calculation: Reference area A_ref of the drag D = ½·ρ·v²·C_D·A_ref; it must be the area your drag coefficient is referred to (for a rocket, the body cross-section). Entered in square millimetres and converted to square metres before it is sent to the solver. Form range 100-10000000 mm^2. *If blank:* Empty: Calculate Trajectory refuses and names the field.

Launch Angle	`launch_angle`	degrees	0 – 90	90

Effect on the calculation: Launch angle from the horizontal (degrees): 90° = vertical launch; accepted 0-90. *If blank:* Empty or not a number: the trajectory is not computed and the field is named (no angle is assumed).

Wind Speed	`wind_speed`	m/s	0 – 50	0

Effect on the calculation: Horizontal wind speed affecting the trajectory (m/s); must be ≥ 0 and below the ISA sea-level speed of sound. *If blank:* Empty or not a number: the trajectory is not computed and the field is named (no wind is assumed).

Launch Altitude	`launch_altitude`	m	0 – 6000	0

Effect on the calculation: Elevation of the launch site above sea level. Sets the starting altitude and the ambient conditions the trajectory begins from.

Wind Direction	`wind_direction`	degrees	0 – 360	0

Effect on the calculation: Compass direction the wind blows towards, measured clockwise from the launch azimuth. 0° = tailwind along the launch heading.

Parachute Area	`parachute_area`	m2	≥ 0	(blank)

Effect on the calculation: Projected (canopy) area of the recovery parachute (m^2). Descent velocity scales as 1/sqrt(area). When given, the descent under the parachute is solved and the landing speed is reported. Must be greater than 0; otherwise it is rejected with the reason. *If blank:* Leave empty for no parachute: the descent is not solved and the results say so (no area is assumed).

Parachute Cd	`parachute_cd`		≥ 0	(blank)

Effect on the calculation: Drag coefficient of the parachute referenced to its projected area (1.4 is the classic flat-circular value). Read only when a parachute area is given. Must be greater than 0. *If blank:* Leave empty to keep the solver assumption (reported as an assumption).

Deploy Delay after Apogee	`parachute_deploy_delay`	s	≥ 0	(blank)

Effect on the calculation: Time between apogee and canopy inflation (s). The vehicle falls ballistically during this delay. Read only when a parachute area is given. Must not be negative. *If blank:* Leave empty to keep the solver assumption.

Table 9.11: Hybrid page: Design Configuration window

Input	Identifier	Unit	Form limit	Form default
Wall Thickness	`wall_thickness`	mm	2 – 20	5

Effect on the calculation: Thickness of chamber wall. Must withstand internal pressure with safety factor. Thicker walls are stronger but heavier. Calculated based on hoop stress analysis and material yield strength.

Safety Factor, FoS	`safety_factor`		2 – 6	4.0

Effect on the calculation: Design safety-factor target of the structural sizing (dimensionless): the wall, closure and nozzle sections are sized so that their safety factor reaches this target. Applied only when you change it (or it comes from a project file); otherwise the target of the selected material record is used and declared. A higher value gives thicker walls and more mass. Accepted by the server 1.05-10; form range 2-6.

continued on next page

Input	Identifier	Unit	Form limit	Form default
Chamber Length Override	`chamber_length_override`	mm		(blank)
Effect on the calculation: Chamber length you set (mm; converted to m for the solver). It replaces the chamber length the solver derives from L*; the post-combustion chamber takes the remaining length (override minus grain minus pre-combustion chamber). Accepted: above 0 and at most 20000 mm (20 m); a larger value is not applied and a warning names it. It must be at least grain length plus pre-combustion chamber length; a shorter value is not applied, the automatic length is used and the warning gives the minimum. *If blank:* Empty or 0: the length is derived from L* (automatic).				
Pressure Drop Override	`pressure_drop_percent`	%	10 – 50	20
Effect on the calculation: Injector pressure drop as a percentage of chamber pressure. Applied only when you enter a value yourself: the injector plan then uses ΔP = (value / 100) · Pc. Otherwise the injector circuit solver's ΔP is used, which comes from the feed (for self-pressurised N2O: saturation pressure minus chamber pressure). Form range 10-50 %.				
Number of Holes Override	`n_holes_override`		1 – 200	(blank)
Effect on the calculation: Override automatic hole count calculation. Manual entry allows custom injector patterns but must satisfy flow and pressure requirements. *If blank:* Leave empty for auto-calculation based on hole size constraints.				
Hole Pattern	`hole_pattern`			Circular Rings
Effect on the calculation: Options: Circular Rings; Hexagonal; Square Grid.				
Include Cooling Channels	`cooling_channels`			None
Effect on the calculation: Options: None; Radial Channels; Spiral Channels.				
Ambient Temperature, T_a	`ambient_temp`	K	180 – 340	(blank)
Effect on the calculation: Ambient (initial) temperature of the motor hardware and its surroundings. The heat-transfer chain (wall temperatures, heat-sink sizing) and the structural chain (thermal stress, derating) both use this single value. *If blank:* Leave empty to use the solver default of 293.15 K, which is declared in the results instead of being silently assumed.				

9.4 Outputs

Table 9.12: Outputs of the hybrid page

Output	Unit	Meaning	Source / model
ISP (DESIGN POINT, AT DESIGN-ALTITUDE AMBIENT)	s	Delivered specific impulse at the design point.	$c^* C_F / g_0$; $c^* = \eta_{c^*} \times$ equilibrium c^* (Cantera).
THRUST, CHAMBER PRESSURE	N, bar	Design point; inputs.	Form.
MASS FLOW RATE	kg/s	Total propellant flow.	$F/(I_{sp}\, g_0)$.
STRUCTURAL MIN SF	–	Verdict of the structural chain and the minimum earned safety factor.	Wall check (Section 4.5).
Throat / exit diameter	mm	Nozzle dimensions.	Throat: $\dot{m}\, c^*/(C_d\, P_c)$; exit: ε.
Expansion ratio	–	A_e/A_t.	Input or the $P_e = P_a$ solution.
Chamber diameter, length, volume	mm, cm^3	Combustion chamber dimensions.	Fuel-grain sizing and L*.

Output	Unit	Meaning	Source / model
Characteristic velocity, thrust coefficient	m/s, –	Delivered c^* and C_F.	Equilibrium thermochemistry, isentropic nozzle, divergence efficiency λ.
L* requested / achieved	m	Characteristic length.	Chamber and port volume / throat area.
Port diameters, regression rate	mm, mm/s	Port growth.	$r = a\,G^n$; HRMA default is the total mass flux G.
Injector rows	–	Type, exit velocity, Reynolds and Weber numbers, pressure drop, hole count and diameter, L/D, C_d, injection area and their bases.	Injector circuit solution.
Propellant rows	kg/s, kg	Flows; Total Propellant Mass (loaded); oxidizer mass (for a self-pressurised tank the mass in the tank at ignition: liquid and ullage vapour); burned, unburned and loaded fuel; residual oxidizer.	Design point; for self-pressurised N_2O the tank blowdown solution.
Nozzle and fuel-grain rows	mm, °	Nozzle type and angles; grain length, outer diameter, web thickness, L/D.	Solver geometry.
Design Summary	mm, kg, N · s, s	Total length (without and with the forward closure), motor envelope diameter (the larger of case and drawn nozzle), dry and total mass, total impulse, burn time.	Dry mass from the structural chain: chamber shell, end closures, drawn nozzle and injector plate; impulse and burn time are the delivered values (Section 4.4); basis lines below the summary.
Wall zones	–	Three-zone wall verdict, liner thickness.	Table 4.3.
Blank-field assumptions	–	Values used for fields left empty.	The server's declaration list.
Charts	–	Regression rate and port growth, motor cross-section, performance dashboard, thrust and chamber pressure vs time, injector schematic, altitude performance, total impulse vs altitude, combustion products through the nozzle, combustion analysis, operating-point dashboard.	Time-stepped solution and equilibrium calculation.

Output	Unit	Meaning	Source / model
Side panels	–	Transient Analysis, Tank Blowdown, Structural FEA, Thermal FEA, Flight Dynamics (6-DOF), Injector Design.	Tables 13.17–13.22.
NASA CEA Comparison	–	HRMA equilibrium and RocketCEA (NASA CEA) at the same operating point.	Section 12.



Figure 9.7: Performance analysis dashboard: flows, pressure distribution, regression rate and port growth, injection velocity.

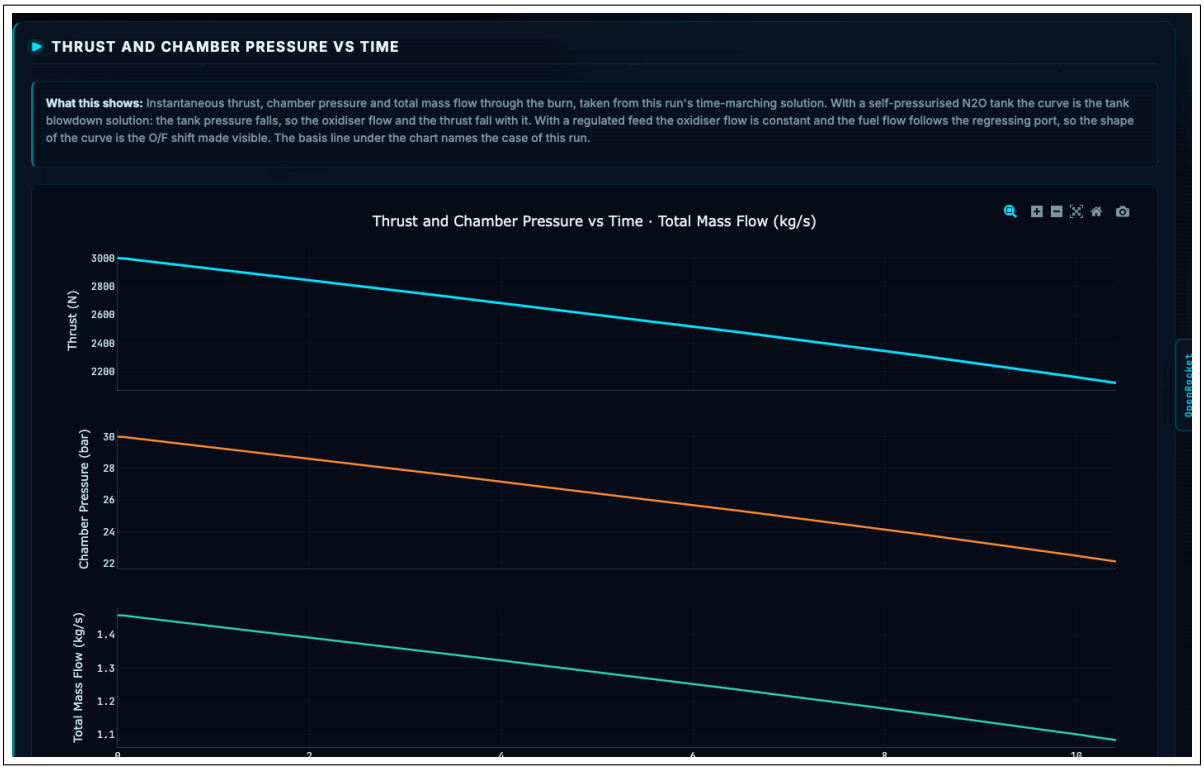


Figure 9.8: Thrust and chamber pressure against time (separate subplots) and total mass flow.

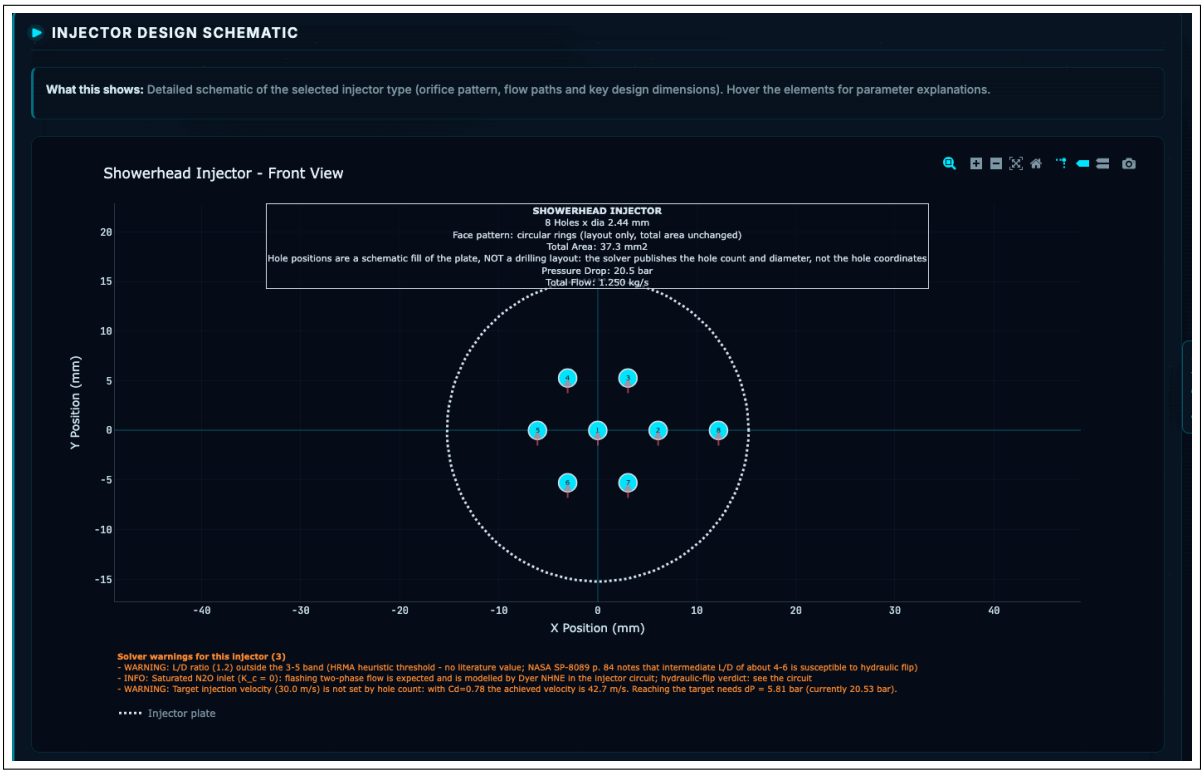


Figure 9.9: Injector design schematic: hole pattern and plate dimensions.

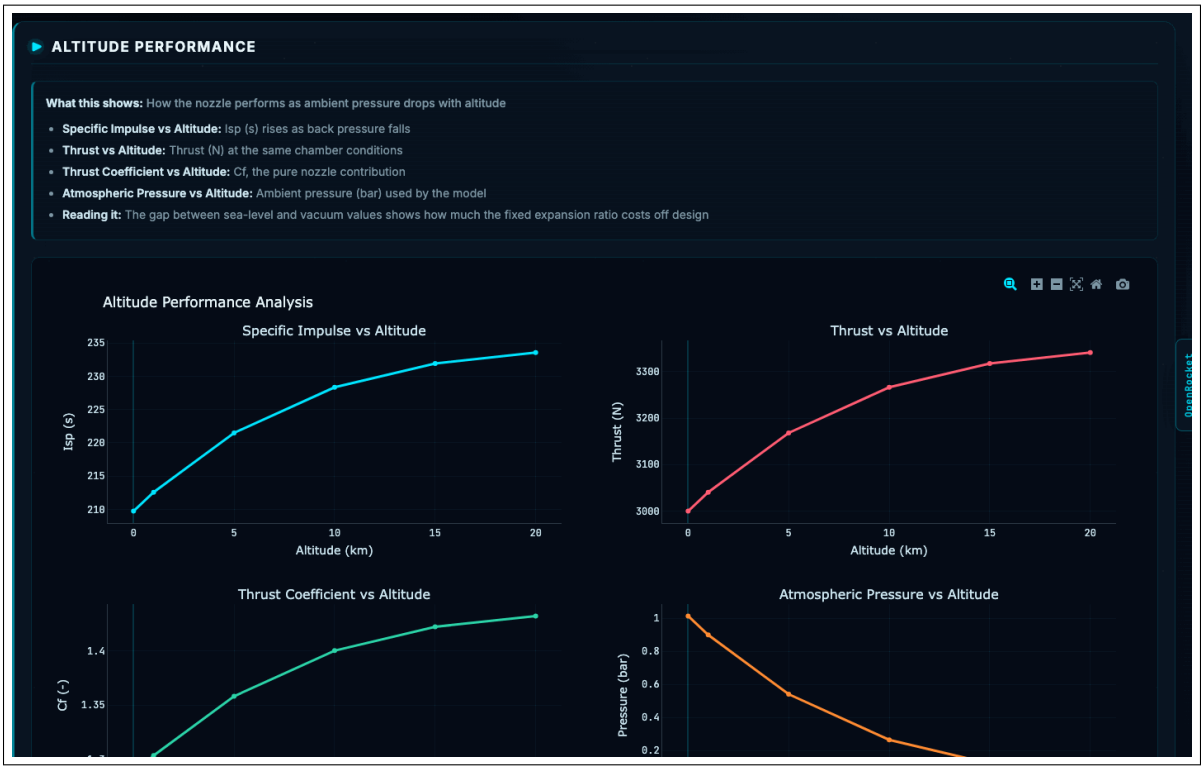


Figure 9.10: Altitude performance: how Isp and thrust change as ambient pressure falls.

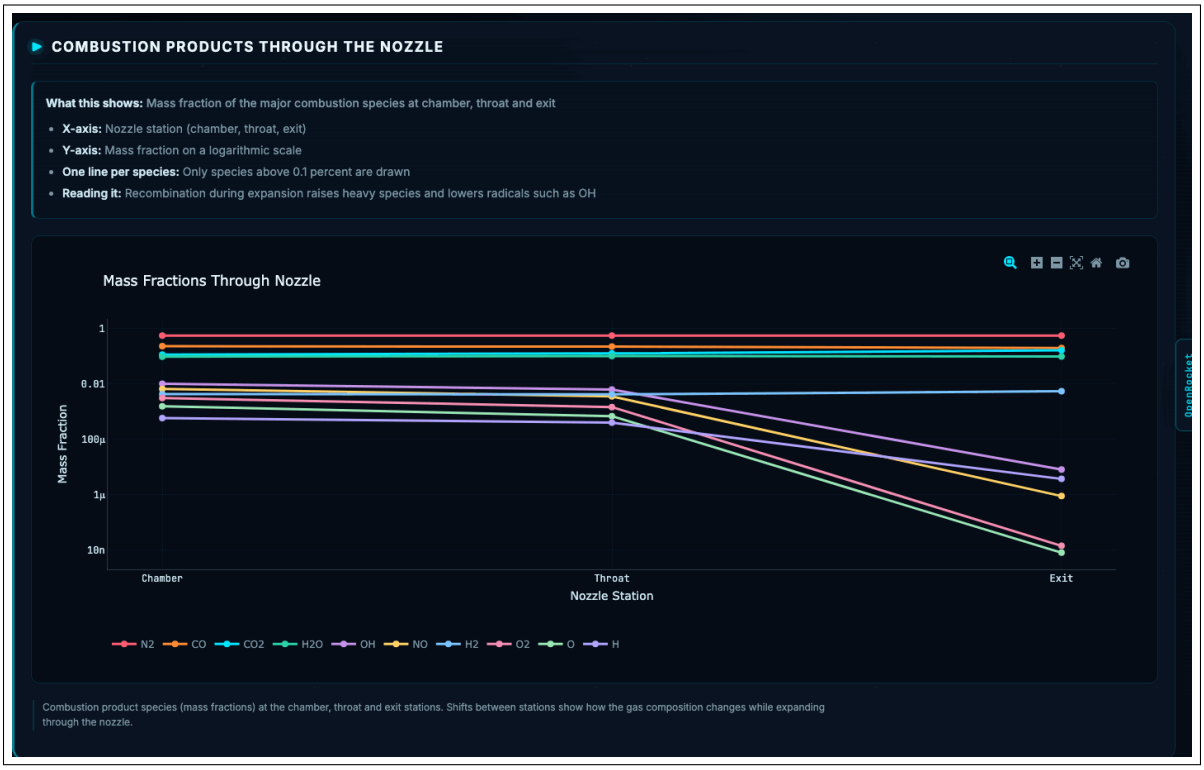


Figure 9.11: Combustion products through the nozzle.



Figure 9.12: Combustion analysis.

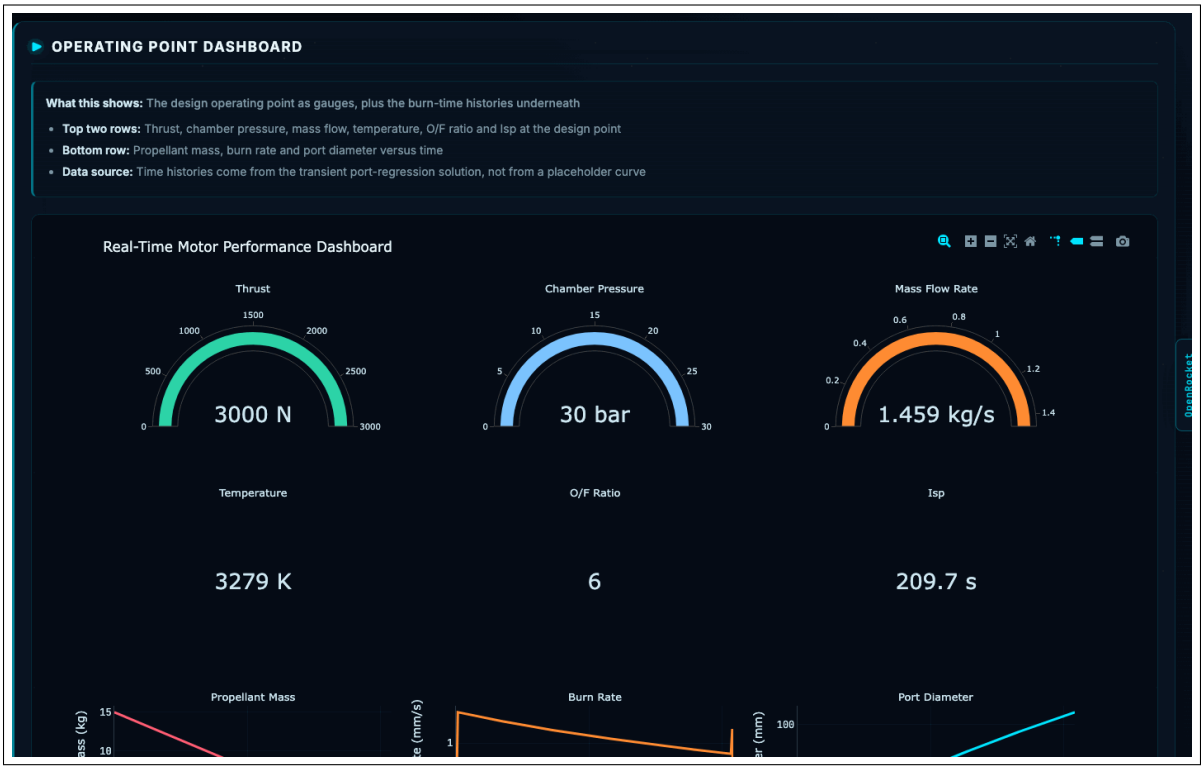


Figure 9.13: Operating-point dashboard.



Figure 9.14: The 3D motor model built from the solver geometry and the viewer toolbar (Table A.3).

9.5 Verdicts and warnings

Warning classes and verdict codes are in Section 17. Lines specific to the hybrid page:

Table 9.13: Hybrid-specific verdicts and declarations

On screen	Meaning	Action
Regulated design-point line (end: OXIDIZER DEPLETED)	With a self-pressurised tank the true impulse and duration differ from the headline.	Use the Tank Blowdown panel's number.
LINER REQUIRED	The bare wall exceeds its limit in the pre- or post-combustion chamber.	Carry the liner thickness into CAD and mass.
PROTECTED DURING BURN	The fuel grain covers the case.	With residual web 0, consider the end-of-burn exposure separately.
HTPB/ABS regression bias	Measured bias of the built-in coefficients.	Enter your own a, n.
Grain and chamber L/D threshold	Long, thin motor for a single port.	Increase the port count or the port diameter.
IDEAL c* line (no c* (combustion) efficiency applied)	No measured c^* efficiency exists for this propellant pair; 1.00 is used.	Enter a measured efficiency.

9.6 Limits

- The regression law is of the form $r = a\,G^n$; scale effects, combustion instability and ignition transients are not modelled.
- In user tank-pressure mode, regulator lock-up and thermal drift of the tank are not modelled; the tank structural mass is not computed.
- H_2O_2 in the hybrid is taken as 100 % pure.

9.7 Related sections

Example 1 (Section 4); Analysis Dock (Section 13); exports (Section 16); canonical bands (Appendix E).

10. Solid Motor Page

10.1 Purpose

Computes the burning area from the grain geometry, the internal ballistics (chamber pressure, thrust curve), the nozzle, the case and closure thickness and the mass breakdown of a solid propellant motor.

10.2 Screen

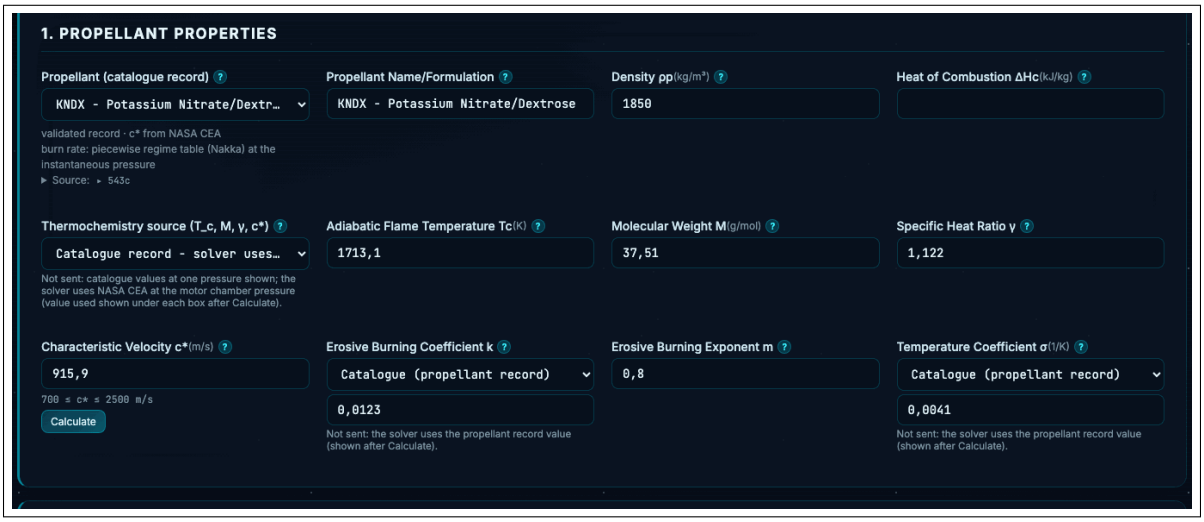


Figure 10.1: Solid page: 1. PROPELLANT PROPERTIES. Note the Thermochemistry source selector and the fields that open with the grain type.

10.3 Inputs

The default of an empty solid page is KNDX propellant with three BATES grains (outer diameter 100 mm, core 48 mm, total length 400 mm). With Thermochemistry source (T_c, M, γ, c*) set to Catalogue record the flame temperature, molar mass, γ and c^* boxes are read-only and not sent; the solver takes these values from NASA CEA at the motor's chamber pressure and, after the calculation, writes the value used below each box. With My value the four boxes are sent and the run is declared as user input.

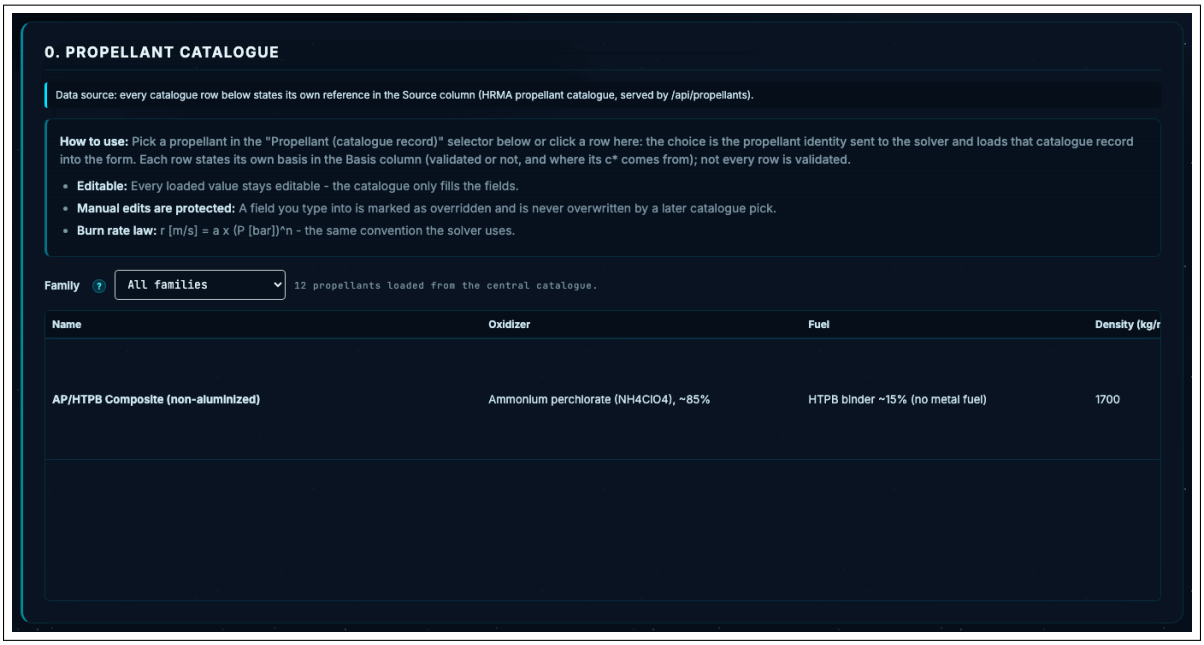


Figure 10.2: 0. PROPELLANT CATALOGUE: family filter, records and the source column of each record.

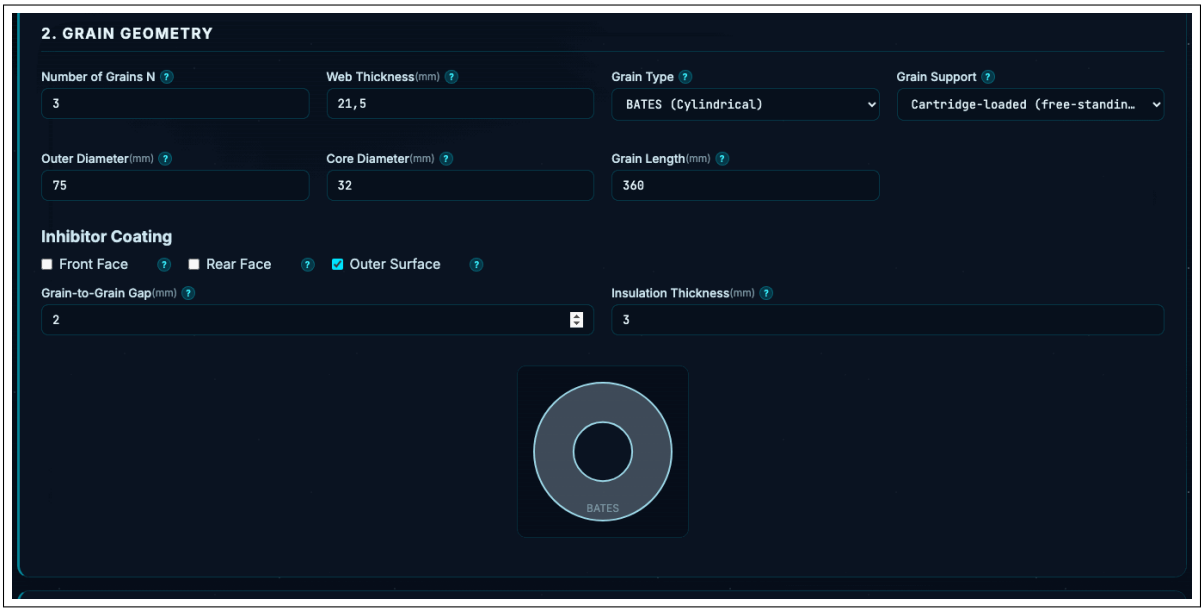


Figure 10.3: 2. GRAIN GEOMETRY with the example project loaded: 3 BATES grains, outer diameter 75 mm, core 32 mm, total length 360 mm. The default of an empty page is 100 / 48 / 400 mm.

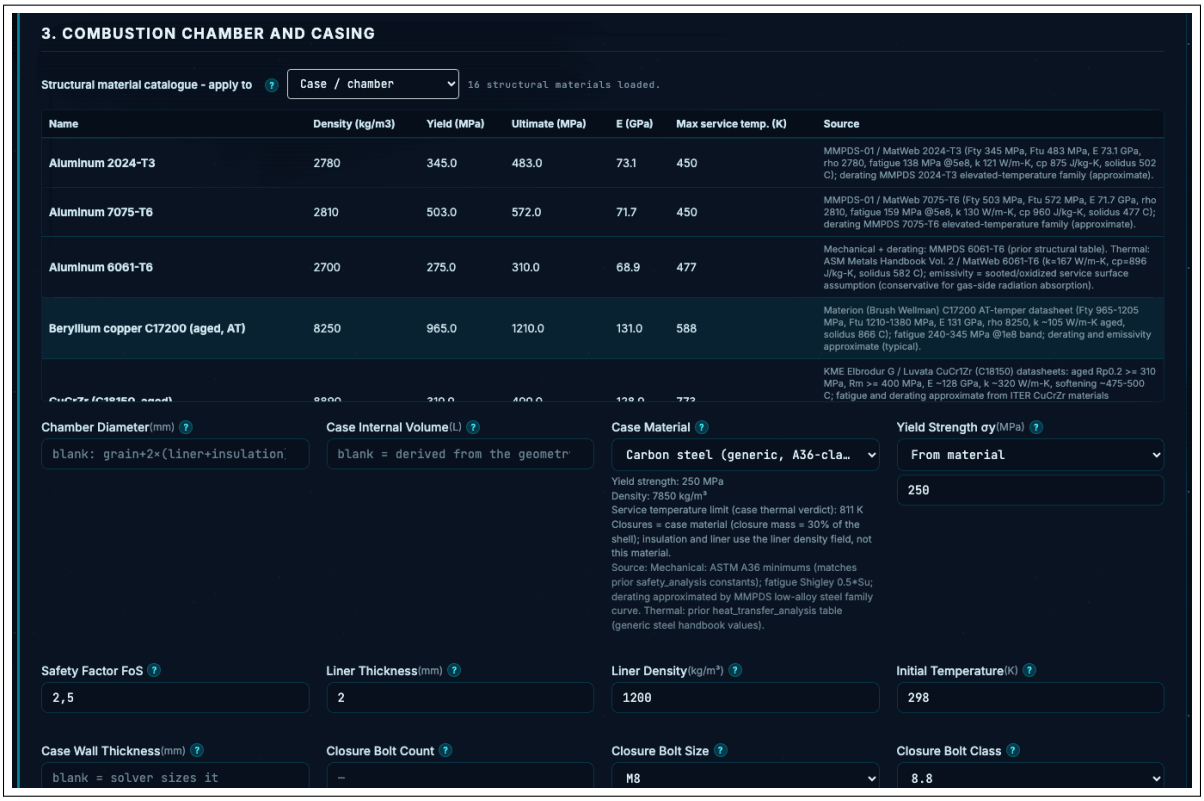


Figure 10.4: 3. COMBUSTION CHAMBER AND CASING: case material, yield-strength source, wall, liner and insulation.

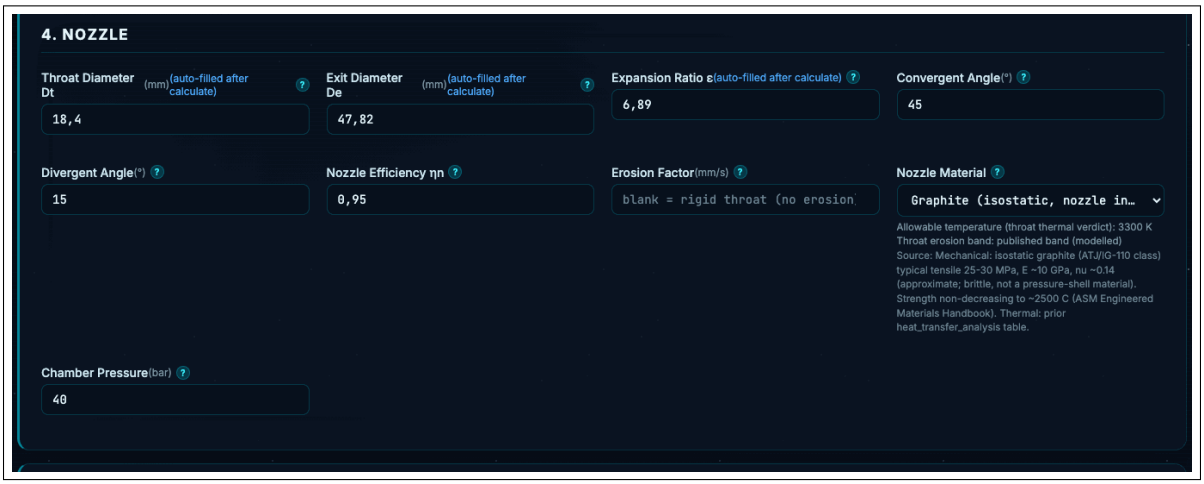


Figure 10.5: 4. NOZZLE: throat and exit diameter, nozzle material, erosion and design pressure.



Figure 10.6: “?” note: Initial Temperature.

Table 10.1: Solid page: 0. Propellant Catalogue

Input	Identifier	Unit	Form limit	Form default
Family	`propellant_family_filter`			All families

Effect on the calculation: Options: All families; Composite; Sugar (KN-based); Double base; Other.

Table 10.2: Solid page: 1. Propellant Properties

Input	Identifier	Unit	Form limit	Form default
Propellant (catalogue record)	`propellant_type`			KNDX - Potassium Nitrate/Dextrose 65/35

Effect on the calculation: The propellant identity sent to the solver: grain mechanics, two-phase loss, burn-rate regime table and the NASA CEA composition all come from this catalogue record. Changing it loads the record into the form, switches the Burn Rate Preset to this propellant's preset (or to Custom with the record's a-n pair when it has none) and keeps only the fields you typed yourself. The line below states the record's basis (validated or not, c* basis) and source. ”Not selected” appears only for a project saved before v3.0.5 whose file does not name a catalogue record; the solver then derives the identity from the name and says so in the warnings. Options: APCP - Ammonium Perchlorate Composite (AP/Al/HTPB); HTPB/AP/Al Composite (aluminized); AP/HTPB Composite (non-aluminized); PBAN/AP/Al Composite (Shuttle SRB class); Fast Commercial Composite (Blue Thunder class); KNDX - Potassium Nitrate/Dextrose 65/35; KNSB - Potassium Nitrate/Sorbitol 65/35; KNSU - Potassium Nitrate/Sucrose 65/35 (ideal/theoretical maximum density); KNSU - Potassium Nitrate/Sucrose 65/35 (cast bulk density); KNER - Potassium Nitrate/Erythritol 65/35; Double Base (Nitrocellulose/Nitroglycerin); Black Powder (KNO3/Charcoal/Sulfur); Not selected (derived from the name by the solver).

Propellant Name/Formulation	`propellant_name`			KNDX - Potassium Nitrate/Dextrose 65/35

Effect on the calculation: Label only: a free-text name for this propellant that is printed in the results, reports and the saved project. It does NOT select the propellant and changes no number - the identity is the ”Propellant (catalogue record)” selector above. Picking a catalogue record fills it with the record name (unless you typed your own label). *If blank:* The results show no propellant label.

Density ρp, ρ_p	`density`	kg/m^3		1850

Effect on the calculation: Bulk density ρp of the cast propellant in kg/m^3. It sets the propellant mass of the grain, so the burn time, total impulse and the mass flow that sizes the throat change with it. Picking a catalogue record fills it with the record value (see the catalogue row); the value you type here is sent and used. Accepted range 500-3000 kg/m^3. *If blank:* The selected record's density is used.

continued on next page

Input	Identifier	Unit	Form limit	Form default
Heat of Combustion ΔHc	`heat_value`	kJ/kg	500 – 10000	(blank)

Effect on the calculation: Heat of combustion ΔHc (kJ/kg) - informational only: no calculation reads it (the solver lists it among the informational inputs it does not use); c* and T_c come from the selected catalogue record and NASA CEA at the motor chamber pressure. The catalogue has no ΔHc for any propellant, so picking a record clears this box unless you typed your own value. The box accepts 500-10000 kJ/kg; the solver does not use it. *If blank:* Nothing is sent and no result changes.

Thermochemistry source (T_c, M, γ, c*)	`thermo_source`			Catalogue record - solver uses NASA CEA at motor Pc

Effect on the calculation: Where T_c, M, γ and c* come from. "Catalogue record" (default): the four boxes below show the selected record's values at ONE reference pressure, are read-only and are NOT sent; for a propellant with a known composition the solver computes them with NASA CEA at the motor chamber pressure (otherwise it uses the record values and says so in the warnings), and after Calculate each box shows the value the solver used underneath. "My values": the four boxes are sent and replace the thermochemistry; NASA CEA is then not applied and the result names your values as the source. Options: Catalogue record - solver uses NASA CEA at motor Pc; My value. *If blank:* A blank box in "My values" is not sent (the record value is used for it).

Adiabatic Flame Temperature Tc, T_c	`flame_temp`	K		1713.1

Effect on the calculation: Adiabatic flame (chamber) temperature T_c in K. It sets the gas enthalpy behind c* and Isp and the throat heat flux. "Catalogue record" mode: shows the selected record's value at one pressure (see the catalogue row) and is not sent; the value the solver used (NASA CEA at the motor chamber pressure) appears below after Calculate. "My values" mode: this number is sent and used. Accepted range 1000-4500 K.

Molecular Weight M, $\mathcal{M}$	`molecular_weight`	g/mol		37.51

Effect on the calculation: Mean molar mass M of the combustion products in g/mol (= kg/kmol). It enters c* and Isp (lower M gives higher values) and the gas properties of the heat-flux model. "Catalogue record" mode: shows the selected record's value at one pressure and is not sent; the solver's value (NASA CEA at the motor chamber pressure, mixture mass per mole of gas, condensed phase included) appears below after Calculate - it can differ from the catalogue value because the catalogue rows use different molar conventions. "My values" mode: this number is sent and used. Accepted range 2-100 g/mol.

Specific Heat Ratio γ, γ	`gamma`		1.05 – 1.5	1.122

Effect on the calculation: Ratio of specific heats γ = Cp/Cv of the chamber gas. It enters c* and the nozzle expansion (exit pressure, thrust coefficient). "Catalogue record" mode: shows the selected record's value at one pressure and is not sent; the solver's value (NASA CEA at the motor chamber pressure, shifting equilibrium) appears below after Calculate. "My values" mode: this number is sent and used. Accepted range 1.05-1.5.

Characteristic Velocity c*, c^*	`char_velocity`	m/s	700 – 2500	915.9

Effect on the calculation: Theoretical characteristic velocity c* in m/s, before the combustion efficiency. It sizes the throat (A_t = $\dot{m}$·c*/Pc) and the thrust. "Catalogue record" mode: shows the selected record's value at one pressure (see the catalogue row) and is not sent; the solver's value (NASA CEA at the motor chamber pressure) appears below after Calculate. "My values" mode: this number is sent and replaces the thermochemistry (NASA CEA is not applied). "Calculate" computes c* from γ, T_c and M (Sutton & Biblarz Eq. 3-32) and switches the source to "My values". Accepted range 700-2500 m/s; outside it the value is ignored with a warning.

Erosive Burning Coefficient k	`erosive_k_source`			Catalogue (propellant record)

Effect on the calculation: Erosive burning coefficient k of the mass-flux law (dimensionless here; the solver multiplies the base rate by 1 + k·(G/G_ref)^m with m from the field below). It raises the burn rate where the port gas mass flux G is high, so it changes the start of the thrust curve, the peak pressure and the burn time. Source selector: "Catalogue" uses the value calibrated in the propellant record (this box is then read-only and shows the value the solver used after Calculate); "My value" sends this number and the solver applies it. Accepted range 0-1. Options: Catalogue (propellant record); My value. *If blank:* In "My value" mode: nothing is sent and the catalogue value is used.

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Input	Identifier	Unit	Form limit	Form default
Erosive Burning Coefficient k, k_e	`erosive_k`		0 – 1	(blank)

Effect on the calculation: Erosive burning coefficient k of the mass-flux law (dimensionless here; the solver multiplies the base rate by 1 + k·(G/G_ref)^m with m from the field below). It raises the burn rate where the port gas mass flux G is high, so it changes the start of the thrust curve, the peak pressure and the burn time. Source selector: "Catalogue" uses the value calibrated in the propellant record (this box is then read-only and shows the value the solver used after Calculate); "My value" sends this number and the solver applies it. Accepted range 0-1. *If blank:* In "My value" mode: nothing is sent and the catalogue value is used.

Erosive Burning Exponent m	`erosive_m`		0.1 – 2	0.8

Effect on the calculation: Mass-flux exponent of the erosive burning law, dimensionless: it sets how strongly the erosive burn-rate increase follows the port mass flux, so it changes the early chamber pressure and thrust of a narrow port. Accepted range 0.1-2.0; a value outside it is rejected with a named warning. *If blank:* 0.8 (Lenoir-Robillard mass-flux exponent, Sutton & Biblarz, 9th ed., Ch. 12).

Temperature Coefficient σ	`temp_coeff_source`	1/K		Catalogue (propellant record)

Effect on the calculation: Temperature sensitivity σp of the burn rate (1/K). The solver uses the exponential law a_T = a·exp(σp·(T0 − T_ref)), where T0 is the grain temperature (Initial Temperature field) and T_ref the record reference temperature; it shifts the burn rate, so chamber pressure, thrust and burn time change when T0 ≠ T_ref. It also sets the cold/hot-day sensitivity table. Source selector: "Catalogue" uses the propellant record value (read-only box, filled after Calculate); "My value" sends this number. Accepted range 0-0.02 1/K. Options: Catalogue (propellant record); My value. *If blank:* In "My value" mode: nothing is sent and the catalogue value is used.

Temperature Coefficient σ**,** σ_p	`temp_coeff`	1/K	0 – 0.02	(blank)

Effect on the calculation: Temperature sensitivity σp of the burn rate (1/K). The solver uses the exponential law a_T = a·exp(σp·(T0 − T_ref)), where T0 is the grain temperature (Initial Temperature field) and T_ref the record reference temperature; it shifts the burn rate, so chamber pressure, thrust and burn time change when T0 ≠ T_ref. It also sets the cold/hot-day sensitivity table. Source selector: "Catalogue" uses the propellant record value (read-only box, filled after Calculate); "My value" sends this number. Accepted range 0-0.02 1/K. *If blank:* In "My value" mode: nothing is sent and the catalogue value is used.

Table 10.3: Solid page: 2. Grain Geometry

Input	Identifier	Unit	Form limit	Form default
Number of Grains N, N	`grain_count`		≥ 1	3

Effect on the calculation: Number of BATES grain segments, whole number. The stack length (Grain Length) is split into this many segments; each segment adds burning end faces, so the count changes the burning area and the thrust profile but not the propellant mass. Accepted range 1-20; a fractional or out-of-range value is rejected and named (it is not rounded or clipped). *If blank:* The solver uses stack length / grain outer diameter, rounded (at least 1).

Web Thickness, w	`web_thickness`	mm		26.0

Effect on the calculation: Derived output, read-only: the propellant web implied by the grain geometry ((outer − core)/2), filled after Calculate. When the outer surface burns as well the web is consumed from both faces, so the thickness actually burnt is half of it. Typing here changes no result.

Grain Type	`grain_type`			BATES (Cylindrical)

Effect on the calculation: Geometric configuration of the solid propellant grain. BATES: cylinder with a central core port. Star: star-shaped core port. Finocyl: core port with fins for a larger burning surface. Wagon Wheel: multiple core ports. End Burner: no core port, burns from one end only. The thrust profile is NOT fixed by the shape alone (it also depends on the inhibitor layout and the port-to-throat ratio); this run's profile is classified by the solver and printed under "Thrust Curve". Options: BATES (Cylindrical); Star; Finocyl; Slotted; Wagon Wheel; End Burner.

continued on next page

Input	Identifier	Unit	Form limit	Form default
Grain Support	`grain_support`			Case-bonded
Effect on the calculation: How the grain is held in the case. Case-bonded: cast and cured inside the case and bonded to its liner; the case restrains the cure and cool-down shrinkage, so the bore carries that thermal strain on top of the pressure strain. Cartridge-loaded: cast in a mold or liner outside the case and inserted, bonded at most at its ends (NASA SP-8073); no shrinkage strain is restrained, and the pressure strain is computed with the case supporting the grain (snug fit, upper bound). Changes only the grain structural results (bore strain, strain safety factor, crack risk), not the ballistics. If the field is not sent, case-bonded is used and declared in the result. Options: Case-bonded; Cartridge-loaded (free-standing).				
Outer Diameter, D_g	`outer_diameter`	mm		100
Effect on the calculation: Outer diameter of the propellant grain, mm. Accepted range 1-5000 mm, and larger than the core. *If blank:* With the core and length it sets the web, the propellant mass and the burning surface; when Chamber Diameter is blank, the case bore is derived from it (grain outer diameter + 2·(liner + insulation)). No grain diameter is assumed; with Chamber Diameter also blank the calculation is refused, naming the missing diameter.				
Core Diameter, d_c	`core_diameter`	mm		48
Effect on the calculation: Initial diameter of the central core hole (port diameter). Determines initial burning surface area and thrust profile. Measured in this solver for a BATES grain with burning ends (10-60 mm cores): the end plateau is below the start plateau at every core (end/start 0.09-0.49) and a larger core falls off less, but the trend is not monotonic (0.11 at 10 mm, 0.09 at 15 mm, 0.21 at 30 mm, 0.49 at 60 mm). At 10 mm the solver classifies the curve as Progressive-Regressive (thrust rises first, then falls), so this is not a single class over the whole core range. The published shape class is the Thrust Profile Type in the result, not this hint.				
Grain Length, L_g	`grain_length`	mm		400
Effect on the calculation: Total axial length of the propellant stack in mm, not of a single segment. The solver splits this length across the segment count (each BATES segment is L / N long), so changing the segment count does not change the propellant mass. It sets the burning surface and the propellant mass: a longer stack carries more propellant and total impulse. Accepted range 50-10000 mm. *If blank:* The calculation is refused and the field is named (no length is assumed).				

Table 10.4: Solid page: grain-type specific fields (Star, Finocyl, Wagon Wheel, Slotted)

Input	Identifier	Unit	Form limit	Form default
Number of Points	`star_points`		≥ 3	6
Effect on the calculation: Number of star points, whole number. More points raise the initial burning area and so the early chamber pressure and thrust. Accepted range 3-12; a fractional or out-of-range value is rejected and named (it is not rounded or clipped). *If blank:* 6.				
Point Radius	`star_radius`	mm	2 – 60	15
Effect on the calculation: Radius of star points from center. Determines burning surface area and progressive characteristics.				
Number of Fins	`fin_count`		≥ 2	4
Effect on the calculation: Number of radial fins extending from the core, whole number. More fins raise the initial burning area and so the early chamber pressure and thrust. Accepted range 2-12; a value outside it is rejected with a named warning. *If blank:* 4 (HRMA default, declared as assumed in the result).				
Fin Width	`fin_width`	mm		8
Effect on the calculation: Circumferential width of each fin slot (mm). It sets the burning area of the finned section and the web left between fins. When you switch the grain type, this box is cleared unless you typed in it. Accepted 0.5-60 mm and not above that limit; a larger value is rejected by name. *If blank:* The solver uses its default width, limited to $\pi \times$ port diameter / (2 $\times$ fin count), and names the value it used in the result.				

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Input	Identifier	Unit	Form limit	Form default
Fin Length	`fin_length`	mm	1 – 200	20
Effect on the calculation: Radial depth of each fin slot, measured from the port wall (mm). It sets how progressive the burn is in the finned section. When you switch the grain type, this box is cleared unless you typed in it. Accepted 1-200 mm and not above that limit; a larger value is rejected by name. *If blank:* The solver uses its default depth, limited to 0.85 × (grain outer radius − port radius), and names the value it used in the result.				
Finned Length Fraction	`finned_length_fraction`		0.05 – 1	(blank)
Effect on the calculation: Fraction of the grain length that carries the fins (dimensionless, 0.05-1); the rest of the grain is a plain cylindrical port. A larger fraction raises the initial burning area, so thrust at start-up and the curve shape change. Accepted range 0.05-1. *If blank:* Nothing is sent and the solver applies 0.4 (HRMA default, declared in the result under "The solver assumed these inputs").				
Number of Outer Holes	`wagon_hole_count`		3 – 12	(blank)
Effect on the calculation: Number of outer port holes around the central hole (whole number). More holes raise the initial burning area and the start-up thrust. Accepted range 3-12; holes that would intersect are rejected with the largest usable hole diameter. *If blank:* The solver uses 6 (HRMA selection, declared in the result).				
Hole Diameter	`wagon_hole_diameter`	mm	≥ 0.1	(blank)
Effect on the calculation: Diameter of every port hole, the central one and the outer ones (mm). It sets the initial port area and burning perimeter, so it changes the port-to-throat ratio, the start-up thrust and the burn time. Accepted: above 0.1 mm and below the limit where holes touch each other or the grain surface (the rejection names that limit). *If blank:* Core diameter / 2 (HRMA selection, declared in the result).				
Pitch Circle Diameter	`wagon_pitch_diameter`	mm	≥ 0.1	(blank)
Effect on the calculation: Diameter of the circle through the centres of the outer holes (mm). It sets how far the outer holes sit from the central hole and from the grain surface, so it changes when the webs between holes burn through. Accepted: 0.1 mm up to the grain outer diameter, with no hole intersecting another or the grain surface. *If blank:* Grain outer diameter / 2 (HRMA selection, declared in the result).				
Number of Slots	`slot_count`		2 – 16	6
Effect on the calculation: Number of longitudinal slots cut radially from the central port, whole number. More slots raise the initial burning area and so the early chamber pressure and thrust. Accepted range 2-16; a value outside it is rejected with a named warning. *If blank:* 6 (HRMA default, declared as assumed in the result).				
Slot Width	`slot_width`	mm	≥ 0.5	4
Effect on the calculation: Circumferential width of each slot (mm). Wider slots burn out sooner and shift the curve towards regressive. When you switch the grain type, this box is cleared unless you typed in it. Accepted 0.5-60 mm and not above that limit; a larger value is rejected by name. *If blank:* The solver uses its default width, limited to π × port diameter / (2 × slot count), and names the value it used in the result.				
Slot Depth	`slot_depth`	mm	≥ 1	20
Effect on the calculation: Radial depth of each slot, measured from the port wall (mm). Deep slots reach the case early and make the curve regressive; shallow slots stay progressive. When you switch the grain type, this box is cleared unless you typed in it. Accepted 1-200 mm and not above that limit; a larger value is rejected by name. *If blank:* The solver uses its default depth, limited to 0.85 × (grain outer radius − port radius), and names the value it used in the result.				

Table 10.5: Solid page: Inhibitor Coating

Input	Identifier	Unit	Form limit	Form default
inhibit_front	`inhibit_front`			unchecked
Effect on the calculation: Prevents burning on the forward end face. In a BATES stack the end faces are left EXPOSED: the growing core area offsets the shrinking end area, and the burn is neutral when each grain is (3D + d)/2 long (D outer, d core diameter; burn area at burnout equals burn area at ignition). Measured at 100/48 mm, 3 × 174 mm: Kn 222.7 -> 223.4, peak +8.3%. The default grain (3 × 133.3 mm) is shorter and burns regressive: Kn 235.0 -> 187.9 (-20.0%). With the outer surface inhibited, coating both ends makes the burn area grow (default grain: Kn 214.0 -> 236.0, peak +13.4%) – tick this only if you want that.				
inhibit_rear	`inhibit_rear`			unchecked
Effect on the calculation: Prevents burning on the aft end face. In a BATES stack the end faces are left EXPOSED: the growing core area offsets the shrinking end area, and the burn is neutral when each grain is (3D + d)/2 long (D outer, d core diameter; burn area at burnout equals burn area at ignition). Measured at 100/48 mm, 3 × 174 mm: Kn 222.7 -> 223.4, peak +8.3%. The default grain (3 × 133.3 mm) is shorter and burns regressive: Kn 235.0 -> 187.9 (-20.0%). With the outer surface inhibited, coating both ends makes the burn area grow (default grain: Kn 214.0 -> 236.0, peak +13.4%) – tick this only if you want that.				
inhibit_outer	`inhibit_outer`			checked
Effect on the calculation: Inhibits the outer cylindrical surface of the grain (case-bonded grain). Checked (default): the outer surface does not burn. Unchecked: the outer surface burns as well, the burning area grows and the web is consumed from both faces. It changes the burning area, the thrust profile and the burn time.				
Grain-to-Grain Gap	`grain_gap`	mm		2
Effect on the calculation: Spacing between individual grains. Allows for thermal expansion and prevents grain-to-grain bonding during storage.				
Insulation Thickness, t_{ins}	`insulation_thickness`	mm	0 – 100	3
Effect on the calculation: Thickness of thermal insulation layer between propellant and case. Protects case from high temperatures and provides thermal barrier.				

Table 10.6: Solid page: 3. Combustion Chamber and Casing

Input	Identifier	Unit	Form limit	Form default
Structural material catalogue - apply to	`material_catalog_target`			Case / chamber
Effect on the calculation: Options: Case / chamber; Nozzle.				
Chamber Diameter, D_c	`chamber_diameter`	mm		(blank)
Effect on the calculation: Case inner diameter in mm - the bore that holds the grain, liner and insulation (the grain outer diameter is its own field). It sizes the case wall, the case mass and the burst margin. A value below that minimum is rejected with the minimum named; the case is never enlarged silently. Accepted range 10-5000 mm. *If blank:* Derived as grain outer diameter + 2·(liner + insulation) and shown after Calculate.				
Case Internal Volume	`chamber_volume`	L	0.01 – 16000	(blank)
Effect on the calculation: Your stated free volume inside the case (case bore minus propellant), in litres. The solver derives this volume from the geometry and uses its own value; the number you type is only compared with it and a named warning appears when they disagree (measured: changing it changes only the warnings, not thrust, pressure or burn time). Accepted range 0.001-100000 L. *If blank:* (default): nothing is sent and no comparison is made.				

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Input	Identifier	Unit	Form limit	Form default
Case Material	`case_material`			Carbon steel (generic, A36-class)

Effect on the calculation: Material of the motor case. The case wall is sized against hoop stress with this material's yield strength (see the yield source below), the case and closure mass come from its density, and the case thermal verdict compares the computed case wall temperature with its service temperature limit (shown below the selection; where no limit is on record the verdict is not evaluated). Options come from the server material records; a name the server cannot resolve is rejected. Options: AISI 4130 low-alloy steel (normalized); Carbon steel (generic, A36-class); Stainless steel 304 (annealed); Stainless steel 316 (annealed); Aluminum 6061-T6; Titanium Ti-6Al-4V; Inconel 718 (aged); Aluminum 7075-T6; Aluminum 2024-T3; Stainless 17-4PH (H900); AISI 4340 low-alloy steel (normalized); Inconel 625 (annealed); Titanium CP Grade 2; Magnesium AZ31B-H24; Filament-wound composite (generic).

Yield Strength σy	`yield_strength_source`	MPa		From material

Effect on the calculation: Yield strength σy of the case material (MPa). It sizes the case wall (hoop stress with the safety factor) and sets the burst pressure and the burst margin. Source selector: "From material" uses the yield of the selected case material record for BOTH the wall sizing and the burst pressure (this box is then read-only and shows that value); "My value" sends this number, which sizes the wall, while the burst pressure still comes from the material record (a named warning says so when the two differ). Accepted range 10-3000 MPa. Options: From material; My value. *If blank:* In "My value" mode: nothing is sent and the material record is used.

Yield Strength σy, σ_y	`yield_strength`	MPa	10 – 3000	250

Effect on the calculation: Yield strength σy of the case material (MPa). It sizes the case wall (hoop stress with the safety factor) and sets the burst pressure and the burst margin. Source selector: "From material" uses the yield of the selected case material record for BOTH the wall sizing and the burst pressure (this box is then read-only and shows that value); "My value" sends this number, which sizes the wall, while the burst pressure still comes from the material record (a named warning says so when the two differ). Accepted range 10-3000 MPa. *If blank:* In "My value" mode: nothing is sent and the material record is used.

Safety Factor FoS, FoS	`safety_factor`		1.1 – 10	2.5

Effect on the calculation: Design safety factor of the case against yield, dimensionless: design pressure = maximum operating pressure × this factor. A larger factor gives a thicker, heavier case. Accepted range 1.1-10; a value outside it is rejected with a named warning. *If blank:* With Case Wall Thickness blank, HRMA sizes the wall to meet it; with a wall entered, the Safety tab checks that wall against it. 3.0 (HRMA default).

Liner Thickness, t_{lin}	`liner_thickness`	mm	0.1 – 100	2

Effect on the calculation: Thickness of the liner layer between grain and insulation, mm. Accepted range 0.1-100 mm; a value outside it is rejected with a named warning. *If blank:* It sets the liner ring mass (Liner Mass) and the thermal resistance between the hot gas and the case, so it changes the dry mass and the case temperature; with Chamber Diameter blank it also enlarges the derived case bore. 2 mm (HRMA default).

Liner Density	`liner_density`	kg/m^3	200 – 3000	1200

Effect on the calculation: Density of liner material for mass calculation. Phenolic: 1200-1400 kg/m^3, EPDM: 1200-1300 kg/m^3. Affects overall motor mass.

Initial Temperature, T_0	`initial_temp`	K	200 – 350	298

Effect on the calculation: Grain (propellant conditioning) temperature T0 at ignition, in K - not the chamber gas temperature. The burn rate is shifted with the exponential law a_T = a·exp(σp·(T0 − T_ref)) (σp: Temperature Coefficient field, T_ref: the record reference temperature), so chamber pressure, thrust, burn time and Isp change with it (measured on the KNDX example: +20 K changes all four). It also anchors the cold/hot-day sensitivity table. Accepted range 200-350 K. *If blank:* T0 = T_ref (no shift).

Case Wall Thickness, t_{case}	`case_thickness`	mm	2 – 100	(blank)

Effect on the calculation: Wall thickness of motor case. Must be sufficient to withstand internal pressure with adequate safety factor. Calculated from hoop stress formula or specified directly. *If blank:* Left blank, the solver sizes the wall itself from the Barlow thin-wall relation and declares that it did so.

Closure Bolt Count	`closure_bolt_count`		1 – 200	(blank)

Effect on the calculation: Number of bolts holding each closure against chamber pressure. Drives the separation and proof safety factors of the closure joint. *If blank:* Leave empty if unknown - the joint is then reported as "not sized" instead of guessed.

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Input	Identifier	Unit	Form limit	Form default
Closure Bolt Size	`closure_bolt_size`			M8
Effect on the calculation: Metric coarse thread size of the closure bolts (ISO 898-1 stress areas). Only used when a bolt count is given. Options: M4; M5; M6; M8; M10; M12; M14; M16; M20; M24.				
Closure Bolt Class	`closure_bolt_class`			8.8
Effect on the calculation: Bolt property class: 8.8 / 10.9 / 12.9 (ISO 898-1) or A2-70 stainless (ISO 3506-1). Sets proof strength and therefore the preload. Options: 8.8; 10.9; 12.9; A2-70.				

Table 10.7: Solid page: 4. Nozzle

Input	Identifier	Unit	Form limit	Form default
Throat Diameter Dt ((auto-filled after calculate)), D_t	`throat_diameter`	mm	1 – 500	(blank)
Effect on the calculation: Nozzle throat diameter (mm), computed by the solver; the box is read-only and is filled after Calculate. The effective throat area follows from the grain mass flow at the design chamber pressure, A_t = ρ_p·A_b·r(Pc)·c*/Pc, and the drawn (geometric) throat is A_t/Cd. This version does not take the throat as an input: to change it, change the chamber pressure or the grain.				
Exit Diameter De ((auto-filled after calculate)), D_e	`exit_diameter`	mm	1.5 – 2500	(blank)
Effect on the calculation: Nozzle exit diameter (mm), computed by the solver; the box is read-only and is filled after Calculate: A_e = ε·A_t, with ε the optimum expansion ratio at the design ambient pressure (isentropic, exit pressure = ambient). This version does not take the exit diameter as an input.				
Expansion Ratio ε **(auto-filled after calculate),** ε	`expansion_ratio`		1 – 25	(blank)
Effect on the calculation: Ratio of exit area to throat area (Ae/At), computed by the solver; the box is read-only and is filled after Calculate. Higher ratios give more thrust at altitude but risk flow separation at sea level. This version does not take the ratio as an input.				
Convergent Angle, α_c	`convergent_angle`	°	10 – 80	45
Effect on the calculation: Half-angle of the convergent section ahead of the throat, degrees. It shapes the drawn nozzle (CAD/STEP contour and nozzle mass); the performance solution does not use it. Accepted range 10-80°; a value outside it is rejected with a named warning. *If blank:* 30° (common default, Sutton & Biblarz, 9th ed.).				
Divergent Angle, α_d	`divergent_angle`	°	5 – 45	15
Effect on the calculation: Half-angle of the conical divergent section after the throat, degrees. It shapes the drawn nozzle and corrects the divergence efficiency against the 15° reference cone, $\lambda = (1 + \cos\alpha)/2$ (Sutton & Biblarz, 9th ed., Ch. 3), so it changes Isp and thrust. Accepted range 5-45°; a value outside it is rejected with a named warning. *If blank:* 15° (the reference cone, no correction).				
Nozzle Efficiency η**n,** η_n	`nozzle_efficiency`		0.8 – 1	0.95
Effect on the calculation: Overall nozzle efficiency, dimensionless: the base nozzle efficiency, covering divergence, friction and heat-transfer losses. It multiplies the thrust coefficient, so it lowers Isp and thrust; it takes precedence over Nozzle Discharge Efficiency (Cf). Accepted range 0.80-1.0; a value outside it is rejected with a named warning. *If blank:* Nozzle Discharge Efficiency (Cf) is used if filled, otherwise the propellant record's base nozzle efficiency.				
Erosion Factor	`erosion_factor`	mm/s	0 – 5	(blank)
Effect on the calculation: Throat erosion reference rate in mm/s at the erosion law reference pressure (the empirical model scales it to the chamber pressure; the reference pressure is named in the erosion result). A value above 0 is sent and the solver erodes the throat with it during the burn; 0 means no erosion (rigid throat by choice). There is no single typical value - see the nozzle material properties line. Accepted range 0-5 mm/s. *If blank:* (default): the solver keeps a rigid throat and, for a nozzle material with a published band (graphite), estimates the erosion and prints the rate it found at the chamber pressure in the warnings; for a material without a published band no erosion number is published.				

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Input	Identifier	Unit	Form limit	Form default
Nozzle Material	`nozzle_material`			Graphite (isostatic, nozzle insert grade)

Effect on the calculation: Material of the nozzle throat insert. The throat thermal check compares the adiabatic wall temperature at the throat (uncooled throat, worst case T_wall → T_aw) with the allowable temperature of this material record; above it the verdict is EXCEEDS_ALLOWABLE (critical warning). Throat erosion uses a published coefficient band only where one exists (shown below); otherwise a rigid throat is assumed and stated. Options come from the server material records; a name the server cannot resolve is rejected (no verdict is derived from another material). Phenolic is not offered: there is no sourced throat record and no ablation model. Options: Copper (OFHC, annealed); CuCrZr (C18150, aged); Graphite (isostatic, nozzle insert grade); Molybdenum TZM (stress-relieved); Tungsten (wrought, stress-relieved); Carbon-carbon (2D/3D C-C).

Chamber Pressure, P_c	`chamber_pressure`	bar		40

Effect on the calculation: Design chamber pressure, bar. The throat is sized so the grain burns at this pressure (A_t = ρ_p·A_b·r(Pc)·c*/Pc); it also sets the burn rate, the case wall loads and the nozzle expansion. A higher pressure raises Isp but needs a thicker, heavier case. Accepted range 1-500 bar. *If blank:* The calculation is refused and the field is named (no pressure is assumed).

Table 10.8: Solid page: 5. Efficiency and Empirical Coefficients

Input	Identifier	Unit	Form limit	Form default
Burn Rate Preset	`burn_rate_preset`			KNDX – KNO3/Dextrose 65/35 (Nakka fit)

Effect on the calculation: Burn-rate law of the selected propellant. KNDX/KNSB: the piecewise regime table (Nakka 1999/2001 fits) is read at the instantaneous chamber pressure, and a, n show the regime at the current design pressure. Custom: the single a-n pair in the boxes below is used (for a propellant without a regime table this is its catalogue pair). The preset follows the "Propellant (catalogue record)" selector; choosing another propellant's preset switches the selector to it, so a regime table can never belong to a different propellant (the solver rejects such a mix). Editing a or n manually switches to Custom. Options: Custom (manual a, n); KNDX – KNO3/Dextrose 65/35 (Nakka fit); KNSB – KNO3/Sorbitol 65/35 (Nakka fit).

Burn Rate Coefficient (a), a	`burn_rate_a`	m/s/ bar^n		0.0007876

Effect on the calculation: Burn-rate coefficient a of r [m/s] = a × (P [bar])^n. With the Burn Rate Preset on a propellant's regime table (KNDX, KNSB) the solver reads the table at the instantaneous pressure and this box only shows the coefficient of the current regime; with Custom it is the a used in every step. Picking a catalogue record fills it (see the catalogue row for each propellant's value); typing here switches the preset to Custom. *If blank:* The selected record's a is used.

Burn Rate Exponent (n), n	`burn_rate_n`		-0.5 – 1	0.688

Effect on the calculation: Pressure exponent n of the burn-rate law r = a·P^n, dimensionless. It sets how the burn rate, and so the chamber pressure, responds to pressure; as n approaches 1 the chamber pressure becomes increasingly sensitive to the burning area. KN-sugar plateau/mesa regimes can have a NEGATIVE n (burn rate drops with pressure) – this is physical, from published fits. Accepted range -0.5 to 1.0. *If blank:* The selected propellant's catalogue value (or its pressure-regime table) is used.

Combustion Efficiency η**c,** η_{c^*}	`combustion_efficiency`		0.5 – 1	0.95

Effect on the calculation: c* efficiency, dimensionless: the delivered c* is the theoretical c* (NASA CEA or the catalogue record) times this factor; it stands for incomplete combustion and heat loss. It lowers c* and with it Isp; the throat is sized with this c*. Accepted range 0.50-1.0; a value outside it is rejected with a named warning and factor 1 is used. *If blank:* Factor 1 (no combustion loss applied).

continued on next page

Input	Identifier	Unit	Form limit	Form default
Nozzle Discharge Efficiency (Cf), η_{C_F}	`cf_efficiency`		0.8 – 1	0.98
Effect on the calculation: Cf efficiency, dimensionless: the base nozzle efficiency, covering divergence, friction, heat-transfer and flow non-uniformity losses. It multiplies the thrust coefficient, so it lowers Isp and thrust. When Nozzle Efficiency ηn is filled, that value is used instead. Accepted range 0.80-1.0; a value outside it is rejected with a named warning. *If blank:* The propellant record's base nozzle efficiency is used.				
Overall Thrust Efficiency η**overall**	`overall_efficiency`		0.5 – 1	0.92
Effect on the calculation: Combined efficiency of entire propulsion system including combustion, nozzle, and two-phase flow losses. Product of all individual efficiencies.				
Discharge Coefficient Cd, C_d	`discharge_coeff`		0.7 – 1	0.98
Effect on the calculation: Throat discharge coefficient Cd, dimensionless: the drawn (geometric) throat is the effective throat divided by Cd. Chamber pressure and thrust are solved on the effective throat, so Cd changes only the reported geometric throat diameter and the thrust coefficient referred to it. Accepted range 0.70-1.0; a value outside it is rejected with a named warning. *If blank:* Cd = 1 (geometric throat = effective throat).				
Kinetic Efficiency, η_{kin}	`kinetic_efficiency`		0.8 – 1	0.97
Effect on the calculation: Kinetic efficiency, dimensionless: a separate factor for incomplete expansion and non-ideal gas effects in the nozzle. It multiplies the nozzle efficiency, so it lowers Isp and thrust. Accepted range 0.80-1.0; a value outside it is rejected with a named warning and factor 1 is used. *If blank:* Factor 1 (no kinetic loss applied).				
Divergence Loss Factor	`divergence_loss`		0 – 0.2	(blank)
Effect on the calculation: Divergence loss as a fraction (0.02 = 2%): thrust loss from the non-axial flow in the divergent cone. It is reported in the Isp loss breakdown only and is not applied again, because the base nozzle efficiency already contains it. Accepted range 0-0.20; a value outside it is rejected with a named warning. *If blank:* The breakdown derives it from the divergent half-angle of the cone.				
Two-Phase Loss Factor	`two_phase_loss`		0.8 – 1	(blank)
Effect on the calculation: Two-phase (particle lag) efficiency factor, dimensionless (1 = no loss): solid particles (alumina) in the exhaust expand less efficiently than the gas. A value you enter multiplies the total nozzle efficiency, so it lowers the delivered Isp and thrust. Accepted range 0.80-1.0; a value outside it is rejected with a named warning and no factor is applied. *If blank:* The solver applies no user two-phase factor (factor 1) and says so in the Isp loss breakdown; the two-phase estimate shown in the results is a diagnostic and is not applied.				

Table 10.9: Solid page: 6. Initial and Environmental Conditions

Input	Identifier	Unit	Form limit	Form default
Ignition Delay	`ignition_delay`	ms	1 – 1000	50
Effect on the calculation: Ignition delay (ms): time from the ignition signal to a significant pressure rise (igniter response and flame spreading). Reference only: no calculation reads it in this version, because the start-up transient is not modelled; the box is kept so that the value is saved with your project. The box accepts 1-1000 ms. *If blank:* Nothing is used.				
Igniter Target Pressure (P/Pc)	`igniter_pressure_fraction`		0.02 – 0.5	0.15
Effect on the calculation: Fraction of chamber pressure the igniter must reach in the free volume before the grain takes over (NASA SP-8051 free-volume criterion). This is a design choice, not a computed value; it sets the igniter charge mass.				
Atmospheric Pressure, P_a	`atm_pressure`	kPa	1 – 200	101.325
Effect on the calculation: Ambient pressure in kPa for which the nozzle expansion ratio is sized (exit pressure = ambient pressure at the design chamber pressure) and at which the thrust curve, specific impulse and the Isp at test ambient are solved. The Sea Level Isp card is always this nozzle at 101.325 kPa. Disabled while Test Altitude is above 0: the ISA pressure of that altitude is used instead and is shown under this field after Calculate. Accepted range 1-200 kPa. *If blank:* 101.325 kPa.				

continued on next page

Input	Identifier	Unit	Form limit	Form default
Test Altitude	`test_altitude`	m	0 – 30000	0
Effect on the calculation: Test or design altitude in m. Above 0 it sets the ambient pressure from the ISA (US Standard Atmosphere 1976): the nozzle expansion ratio is sized for that pressure and the thrust curve, specific impulse and the Isp at test ambient are solved there; the pressure used is shown under Atmospheric Pressure after Calculate. The Sea Level Isp card is always this nozzle at 101.325 kPa. Accepted range 0-30000 m. *If blank:* 0 or blank: the Atmospheric Pressure field is used.				
Ambient Temperature, T_a	`ambient_temp`	K	180 – 340	298
Effect on the calculation: Ambient air temperature in K for the case heat-transfer and structural temperature checks (outer convection). It does NOT change the burn rate, thrust, burn time or Isp - the grain temperature is the Initial Temperature field (measured on the KNDX example: +20 K changes only thermal/structural results). Accepted range 180-340 K (the server value gate rejects anything outside it by name). *If blank:* The solver's reference ambient 298 K (SOLID_THERMAL).				
Humidity	`humidity`	%	0 – 100	60
Effect on the calculation: Relative humidity of the ambient air (%) - informational only: no calculation reads it in this version (the solver lists it among the informational inputs it does not use; measured: changing it changes no result). Moisture uptake of hygroscopic KN propellants during storage is not modelled. *If blank:* Nothing is sent.				
Wind Speed	`wind_speed`	m/s	0 – 100	0
Effect on the calculation: Environmental wind speed for flight trajectory calculations. Affects rocket stability and trajectory but not motor performance. Used for launch safety analysis.				

Table 10.10: Solid page: 7. Mass and Weight Breakdown

Input	Identifier	Unit	Form limit	Form default
Propellant Mass	`propellant_mass`	kg	0 – 60000	4.473
Effect on the calculation: Total mass of solid propellant. Automatically calculated from grain geometry and propellant density. Critical for performance calculations and thrust-to-weight ratio.				
Case Mass	`case_mass`	kg	0.001 – 30000	2.5
Effect on the calculation: Mass of the case cylinder shell (kg), filled by the solver after Calculate: case wall thickness (the one you entered, or the one HRMA sized from the hoop stress) × case material density over the case inner length and diameter. The forward and aft closures are not included; they are in Closure Mass. Output only: the solver does not read this field; an edited value appears as a difference in the mass reconciliation table.				
Nozzle Mass	`nozzle_mass`	kg	0.001 – 10000	0.5
Effect on the calculation: Nozzle mass in kg, filled by the solver after Calculate: the drawn nozzle (converging section, throat and exit cone) with the same contour, wall thickness and material as the nozzle in the CAD (STEP/STL) package, integrated as frustum-annulus volume times the density of the selected Nozzle Material. It changes with the nozzle size and material, not with the case material. Mounting hardware is not included. Output only: the solver does not read this field; an edited value appears as a difference in the mass reconciliation table.				
Insulation Mass	`insulation_mass`	kg	0 – 12000	0.3
Effect on the calculation: Mass of the insulation ring (kg), filled by the solver after Calculate: a ring outside the liner with the Insulation Thickness you entered and the density in the Liner Density field (one density for insulation and liner). The liner is a separate field (Liner Mass). Output only: the solver does not read this field; an edited value appears as a difference in the mass reconciliation table.				
Liner Mass	`liner_mass`	kg		(blank)
Effect on the calculation: Mass of the liner ring between grain and insulation (kg), computed by the solver from the liner thickness and liner density fields. Output only: it is filled after Calculate and is not sent to the solver. Part of the motor dry mass.				
Avionics Mass	`avionics_mass`	kg	0 – 500	0.2
Effect on the calculation: Mass of the vehicle electronics (flight computer, sensors, telemetry, recovery) in kg. It is a vehicle item, not part of the motor: the motor solver does not read it and it is not added to the motor dry mass; the page shows it as its own row in the mass reconciliation table. The box accepts 0-500 kg. *If blank:* The row shows no value.				

continued on next page

Input	Identifier	Unit	Form limit	Form default
Closure Mass	`closure_mass`	kg	0.001 – 10000	0.8
Effect on the calculation: Closure mass (kg), filled by the solver after Calculate. It is the sum of two items: the forward and aft closures, taken as 30% of the case shell mass (a fixed HRMA ratio, not a closure-head analysis), and the igniter and mounting hardware, taken as 5% of the case structure (shell + closures; also an HRMA ratio). Output only: the solver does not read this field; an edited value appears as a difference in the mass reconciliation table.				
Total Dry Mass	`dry_mass`	kg	0.1 – 65000	4.100
Effect on the calculation: Motor dry mass (kg): sum of the motor component fields above (case + nozzle + insulation + liner + closures, the closure field including igniter and mounting). With the fields as filled by the solver it equals the solver motor dry mass. Avionics are a VEHICLE item, NOT part of the motor, and are not included. Output only: not used by the solver. The reconciliation table below compares it with the solver, component by component.				
Total Wet Mass	`wet_mass`	kg	0.1 – 130000	8.573
Effect on the calculation: Total loaded mass including all propellant (dry mass + propellant mass). Initial vehicle mass at ignition. Used for thrust-to-weight calculations.				

Table 10.11: Solid page: 8. Test Stand Planning (not used in the motor calculation)

Input	Identifier	Unit	Form limit	Form default
Pressure Sensor Range	`pressure_sensor_range`	MPa	0.1 – 100	10
Effect on the calculation: Full-scale range of the pressure transducer on your static-test stand (MPa). Test-stand planning only: no motor calculation reads it and no result changes. The box accepts 0.1-100 MPa. *If blank:* Nothing is used.				
Sampling Frequency	`sampling_freq`	Hz	1 – 100000	1000
Effect on the calculation: Data-acquisition sampling rate of your static-test stand (Hz); by the sampling theorem it must be more than twice the highest frequency you want to record. Test-stand planning only: no motor calculation reads it and no result changes. The box accepts 1-100000 Hz. *If blank:* Nothing is used.				
Load Cell Capacity	`load_cell_capacity`	N	1 – 20000000	5000
Effect on the calculation: Full-scale capacity of the thrust load cell on your static-test stand (N). Test-stand planning only: no motor calculation reads it and no result changes. The box accepts 1-20000000 N. *If blank:* Nothing is used.				
Calibration Factor	`calibration_factor`		0.1 – 10	1.0
Effect on the calculation: Calibration coefficient (dimensionless) that converts the raw sensor reading of your static-test stand to engineering units; it is found by calibrating with known loads or pressures. Test-stand planning only: no motor calculation reads it and no result changes. The box accepts 0.1-10. *If blank:* Nothing is used.				
Data Collection Time	`data_collection_time`	s	0.1 – 3600	10
Effect on the calculation: Total recording time of your static-test stand (s), including the baseline before ignition and the tail after burnout. Test-stand planning only: no motor calculation reads it and no result changes. The box accepts 0.1-3600 s. *If blank:* Nothing is used.				
Filter Cutoff Frequency	`filter_cutoff`	Hz	1 – 50000	100
Effect on the calculation: Cut-off frequency of the low-pass filter applied to your static-test stand signals (Hz). Test-stand planning only: no motor calculation reads it and no result changes. The box accepts 1-50000 Hz. *If blank:* Nothing is used.				
Uncertainty Level	`uncertainty_level`	%	0 – 100	2
Effect on the calculation: Stated measurement uncertainty of your static-test stand (%). Test-stand planning only: no calculation reads it, and the Monte Carlo analysis uses its own input tolerances (measured: changing it changes no result). The box accepts 0-100 %. *If blank:* Nothing is used.				

Table 10.12: Solid page: Trajectory Analysis inputs

Input	Identifier	Unit	Form limit	Form default
traj_initial_mass	`traj_initial_mass`			12
Effect on the calculation: Vehicle mass at lift-off (kg), the starting mass of the trajectory calculation. After a motor calculation it is filled with the motor's total mass from the solver unless you typed a value. Initial − final mass must match the propellant the motor burns (within the trajectory calculation's tolerance), otherwise no trajectory is computed and both masses are named. *If blank:* Empty or not a number: no trajectory is computed and the field is named (no value is assumed).				
traj_final_mass	`traj_final_mass`			5
Effect on the calculation: Vehicle mass after burnout (kg). After a motor calculation it is filled with the motor dry mass unless you typed a value; add airframe and payload to both masses to fly a real vehicle. Avionics, wiring and the recovery system belong in this mass: the engine dry (flying inert) mass does not contain them. *If blank:* Empty or not a number: the field is not sent and the trajectory refuses the run, naming the field (no value is assumed).				
traj_drag_coeff	`traj_drag_coeff`			0.5
Effect on the calculation: Vehicle drag coefficient (dimensionless) on the reference area; it scales the aerodynamic drag over the whole flight. Not filled from the motor calculation. *If blank:* Empty or not a number: no trajectory is computed and the field is named (no value is assumed).				
traj_ref_area	`traj_ref_area`			0.008
Effect on the calculation: Drag reference area (m^2). After a motor calculation it is filled with the motor case outer cross-section $\pi(D/2)^2$ unless you typed a value. *If blank:* Empty or not a number: no trajectory is computed and the field is named (no value is assumed).				
traj_burn_time	`traj_burn_time`			(blank)
Effect on the calculation: Burn time of this design, in seconds: the last time point of the thrust curve from the motor calculation. Not an input: the trajectory is flown on that whole thrust curve (thrust and propellant use follow F(t)), so this number only shows which burn is flown. *If blank:* Empty until a motor calculation has run; without one the trajectory is not computed.				
parachute_area	`parachute_area`		≥ 0	(blank)
Effect on the calculation: Projected (canopy) area of the recovery parachute (m^2); descent speed scales as $1/\sqrt{\text{area}}$. When given, the descent under the parachute is solved and the landing speed is reported. Accepted: greater than 0; 0 or a negative area is rejected with the reason. *If blank:* Empty: no parachute – the descent is not solved and the results say so (no area is assumed).				
parachute_cd	`parachute_cd`		≥ 0	(blank)
Effect on the calculation: Parachute drag coefficient referenced to its projected area (dimensionless). Read only when a parachute area is given. Accepted: greater than 0; otherwise it is rejected with the reason. *If blank:* Empty: the solver uses 1.4 (Hoerner, hemispherical canopy) and marks it as assumed.				
parachute_deploy_delay	`parachute_deploy_delay`		≥ 0	(blank)
Effect on the calculation: Time from apogee to canopy inflation (s); the vehicle falls ballistically during this delay. Read only when a parachute area is given. Accepted: 0 or more; a negative delay is rejected with the reason. *If blank:* Empty: the solver uses 2.0 s and marks it as assumed.				

10.4 Outputs

Table 10.13: Outputs of the solid page

Output	Unit	Meaning	Source / model
SEA LEVEL ISP, VACUUM ISP	s	Delivered specific impulse. The sea-level Isp is the same nozzle at 101.325 kPa; if a test altitude is given, the Isp in that ambient is shown on a separate card (Isp at test ambient, with altitude and pressure).	Internal-ballistics solution and the efficiency chain (5. EFFICIENCY AND EMPIRICAL COEFFICIENTS).
AVERAGE THRUST, PEAK THRUST	N	Average = total impulse / burn time.	Thrust curve.
PEAK CHAMBER PRESSURE	bar	Largest pressure of the curve.	Internal-ballistics equilibrium.
BURN TIME	s	From ignition to web burnout.	The full thrust curve.
TOTAL IMPULSE	kN · s	Integral of the thrust curve.	
C* EFFICIENCY, THRUST COEFFICIENT, PROPELLANT MASS	%, –, kg		Efficiency inputs; nozzle; grain volume $\times\rho_p$.
Throat diameter (machined, effective)	mm	Machined diameter and the effective sonic-area diameter with discharge coefficient.	Design pressure.
Case inner diameter, case length, total motor length	mm	The case bore is derived from the grain diameter (grain + 2 × liner).	Geometry.
Kn (initial, final)	–	Burning area / throat area.	Grain geometry.
Wall, closure	mm	Thickness sized by HRMA and the manufacturing floor.	Hoop stress, safety factor.
Charts	–	Motor cross-section, performance dashboard, thrust curve, chamber pressure, 3D CAD view, altitude performance.	Time-stepped internal ballistics.
Monte Carlo (manufacturing tolerances)	–	Performance distribution with a, n, density and c^* tolerances; Sample Count field (form default 300, accepted range 20–2000, rejected by name outside it).	Run Analysis. When the motor is recalculated the previous distribution is cleared and this is stated.
Side panels	–	Grain Section FEA, Trajectory Analysis, Flight Dynamics (6-DOF).	
NASA CEA Comparison	–	For catalogue propellants of known composition (APCP, KNDX, KNSB, KNSU).	Section 12.



MOTOR DESIGN

Parameter	Value	Unit
Throat Diameter (machined)	18.40	mm
Throat Diameter (effective sonic area, Cd)	18.22	mm
Exit Diameter	47.82	mm
Expansion Ratio	6.89	
Case Inner Diameter	85.0	mm
Case Length (chamber)	464.0	mm
Total Motor Length	547.6	mm
Grain Outer Diameter	75.0	mm
Core Diameter	32.0	mm
Grain Length	360.0	mm
Web Burnt	21.5	mm
Web Thickness (geometric)	21.5	mm
Kn (initial)	222	
Kn (final)	210	
C*	869	m/s
Nozzle Length	83.2	mm
Convergent Half Angle	45.0	°
Divergent Half Angle	15.0	°
Case wall	2.0 sized by HRMA (hoop stress requires 1.7 mm), raised to the manufacturing floor of 2.0 mm	mm
Closure head	analysis: dished 1.71 mm · manufacturing solid: flat 9.44 mm	

Burn time and average thrust use the whole computed curve: from ignition to web burnout (last curve point), including any part of the burn where the throat is no longer choked (see the warnings); average thrust = total impulse / burn time. The tail-off after burnout is not modelled (the curve drops to zero at burnout).

Figure 10.7: Solid MOTOR DESIGN table: case and total length, machined and effective throat diameter, burn-window declaration.



Figure 10.8: Solid performance analysis dashboard: flow, pressure distribution, thrust and chamber pressure, burning area and Kn.

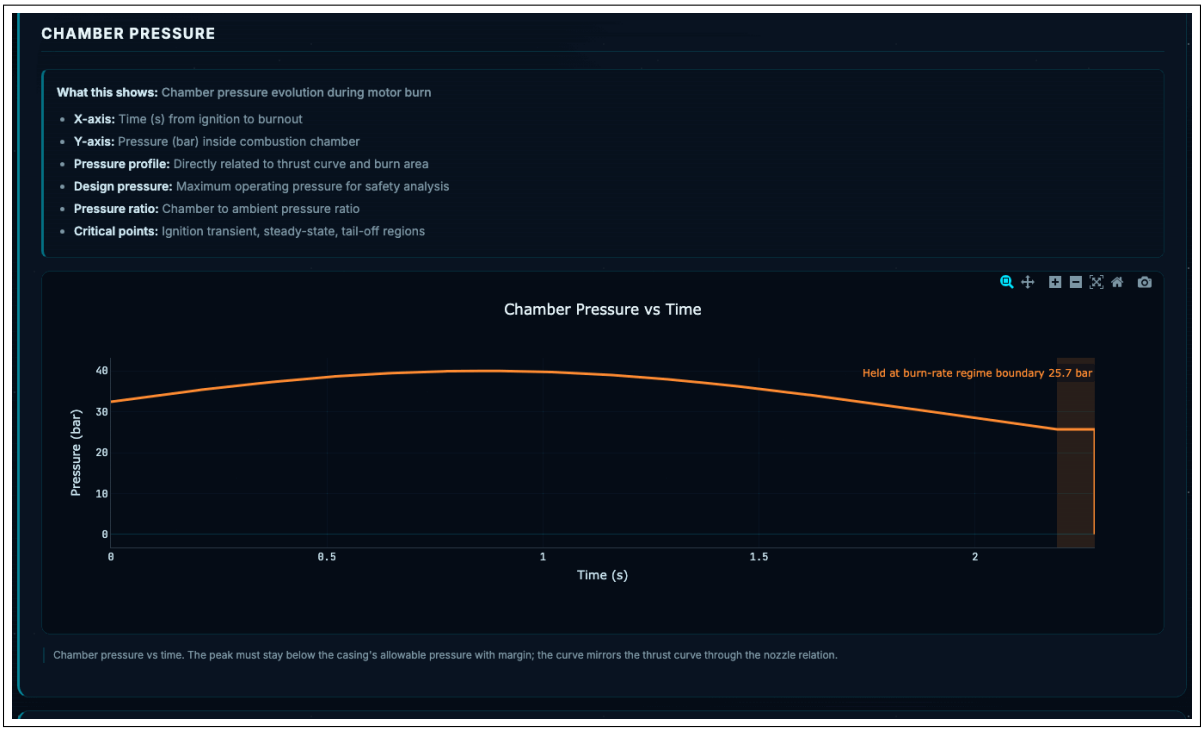


Figure 10.9: Chamber pressure curve.

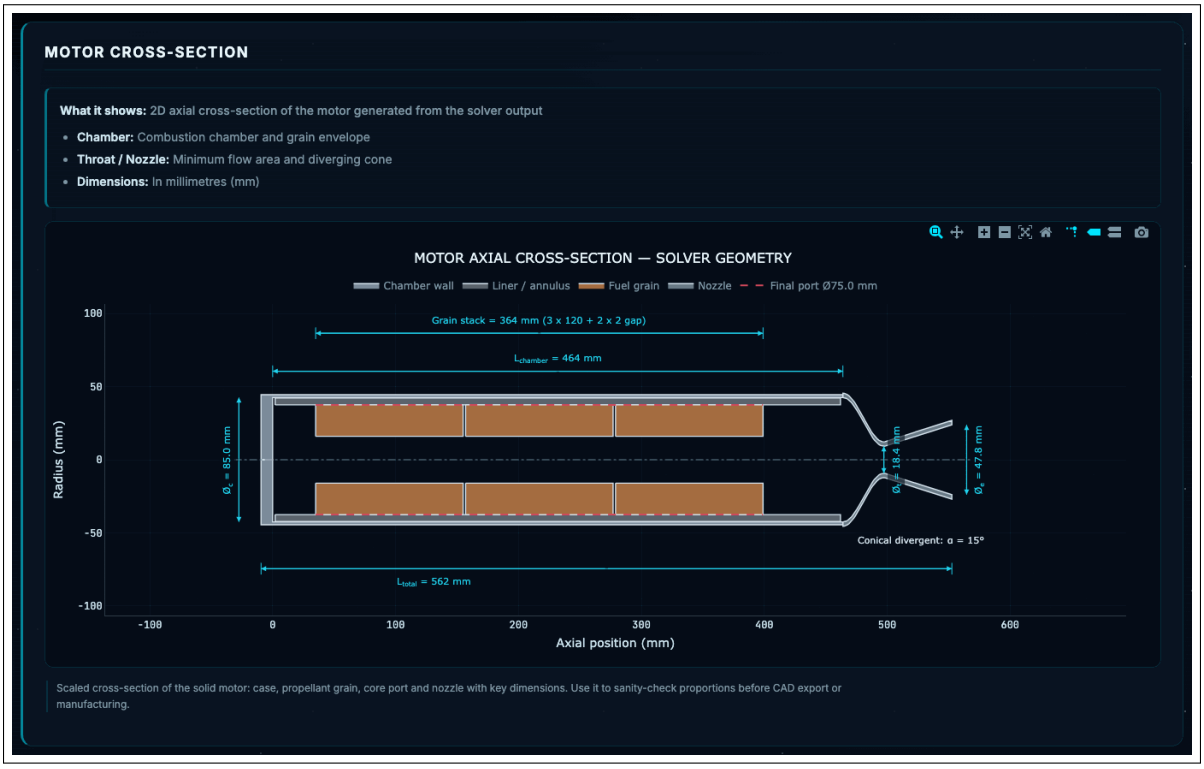


Figure 10.10: Solid motor cross-section.



Figure 10.11: Motor Specifications tabs: technical specifications.



Figure 10.12: Trajectory analysis (with the solid thrust curve).

10.5 Verdicts and warnings

On the solid page every non-critical line is printed with the CHECK tag (Table 17.1). Example lines and their reading are in Table 5.2. Gate rejections: a grain that does not fit the case, a core larger than the grain, or a port area not larger than the throat area is rejected by name; where applicable the smallest case that fits is stated.

10.6 Limits

- Below about 75 mm the delivered Isp can be optimistic; two-phase flow, heat loss and short-L* scale effects are not modelled.
- The two-phase correction is not verified and its status is declared.

- Throat erosion is estimated, not applied to the thrust curve.
- Star and finocyl grains are not exported to CAD; the export is refused with its reason.

10.7 Related sections

Example 2 (Section 5); Grain Section FEA (Section 13); canonical bands (Appendix E).

11. Liquid Engine Page

11.1 Purpose

Sizes the thermochemistry, chamber, nozzle, injector, feed cycle (Pressure-Fed, Gas Generator, Staged Combustion, Full-Flow Staged Combustion, Expander Cycle, Tap-Off Cycle), regenerative or film cooling and the wall of a bipropellant liquid engine; gives the throttle map and the feed-line pressure budget.

11.2 Screen

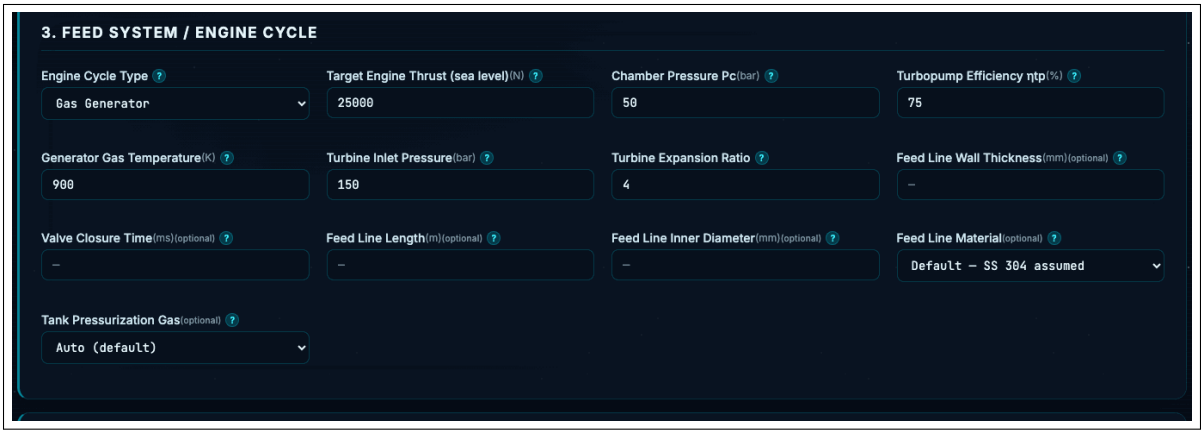


Figure 11.1: Liquid page: 3. FEED SYSTEM / ENGINE CYCLE. Note that the feed-pressure field is hidden in a turbopump cycle.

11.3 Inputs

In a regenerative chamber the wall has two layers: a channelled liner (Chamber Material and wall thickness) and a structural closeout shell (closeout material and thickness; HRMA sizes it if empty). The wall temperature field is a design target; the limit verdict uses the temperature the solver reaches.

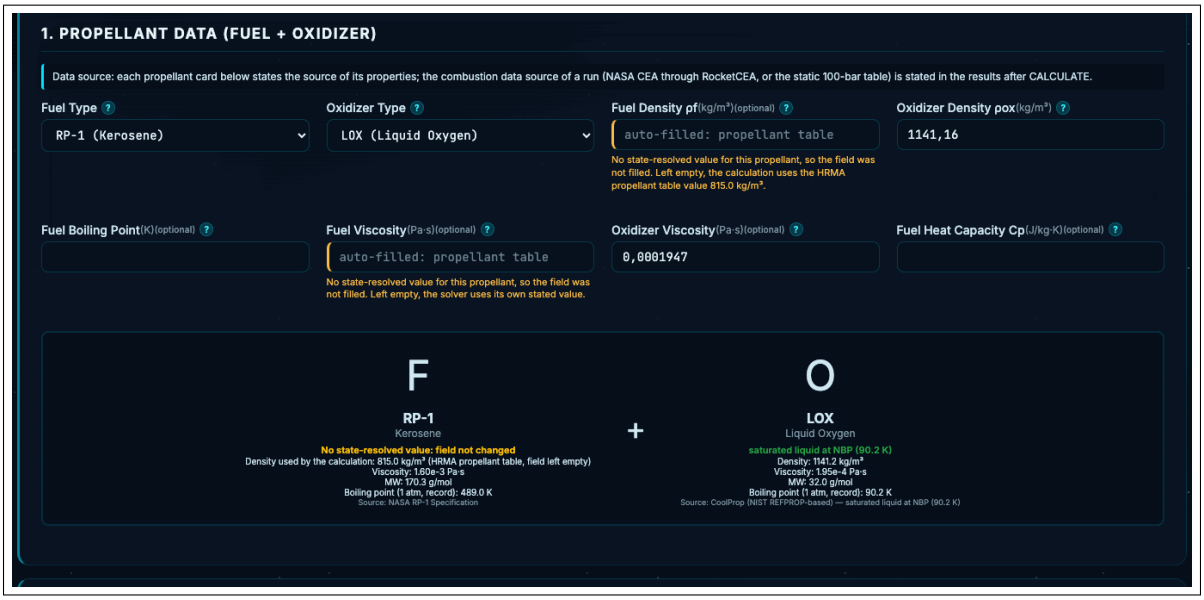


Figure 11.2: 1. PROPELLANT DATA (FUEL + OXIDIZER).

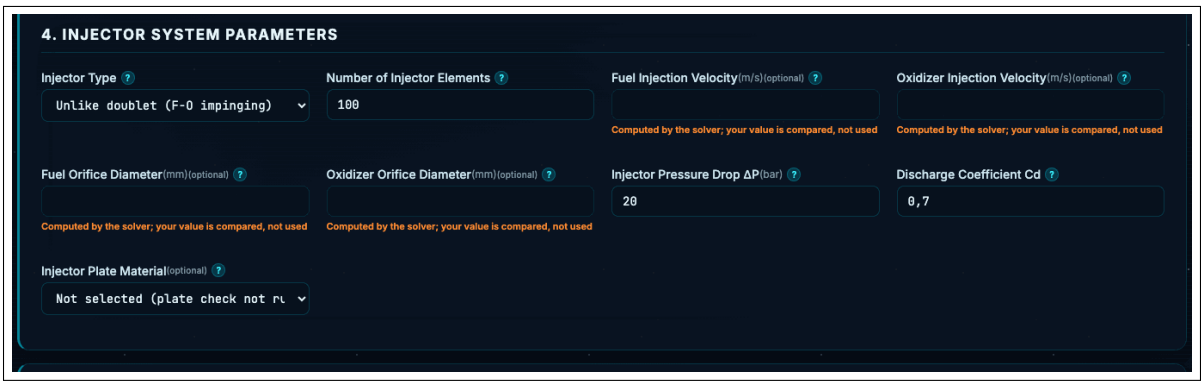


Figure 11.3: 4. INJECTOR SYSTEM PARAMETERS.



Figure 11.4: 5. COMBUSTION CHAMBER DESIGN: chamber material and wall (the liner in a regenerative chamber).



Figure 11.5: 6. NOZZLE & COOLING SYSTEM: closeout material and thickness, film cooling.



Figure 11.6: "?" note: closeout thickness (two-layer regenerative wall).

Table 11.1: Liquid page: 1. Propellant Data

Input	Identifier	Unit	Form limit	Form default
Fuel Type	`fuel_type`			RP-1 (Kerosene)
Effect on the calculation: The liquid fuel component. RP-1: refined kerosene (810 kg/m^3), high density, room temperature storable. LH2: cryogenic liquid hydrogen (71 kg/m^3), highest specific impulse, complex handling. Methane: storable at moderate pressure (422 kg/m^3), Mars missions. MMH/UDMH: hypergolic fuels for spacecraft. Options: RP-1 (Kerosene); LH2 (Liquid Hydrogen); Methane (CH4); MMH (Monomethylhydrazine); UDMH (Unsymmetrical Dimethylhydrazine).				
Oxidizer Type	`oxidizer_type`			LOX (Liquid Oxygen)
Effect on the calculation: The liquid oxidizer. LOX: cryogenic liquid oxygen. N2O4: storable nitrogen tetroxide, hypergolic with hydrazine fuels. Selects the propellant pair, so the combustion solution, the propellant densities and the tank sizing. Options: LOX (Liquid Oxygen); N2O4 (Nitrogen Tetroxide).				
Fuel Density ρf, ρ_f	`fuel_density`	kg/m^3	20 – 2500	(blank)
Effect on the calculation: Fuel density at operating temperature, kg/m^3. *If blank:* Left blank, the value is auto-filled from the propellant table when the state solution resolves; if it does not resolve, the field stays empty.				
Oxidizer Density ρox, ρ_{ox}	`oxidizer_density`	kg/m^3	20 – 2500	1141.16
Effect on the calculation: Oxidizer density at operating temperature. LOX: 1141, N2O4: 1443. Higher density oxidizers reduce tank volume and vehicle size, critical for spacecraft design. *If blank:* Left blank, the value is auto-filled from the propellant table when the state solution resolves; if it does not resolve, the field stays empty.				
Fuel Boiling Point	`fuel_boiling_point`	K	20 – 900	(blank)
Effect on the calculation: Fuel boiling point at 1 atm (K). Used only for the coolant-boiling check: if the regenerative coolant exit temperature exceeds it, a warning says the single-phase cooling model no longer applies. The propellant card above shows the property record's value for the selected fuel. A value left over from another fuel is cleared when the fuel changes. Accepted 20-900 K. *If blank:* Leave it EMPTY and the check is not run; the results say so.				

continued on next page

Input	Identifier	Unit	Form limit	Form default
Fuel Viscosity	`fuel_viscosity`	Pa·s	1e-07 – 10	(blank)
Effect on the calculation: Fuel dynamic viscosity at operating temperature. Affects injection pressure drop and atomization. RP-1: 0.0012, LH2: 0.000013, Methane: 0.00011, MMH: 0.00085. Lower viscosity improves injection. Accepted range: 1e-7 to 10 Pa·s (server band; outside it the request is rejected with a named 422). *If blank:* Left blank, the value is auto-filled from the propellant table when the state solution resolves; if it does not resolve, the field stays empty.				
Oxidizer Viscosity, μ_{ox}	`oxidizer_viscosity`	Pa·s	1e-07 – 10	0.0001947
Effect on the calculation: Oxidizer dynamic viscosity at operating temperature. LOX: 0.00019, N2O4: 0.00042. Higher viscosity increases injection pressure requirements and affects atomization quality. Accepted range: 1e-7 to 10 Pa·s (server band; outside it the request is rejected with a named 422). *If blank:* Left blank, the value is auto-filled from the propellant table when the state solution resolves; if it does not resolve, the field stays empty.				
Fuel Heat Capacity Cp	`fuel_heat_capacity`	J/kg·K	500 – 20000	(blank)
Effect on the calculation: Liquid fuel specific heat (J/kg·K). Used by the regenerative-jacket bulk coolant temperature rise. A value left over from another fuel is cleared when the fuel changes. Accepted 500-20000 J/kg·K. *If blank:* Leave it EMPTY and HRMA uses its propellant table value for the selected fuel (RP-1 2090, LH2 14300, methane 3480 J/kg·K; other fuels a declared 2000 J/kg·K fallback) and states the source in the results.				

Table 11.2: Liquid page: 2. O/F Optimization Data

Input	Identifier	Unit	Form limit	Form default
Target Mixture Ratio (O/F), O/F	`mixture_ratio`		0.5 – 20	(blank)
Effect on the calculation: Oxidizer-to-fuel mass flow ratio of the design point; used as entered. The ratio of maximum specific impulse depends on the propellant pair and the chamber pressure: HRMA solves it with NASA CEA at your chamber pressure and shows it under "Optimal O/F (computed)" after the run. Accepted 0.5-20. *If blank:* Required: left blank, the request is refused by name.				
O/F Range Minimum	`of_min`		0.2 – 20	2.0
Effect on the calculation: Lower end of the mixture-ratio (O/F) sweep of the performance map; the engine is solved again at each sweep point. It does not change the design point. Accepted 0.2-20. *If blank:* Empty: optimum O/F × 0.55 (at least 0.2), stated among the defaults applied.				
O/F Range Maximum	`of_max`		0.2 – 20	3.2
Effect on the calculation: Upper end of the mixture-ratio (O/F) sweep of the performance map; the engine is solved again at each sweep point. It does not change the design point. Accepted 0.2-20. *If blank:* Empty: optimum O/F × 1.45, stated among the defaults applied.				
Combustion Efficiency η**c,** η_{c^*}	`combustion_efficiency`	%	50 – 100	97
Effect on the calculation: Percentage of theoretical combustion achieved. Accounts for incomplete mixing, finite reaction time, heat losses. Well-designed engines: 95-99%. Affects c* and Isp directly.				

Table 11.3: Liquid page: 3. Feed System / Engine Cycle

Input	Identifier	Unit	Form limit	Form default
Engine Cycle Type	`engine_cycle`			Pressure-Fed
Effect on the calculation: Engine cycle: how the propellants reach the chamber. Pressure-fed: the tank pressure feeds the chamber, there is no turbopump. Gas generator and tap-off: a turbine branch drives the pumps and is exhausted separately (open cycle). Staged combustion, full-flow staged combustion and expander: the turbine flow goes on into the main chamber (closed cycle). Changes the feed system, turbopump and power-balance solution. Options: Pressure-Fed; Gas Generator; Staged Combustion; Full-Flow Staged Combustion; Expander Cycle; Tap-Off Cycle.				

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Input	Identifier	Unit	Form limit	Form default
Preburner Type	`preburner_type`			Fuel-Rich
Effect on the calculation: Staged combustion runs its preburner either fuel-rich (RS-25, RD-0120) or oxidizer-rich (RD-180, NK-33). This sets the turbine drive gas and the pump pressure ladder. Full-flow staged combustion uses both and needs no choice here. Options: Fuel-Rich; Oxidizer-Rich.				
Target Engine Thrust (sea level), F	`thrust`	N	100 – 10000000	(blank)
Effect on the calculation: Engine thrust at the sea-level design point, in N: ambient pressure 1.01325 bar (ISA sea level). For an open cycle (gas generator, tap-off) it is the ENGINE thrust = main chamber + turbine exhaust; the propellant flow, the throat and the tanks are sized from it. A vacuum thrust cannot be entered as the target: the vacuum thrust is computed from this design point and published separately. Accepted range: 100 N - 10 MN. *If blank:* This field is a required design point: left blank, the request is refused by name.				
Chamber Pressure Pc, P_c	`chamber_pressure`	bar	10 – 500	(blank)
Effect on the calculation: Combustion chamber operating pressure. Higher pressure improves Isp but increases complexity. Pressure-fed: 10-50 bar, Gas Generator: 50-150 bar, Staged Combustion: 150-300 bar. Accepted range: 10-500 bar. *If blank:* This field is a required design point: left blank, the request is refused by name.				
Feed System Pressure, P_{feed}	`feed_pressure`	bar	1 – 1000	130
Effect on the calculation: Pressure-fed cycle: the tank pressure (bar). HRMA sizes the tanks and the feed-line and valve budget from it and checks it against the pressure the chamber needs (Pc + injector drop + cooling-jacket and line losses); a lower value is refused by name and the required minimum is stated. Not used in pump-fed (turbopump) cycles: the field is hidden there and not sent. Accepted 1-1000 bar. *If blank:* Left blank in a pressure-fed cycle, the request is refused by name (no tank pressure is assumed).				
Turbopump Efficiency η**tp,** η_{tp}	`turbopump_efficiency`	%	20 – 95	75
Effect on the calculation: Combined efficiency of turbine and pump. Modern turbopumps: 65-85%. Higher efficiency reduces power requirements and improves overall engine performance.				
Generator Gas Temperature, T_{GG}	`generator_gas_temp`	K	400 – 2500	1000
Effect on the calculation: Turbine inlet gas temperature (gas generator or preburner), K. With the turbine pressure ratio it sets the turbine specific work, so the turbine flow needed for the pump power. Not used in the pressure-fed cycle. Accepted 400-2500 K. *If blank:* Empty: an assumed 1200 K is used and the turbopump result names it as assumed.				
Turbine Inlet Pressure, $P_{t,in}$	`turbine_inlet_pressure`	bar	1 – 1000	150
Effect on the calculation: Turbine inlet pressure (bar), turbopump cycles only. When the turbine expansion ratio is also entered, the solver uses that ratio and only compares this value with the inlet pressure its cycle balance needs (a warning names both). Accepted 1-1000 bar. *If blank:* Leave the expansion ratio empty and the ratio is solved from this pressure.				
Turbine Expansion Ratio, π_t	`turbine_expansion_ratio`		1.2 – 60	4
Effect on the calculation: Turbine pressure ratio (inlet / exit). With the turbine inlet temperature it sets the turbine specific work, so the turbine flow needed for the pump power. Not used in the pressure-fed cycle. Accepted 1.2-60. *If blank:* Empty: an assumed 8.5 is used.				
Feed Line Wall Thickness	`feed_line_wall_thickness`	mm	0.1 – 50	(blank)
Effect on the calculation: Feed-line pipe wall thickness. HRMA sizes the line diameter for flow velocity but computes no wall thickness anywhere, so this is a user input. Needed for the elastic-pipe wave speed and the pipe pressure-rating check. *If blank:* Leave empty if unknown - the water-hammer transient is then reported as NOT MODELLED instead of guessed.				

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Input	Identifier	Unit	Form limit	Form default
Valve Closure Time	`valve_closure_time_ms`	ms	0.1 – 600000	(blank)

Effect on the calculation: Main valve closure time for the water-hammer transient. Closure slower than the pipe period reduces the surge (Michaud). *If blank:* Leave empty - the analysis then assumes instantaneous closure (full Joukowsky surge, the conservative upper bound) and declares that assumption.

Feed Line Length	`feed_line_length_m`	m	0.05 – 50	(blank)

Effect on the calculation: Feed-line length from the tank outlet to the injector manifold. Together with the inner diameter it sets the feed-line inertance, which switches the chug (low-frequency) stability loop to its second-order form. *If blank:* Leave empty if unknown - the chug analysis then runs without line inertia and declares that assumption instead of guessing a layout.

Feed Line Inner Diameter	`feed_line_diameter_mm`	mm	0.5 – 500	(blank)

Effect on the calculation: Inner diameter of the feed line. The flow area for the feed-line inertance is derived from it as a circular duct. Accepted 0.5-500 mm. *If blank:* Leave empty if unknown - the chug analysis then runs without line inertia and declares that assumption.

Feed Line Material (optional)	`feed_line_material`			Default – SS 304 assumed

Effect on the calculation: Pipe material of the feed lines: sets the elastic modulus for the water-hammer wave speed and the strength for the pipe pressure-rating check. Leave on default - stainless 304 is then assumed and the assumption is declared in the output. Options: Default – SS 304 assumed; Stainless Steel 304; Stainless Steel 316; Aluminum 6061; Titanium Ti-6Al-4V; Inconel 718.

Tank Pressurization Gas (optional)	`pressurization_type`			Auto (default)

Effect on the calculation: Tank pressurization gas choice. Sets the relieving-gas identity of the tank relief-valve sizing (helium unless nitrogen is chosen). Autogenous pressurization is sized numerically only for methane/LOX and LH2/LOX turbopump configurations; elsewhere the choice is echoed with its declared limits. Leave on auto for the standard stored-helium assumption. Options: Auto (default); Helium; Nitrogen; Autogenous (propellant vapor).

Table 11.4: Liquid page: 4. Injector System Parameters

Input	Identifier	Unit	Form limit	Form default
Injector Type	`injector_type`			Unlike doublet (F-O impinging)

Effect on the calculation: Injector element family; it selects the injector circuit model and so the hole plan, the impingement or spray geometry and the atomisation correlation. Unlike doublet (F-O impinging): each fuel jet hits an oxidizer jet (fuel and oxidizer holes in pairs, momentum ratio checked). Like-on-like doublet (F-F / O-O): fuel jets hit fuel jets and oxidizer jets hit oxidizer jets; each circuit has an even hole count and the two fans mix afterwards (no momentum-ratio criterion). Coaxial swirl: oxidizer pressure-swirl element with a concentric fuel stream. Showerhead: parallel axial jets, no impingement. Pintle: one central element, fuel through radial tip holes and oxidizer through the annulus around the post. The field always carries a choice; an API request without it is solved as the unlike doublet and says so. Options: Unlike doublet (F-O impinging); Like-on-like doublet (F-F / O-O); Coaxial Swirl; Showerhead; Pintle; Unlike doublet (F-O impinging).

Number of Injector Elements, N_{el}	`injector_elements`		1 – 20000	100

Effect on the calculation: Number of injection elements (count). It fixes the hole count of the injector circuits; the hole diameters then follow from the flow areas. Unlike doublet, showerhead, coaxial: holes per circuit (one fuel + one oxidizer hole per element). Like-on-like doublet: same-fluid PAIRS per circuit (two holes each), 1-60 because the circuit planner allows at most 120 holes per circuit; more is rejected, not clamped. Not used for the pintle (a single element; the field is hidden). Accepted band 1-20000. *If blank:* Empty: each circuit plans the fewest holes inside its 0.5-2.5 mm diameter band and the answer says so.

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Input	Identifier	Unit	Form limit	Form default
Fuel Injection Velocity, v_f	`fuel_injection_velocity`	m/s	0.5 – 300	(blank)

Effect on the calculation: Fuel injection velocity (m/s). A value you enter changes no calculation: it is only compared with the solver value (deviation above 20 % is flagged). Accepted 0.5-300 m/s. *If blank:* HRMA computes it from the injector pressure drop, Cd and flow (orifice equation) and publishes it, so the field opens empty.

Oxidizer Injection Velocity, v_{ox}	`oxidizer_injection_velocity`	m/s	0.5 – 300	(blank)

Effect on the calculation: Oxidizer injection velocity (m/s). A value you enter changes no calculation: it is only compared with the solver value (deviation above 20 % is flagged). Accepted 0.5-300 m/s. *If blank:* HRMA computes it from the injector pressure drop, Cd and flow (orifice equation) and publishes it, so the field opens empty.

Fuel Orifice Diameter	`fuel_orifice_diameter`	mm	0.05 – 50	(blank)

Effect on the calculation: Fuel orifice diameter (mm). A value you enter changes no calculation: it is only compared with the solver value (deviation above 20 % is flagged). Accepted 0.05-50 mm. *If blank:* HRMA sizes the orifices from flow, pressure drop, Cd and element count and publishes them, so the field opens empty.

Oxidizer Orifice Diameter	`oxidizer_orifice_diameter`	mm	0.05 – 50	(blank)

Effect on the calculation: Oxidizer orifice diameter (mm). A value you enter changes no calculation: it is only compared with the solver value (deviation above 20 % is flagged). Accepted 0.05-50 mm. *If blank:* HRMA sizes the orifices from flow, pressure drop, Cd and element count and publishes them, so the field opens empty.

Injector Pressure Drop ΔP, ΔP_{inj}	`injector_pressure_drop`	bar	0.2 – 200	20

Effect on the calculation: Absolute pressure drop across the injector, in bar. A number you type is used as bar and labelled "user input (injector pressure drop)". Accepted 0.2-200 bar: a value outside that band is refused by name and no design is published. A drop with dP/Pc outside 0.05-0.40 is calculated but warned (SP-8089 guideline). A larger drop improves mixing and stability but raises the required feed pressure. *If blank:* Leave it EMPTY and HRMA sets the drop as chamber pressure x a dP/Pc factor of the injector type (impinging 0.22, coaxial 0.18, showerhead 0.28, pintle 0.15, other types 0.20); the result labels it "NASA SP-8089 dP/Pc ratio for the element type".

Discharge Coefficient Cd, C_d	`discharge_coefficient`		0.2 – 1	0.7

Effect on the calculation: Accounts for real flow effects in orifices. Sharp-edged orifices: 0.6-0.65, Well-rounded: 0.8-0.85. Higher Cd reduces required pressure drop for given flow rate.

Injector Plate Material (optional)	`injector_material`			Not selected (plate check not run)

Effect on the calculation: Material of the injector face plate. Used in the injector plate structural check: a clamped circular plate (Roark Table 11.2, case 10b) of the chamber diameter under the injector pressure drop, with ASME PG-52 ligament efficiency from the fuel + oxidizer hole plan (all holes at the larger diameter, conservative). It gives the plate thickness that meets the Structural Safety Factor from the room-temperature yield strength; the page has no plate-thickness input. That sized plate is also the injector mass in the engine dry mass and the plate the CAD draws (one plate, one number). Not run for pintle, coaxial and swirl elements (no drilled-hole plan; the result says why). An unknown name is rejected. Options: Not selected (plate check not run); AISI 4130 low-alloy steel (normalized); Carbon steel (generic, A36-class); Stainless steel 304 (annealed); Stainless steel 316 (annealed); Aluminum 6061-T6; Titanium Ti-6Al-4V; Inconel 718 (aged); CuCrZr (C18150, aged); Aluminum 7075-T6; Aluminum 2024-T3; Stainless 17-4PH (H900); AISI 4340 low-alloy steel (normalized); Inconel 625 (annealed); Titanium CP Grade 2; Beryllium copper C17200 (aged, AT); Brass C36000 (free-cutting, H02); Magnesium AZ31B-H24. *If blank:* Empty: the check is not run and the injector is not weighed (not in the dry mass).

Pintle Post Diameter D_p	`pintle_diameter`	mm	≥ 0	(blank)

Effect on the calculation: Outer diameter of the pintle post at its tip (D_p, mm); the fuel tip holes are drilled into it and the oxidizer flows through the annulus around it. With the annulus area fixed by the oxidizer circuit, D_p sets the annular gap t, and through the blockage factor the tip-hole count. Must be a finite positive length; the annulus bore D_p + 2t must fit inside the chamber. *If blank:* Empty: sized from the oxidizer flow with the annulus rule t = 0.05·D_p (HRMA choice, no literature value).

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Input	Identifier	Unit	Form limit	Form default
Oxidizer Annular Gap t	`annular_gap`	mm	≥ 0	(blank)
Effect on the calculation: Radial gap of the oxidizer annulus around the pintle post (t, mm). The annulus area pi·(D_p + t)·t must equal the oxidizer circuit flow area: given alone, D_p is solved from it; given with D_p, the two must agree within 5% (HRMA tolerance) or the design is rejected with the mismatch. *If blank:* Empty: solved from the oxidizer flow area.				
Fuel Tip Hole Count n (optional)	`pintle_radial_hole_count`		≥ 1	(blank)
Effect on the calculation: Number of radial fuel holes in the single row on the pintle tip (whole number ≥ 1). The holes carry the fuel circuit flow area n·pi·d^2/4, so the hole diameter follows from n. *If blank:* Empty: from the blockage factor n·d/(pi·D_p) = 0.58 (HRMA choice, no literature value).				
Fuel Tip Hole Diameter d	`pintle_radial_hole_diameter`	mm	≥ 0	(blank)
Effect on the calculation: Diameter of each radial fuel tip hole (d, mm). Given alone, the hole count is the nearest whole number for this diameter and the published diameter is the one that delivers the fuel circuit area exactly with that count; a plan more than 5% off (also with a given count) is rejected with the nearest valid plans. *If blank:* Empty: solved from the fuel flow area.				
Skip Distance L_s	`skip_distance`	mm	≥ 0	(blank)
Effect on the calculation: Axial distance from the annulus exit (injector face) to the axis of the tip-hole row (L_s, mm; definition Dressler & Bauer, AIAA 2000-3871). Geometry and CAD only: no performance model uses it. Must be shorter than the chamber. *If blank:* Empty: L_s = D_p (HRMA choice, no literature value).				

Table 11.5: Liquid page: 5. Combustion Chamber Design

Input	Identifier	Unit	Form limit	Form default
Chamber Diameter Dc, D_c	`chamber_diameter`	mm	10 – 5000	(blank)
Effect on the calculation: Inner diameter of the combustion chamber, mm. Takes priority over the Contraction Ratio: with the solved throat it fixes the chamber-to-throat area ratio, which must fall within 1.5-60, otherwise the design is refused naming this field. Accepted 10-5000 mm. *If blank:* Empty: the chamber diameter comes from the Contraction Ratio or, if that is also empty, from an estimate based on the solved throat diameter.				
Characteristic Length L*, L^*	`characteristic_length`	m	0.3 – 5	1.2
Effect on the calculation: L* = Vc/At, where Vc is chamber volume and At is throat area. Indicates combustion residence time. Higher L* improves combustion efficiency.				
Contraction Ratio εc, ε_c	`contraction_ratio`		1.5 – 60	4
Effect on the calculation: Chamber-to-throat area ratio (Ac/At). Sets the chamber diameter from the solved throat (Dc = Dt·$\sqrt{}$(Ac/At)), so the chamber gas velocity and the chamber length needed for L*. If Chamber Diameter is also entered, the diameter wins and this value is only compared. Accepted 1.5-60. *If blank:* Empty: estimated from the solved throat diameter (a warning says so if the estimate is limited).				
Chamber Wall Thickness, t_w	`chamber_wall_thickness`	mm	0.2 – 500	(blank)
Effect on the calculation: Chamber wall thickness, mm. With Regenerative cooling this is the LINER: the hot-gas wall from the gas side to the channel base (the hot wall of the thermal chain and of the liner stress check); the coolant pressure is carried by the closeout (Closeout Thickness). With other cooling types it is the single structural wall that carries the chamber pressure. Server accepted range 0.2-500 mm. *If blank:* Empty: Regenerative - HRMA uses a 1 mm liner and says so; other types - HRMA sizes the wall for the chamber pressure with your safety factor (next standard plate).				

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Input	Identifier	Unit	Form limit	Form default
Chamber Material	`chamber_material`			Stainless steel 304 (annealed)
Effect on the calculation: Chamber wall (liner) material. The options and the properties shown under the selector come from the HRMA materials record that the calculation reads: the chamber structural check (yield strength derated at the wall temperature) and the cooled-wall station march (conductivity, melting point) use this record. The same material is used along the cooled nozzle; a separate nozzle material analysis is not in this version. Options: Stainless steel 304 (annealed); Stainless steel 316 (annealed); Inconel 718 (aged); Inconel 625 (annealed); Aluminum 6061-T6; Copper (OFHC, annealed); CuCrZr (C18150, aged); AISI 4130 low-alloy steel (normalized).				
Chamber Surface Roughness, R_a	`chamber_roughness`	μm Ra	0.05 – 100	3.2
Effect on the calculation: Internal surface roughness affects heat transfer and flow. Smooth surfaces (Ra < 1.6) reduce heat transfer but improve flow. Rougher surfaces enhance cooling but increase losses.				

Table 11.6: Liquid page: 6. Nozzle and Cooling System

Input	Identifier	Unit	Form limit	Form default
Cooling Type	`cooling_type`			Regenerative (Fuel Cooling)
Effect on the calculation: Method for managing high chamber temperatures (3000K+). Regenerative: fuel/coolant channels, best performance, complex. Ablative: sacrificial liner, simpler, limited reuse. Film: propellant film cooling. Radiation: space engines only. Options: Regenerative (Fuel Cooling); Ablative Liner; Film Cooling; Radiation Cooling; Dump Cooling.				
Nozzle Expansion Ratio εe, ε	`nozzle_expansion_ratio`		1.5 – 300	(blank)
Effect on the calculation: Exit area to throat area ratio (Ae/At). It sets the nozzle exit and with it the thrust coefficient: a larger ratio raises the vacuum specific impulse but over-expands the nozzle at sea level, and at a low chamber pressure the flow separates and the sea-level thrust the engine is sized for collapses (the results warn about separation). Accepted 1.5-300. *If blank:* Leave it empty: the solver matches the exit pressure to sea-level ambient pressure (ambient-matched; the results name the source of the ratio).				
Nozzle Type	`nozzle_type`			Conical (15° half-angle)
Effect on the calculation: Nozzle geometry type. Conical: simple, 15° half-angle. Bell: optimized contour, shorter. Aerospike: altitude compensation, complex. Dual-bell: two expansion ratios. Extendable: deployable for space. Options: Conical (15° half-angle); Bell Nozzle (80% length); Bell Nozzle (60% length); Aerospike; Dual-Bell.				
Nozzle Throat Diameter, D_t	`throat_diameter`	mm	1 – 3000	(blank)
Effect on the calculation: Nozzle throat diameter (mm). A value you enter changes no calculation: it is only compared with the solved throat (the result shows the solver value next to your entry and flags a deviation above 20 %). Accepted 1-3000 mm. *If blank:* HRMA sizes the throat itself from thrust, chamber pressure and c* (mass balance) and publishes it in the results, so the field opens empty.				
Cooling Channel Count (optional)	`cooling_channels`		4 – 2000	(blank)
Effect on the calculation: Number of regenerative cooling channels around the chamber. Changes the flow area per channel, so the coolant velocity, the coolant-side heat transfer and the jacket pressure drop. Accepted 4-2000. *If blank:* Empty: computed from the geometry (as many channels of the channel width plus land as fit around the channel-base circumference at the throat).				

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Input	Identifier	Unit	Form limit	Form default
Closeout Material (optional)	`closeout_material`			HRMA default (Inconel 718)

Effect on the calculation: Material of the closeout: the outer shell that covers the cooling channels of a regenerative wall and carries the coolant pressure (NASA SP-8087: electroformed or brazed closure). Shown with Regenerative cooling only. Options: HRMA default (Inconel 718); Inconel 718; Inconel 625; Stainless Steel 304; Stainless Steel 316. *If blank:* Changes the closeout hoop check (yield at the closeout temperature, the highest coolant temperature) and, when Closeout Thickness is empty, the sized closeout thickness and the chamber mass; it does not change the thermal chain (the liner, Chamber Material, is the hot-gas wall). Empty: HRMA default Inconel 718, stated in the result.

Closeout Thickness	`closeout_thickness`	mm	0.2 – 500	(blank)

Effect on the calculation: Closeout shell thickness, mm. Changes the closeout hoop stress (coolant pressure x outer radius / thickness) and its safety factor, the drawn wall stack (liner + channel depth + closeout) and the chamber mass. Accepted range 0.2-500 mm. Shown with Regenerative cooling only. *If blank:* Empty: HRMA sizes it - hoop under the jacket coolant pressure at the largest jacket radius, yield at the closeout temperature / safety factor, 2 mm manufacturing floor, next standard plate - and states 'sized by HRMA'.

Coolant Flow Rate (% of fuel flow)	`coolant_flow_percent`		5 – 100	100

Effect on the calculation: Share of the fuel flow routed through the regenerative cooling channels, %. 100 % sends all fuel through the channels; a lower value sends the rest around the jacket. Changes the coolant mass flow, so the coolant velocity, its temperature rise and the jacket pressure drop. Accepted 5-100 %. *If blank:* Empty: 100 %.

Film Cooling (% of fuel flow)	`film_cooling_percent`		0 – 30	0

Effect on the calculation: Film coolant flow as a percentage of the TOTAL fuel flow (burning core fuel + film). The film is fuel bled before the injector and laid along the chamber wall; it does not burn. Accepted range 0-30 %. Effect: at constant thrust the film flow is ADDED to the core flow, so the total mass flow rises and the delivered Isp drops (Isp_core x core flow / (core + film); the film's thrust contribution is neglected, a conservative lower bound). The film also absorbs part of the chamber wall heat. 0 % with the Film Cooling type would leave the wall without any coolant and is rejected by name (no design is computed). *If blank:* Empty: with the Film Cooling type HRMA applies a declared 5 % default; with other cooling types there is no film.

Coolant Inlet Temperature, $T_{cool,in}$	`coolant_inlet_temp`	K	20 – 700	300

Effect on the calculation: Coolant temperature at the cooling jacket inlet, K. Sets the coolant properties along the channels and its temperature rise budget. Accepted 20-700 K. *If blank:* Empty: 115 K for methane and 25 K for LH2 (normal boiling point plus pump heating, NIST WebBook), 300 K for the other fuels; the value used is stated.

Maximum Wall Temperature, $T_{w,max}$	`max_wall_temp`	K	300 – 3000	800

Effect on the calculation: Maximum allowable hot-side wall temperature. Limited by material properties and coking (for hydrocarbon fuels). Steel: 800K, Inconel: 1200K, Copper: 700K. Critical for engine life.

Table 11.7: Liquid page: 7. Throttling and Burn Duration

Input	Identifier	Unit	Form limit	Form default
Minimum Throttle (% of max thrust)	`min_throttle`		5 – 100	40

Effect on the calculation: Minimum throttle level as percentage of maximum thrust. Deep throttling enables precision maneuvers and landing. Gas generators: 40-100%, Staged combustion: 20-100%. Limited by combustion stability.

Maximum Burn Duration, t_b	`max_burn_duration`	s	0.1 – 100000	(blank)

Effect on the calculation: Burn time (s). Sets the propellant mass, the tank sizing, the total impulse and the OpenRocket .eng file, and is passed to the analysis panels (thermal, safety, structural, performance). Accepted 0.1-100000 s. *If blank:* Leave it empty if unknown: none of those is computed, no burn time is assumed, and a panel that needs it refuses, naming the missing burn time or propellant mass.

Table 11.8: Liquid page: 8. Mass and Structural Parameters

Input	Identifier	Unit	Form limit	Form default
Target Thrust-to-Weight T/W (optional)	`target_thrust_to_weight`		1 – 500	(blank)
Effect on the calculation: Target engine thrust-to-weight. A value you enter changes no calculation: it is only compared with the computed T/W (deviation above 20 % is flagged). Accepted 1-500. *If blank:* HRMA computes the engine T/W from its own dry-mass estimate and publishes it, so the field opens empty.				
Structural Safety Factor, FoS	`safety_factor`		1.1 – 10	2.5
Effect on the calculation: Design margin for structural loads. Higher factors increase reliability but add mass. Human-rated: 4.0, Unmanned: 2.5-3.0, Expendable: 1.4-2.0. Covers pressure, thermal, and dynamic loads.				
Closure Bolt Count (optional)	`closure_bolt_count`		1 – 200	(blank)
Effect on the calculation: Number of bolts holding the injector/closure flange against chamber pressure. Drives the separation and proof safety factors of the closure joint. *If blank:* Leave empty if unknown - the joint is then reported as "not sized" instead of guessed.				
Closure Bolt Size	`closure_bolt_size`			M8
Effect on the calculation: Metric coarse thread size of the closure bolts (ISO 898-1 stress areas). Only used when a bolt count is given. Options: M4; M5; M6; M8; M10; M12; M14; M16; M20; M24.				
Closure Bolt Class	`closure_bolt_class`			8.8
Effect on the calculation: Bolt property class: 8.8 / 10.9 / 12.9 (ISO 898-1) or A2-70 stainless (ISO 3506-1). Sets proof strength and therefore the preload. Only used when a bolt count is given. Options: 8.8; 10.9; 12.9; A2-70.				
Engine Mount Type	`engine_mount`			2-Axis Gimbal Mount
Effect on the calculation: How the engine is attached to the vehicle. Used only by the "Gimbal Mount Loads" panel in the results (a fixed mount gives a not-applicable verdict there); the engine calculation does not read it. Options: 2-Axis Gimbal Mount; 1-Axis Gimbal Mount; Fixed Mount; Flexible Joint.				
Gimbal Range	`gimbal_range`	± degrees	0 – 45	8
Effect on the calculation: Maximum gimbal deflection angle (± degrees). Used only by the "Gimbal Mount Loads" panel in the results (side force = thrust x sin(angle)); the engine calculation does not read it. *If blank:* Left blank, the panel states the missing input instead of assuming one.				
Gimbal Actuator Arm	`gimbal_actuator_arm`	m	≥ 0	(blank)
Effect on the calculation: Perpendicular distance from the gimbal pivot axis to the actuator line of action (moment arm). Sets the actuator stroke and the force needed to react a given moment. *If blank:* Leave blank if unknown - no default is assumed and the load chain is not computed.				
Mount Ring Offset	`gimbal_ring_offset`	m	≥ 0	(blank)
Effect on the calculation: Axial distance from the gimbal pivot to the mount ring plane. The side force acting over this arm bends the ring. *If blank:* Leave blank for a ring in the pivot plane (h = 0), where the bending moment is zero by construction.				
Mount Bolt Circle Diameter	`gimbal_bolt_circle`	mm	≥ 0	(blank)
Effect on the calculation: Diameter of the bolt circle carrying the engine on the mount ring. Together with the bolt count it converts the ring moment into a per-bolt load. *If blank:* Leave blank to skip the per-bolt loads.				
Mount Bolt Count (optional)	`gimbal_bolt_count`		≥ 3	(blank)
Effect on the calculation: Number of equally spaced bolts on the bolt circle. The bolt-circle formula needs at least 3; fewer are rejected rather than extrapolated. *If blank:* Leave blank to skip the per-bolt loads.				
Expected Engine Life	`engine_life_cycles`	cycles	1 – 100000	10
Effect on the calculation: Number of start/stop cycles before refurbishment. Reusable engines need higher life. Expendable: 1-2 cycles, Reusable booster: 10-100 cycles, Spacecraft: 50-1000 cycles.				

Table 11.9: Liquid page: Trajectory Inputs

Input	Identifier	Unit	Form limit	Form default
traj_initial_mass	`traj_initial_mass`			5000
Effect on the calculation: Vehicle mass at lift-off, in kg. After an engine calculation (and again when the trajectory is computed), unless you typed in this field, it is filled from the flown vehicle: the engine flying inert mass (engine dry + propellant tank structure + reserve propellant, the same number the Launch Site flies) + the propellant mass. If the engine publishes no propellant mass (no Maximum Burn Duration entered) the field is emptied and the trajectory asks for it by name; no mass is assumed. Initial minus final mass is the propellant burned; the burn time is that mass divided by the engine mass flow. Must be greater than the final mass, otherwise no trajectory is computed.				
traj_final_mass	`traj_final_mass`			2000
Effect on the calculation: Vehicle mass after burnout, in kg. After an engine calculation (and again when the trajectory is computed), unless you typed in this field, it is filled with the engine flying inert mass (engine dry + propellant tank structure + reserve propellant; the same number the Launch Site and the 6-DOF panel use). Add airframe and payload to both masses to fly a real vehicle; a value you type is kept. Avionics, wiring and the recovery system belong in this mass: the engine dry (flying inert) mass does not contain them. Must be smaller than the initial mass. *If blank:* Empty: the trajectory asks for it by name (no mass is assumed).				
traj_drag_coeff	`traj_drag_coeff`			0.5
Effect on the calculation: Vehicle drag coefficient (dimensionless) on the reference area; it scales the aerodynamic drag over the whole flight. Not filled from the engine calculation. *If blank:* Empty or not a number: no trajectory is computed and the field is named (no value is assumed).				
traj_ref_area	`traj_ref_area`			0.05
Effect on the calculation: Drag reference area of the vehicle (m^2). Not filled from the engine calculation. *If blank:* Empty or not a number: no trajectory is computed and the field is named (no value is assumed).				
traj_burn_time	`traj_burn_time`			(blank)
Effect on the calculation: Burn time flown by the trajectory, in seconds: (initial mass - final mass) divided by the engine total mass flow from the engine calculation. *If blank:* Not an input: it is written here when the trajectory is computed and is empty before that.				

11.4 Outputs

Table 11.10: Outputs of the liquid page

Output	Unit	Meaning	Source / model
SOLVER INPUT SCOPE	–	Which form fields were read in this run.	Server declaration.
ENGINE ISP — PER TOTAL PROPELLANT CONSUMPTION (sea level, vacuum)	s	Delivered specific impulse of the engine: thrust of main chamber and turbine exhaust / total propellant flow. Equal to the main-chamber Isp for pressure-fed and closed cycles.	Open-cycle flow ledger (including the gas-generator or tap-off flow).
MAIN THRUST CHAMBER + NOZZLE ISP — PER MAIN-CHAMBER FLOW (sea level, vacuum)	s	Delivered specific impulse of the main chamber alone.	CEA (at the engine's P_c and O/F) and the loss chain.
SPACE OPTIMIZED ISP	s	Same chamber with the vacuum expansion ratio.	
VACUUM THRUST	kN		

Output	Unit	Meaning	Source / model
Engine cycle card	bar, s	Power balance, pump discharge pressures, open-cycle Isp loss.	Cycle calculation (not a pump design).
Throttle map	%, bar, s	P_c, vacuum Isp, $\Delta P/P_c$, ratio-rule and chug-loop verdict at each throttle point; feed pressure feasibility row.	Steady-state solutions; the loop model is the feed-coupled chug loop of the Analysis Center.
Motor design rows	mm, m/s	Throat, exit, expansion ratios, chamber diameter and length, nozzle length and angles, c^*, T/W, engine mass estimate.	Solver geometry.
Performance dashboard	kg/s, bar, m/s	Flows, feed and injector pressure distribution, feed pressure budget per circuit, injection velocities.	Darcy–Weisbach and local loss coefficients (Crane TP-410).
Gimbal Mount Loads	–	Gimbal loads; missing inputs are requested by name.	
Engine Specifications	–	Propellants, mixture ratio and optimum, mixture-ratio efficiency, chamber temperature, total impulse, data sources.	
Side panels	–	Flight Dynamics (6-DOF), Injector Design, Structural FEA.	Tables 13.19, 13.20, 13.21.
NASA CEA Comparison	–	CEA ideal values and HRMA delivered values.	Section 12.

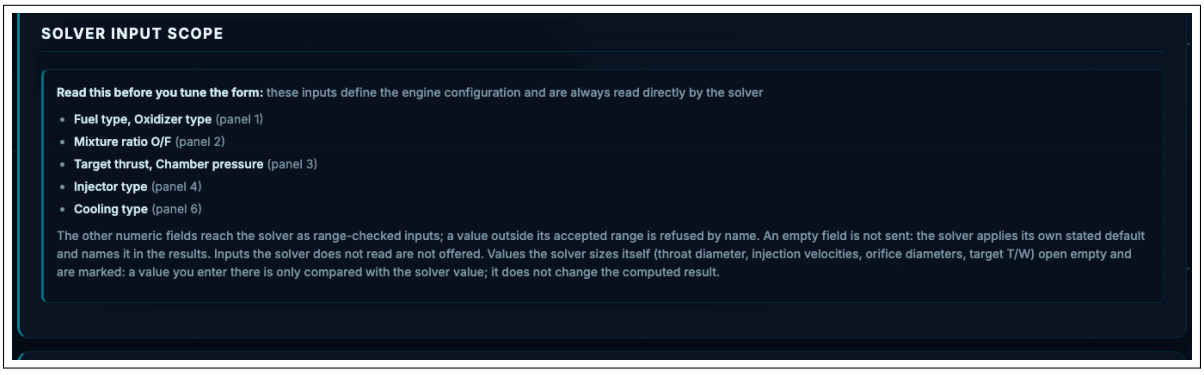
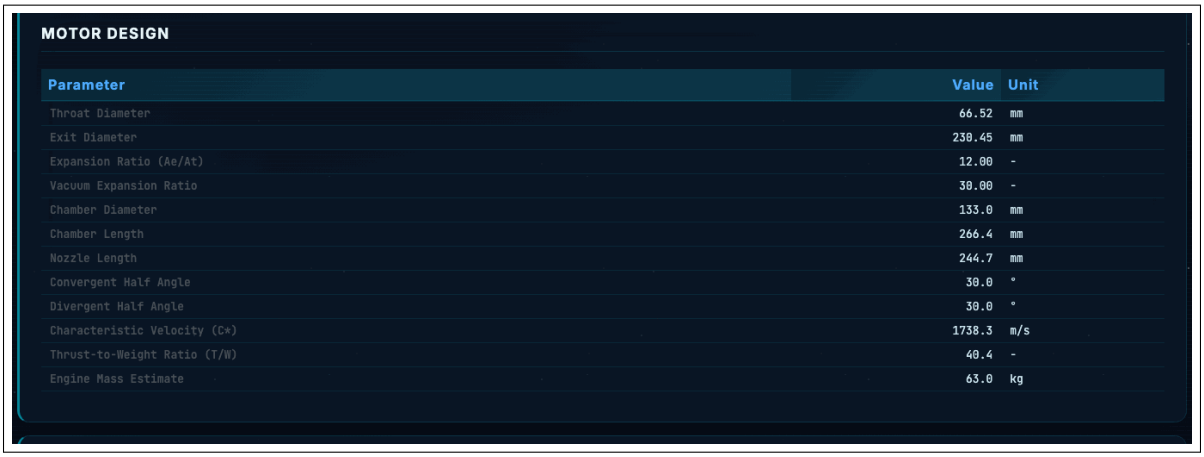


Figure 11.7: SOLVER INPUT SCOPE: the fields read in this run.



MOTOR DESIGN

Parameter	Value	Unit
Throat Diameter	66.52	mm
Exit Diameter	230.45	mm
Expansion Ratio (Ae/At)	12.00	-
Vacuum Expansion Ratio	30.00	-
Chamber Diameter	133.0	mm
Chamber Length	266.4	mm
Nozzle Length	244.7	mm
Convergent Half Angle	30.0	°
Divergent Half Angle	30.0	°
Characteristic Velocity (C*)	1738.3	m/s
Thrust-to-Weight Ratio (T/W)	40.4	-
Engine Mass Estimate	63.0	kg

Figure 11.8: Liquid MOTOR DESIGN table.



Figure 11.9: Liquid performance analysis dashboard: flows, pressure distribution and the feed pressure budget per circuit.

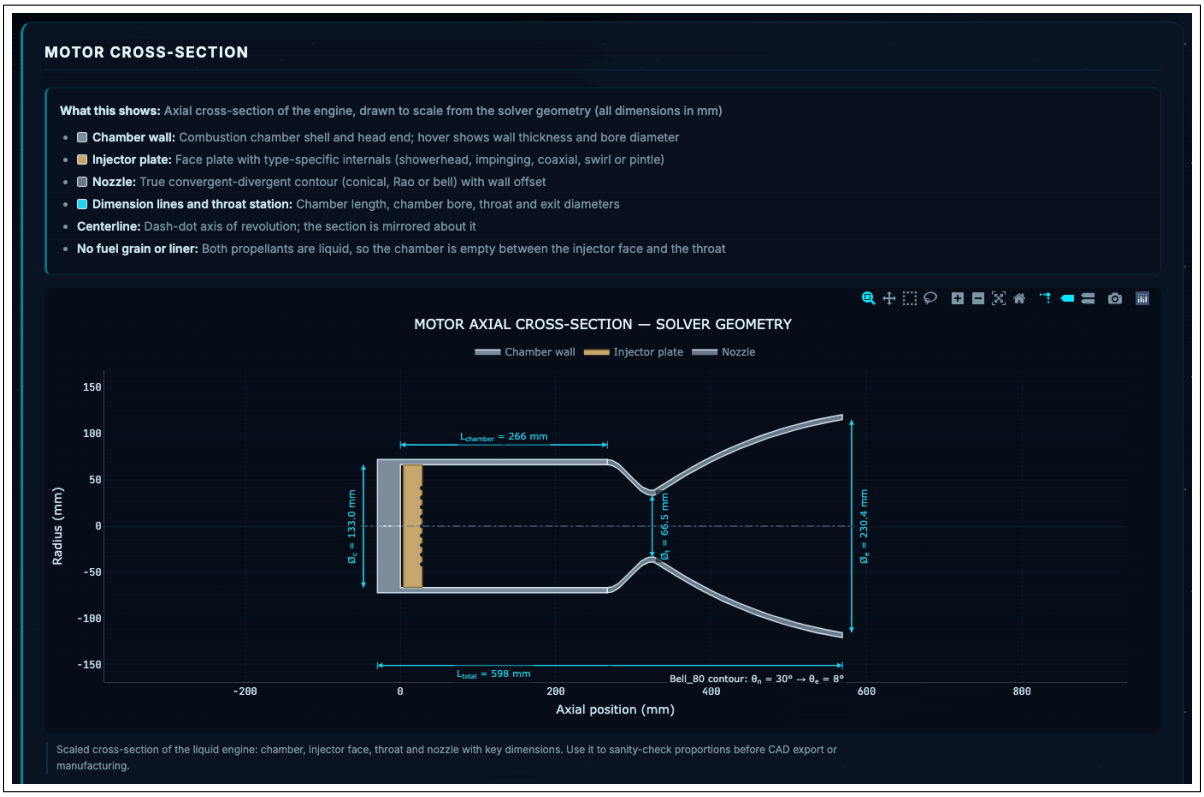


Figure 11.10: Liquid engine cross-section.

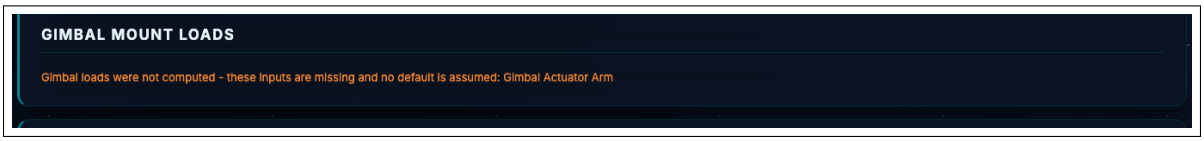


Figure 11.11: Gimbal Mount Loads: the missing input is requested by name and no default is used.

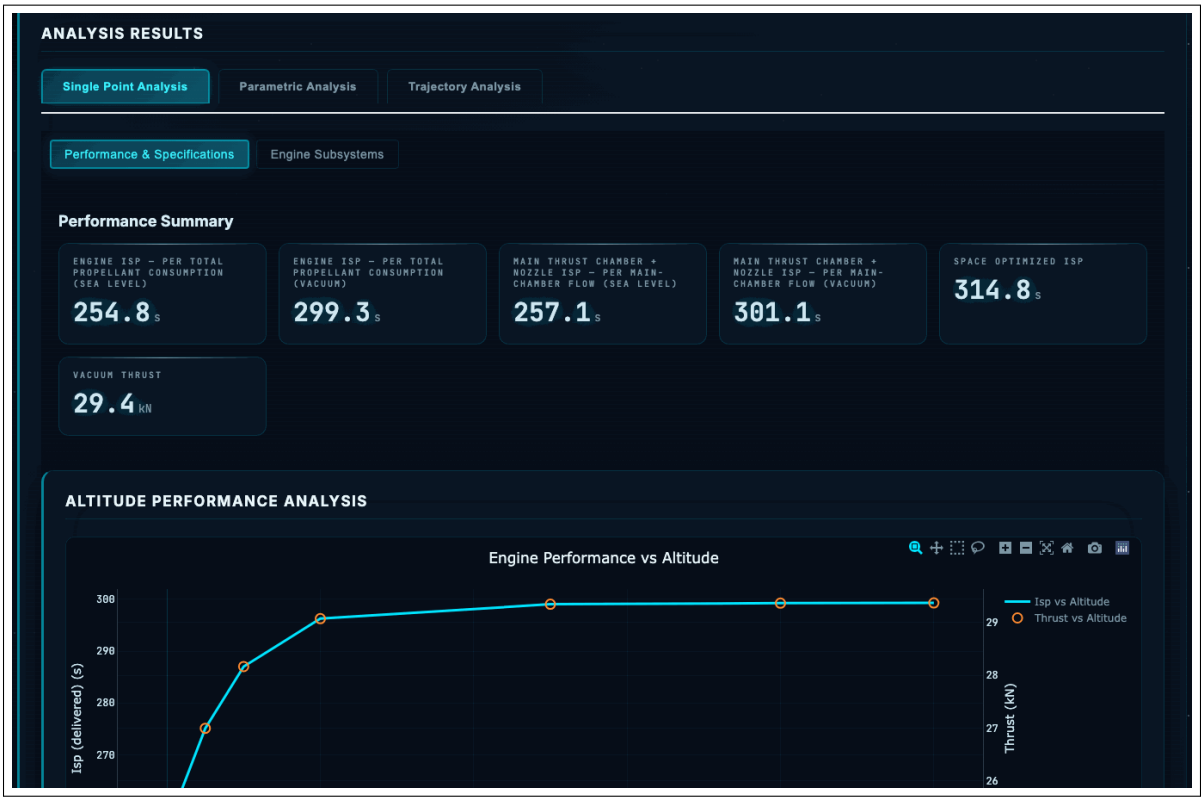


Figure 11.12: Analysis results: the Performance & Specifications sub-tab.

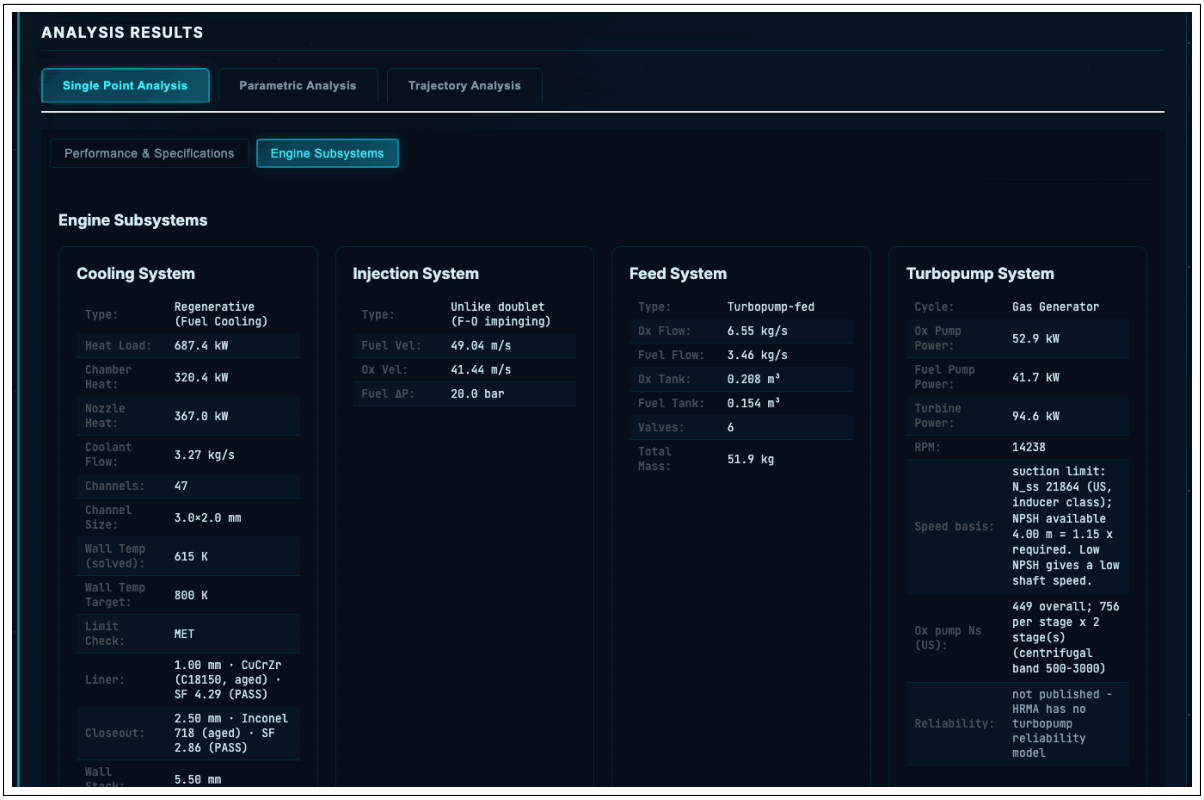


Figure 11.13: Analysis results: the Engine Subsystems sub-tab.

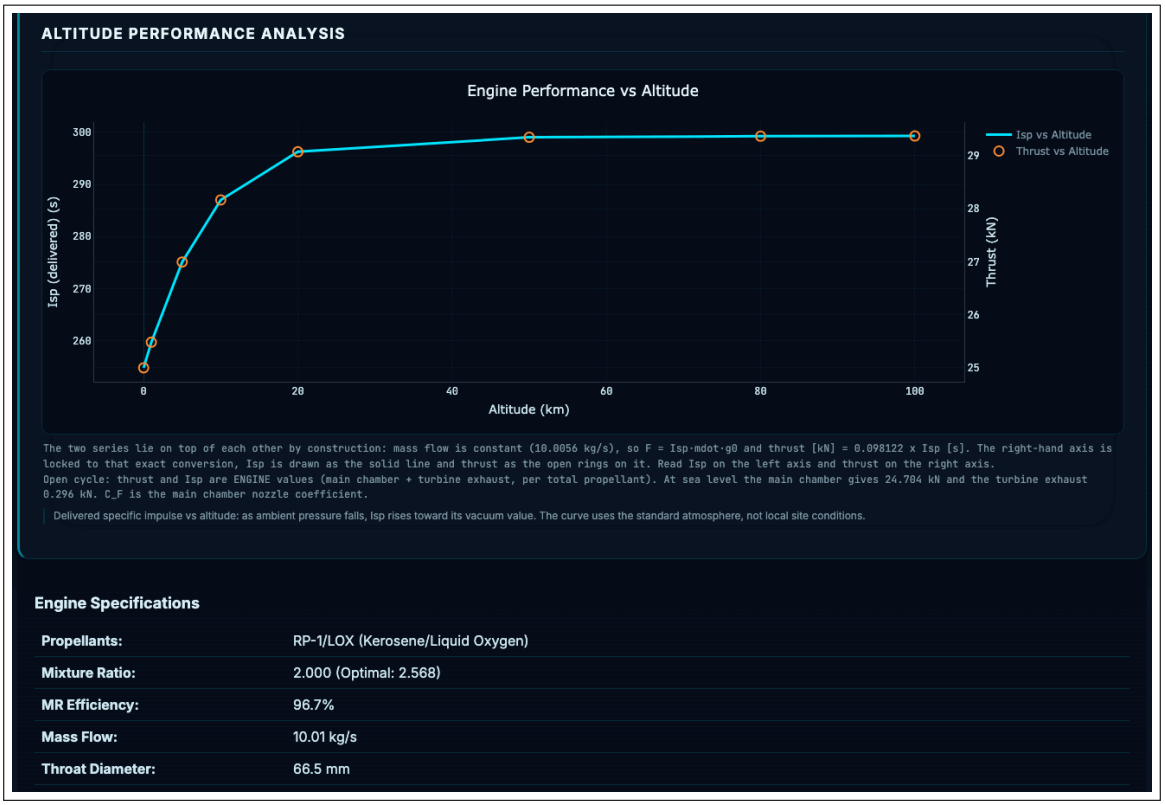


Figure 11.14: Altitude performance analysis.

11.5 Verdicts and warnings

Table 11.11: Liquid-specific verdicts and declarations

On screen	Meaning	Action
CARBON DEPOSIT	RP-1 gas-side deposit resistance applied; hot-wall temperature with the deposit and where it spalls.	Compare the spalled-zone temperature with the melting limit.
COKING RISK	Coolant-side wall above the 561 K warning level; 728 K (NASA SP-8087) is the critical limit.	Lower the coolant-side temperature.
PROPERTY RANGE (WALL)	Outside the coolant property table band; the coolant-side coefficient is a conservative lower bound.	
INPUT lines	Input consistency or a missing input (boiling point, turbine pressures).	Complete or correct the input.
Chug verdict MARGINAL (pintle)	The two atomisation scales give different stability verdicts.	Raise the stability margin with the injector pressure drop or the chamber volume.
Throttle note (separation)	Below a given throttle the nozzle separates at sea level; the sea-level Isp of those points is not published.	

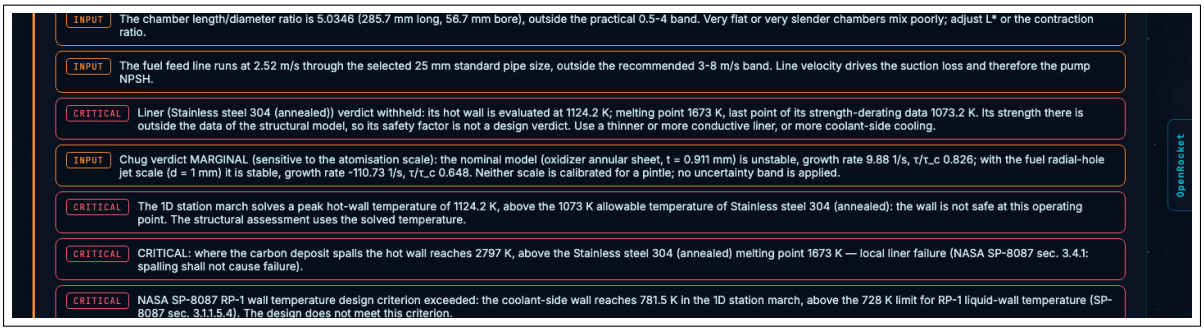


Figure 11.15: With the pintle selected, the chug verdict is MARGINAL: the two atomisation scales give different verdicts.

Gate rejections: out-of-band inputs, a pressure-fed tank below the required pressure and a turbopump cycle whose power balance does not close are rejected by name. With film cooling there is no regenerative jacket and the film flow is subtracted from the delivered Isp; 0 % film is rejected.

11.6 Limits

- The delivered Isp is a preliminary-design estimate (92.9–93.9 % of the CEA ideal Isp over the CEA comparison grid).
- In the two-layer wall, low-cycle fatigue, creep, liner–shell thermal mismatch, bond strength and buckling are not modelled and are named in the response. CAD draws both layers, not the cooling channels.
- Turbopump maps, cavitation, rotordynamics, start-up and shut-down transients are not included.
- Channel sizing applies the 40 m/s coolant velocity target (NASA SP-8087) also to supercritical hydrogen; the result errs on the safe side (hotter predicted wall).

11.7 Related sections

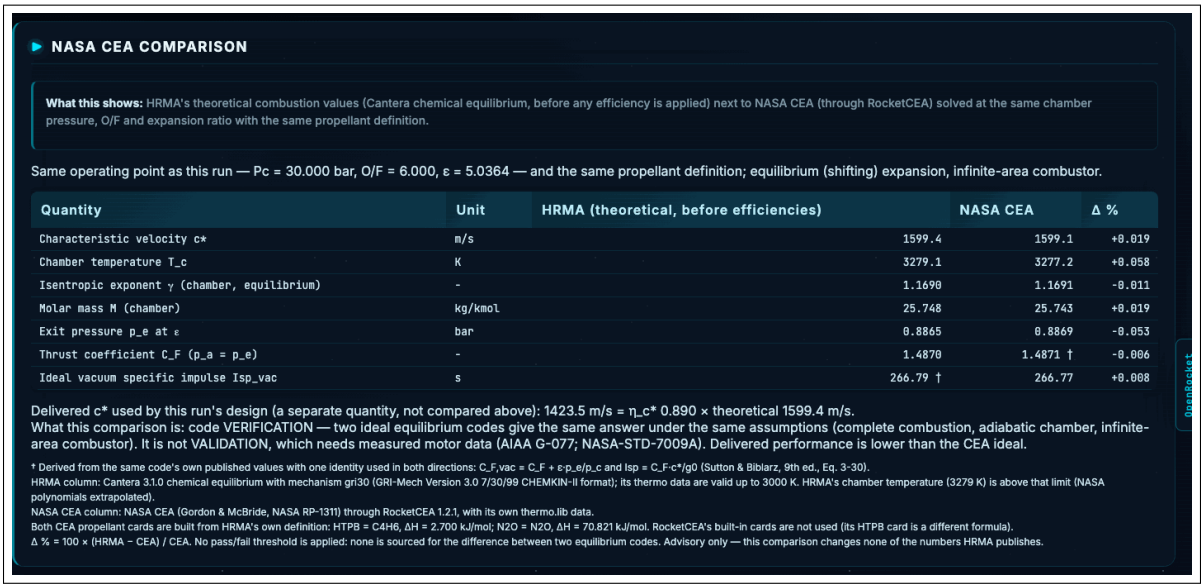
Example 3 (Section 6); Regenerative Cooling panel (Table 13.5); canonical bands (Appendix E).

12. NASA CEA Comparison

The three designer pages show a NASA CEA comparison block at the end of the results. RocketCEA (NASA CEA, Gordon and McBride, NASA RP-1311) is run at the design point of the run: same chamber pressure, O/F and expansion ratio.

Table 12.1: Content of the CEA comparison by motor type

Motor	Content
Hybrid	HRMA's own equilibrium (Cantera) against CEA; delivered c^* on a separate row. H_2O_2 is taken as 100 % pure and the block says so.
Solid	For catalogue propellants of known composition, catalogue values and the ideal nozzle next to CEA at the motor's own pressure. For a propellant of uncertain composition the block gives the reason and no number.
Liquid	CEA ideal values next to HRMA's delivered values; the efficiency that explains each difference is stated (Table 6.3).



NASA CEA COMPARISON

What this shows: HRMA's theoretical combustion values (Cantera chemical equilibrium, before any efficiency is applied) next to NASA CEA (through RocketCEA) solved at the same chamber pressure, O/F and expansion ratio with the same propellant definition.

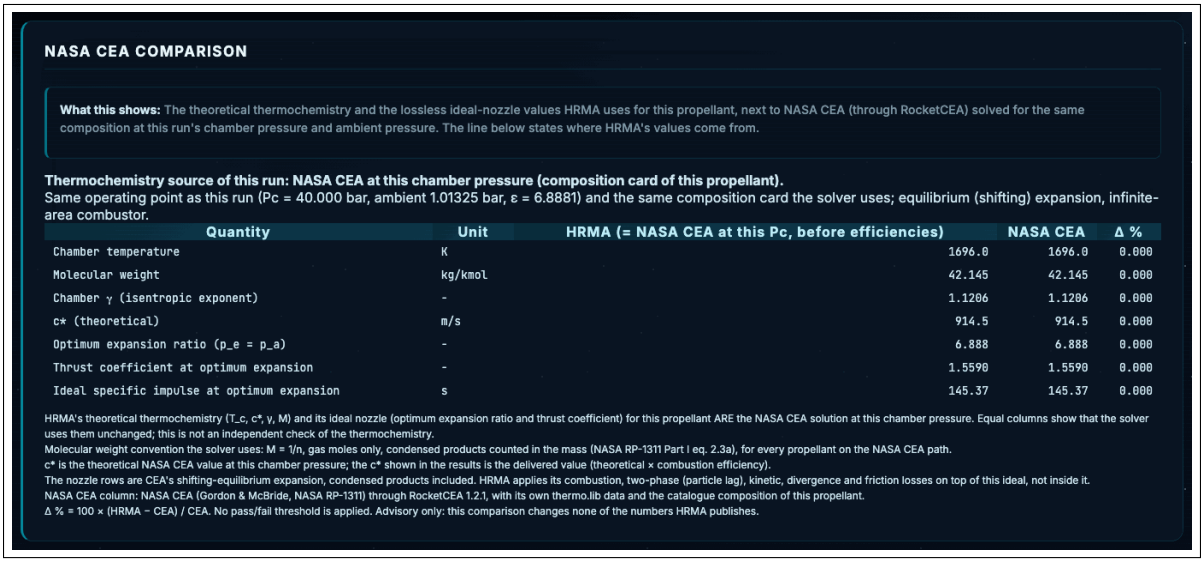
Same operating point as this run — Pc = 30.000 bar, O/F = 6.000, ε = 5.0364 — and the same propellant definition; equilibrium (shifting) expansion, infinite-area combustor.

Quantity	Unit	HRMA (theoretical, before efficiencies)	NASA CEA	Δ %
Characteristic velocity c*	m/s	1599.4	1599.1	+0.019
Chamber temperature T_c	K	3279.1	3277.2	+0.058
Isentropic exponent γ (chamber, equilibrium)	-	1.1690	1.1691	-0.011
Molar mass M (chamber)	kg/kmol	25.748	25.743	+0.019
Exit pressure p_e at ε	bar	0.8865	0.8869	-0.053
Thrust coefficient C_F (p_a = p_e)	-	1.4870	1.4871 †	-0.006
Ideal vacuum specific impulse Isp_vac	s	266.79 †	266.77	+0.008

Delivered c* used by this run's design (a separate quantity, not compared above): 1423.5 m/s = η_c* 0.890 × theoretical 1599.4 m/s.
What this comparison is: code VERIFICATION — two ideal equilibrium codes give the same answer under the same assumptions (complete combustion, adiabatic chamber, infinite-area combustor). It is not VALIDATION, which needs measured motor data (AIAA G-077; NASA-STD-7009A). Delivered performance is lower than the CEA ideal.

† Derived from the same code's own published values with one identity used in both directions: C_F,vac = C_F + ε·p_e/p_c and Isp = C_F·c*/g0 (Sutton & Biblarz, 9th ed., Eq. 3-30).
HRMA column: Cantera 3.1.0 chemical equilibrium with mechanism gri30 (GRI-Mech Version 3.0 7/30/99 CHEMKIN-II format); its thermo data are valid up to 3000 K. HRMA's chamber temperature (3279 K) is above that limit (NASA polynomials extrapolated).
NASA CEA column: NASA CEA (Gordon & McBride, NASA RP-1311) through RocketCEA 1.2.1, with its own thermo.lib data.
Both CEA propellant cards are built from HRMA's own definition: HTPB = C4H6, ΔH = 2.700 kJ/mol; N2O = N2O, ΔH = 70.821 kJ/mol. RocketCEA's built-in cards are not used (its HTPB card is a different formula).
Δ % = 100 × (HRMA − CEA) / CEA. No pass/fail threshold is applied: none is sourced for the difference between two equilibrium codes. Advisory only — this comparison changes none of the numbers HRMA publishes.

Figure 12.1: NASA CEA Comparison in the hybrid: HRMA equilibrium, CEA and difference.



NASA CEA COMPARISON

What this shows: The theoretical thermochemistry and the lossless ideal-nozzle values HRMA uses for this propellant, next to NASA CEA (through RocketCEA) solved for the same composition at this run's chamber pressure and ambient pressure. The line below states where HRMA's values come from.

Thermochemistry source of this run: NASA CEA at this chamber pressure (composition card of this propellant).
Same operating point as this run (Pc = 40.000 bar, ambient 1.01325 bar, ε = 6.8881) and the same composition card the solver uses; equilibrium (shifting) expansion, infinite-area combustor.

Quantity	Unit	HRMA (= NASA CEA at this Pc, before efficiencies)	NASA CEA	Δ %
Chamber temperature	K	1696.0	1696.0	0.000
Molecular weight	kg/kmol	42.145	42.145	0.000
Chamber γ (isentropic exponent)	-	1.1206	1.1206	0.000
c* (theoretical)	m/s	914.5	914.5	0.000
Optimum expansion ratio (p_e = p_a)	-	6.888	6.888	0.000
Thrust coefficient at optimum expansion	-	1.5590	1.5590	0.000
Ideal specific impulse at optimum expansion	s	145.37	145.37	0.000

HRMA's theoretical thermochemistry (T_c, c*, γ, M) and its ideal nozzle (optimum expansion ratio and thrust coefficient) for this propellant ARE the NASA CEA solution at this chamber pressure. Equal columns show that the solver uses them unchanged; this is not an independent check of the thermochemistry.
Molecular weight convention the solver uses: M = 1/n, gas moles only, condensed products counted in the mass (NASA RP-1311 Part I eq. 2.3a), for every propellant on the NASA CEA path.
c* is the theoretical NASA CEA value at this chamber pressure; the c* shown in the results is the delivered value (theoretical × combustion efficiency).
The nozzle rows are CEA's shifting-equilibrium expansion, condensed products included. HRMA applies its combustion, two-phase (particle lag), kinetic, divergence and friction losses on top of this ideal, not inside it.
NASA CEA column: NASA CEA (Gordon & McBride, NASA RP-1311) through RocketCEA 1.2.1, with its own thermo.lib data and the catalogue composition of this propellant.
Δ % = 100 × (HRMA − CEA) / CEA. No pass/fail threshold is applied. Advisory only: this comparison changes none of the numbers HRMA publishes.

Figure 12.2: NASA CEA Comparison in the solid: catalogue values and the ideal nozzle next to CEA at the motor's own pressure.

The comparison is a code verification, not a validation. It gives the same input to two codes and shows that the two codes agree; it does not show how a real motor will burn. Measured agreement over a grid run with CEA2 (RocketCEA 1.2.1) and NASA CEA v3 (3.3.4) (the two references differ by at most 0.001 %): in the hybrid 4532 of 4536 rows within 0.5 % (T_c within 0.12 %, c^* within 0.04 %), in liquid thermochemistry 1559 of 1559 rows, in the solid 221 of 224 rows. $\Delta\% = 100\,(\text{HRMA} - \text{CEA})/\text{CEA}$; no pass/fail threshold is applied and the comparison changes no design number.

13. Analysis Dock and Analysis Center

13.1 Purpose

Runs detailed analyses on the motor of the main calculation. Panel inputs are filled from the last calculation; every field can be changed and the request is always built from the fields visible on screen.

13.2 Screen

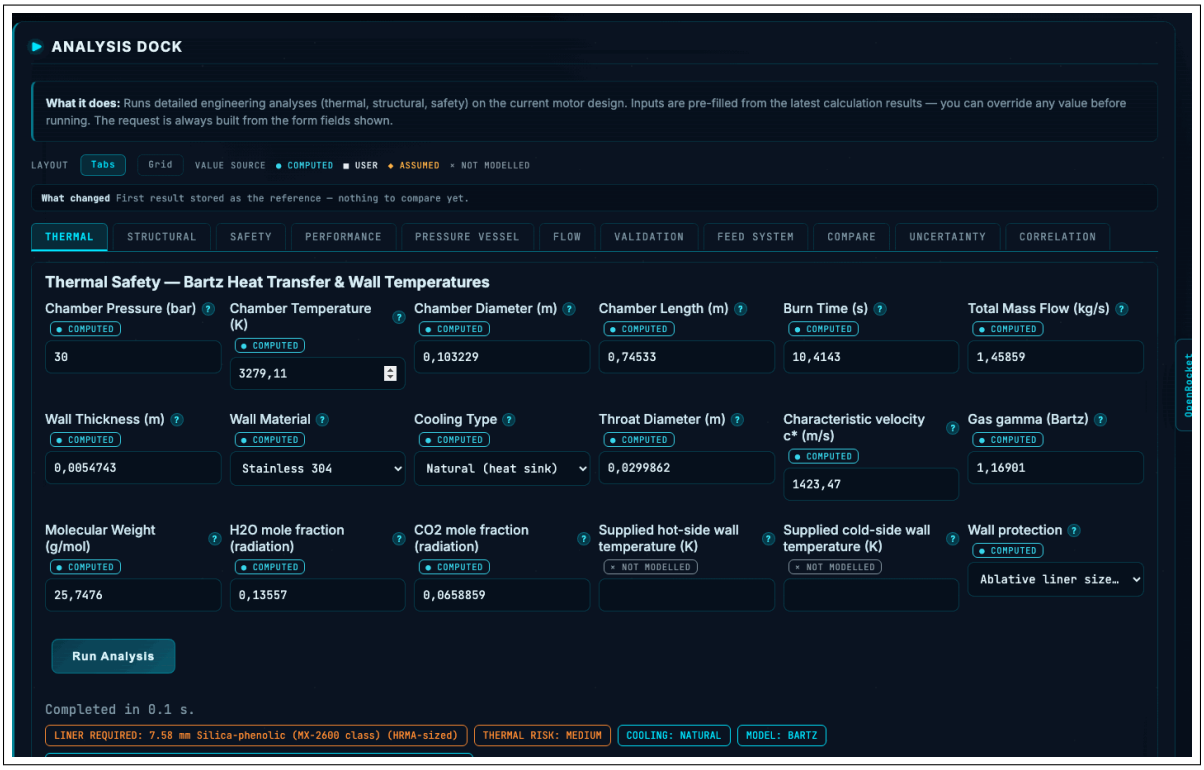


Figure 13.1: ANALYSIS DOCK: category tabs, the prefilled inputs of the selected panel and the value-source badges. The ANALYSIS CENTER is below this section.

13.3 Common contract

The dock carries panels in eleven categories: THERMAL, STRUCTURAL, SAFETY, PERFORMANCE, PRESSURE VESSEL, FLOW, VALIDATION, FEED SYSTEM, COMPARE, UNCERTAINTY, CORRELATION. The layout is chosen as Tabs or Grid. Each panel has a Run Analysis button, value-source badges, a listing of the fields the panel sends and a What changed block listing the differences between two consecutive runs.

Table 13.1: Value-source badges

Badge	Meaning
COMPUTED	The value was filled from the main calculation's solution for this design.
USER	The user entered or changed the value; later prefills do not overwrite it.
ASSUMED	The value was not solved here: a default, a literature value or a choice.
NOT MODELLED	No value was computed for this quantity.

If a required field is empty the request is not sent and the screen says Required fields are empty: … The request was not sent. If the material used by the solver is not in the panel's material list the

request is not sent either: a safety verdict is not issued on a material other than the one the motor was computed with.

A panel result is read in this order:

1. The status line: was a verdict issued or withheld; if withheld, the reason.
2. The input badges: ASSUMED inputs show that part of the result comes from an assumption.
3. The numbers and their bands. Numbers without a threshold are printed without colour; they are measurements, not verdicts.
4. The list of items not modelled: the physics outside the panel's scope.

13.4 Dock panels

Table 13.2: Analysis Dock panels

Category	Panel	Outputs	Verdict and limits
THERMAL	Thermal Safety (Bartz heat transfer and wall temperatures)	Gas-side heat-transfer coefficient, throat and chamber heat flux, wall temperatures, material-limit chart.	The wall temperature is compared with the material's allowable temperature. The γ input is accepted in the band 1.05–1.67; outside it is rejected, not clipped. If empty, the repository default 1.20 is used and the result states `gamma_source: repo_default`. An ambient temperature outside 180–340 K is rejected.

Category	Panel	Outputs	Verdict and limits
THERMAL	Thermal Protection (ablative, heat sink, radiation cooled)	Three modes: ablative liner (Q*), heat-sink wall (1D), radiation-cooled extension. In ablative mode: recession rate, total recession, required liner thickness, Q* used, mean heat flux and total heat load.	A steady Q* model (SIMPLIFIED MODEL). No thickness is published, and the reason is given, in two cases: with no net heating (recovery temperature below the ablation temperature; solid example) and when the recession rate exceeds the model's validity ceiling of 0.35 mm/s (MODEL NOT VALID; 1.200 mm/s in the hybrid example). In the liquid example the panel reads the motor's gas-side heat transfer coefficient and recovery temperature pair; the recession rate is 0.160 mm/s and the required liner thickness of 7.18 mm is published. Zero recession is not zero thickness.

Category	Panel	Outputs	Verdict and limits
THERMAL	Regenerative Cooling (1D station march, Bartz and Dittus–Boelter), liquid only	Gas- and coolant-side temperatures, heat flux and pressure drop station by station.	The coolant-side film coefficient is taken from a literature band. The jacket pressure drop is one number: the drop solved by the station march also enters the feed budget; for a pressure-sensitive coolant (methane, LH_2) the inlet pressure is found by fixed-point iteration. The panel runs with the motor's own march inputs; the jacket pressure drop equals the motor's (example 6.13 bar). In staged and expander cycles the jacket inlet is the cycle's fuel pump discharge. If the model cannot build a consistent circuit for the engine (F-1 class) the jacket carries the model limit label; the cooling-dependent feed, cycle, engine thrust, engine Isp and mass numbers are withheld with their reason. With dump cooling no jacket numbers are published.

Category	Panel	Outputs	Verdict and limits
STRUCTURAL	Structural Safety (pressure vessel, buckling, fatigue)	Stress components and safety factors, buckling, fatigue, thermal stress.	If HRMA sized the wall the verdict is withheld. A wall entered by the user that misses the target receives a verdict (UNSAFE); one that meets it does not make the response as a whole accepted, because the closure and nozzle stations are not verified in this panel (Section 5). Without a declared external-pressure load case buckling is NOT_EVALUATED. The gas temperature is filled only if the motor published a solved wall-temperature pair; otherwise the thermal path does not run, strength is read at 293.15 K and the panel says so. The panel gives the motor result's own check of the same wall (solid: case; liquid: chamber or two-layer stack safety factor) next to its own hoop safety factor; any difference is named with its reason (load: chamber pressure $\leftrightarrow$ 1.5 $\times$ design pressure; for two layers, which layer carries the coolant pressure; strength temperature; wall thickness). Burn time is not used by this model, and this is declared.

Category	Panel	Outputs	Verdict and limits
STRUCTURAL	Bolted Joint (Shigley)	Recommended tightening torque, preload and its ±25 % scatter, proof safety factor, overload factor n_L, separation factor n_0, load sharing.	Proof and yield safety are evaluated at the upper end of the preload scatter and separation at the lower end (assumptions on screen: Shigley Ch. 8, NASA-STD-5020A). Member crushing under the bolt head is checked; it is flagged in the hybrid example (aluminium member). Bolt count, size and grade are the panel defaults (ASSUMED badge, 8 × M8, 8.8): the result is an example joint until the motor's actual joint is entered.

Category	Panel	Outputs	Verdict and limits
SAFETY	Comprehensive Safety (risk assessment and pressure vessel)	Two separate verdicts: MOTOR DESIGN RISK and FACILITY / SITING HAZARD. Risk score on six axes (1–5: structural, pressure, thermal, explosive, toxic, fire), 5×5 risk matrix, pressure vessel summary, TNT equivalent and facility distances.	An axis that cannot be evaluated is shown as NOT MODELLED and left out of the score. If the motor basket is partial and the evaluated axes come out low or medium, the motor verdict is NOT EVALUATED, the number of evaluated axes is stated and the score is given as SCORE OF EVALUATED AXES, a lower bound only (a missing evaluation is not a low risk); an axis that comes out high or critical still sets the verdict. Examples: in all three examples HRMA sized the wall, so the structural axis is not evaluated (for the hybrid the thermal axis too) and the motor design risk is NOT EVALUATED (score of evaluated axes 2.00 / 5); the facility hazard is NOT EVALUATED for the hybrid and the solid and HIGH for the liquid. For liquid pairs the TNT equivalent is taken from 14 CFR Part 420, Appendix E, Table E-5 (10 % of the propellant mass for a test stand in the liquid example, 29.745 kg); a liquid pair not in the table uses 0.35, declared as an HRMA choice. In the hybrid and KNDX solid examples the explosive class is Not modelled (no sourced TNT equivalent). Without the required dimensions no safety verdict is produced. Regulatory compliance is not assessed.

Category	Panel	Outputs	Verdict and limits
PERFORMANCE	Advanced Performance (3D surface in hybrid and liquid; Mach contour and heat flux in solid)	Isp–P_c–O/F surface, Mach contour, heat flux.	No 3D performance surface in the solid: O/F is not a design variable.
PRESSURE VESSEL	Pressure Vessel (sizing, MAWP, burst pressure)	Status, design code (AIAA S-080, ASME VIII Div 1 or NASA-STD-5012C), required and used wall, MAWP, predicted burst pressure and margin, pressure ladder (MEOP or MDP, proof test = hydrostatic test, with NASA-STD-5012C also the qualification test, required burst), head thicknesses (ASME UG-32), temperature derating.	With the wall empty, the thickness meeting both the code minimum and a 1.2 margin against the required burst pressure is sized and stated in a warning. The burst pressure is the smaller of the Faupel thick-wall and the thin-wall plastic-limit relations; it is a model estimate, not a test. The material is the case material on the solid page and the chamber material on the hybrid page (COMPUTED); if the result publishes no material, the panel default (Aluminum 6061-T6) is used and stated by name. On the liquid page the chamber is engine hardware: the code NASA-STD-5012C is selected; pressure, inner diameter, wall, temperature and material come from the engine's closeout layer (MDP); if the closeout is not published, the chamber values are shown with the reason. AIAA S-080 remains the code for propellant tanks and lines.

Category	Panel	Outputs	Verdict and limits
FLOW	Nozzle Flow (quasi-1D compressible)	Regime, pressure and Mach along the axis; the IDEAL cards (c^* efficiency and discharge coefficient 1: C_F, thrust, mass flow, c^*); effective C_F and 1D thrust; the thrust, mass flow, c^* and C_F the motor delivers in a separate block.	With any of the six required fields empty the request is rejected. The effective C_F is compared with the C_F of the main result; the thrust cross-check is stated as well (Section 4).
VALIDATION	User Data Validation (static-fire CSV)	Differences in total impulse, peak and average thrust, burn time; NRMSE.	Relative difference (prediction − measurement)/measurement. The score is the larger of the impulse difference and NRMSE; if the server sends no band the card is not coloured.
FEED SYSTEM	Feed System (slosh, pressurant, water hammer)	Slosh frequency, pressurant mass, water-hammer peak pressure.	The fields are filled only from the motor's own result (they stay empty for a solid motor); the value used for an optional field left empty is declared under `defaults_applied`, and an empty required field rejects the request by name. The water-hammer map covers LOX, LH2, methane, MMH, UDMH, N_2O_4 and N_2O; no block is produced for H_2O_2.
COMPARE	Comparative Analysis	Saved configurations side by side, two designs overlaid.	Only relative changes above the threshold are marked.

Category	Panel	Outputs	Verdict and limits
UNCERTAINTY	Uncertainty Quantification (Monte Carlo)	Analysis level (the level name gives the sample count: Fast Screening (200 samples), Engineering (1000 samples), Extended (3000 samples)) and seed; during the run a real progress counter (completed samples, measured time per sample, estimated time remaining) and Cancel Analysis; run summary (sample count, failed samples, duration, sampler); sampled input distributions; for each output P50, the [P5, P95] band, the nominal value and σ; distribution plot and sensitivity tornado (Spearman rank correlation).	The band is only the spread produced by the sampled inputs: model-form uncertainty is not included and it is not a confidence interval for the real motor; the screen says so in the How to read the band row. The nominal value is the number of the main result. If the form changes or the motor is recalculated during a run, the result is discarded and the reason is stated. An output insensitive to the sampled inputs gets no distribution: for the hybrid, thrust is a design target, and total impulse is not sampled at these levels and its nominal value is that of the main result. Correlations inside the noise floor are not ranked as effects.
CORRELATION	Experimental Correlation	Deviation of the model against the shipped static-fire database.	Checks the model, not the user's motor.

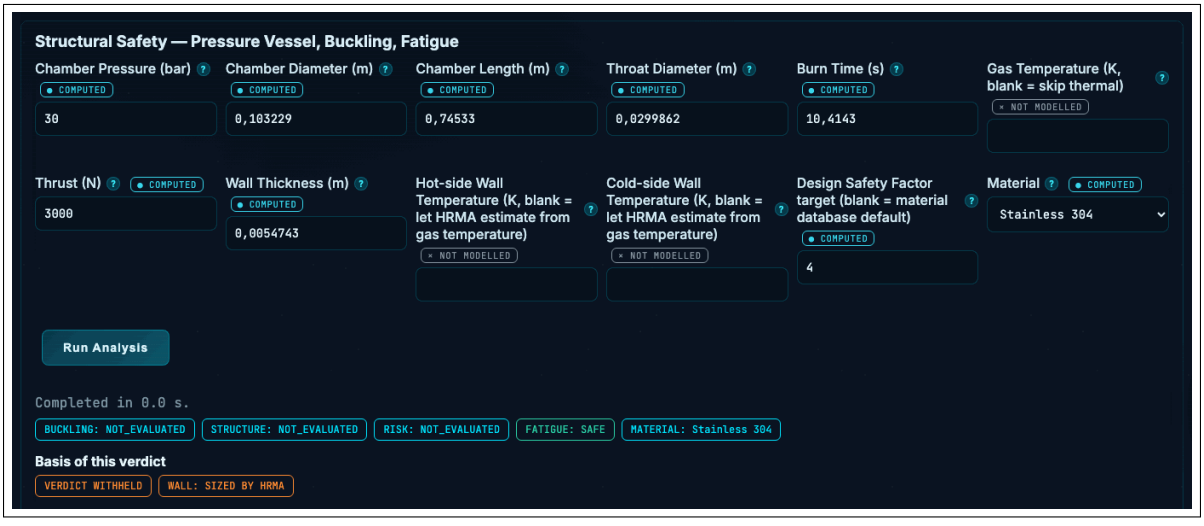


Figure 13.2: Structural Safety panel of the hybrid example after the run: inputs, value-source badges and the verdict line. HRMA sized the wall, so the verdict is VERDICT WITHHELD.

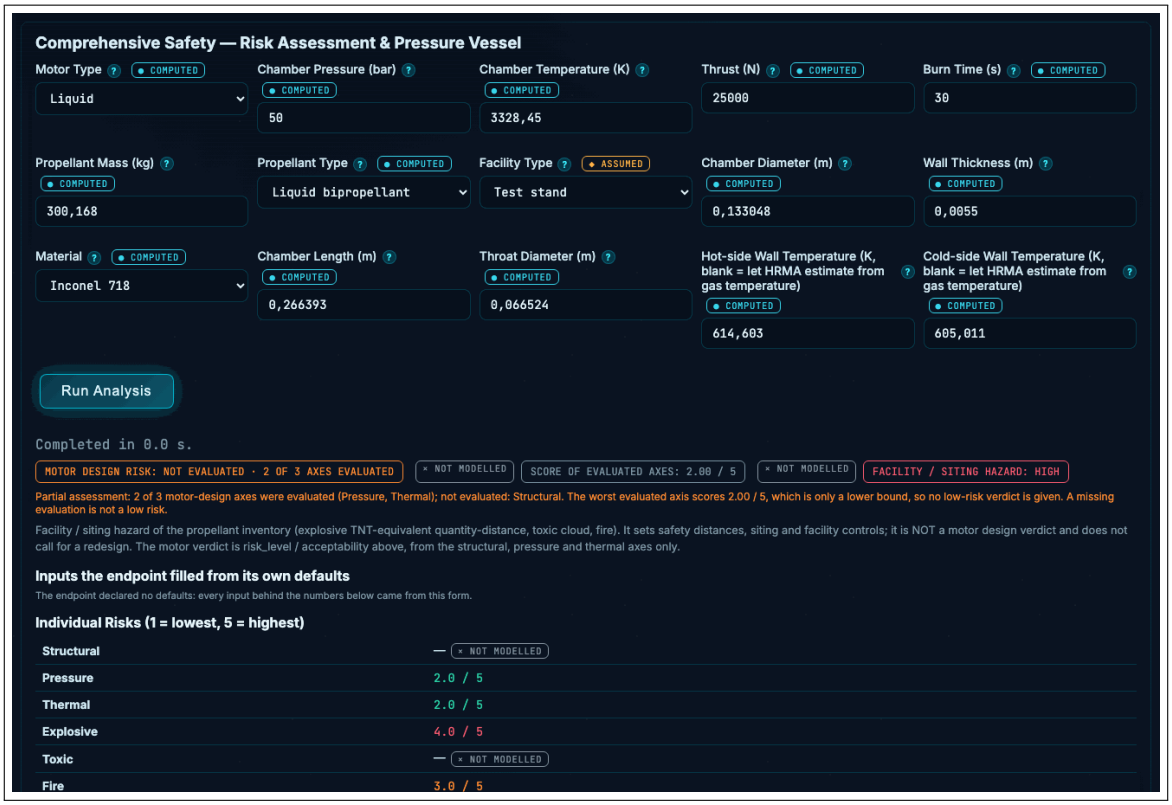


Figure 13.3: Comprehensive Safety panel of the liquid example after the run: MOTOR DESIGN RISK and FACILITY / SITING HAZARD are separate verdicts. Note the NOT MODELLED badge on the axes that could not be evaluated.

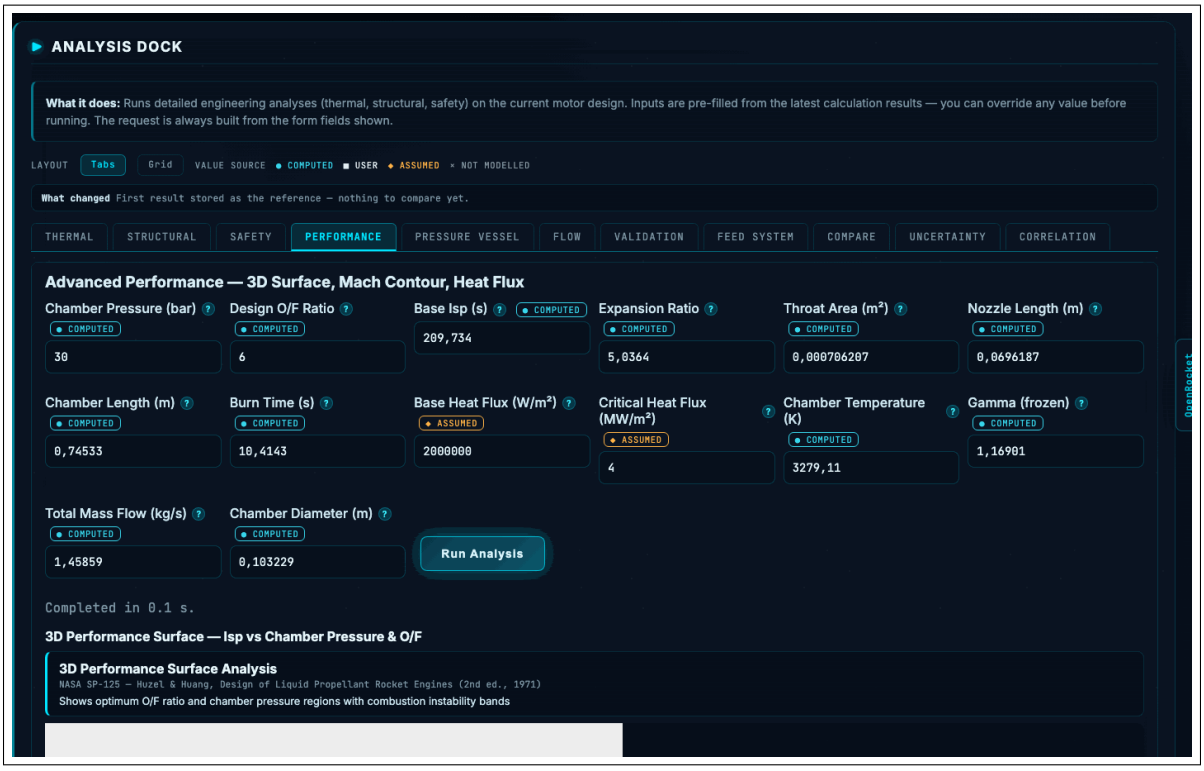


Figure 13.4: Dock, PERFORMANCE tab.

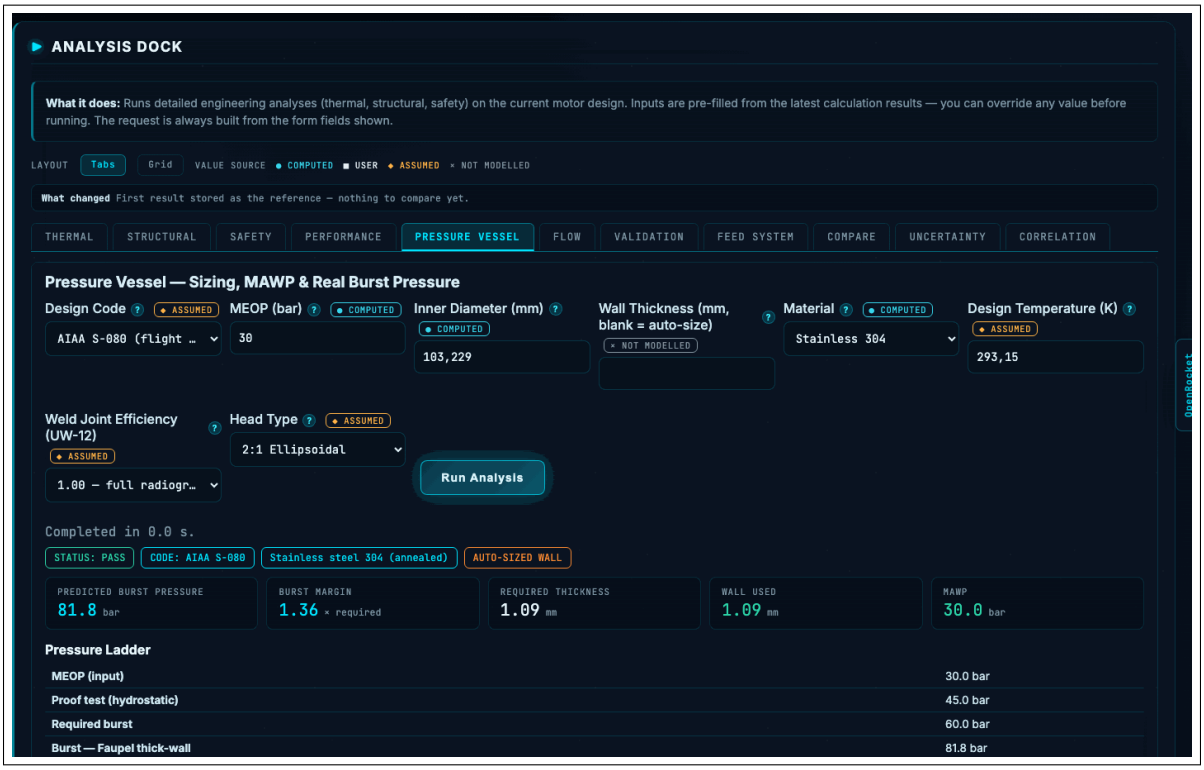


Figure 13.5: Dock, PRESSURE VESSEL tab.

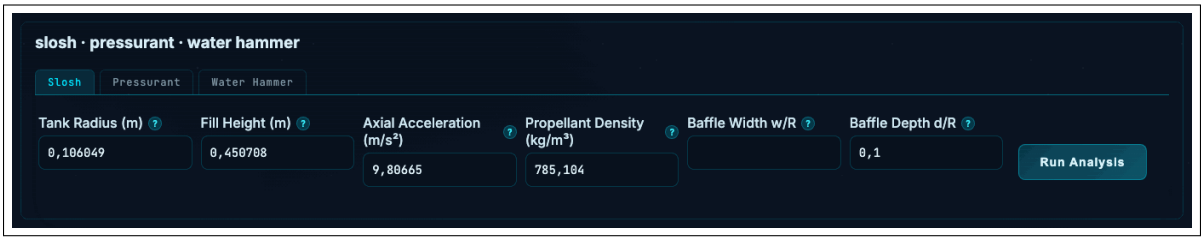


Figure 13.6: Dock, FEED SYSTEM panel before the run: the fields of the Slosh tab are filled from the motor's own result.

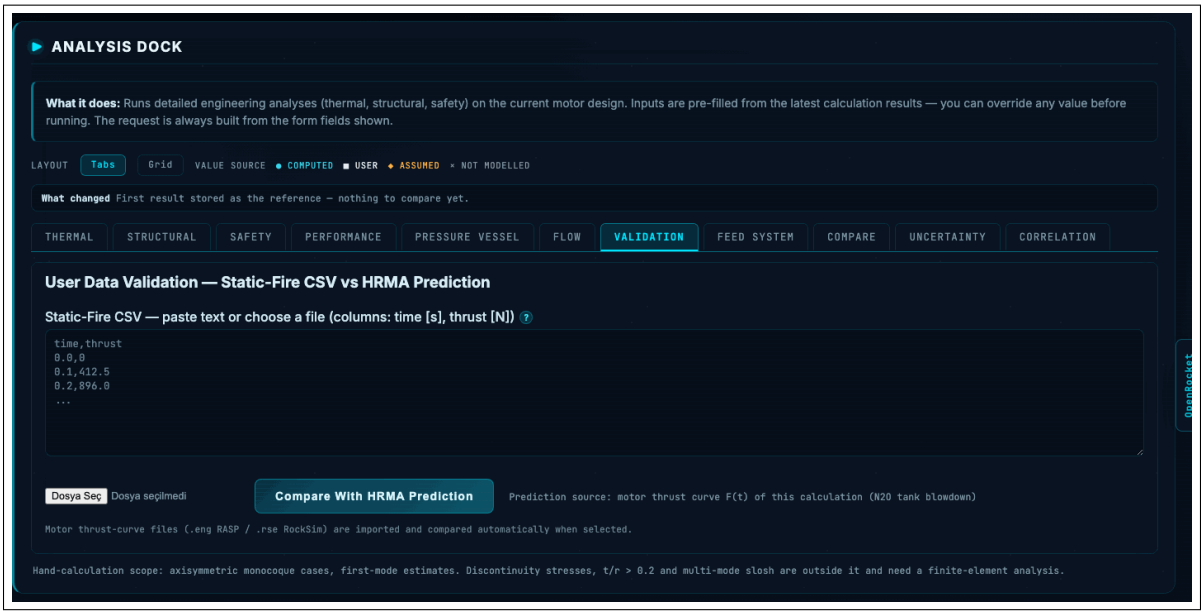


Figure 13.7: Dock, VALIDATION tab: static-fire CSV input.

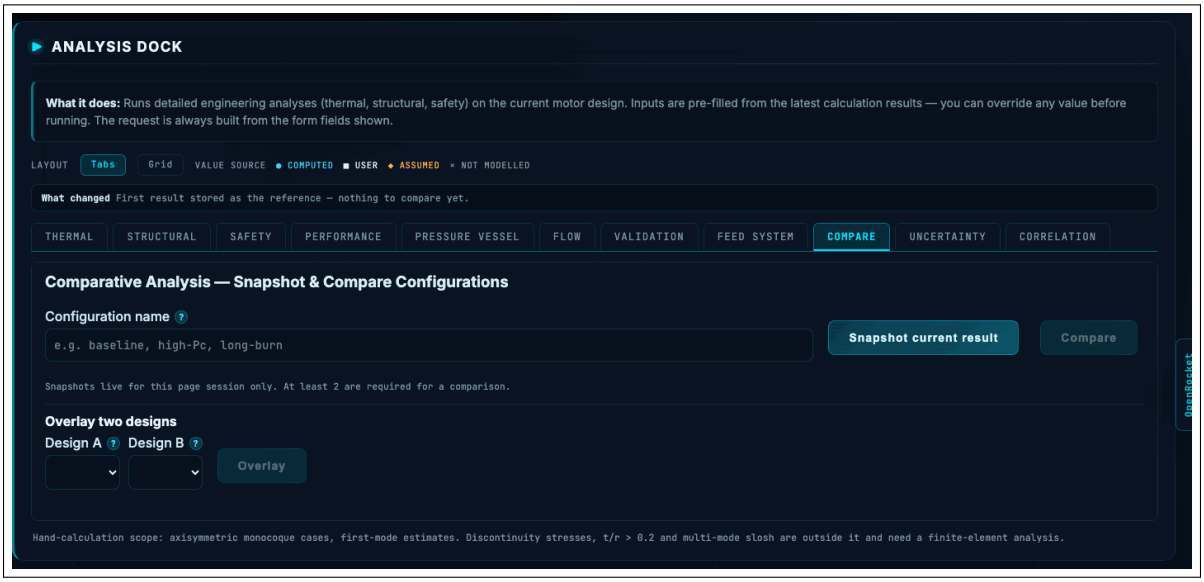


Figure 13.8: Dock, COMPARE tab.

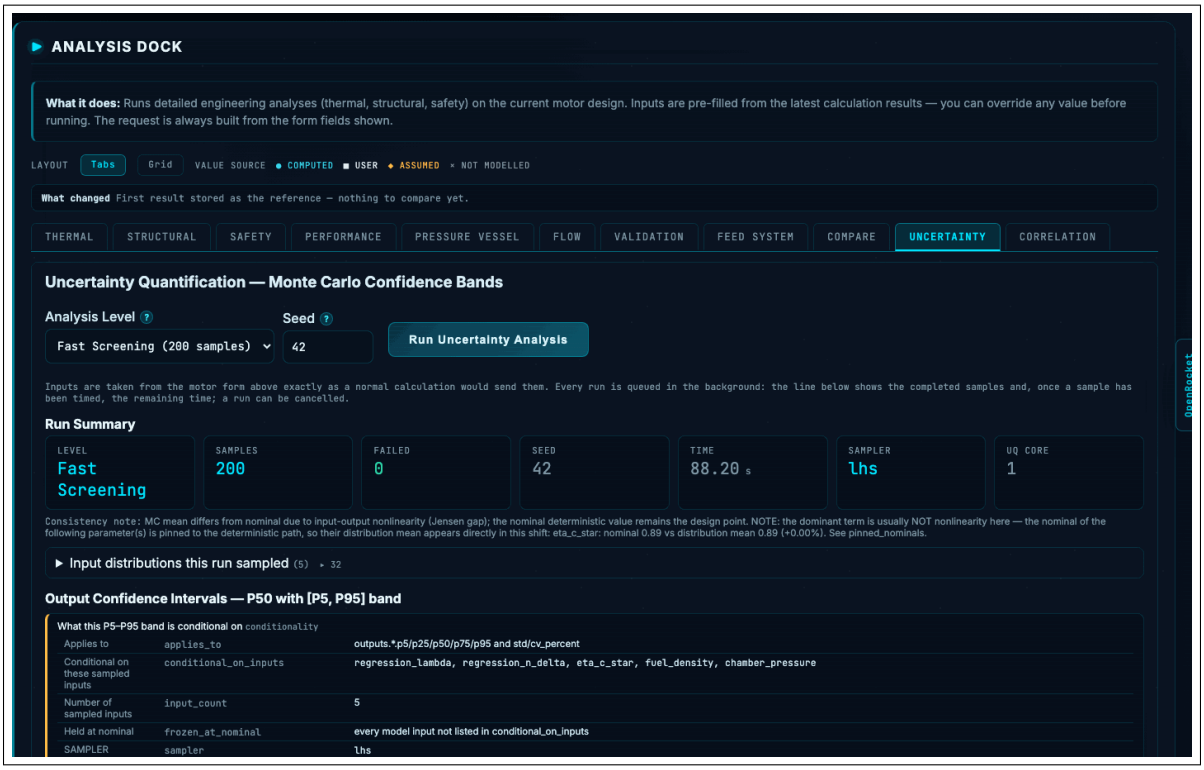


Figure 13.9: Dock, UNCERTAINTY: run result, summary and P5–P95 band (nominal = main design).

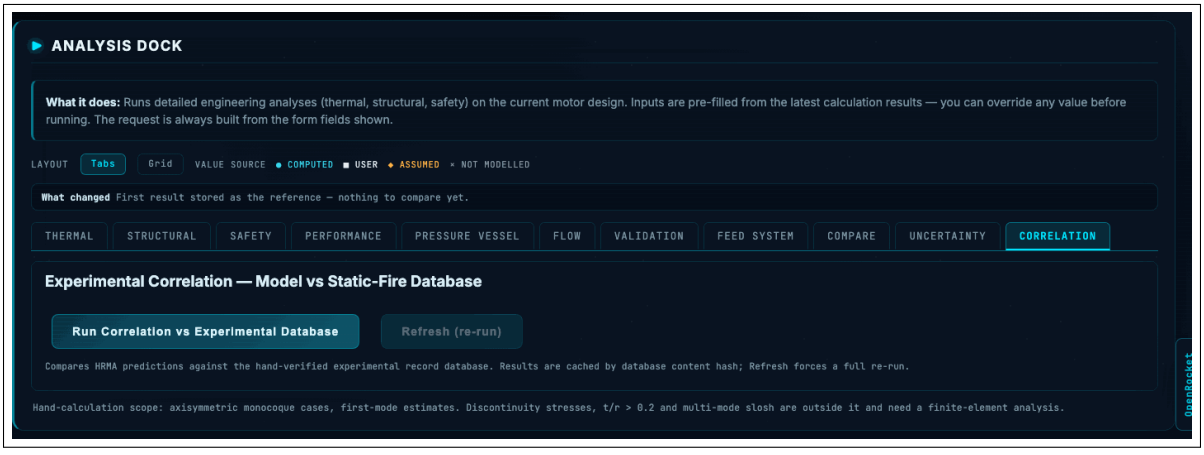


Figure 13.10: Dock, CORRELATION tab.

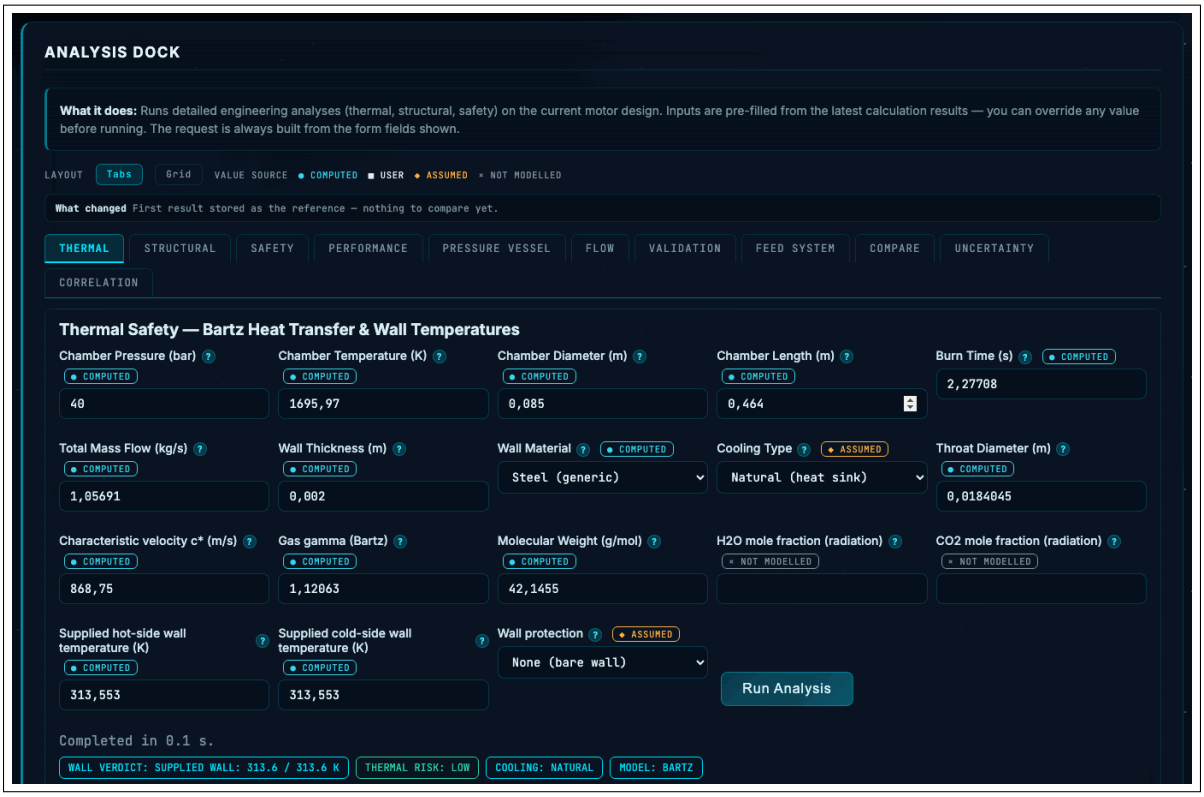


Figure 13.11: Dock, THERMAL in the solid: the case temperature solved by the motor is used.

13.5 Panel inputs

Table 13.3: Dock: Thermal Safety panel inputs

Input	Identifier	Unit	Form limit	Form default
Chamber Pressure	`ad_f_thermal_chamber_pressure`	bar		40
Effect on the calculation: Chamber pressure, bar. The thermal safety check uses it in the Bartz gas-side heat transfer (mass flux). Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.				
Chamber Temperature	`ad_f_thermal_chamber_temperature`	K		3000
Effect on the calculation: Chamber (combustion) temperature, K. Sets the recovery temperature that drives the wall heat flux. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.				
Chamber Diameter	`ad_f_thermal_chamber_diameter`	m		0.1
Effect on the calculation: Chamber inner diameter, m. Sets the wetted area and mass flux of the thermal check. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.				
Chamber Length	`ad_f_thermal_chamber_length`	m		0.5
Effect on the calculation: Chamber length, m. Sets the heated (wetted) area of the thermal check. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.				

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Input	Identifier	Unit	Form limit	Form default
Burn Time	`ad_f_thermal_burn_time`	s		(blank)

Effect on the calculation: Burn time, s (> 0). Sets the total heat energy, the heat-sink mass and the liner sizing of the thermal check; the steady bare-wall equilibrium temperature does not depend on it. Filled from the motor result after a calculation (solid: the solved burn; hybrid: the burn time input; liquid: Maximum Burn Duration on the liquid form), unless you edit it. *If blank:* The request is not sent and this field is named; enter the burn time here, or set the motor's burn duration and recalculate (no duration is assumed).

Total Mass Flow	`ad_f_thermal_mdot_total`	kg/s		1

Effect on the calculation: Total propellant mass flow, kg/s. Sets the gas mass flux in the Bartz correlation. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

Wall Thickness	`ad_f_thermal_wall_thickness`	m		0.005

Effect on the calculation: Thickness of the hot-gas wall, m: the wall the heat flux crosses. Sets the through-wall conduction and the heat-sink mass (changes the result). Filled from the motor result after a calculation, unless you edit it; on a two-layer regenerative liquid wall it is the liner thickness, not the liner + channel + closeout stack. Accepted: a positive value below the chamber radius. *If blank:* The request is not sent and the field is named.

Wall Material	`ad_f_thermal_material`			Steel (generic)

Effect on the calculation: Material of the hot-gas wall. Supplies its conductivity, allowable temperature and melting point to the thermal check (changes the result and the limits shown). Filled from the motor result after a calculation, unless you edit it; on a two-layer regenerative liquid wall it is the liner material, while the closeout is judged in the engine layer verdicts. If the motor result gives no material, the field keeps its default (Steel, generic). Accepted: an entry of the list. Options: Steel (generic); Steel AISI 4130; Steel AISI 4340; Stainless 304; Stainless 316; 17-4PH (H900); Aluminum; Aluminum 6061-T6; Aluminum 7075-T6; Aluminum 2024-T3; Copper; CuCrZr (C18150); BeCu C17200; Inconel; Inconel 718; Inconel 625; Titanium Ti-6Al-4V; Titanium CP Grade 2; \ldots.

Cooling Type	`ad_f_thermal_cooling_type`			Natural (heat sink)

Effect on the calculation: Cooling type: natural (heat sink), forced convection or regenerative. Selects the coolant-side coefficient of the bare-wall check. Filled from the motor result after a calculation (the motor's cooling type, when it is one of these three), unless you edit it. Options: Natural (heat sink); Forced convection; Regenerative.

Throat Diameter	`ad_f_thermal_throat_diameter`	m		(blank)

Effect on the calculation: Throat diameter, m (> 0, at most 2 m: the input check band). Sets the throat area of the Bartz design station. Filled from the motor result after a calculation, unless you edit it. *If blank:* Not sent; the thermal check derives the throat area from the mass flow and c*.

Characteristic velocity c*	`ad_f_thermal_c_star`	m/s		(blank)

Effect on the calculation: Characteristic velocity c*, m/s (> 0). Sets the Bartz mass flux (Pc/c*) and, without a throat diameter, the throat area. Filled from the motor result after a calculation, unless you edit it. *If blank:* Not sent; the thermal check derives c* from γ, the gas constant and the chamber temperature.

Gas gamma (Bartz)	`ad_f_thermal_gamma`			(blank)

Effect on the calculation: Specific-heat ratio γ of the combustion gas used by the Bartz chain ($1.05 < \gamma < 1.67$, the heat-transfer module band; outside it the request is refused and the field is named). Sets the gas Prandtl number and cp. Filled from the motor result after a calculation (solid: the motor's own Bartz gas pair), unless you edit it. *If blank:* Not sent; the module uses its declared default $\gamma = 1.20$.

Molecular Weight	`ad_f_thermal_molecular_weight`	g/mol		(blank)

Effect on the calculation: Gas molecular weight, g/mol (> 0). Sets the gas constant of the Bartz chain. Filled from the motor result after a calculation (solid: the motor's own Bartz gas pair), unless you edit it. *If blank:* Not sent; the module uses its declared default of 24.0 g/mol.

continued on next page

Input	Identifier	Unit	Form limit	Form default
H2O mole fraction (radiation)	`ad_f_thermal_x_H2O`			(blank)

Effect on the calculation: H2O mole fraction of the combustion gas (0 to 1; x_H2O + x_CO2 at most 1). Sets the gas radiation part of the wall heat flux (Leckner emissivity). Filled from the motor result only when the motor solved its own composition (hybrid), unless you edit it. *If blank:* Not sent; the module uses its declared default composition and says so in its result.

CO2 mole fraction (radiation)	`ad_f_thermal_x_CO2`			(blank)

Effect on the calculation: CO2 mole fraction of the combustion gas (0 to 1; x_H2O + x_CO2 at most 1). Sets the gas radiation part of the wall heat flux (Leckner emissivity). Filled from the motor result only when the motor solved its own composition (hybrid), unless you edit it. *If blank:* Not sent; the module uses its declared default composition and says so in its result.

Supplied hot-side wall temperature	`ad_f_thermal_wall_temperature_hot`	K		(blank)

Effect on the calculation: Supplied hot-side (gas-side) wall temperature, K (> 0, not above the chamber temperature; send it together with the cold side). When given, the thermal verdict is evaluated at this wall and the bare-wall equilibrium is kept as a bound only. Filled after a calculation with the motor result's own SOLVED wall: solid, the case temperature behind the insulation and liner (lumped burn transient); liquid, the regenerative station march. Not filled for a hybrid (its case temperature is declared, not solved). *If blank:* The bare-wall equilibrium gives the verdict.

Supplied cold-side wall temperature	`ad_f_thermal_wall_temperature_cold`	K		(blank)

Effect on the calculation: Supplied cold-side (outer) wall temperature, K (> 0, not above the hot side; send it together with the hot side). Sets the through-wall gradient of the thermal stress. Filled together with the hot side from the motor result (solid: the same case temperature, a lumped case has no gradient). *If blank:* The bare-wall equilibrium gives the verdict.

Wall protection	`ad_f_thermal_wall_protection`			None (bare wall)

Effect on the calculation: Protection of the chamber wall. None: the bare-wall verdict. Ablative liner: when the bare wall fails, HRMA sizes one ablative liner for the chamber wall (Level-1 Q*, the same sizing as the hybrid motor's chamber liner) and the verdict becomes LINER REQUIRED (warning); if no liner can be sized, the bare-wall verdict stands. Filled from the motor result after a calculation: a hybrid whose chamber zones are all protected selects Ablative liner. Ignored when supplied wall temperatures are given. Options: None (bare wall); Ablative liner sized by HRMA (Level-1 Q*).

Table 13.4: Dock: Thermal Protection panel inputs

Input	Identifier	Unit	Form limit	Form default
Analysis Mode	`ad_f_protection_mode`			Ablative liner (Q*)

Effect on the calculation: Analysis mode: ablative liner, heat-sink wall or radiation-cooled extension. A different model runs for each mode; fields tagged [A], [H], [R] belong to that mode only. Not filled from the motor result. Options: Ablative liner (Q*); Heat-sink wall (1-D); Radiative extension.

[A] Net Heat Flux	`ad_f_protection_q_net_W_m2`	W/m^2		2000000

Effect on the calculation: [A] Net wall heat flux, W/m^2 (must be ≥ 0). Ablative recession load; when h_gas and recovery temperature are both given the surface energy balance is used instead. Filled from the motor's own Bartz throat heat flux when available, unless you edit it. *If blank:* The request is not sent and the field is named.

[A] Ablative Material	`ad_f_protection_material`			Silica-phenolic

Effect on the calculation: [A] Ablative material. Supplies density, Q* band and emissivity of the liner. Used only in ablative mode. Options: Silica-phenolic; Carbon-phenolic; EPDM insulation.

continued on next page

Input	Identifier	Unit	Form limit	Form default
[A] Design Margin	`ad_f_protection_design_margin`	×		1.5
Effect on the calculation: [A] Design margin on the ablative thickness, × (must be ≥ 1.0). Multiplies the required liner thickness. Not filled from the motor result. *If blank:* The request is not sent and the field is named.				
[A] Q* Override (MJ/kg, blank = band)	`ad_f_protection_q_star_MJ_kg`			(blank)
Effect on the calculation: [A] Effective heat of ablation Q*, MJ/kg (must be > 0). Overrides the material's Q*; changes recession rate and required thickness. Not filled from the motor result. *If blank:* Not sent; the conservative end of the material's Q* band is used.				
[A/H] Burn Time	`ad_f_protection_burn_time_s`	s		10
Effect on the calculation: [A/H] Exposure (burn) time, s (must be > 0). Sets the total heat load, recession and required thickness. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.				
[H/R] h_gas – Bartz	`ad_f_protection_h_gas_W_m2K`	$W/m^2 \cdot K$		3000
Effect on the calculation: [H/R] Gas-side heat transfer coefficient (Bartz), $W/m^2 \cdot K$ (≥ 0). Drives the convective heating; in ablative mode it is used together with the recovery temperature. Filled from the motor's Bartz result when available, unless you edit it. *If blank:* The request is not sent and the field is named.				
[H/R] Recovery Temperature	`ad_f_protection_T_recovery_K`	K		3200
Effect on the calculation: [H/R] Recovery (adiabatic wall) temperature, K (> 0). Driving gas temperature of the convective heating; must be given together with h_gas. Filled from the motor's adiabatic wall temperature when available, unless you edit it. *If blank:* The request is not sent and the field is named.				
[H] Wall Thickness	`ad_f_protection_wall_thickness_m`	m		0.005
Effect on the calculation: [H] Wall thickness, m (> 0). Heat-sink capacity of the wall. Used only in heat-sink mode. *If blank:* The request is not sent and the field is named.				
[H] Wall Material	`ad_f_protection_wall_material`			Steel (generic)
Effect on the calculation: [H] Wall material. Supplies the material properties of the heat-sink wall. Used only in heat-sink mode. Options: Steel (generic); Steel AISI 4130; Stainless 304; Stainless 316; Aluminum 6061-T6; Copper (OFHC); CuCrZr; Inconel 718; Titanium Ti-6Al-4V; Graphite.				
[H] Initial Temperature	`ad_f_protection_T_initial_K`	K		300
Effect on the calculation: [H] Initial wall temperature, K (> 0). Starting temperature of the transient heat-sink calculation. Used only in heat-sink mode. *If blank:* The request is not sent and the field is named.				
[R] Extension Material	`ad_f_protection_radiation_material`			Nb C-103 (silicide)
Effect on the calculation: [R] Radiation-cooled extension material. Supplies the material properties (incl. emissivity) for the radiation-equilibrium check. Used only in radiation-equilibrium mode. Options: Nb C-103 (silicide); Carbon-carbon (C-C); Stainless 316; Inconel 718.				
[R] Emissivity Override (blank = material)	`ad_f_protection_emissivity`			(blank)
Effect on the calculation: [R] Surface emissivity, 0–1. Overrides the material's emissivity in the radiation balance. Used only in radiation-equilibrium mode. *If blank:* Not sent; the material's own emissivity is used.				

Table 13.5: Dock: Regenerative Cooling panel inputs (liquid)

Input	Identifier	Unit	Form limit	Form default
Chamber Pressure	`ad_f_cooling_chamber_pressure`	bar		50
Effect on the calculation: Chamber pressure, bar. Used in the gas-side heat flux along the channel march. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.				
Chamber Temperature	`ad_f_cooling_chamber_temperature`	K		3400
Effect on the calculation: Chamber temperature, K. Drives the gas-side heat flux. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.				
Throat Diameter	`ad_f_cooling_throat_diameter`	m		0.05
Effect on the calculation: Throat diameter, m. Sets the contour and the throat heat-flux peak. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.				
Expansion Ratio	`ad_f_cooling_expansion_ratio`			8
Effect on the calculation: Nozzle area ratio Ae/At. Sets the cooled nozzle length and contour. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.				
Gamma (frozen)	`ad_f_cooling_gamma`			1.2
Effect on the calculation: Frozen ratio of specific heats γ. Enters the gas-side heat transfer. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.				
Molecular Weight	`ad_f_cooling_molecular_weight`	g/mol		22
Effect on the calculation: Gas molecular weight, g/mol. Enters the gas properties of the heat transfer. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.				
Coolant	`ad_f_cooling_coolant`			– select –
Effect on the calculation: Coolant fluid. Selects the coolant property model. Filled from the motor result after a calculation (when the result provides it), unless you edit it. Options: – select –; Water; RP-1 (kerosene); Methane (LCH4); Hydrogen (LH2). *If blank:* Required – the request is not sent and the field is named.				
Coolant Mass Flow	`ad_f_cooling_coolant_mdot`	kg/s		2
Effect on the calculation: Coolant mass flow, kg/s. Sets coolant velocity, heat pickup and pressure drop. Filled with the motor's fuel flow after a calculation, unless you edit it. *If blank:* The request is not sent and the field is named.				
Coolant Inlet T	`ad_f_cooling_coolant_inlet_temp`	K		(blank)
Effect on the calculation: Coolant inlet temperature, K. Starting temperature of the coolant march. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* Required – the request is not sent and the field is named.				
Coolant Inlet P	`ad_f_cooling_coolant_inlet_pressure`	bar		(blank)
Effect on the calculation: Coolant inlet pressure, bar. Starting pressure of the march; if it falls to zero along the channels the analysis is refused. Filled from the motor result: the inlet pressure of the engine's own station march (RP-1: the thermal-protection station march; methane/LH2: the supercritical march). *If blank:* Required – the request is not sent and the field is named.				

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Input	Identifier	Unit	Form limit	Form default
Number of Channels	`ad_f_cooling_n_channels`			(blank)

Effect on the calculation: Number of cooling channels (positive whole number). Splits the coolant flow (changes velocity, heat transfer, pressure drop). Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* Required – the request is not sent and the field is named.

Channel Width	`ad_f_cooling_channel_width`	mm		(blank)

Effect on the calculation: Channel width, mm (> 0). Sets the channel cross-section and hydraulic diameter. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* Required – the request is not sent and the field is named.

Channel Height	`ad_f_cooling_channel_height`	mm		(blank)

Effect on the calculation: Channel height, mm (> 0). Sets the channel cross-section and hydraulic diameter. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* Required – the request is not sent and the field is named.

Liner Thickness	`ad_f_cooling_wall_thickness`	mm		(blank)

Effect on the calculation: Hot-gas liner thickness, mm (> 0). Sets the conduction resistance between gas and coolant. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* Required – the request is not sent and the field is named.

Liner Material	`ad_f_cooling_wall_material`			Copper (OFHC)

Effect on the calculation: Liner material. Supplies the liner material properties (changes the result). Not filled from the motor result. Options: Copper (OFHC); CuCrZr; Inconel 718; Stainless 304; Steel AISI 4130.

Flow Direction	`ad_f_cooling_flow_direction`			Counterflow (exit → chamber)

Effect on the calculation: Coolant flow direction: counterflow (exit → chamber) or coflow. Changes where the coolant is coldest along the contour. Not filled from the motor result. Options: Counterflow (exit → chamber); Co-flow (chamber → exit).

Table 13.6: Dock: Structural Safety panel inputs

Input	Identifier	Unit	Form limit	Form default
Chamber Pressure	`ad_f_structural_chamber_pressure`	bar		40

Effect on the calculation: Chamber design pressure, bar. The wall stresses and safety factors (pressure vessel, buckling, fatigue) are computed at this pressure. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

Chamber Diameter	`ad_f_structural_chamber_diameter`	m		0.1

Effect on the calculation: Chamber inner diameter, m. Sets the vessel radius in the stress and buckling checks. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

Chamber Length	`ad_f_structural_chamber_length`	m		0.5

Effect on the calculation: Chamber cylinder length, m. Enters the structural model with the chamber geometry; changing it changes the structural result. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

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Input	Identifier	Unit	Form limit	Form default
Throat Diameter	`ad_f_structural_throat_diameter`	m		0.02

Effect on the calculation: Nozzle throat diameter, m. Enters the structural model with the chamber geometry; changing it changes the structural result. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

Burn Time	`ad_f_structural_burn_time`	s		(blank)

Effect on the calculation: Burn time, s. Informational here: the analysis validates it but does NOT use it (stresses are steady-state; fatigue counts pressurisation cycles) and says so under the result. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* Not sent.

Gas Temperature (K, blank = skip thermal)	`ad_f_structural_chamber_temperature`			(blank)

Effect on the calculation: Combustion gas temperature, K. Feeds the thermal-stress path of the structural check. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* Not sent and the thermal path is skipped (reported as NOT_MODELLED; no thermal verdict is given).

Thrust	`ad_f_structural_thrust`	N		0

Effect on the calculation: Thrust, N. Axial compressive load for the buckling check; 0 skips the axial load. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

Wall Thickness	`ad_f_structural_wall_thickness`	m		0

Effect on the calculation: Chamber wall thickness, m. A value > 0 is verified as your wall; 0 means HRMA sizes the wall itself (result is a design proposal, not a verification). Must be smaller than the chamber radius, otherwise 422. Filled from the motor result only when the motor verified your own wall; otherwise set to 0. *If blank:* The request is not sent and the field is named.

Hot-side Wall Temperature (K, blank = let HRMA estimate from gas temperature)	`ad_f_structural_wall_temperature_hot`			(blank)

Effect on the calculation: Hot-side (gas-side) wall temperature, K. Used first for the through-wall temperature difference and the material strength at temperature. Filled only when the motor solver declares the wall temperature solved. *If blank:* Not sent and HRMA estimates it from the gas temperature.

Cold-side Wall Temperature (K, blank = let HRMA estimate from gas temperature)	`ad_f_structural_wall_temperature_cold`			(blank)

Effect on the calculation: Cold-side (outer) wall temperature, K. Paired with the hot-side value for the through-wall temperature difference. Filled only when the motor solver declares the wall temperature solved. *If blank:* Not sent and HRMA estimates it from the gas temperature.

Design Safety Factor target (blank = material database default)	`ad_f_structural_safety_factor`			(blank)

Effect on the calculation: Design safety factor target, dimensionless. The structural margins are judged against this target. Filled with the motor's own target when the result has one. *If blank:* Not sent; the material database default is used and declared under the result.

continued on next page

Input	Identifier	Unit	Form limit	Form default
Material	`ad_f_structural_material`			Steel AISI 4130

Effect on the calculation: Chamber wall material. Its yield/ultimate strength scales every structural safety factor; an unknown name is rejected, not substituted. Filled from the motor result after a calculation (when the result provides it), unless you edit it. Options: Steel AISI 4130; Steel AISI 4340; Carbon steel (A36); Stainless 304; Stainless 316; 17-4PH (H900); Aluminum 6061-T6; Aluminum 7075-T6; Aluminum 2024-T3; Titanium Ti-6Al-4V; Titanium CP Grade 2; Inconel 718; Inconel 625; CuCrZr (C18150); BeCu C17200; Magnesium AZ31B.

Table 13.7: Dock: Bolted Joint panel inputs

Input	Identifier	Unit	Form limit	Form default
Chamber Pressure	`ad_f_joint_pressure_bar`	bar		40

Effect on the calculation: Chamber pressure acting on the closure, bar (≥ 0). The analysis turns it into the separating force over the seal area. Filled with the motor's chamber pressure after a calculation, unless you edit it. *If blank:* The request is not sent and the field is named.

Seal / Effective Diameter	`ad_f_joint_seal_diameter_mm`	mm		100

Effect on the calculation: Seal / effective pressure diameter, mm. Area on which the pressure acts; must be positive and within the repository's chamber-diameter band. Filled with the motor's chamber inner diameter after a calculation, unless you edit it. *If blank:* The request is not sent and the field is named.

Bolt Count	`ad_f_joint_bolt_count`			8

Effect on the calculation: Number of bolts (≥ 1). The separating load is shared between them. Not filled from the motor result. *If blank:* The request is not sent and the field is named.

Bolt Size	`ad_f_joint_size`			M8

Effect on the calculation: Metric bolt size. Sets the tensile stress area and preload; unknown sizes are rejected. Not filled from the motor result. Options: M4; M5; M6; M8; M10; M12; M14; M16; M18; M20; M22; M24.

Property Class	`ad_f_joint_property_class`			8.8 (ISO 898-1)

Effect on the calculation: Bolt property class (8.8, 10.9, 12.9, A2-70). Sets the bolt proof/yield strength and preload. Not filled from the motor result. Options: 8.8 (ISO 898-1); 10.9 (ISO 898-1); 12.9 (ISO 898-1); A2-70 stainless (ISO 3506-1).

Grip Length	`ad_f_joint_grip_length_mm`	mm		20

Effect on the calculation: Grip (clamped) length, mm (> 0). Sets bolt and member stiffness, hence the load share. Not filled from the motor result. *If blank:* The request is not sent and the field is named.

Member Material	`ad_f_joint_member_material`			Aluminum 6061-T6

Effect on the calculation: Material of the clamped members. Its Young's modulus sets the member stiffness k_m. Not filled from the motor result. Options: Aluminum 6061-T6; Steel (generic); Steel AISI 4130; Stainless 304; Stainless 316; Titanium Ti-6Al-4V; Inconel 718.

Thread Condition	`ad_f_joint_lubricated`			Dry assembly (K = 0.20)

Effect on the calculation: Thread condition: dry or lubricated. Changes the torque–preload relation. Not filled from the motor result. Options: Dry assembly (K = 0.20); Lubricated (K = 0.15).

Connection Type	`ad_f_joint_reusable`			Reusable – preload 75% of proof

Effect on the calculation: Connection type: reusable or permanent. Changes the target preload fraction of proof load. Not filled from the motor result. Options: Reusable – preload 75% of proof; Permanent – preload 90% of proof.

continued on next page

Input	Identifier	Unit	Form limit	Form default
Extra Axial Load (N, optional)	`ad_f_joint_external_axial_load_n`			(blank)

Effect on the calculation: Extra external axial load, N (total must be ≥ 0). Added to the pressure load on the joint. Not filled from the motor result. *If blank:* Not sent; only the pressure load is used.

Table 13.8: Dock: Comprehensive Safety panel inputs

Input	Identifier	Unit	Form limit	Form default
Motor Type	`ad_f_safety_motor_type`			Hybrid

Effect on the calculation: Motor type: hybrid, liquid or solid. The hazard model (explosive equivalent, stored propellant) is selected by motor type. Set from the page's motor after a calculation. Options: Hybrid; Liquid; Solid.

Chamber Pressure	`ad_f_safety_chamber_pressure`	bar		40

Effect on the calculation: Chamber pressure, bar. Enters the pressure-vessel and stored-energy terms of the safety verdict. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

Chamber Temperature	`ad_f_safety_chamber_temperature`	K		3000

Effect on the calculation: Chamber temperature, K. Enters the thermal part of the safety verdict. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

Thrust	`ad_f_safety_thrust`	N		1000

Effect on the calculation: Thrust, N. Enters the safety verdict (changes the result). Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

Burn Time	`ad_f_safety_burn_time`	s		(blank)

Effect on the calculation: Burn time, s. Feeds the duration-dependent parts of the safety verdict. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* Not sent; the endpoint withholds the duration-dependent parts by name (no duration is assumed).

Propellant Mass	`ad_f_safety_propellant_mass`	kg		(blank)

Effect on the calculation: Propellant mass, kg. Sets the explosive (TNT) equivalent and the evacuation distances. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* Not sent; the analysis is refused with this field named.

Propellant Type	`ad_f_safety_propellant_type`			Composite (APCP/HTPB)

Effect on the calculation: Propellant class. Selects the hazard class used for the explosive equivalent. Set from the motor's own propellant after a calculation. Options: Composite (APCP/HTPB); Double base (NC/NG); Composite double base; Liquid bipropellant; Liquid monopropellant; Solid monopropellant (AP); Not modelled (no sourced TNT equivalent).

Facility Type	`ad_f_safety_facility_type`			Test stand

Effect on the calculation: Facility: test stand, manufacturing, transport or launch site. Selects the facility rules of the safety verdict. Not filled from the motor result. Options: Test stand; Manufacturing; Transport; Launch site.

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Input	Identifier	Unit	Form limit	Form default
Chamber Diameter	`ad_f_safety_chamber_diameter`	m		0.1
Effect on the calculation: Chamber inner diameter, m. Enters the pressure-vessel part of the safety verdict. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.				
Wall Thickness	`ad_f_safety_wall_thickness`	m		0.005
Effect on the calculation: Wall thickness, m. Enters the pressure-vessel part of the safety verdict. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.				
Material	`ad_f_safety_material`			Steel AISI 4130
Effect on the calculation: Wall material. Supplies the strength used by the pressure-vessel part of the verdict. Filled from the motor result after a calculation (when the result provides it), unless you edit it. Options: Steel AISI 4130; Steel AISI 4340; Carbon steel (A36); Stainless 304; Stainless 316; 17-4PH (H900); Aluminum 6061-T6; Aluminum 7075-T6; Aluminum 2024-T3; Titanium Ti-6Al-4V; Titanium CP Grade 2; Inconel 718; Inconel 625; CuCrZr (C18150); BeCu C17200; Magnesium AZ31B.				
Chamber Length	`ad_f_safety_chamber_length`	m		(blank)
Effect on the calculation: Chamber length, m. Needed for the thermal part of the safety verdict. Not filled from the motor result. *If blank:* Not sent and the thermal verdict is withheld.				
Throat Diameter	`ad_f_safety_throat_diameter`	m		(blank)
Effect on the calculation: Nozzle throat diameter, m. Needed for the thermal part of the safety verdict. Not filled from the motor result. *If blank:* Not sent and the thermal verdict is withheld.				
Hot-side Wall Temperature (K, blank = let HRMA estimate from gas temperature)	`ad_f_safety_wall_temperature_hot`			(blank)
Effect on the calculation: Hot-side wall temperature, K. Enters the thermal part of the safety verdict. Filled only when the motor solver declares the wall temperature solved. *If blank:* Not sent; HRMA estimates it from the gas temperature.				
Cold-side Wall Temperature (K, blank = let HRMA estimate from gas temperature)	`ad_f_safety_wall_temperature_cold`			(blank)
Effect on the calculation: Cold-side wall temperature, K. Enters the thermal part of the safety verdict. Filled only when the motor solver declares the wall temperature solved. *If blank:* Not sent; HRMA estimates it from the gas temperature.				

Table 13.9: Dock: Advanced Performance panel inputs

Input	Identifier	Unit	Form limit	Form default
Chamber Pressure	`ad_f_performance_chamber_pressure`	bar		50
Effect on the calculation: Chamber pressure, bar. Used in all three figures (Isp surface design point, Mach contour, heat flux). Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.				

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Input	Identifier	Unit	Form limit	Form default
Design O/F Ratio	`ad_f_performance_optimal_of_ratio`			3.5

Effect on the calculation: Oxidizer-to-fuel mass ratio of the design point. Used only by the 3D Isp surface (hybrid/liquid). Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

Base Isp	`ad_f_performance_base_isp`	s		300

Effect on the calculation: Design-point specific impulse, s. Used only by the 3D Isp surface (hybrid/liquid). Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

Expansion Ratio	`ad_f_performance_expansion_ratio`			16

Effect on the calculation: Nozzle area ratio Ae/At. Used by the nozzle Mach contour and the wall heat-flux figures. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

Throat Area	`ad_f_performance_throat_area`	m^2		0.001

Effect on the calculation: Throat area, m^2. Used by the nozzle Mach contour and the wall heat-flux figures. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

Nozzle Length	`ad_f_performance_nozzle_length`	m		0.1

Effect on the calculation: Divergent nozzle length, m. Used by the nozzle Mach contour and the wall heat-flux figures. Filled from the motor result after a calculation (the divergent length the solver published, design summary), unless you edit it. *If blank:* The request is not sent and the field is named.

Chamber Length	`ad_f_performance_chamber_length`	m		0.5

Effect on the calculation: Chamber length, m. Used only by the wall heat-flux figure. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

Burn Time	`ad_f_performance_burn_time`	s		(blank)

Effect on the calculation: Burn time, s. Used only by the wall heat-flux figure. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* Not sent.

Base Heat Flux	`ad_f_performance_base_heat_flux`	W/m^2		2000000

Effect on the calculation: Reference wall heat flux, W/m^2. Used only by the wall heat-flux figure. Not filled from the motor result. *If blank:* The request is not sent and the field is named.

Critical Heat Flux	`ad_f_performance_critical_heat_flux`	MW/m^2		4

Effect on the calculation: Critical heat-flux line, MW/m^2. Used only by the wall heat-flux figure. Not filled from the motor result. *If blank:* The request is not sent and the field is named.

Chamber Temperature	`ad_f_performance_chamber_temperature`	K		3000

Effect on the calculation: Chamber temperature, K. Used only by the wall heat-flux figure. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

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Input	Identifier	Unit	Form limit	Form default
Gamma (frozen)	`ad_f_performance_gamma`			1.2

Effect on the calculation: Frozen ratio of specific heats γ. Used by the nozzle Mach contour and the wall heat-flux figures. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

Total Mass Flow	`ad_f_performance_mdot_total`	kg/s		1

Effect on the calculation: Total mass flow, kg/s. Informational: sent with the request but none of the three figures changes with it (measured). Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

Chamber Diameter	`ad_f_performance_chamber_diameter`	m		0.1

Effect on the calculation: Chamber inner diameter, m. Used by the nozzle Mach contour and the wall heat-flux figures. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* The request is not sent and the field is named.

Table 13.10: Dock: Pressure Vessel panel inputs

Input	Identifier	Unit	Form limit	Form default
Design Code	`ad_f_vessel_code_mode`			AIAA S-080 (flight hardware)

Effect on the calculation: Design code: AIAA S-080 (metallic pressure vessels: proof 1.5 × MEOP, burst 2.0 × MEOP), ASME VIII Div 1, or NASA-STD-5012C (liquid engine hardware: acceptance test 1.20 × MDP, qualification test 1.50 × MDP, burst 1.50 × MDP). The selected code's factors set the required thickness, the test pressures and the burst margin. Filled from the motor result only on the liquid page (NASA-STD-5012C, the same standard as the safety tab); otherwise AIAA S-080. Options: AIAA S-080 (flight hardware); ASME VIII Div 1 (ground/test); NASA-STD-5012C (liquid engine hardware).

MEOP	`ad_f_vessel_meop_bar`	bar		60

Effect on the calculation: Maximum expected operating pressure (MEOP), bar. Design pressure of the vessel: drives required thickness and burst margins. Must be positive and finite. Filled after a calculation, unless you edit it: with the motor's chamber pressure; on a two-layer regenerative liquid wall with the closeout's maximum design pressure (MDP: the highest coolant pressure in the jacket). *If blank:* The request is not sent and the field is named.

Inner Diameter	`ad_f_vessel_inner_diameter_mm`	mm		150

Effect on the calculation: Vessel inner diameter, mm. Sets the required wall and head thicknesses. Must be positive and finite. Filled after a calculation, unless you edit it: with the motor's chamber diameter; on a two-layer regenerative liquid wall with the inner diameter of the closeout, the layer that carries the coolant pressure (at the largest jacket radius, as the engine's closeout check). *If blank:* The request is not sent and the field is named.

Wall Thickness (mm, blank = auto-size)	`ad_f_vessel_wall_thickness_mm`			(blank)

Effect on the calculation: Actual wall thickness, mm. Given: burst pressure and margins are computed for this wall (must be positive). On a two-layer regenerative liquid wall it is filled with the engine's closeout thickness, unless you edit it; otherwise it is not filled from the motor result. *If blank:* Not sent and the wall is auto-sized (the result says so).

Material	`ad_f_vessel_material`			Aluminum 6061-T6

Effect on the calculation: Vessel material. Supplies the allowable stress (changes thickness, burst pressure and margins). Filled after a calculation, unless you edit it: on the solid page with the case material, on the hybrid page with the chamber material, on a two-layer regenerative liquid wall with the closeout material; otherwise the panel default stays. Options: Aluminum 6061-T6; Steel AISI 4130; Steel (generic); Stainless 304; Stainless 316; Titanium Ti-6Al-4V; Inconel 718.

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Input	Identifier	Unit	Form limit	Form default
Design Temperature	`ad_f_vessel_temperature_K`	K		293.15
Effect on the calculation: Design temperature, K. Derates the material allowable stress. Must be positive and within the repository's physical temperature band. On a two-layer regenerative liquid wall it is filled with the closeout temperature of the engine (the hottest coolant), unless you edit it; otherwise it is not filled from the motor result. *If blank:* The request is not sent and the field is named.				
Weld Joint Efficiency (UW-12)	`ad_f_vessel_weld_efficiency`			1.00 – full radiography
Effect on the calculation: Weld joint efficiency E (ASME VIII Table UW-12): 1.00, 0.85 or 0.70. Multiplies the allowable stress in the thickness equation; other values are rejected. Not filled from the motor result. Options: 1.00 – full radiography; 0.85 – spot radiography; 0.70 – no radiography.				
Head Type	`ad_f_vessel_head_type`			2:1 Ellipsoidal
Effect on the calculation: End-cap (head) shape: 2:1 ellipsoidal, torispherical or hemispherical. Selects the head thickness formula (changes head thickness and margin). Not filled from the motor result. Options: 2:1 Ellipsoidal; Torispherical; Hemispherical.				

Table 13.11: Dock: Nozzle Flow panel inputs

Input	Identifier	Unit	Form limit	Form default
Fidelity Level	`ad_f_flow_fidelity`			Engineering
Effect on the calculation: Flow solution fidelity: fast or engineering. Selects the level of the quasi-1D nozzle solver. Not filled from the motor result. Options: Fast Screening; Engineering.				
Kinetic Model	`ad_f_flow_kinetic_fidelity`			Engineering (JANNAF-style)
Effect on the calculation: Kinetic model level: fast or engineering. Selects the kinetic-efficiency model. Not filled from the motor result. Hybrid motors only: not shown for solid and liquid motors (the kinetics chain has no chemistry for them). Options: Fast (equilibrium reference); Engineering (JANNAF-style); Cantera finite-rate kinetics.				
Chamber Pressure	`ad_f_flow_chamber_pressure`	bar		(blank)
Effect on the calculation: Chamber pressure, bar. Stagnation pressure of the nozzle flow (changes flow and kinetic efficiency). Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* Not sent; the analysis is refused with this field named (no default is assumed).				
Chamber Temperature	`ad_f_flow_chamber_temperature`	K		(blank)
Effect on the calculation: Chamber temperature, K. Stagnation temperature of the nozzle flow. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* Not sent; the analysis is refused with this field named (no default is assumed).				
Gamma (frozen)	`ad_f_flow_gamma`			(blank)
Effect on the calculation: Frozen ratio of specific heats γ. Sets the area–Mach relation (changes flow and kinetic efficiency). Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* Not sent; the analysis is refused with this field named (no default is assumed).				
Molecular Weight	`ad_f_flow_molecular_weight`	g/mol		(blank)
Effect on the calculation: Gas molecular weight, g/mol (> 0). Sets the gas constant of the flow. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* Not sent; the analysis is refused with this field named (no default is assumed).				

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Input	Identifier	Unit	Form limit	Form default
Throat Diameter	`ad_f_flow_throat_diameter`	m		(blank)

Effect on the calculation: Throat diameter, m (> 0). Sets the throat area and mass flow. Filled from the motor result after a calculation (when the result provides it), unless you edit it. *If blank:* Not sent; the analysis is refused with this field named (no default is assumed).

Exit Diameter	`ad_f_flow_exit_diameter`	m		(blank)

Effect on the calculation: Nozzle exit diameter, m (must exceed the throat diameter). Sets the expansion ratio of the solution. Filled from the motor result (or throat $\times \sqrt{}$expansion ratio) after a calculation, unless you edit it. *If blank:* Not sent; the analysis is refused with this field named (no default is assumed).

Ambient Pressure	`ad_f_flow_ambient_pressure`	Pa		(blank)

Effect on the calculation: Ambient (back) pressure, Pa ($\geq$ 0 and below the chamber pressure). Sets the exit regime (over/under-expansion, separation) and thrust. Filled after a calculation with the design ambient pressure the motor's nozzle was solved against (bar $\rightarrow$ Pa), unless you edit it. *If blank:* Not sent; the analysis is refused with this field named (no default is assumed).

Axial Stations (30-60)	`ad_f_flow_n_stations`			45

Effect on the calculation: Number of axial grid stations, whole number (30–60). Resolution of the quasi-1D solution; values outside 30–60 are clamped to that range, a fractional count is rejected. Not filled from the motor result. *If blank:* The request is not sent and the field is named.

O/F Ratio (kinetics)	`ad_f_flow_of_ratio`			(blank)

Effect on the calculation: O/F mass ratio for the kinetics model (hybrid motors only). Changes only the kinetic efficiency. Filled from the motor result after a calculation, unless you edit it. Not shown for solid and liquid motors. *If blank:* The request is not sent and the field is named (no default ratio is assumed).

Fuel (kinetics)	`ad_f_flow_fuel_type`			– select –

Effect on the calculation: Fuel for the kinetics model (hybrid motors only). Selects the fuel of the kinetic-efficiency chain; it does not change the quasi-1D flow solution. Filled from the motor result after a calculation (the motor's fuel), unless you edit it. Only the hybrid fuels the kinetics chain knows are listed; a motor fuel outside the list stops the request and names it. Not shown for solid and liquid motors (the kinetics chain has no chemistry for them). Options: – select –; HTPB; Paraffin; PE; PMMA; ABS; PLA. *If blank:* The request is not sent and the field is named.

Oxidizer (kinetics)	`ad_f_flow_oxidizer_type`			– select –

Effect on the calculation: Oxidizer for the kinetics model (hybrid motors only). Selects the oxidizer of the kinetic-efficiency chain; it does not change the quasi-1D flow solution. Filled from the motor result after a calculation (the oxidizer the motor's combustion chain used), unless you edit it. A motor oxidizer outside the list stops the request and names it. Not shown for solid and liquid motors. Options: – select –; N2O; LOX; H2O2. *If blank:* The request is not sent and the field is named.

Table 13.12: Dock: Feed System panel inputs (slosh, pressurant, water hammer)

Input	Identifier	Unit	Form limit	Form default
Tank Radius	`feed_slosh_radius`	m		(blank)

Effect on the calculation: Tank radius, m (> 0). The sloshing modes are computed with it. After a calculation it is filled from the motor result unless you edit it (hybrid: half the tank diameter of the run's oxidizer-tank sloshing block; liquid: the radius of the run's oxidizer-tank slosh block). *If blank:* Not sent; the analysis rejects the request naming this field (required).

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Input	Identifier	Unit	Form limit	Form default
Fill Height	`feed_slosh_fill_height`	m		(blank)

Effect on the calculation: Liquid fill height, m (> 0). Enters the sloshing frequency and mass. After a calculation it is filled from the motor result unless you edit it (hybrid: the fill height of the run's oxidizer-tank sloshing block; liquid: the fill height of the run's oxidizer-tank slosh block). *If blank:* Not sent; the analysis rejects the request naming this field (required).

Axial Acceleration	`feed_slosh_g_eff`	m/s^2		(blank)

Effect on the calculation: Axial acceleration, m/s^2 (> 0). Restoring acceleration of the slosh modes. Filled from the run's own slosh block (hybrid and liquid: the constant standard 1 g declared there). *If blank:* Not sent; the endpoint uses standard gravity g0 = 9.80665 m/s^2 and lists it below the result as a defaulted field.

Propellant Density	`feed_slosh_fluid_density`	kg/m^3		(blank)

Effect on the calculation: Propellant density, kg/m^3 (> 0). Sets the sloshing mass and forces. After a calculation it is filled from the motor result unless you edit it (hybrid: the liquid density of the run's oxidizer-tank sloshing block; liquid: the oxidizer density of the run, the density its slosh block used). *If blank:* Not sent; the analysis computes the modes without a liquid mass.

Baffle Width w/R	`feed_slosh_baffle_width_ratio`			(blank)

Effect on the calculation: Ring baffle width as a fraction of the tank radius, w/R ($0 < w/R \leq 1$). Sets the ring-baffle damping (Miles 1958). Liquid: filled with the width ratio of the tank's own ring baffle (oxidizer-tank slosh block). Hybrid: not filled. *If blank:* Not sent; the endpoint returns the baffle width needed for its target damping (the hybrid motor's own slosh block runs the same way).

Baffle Depth d/R	`feed_slosh_baffle_depth_ratio`			(blank)

Effect on the calculation: Baffle depth below the free surface as a fraction of the radius, d/R (≥ 0). Sets the ring-baffle damping (Miles 1958). Filled with the depth ratio the run's slosh block used (hybrid: the oxidizer-tank sloshing block; liquid: the tank's own ring baffle). *If blank:* Not sent; the analysis uses d/R = 0.10 and lists it below the result as a defaulted field.

Architecture	`feed_pressurant_mode`			Regulated

Effect on the calculation: Pressurisation architecture: regulated or blowdown. The analysis sizes the pressurant for this architecture; fields of the other mode are ignored. Not filled from the motor result. Options: Regulated; Blowdown.

Pressurant Gas	`feed_pressurant_gas`			Helium

Effect on the calculation: Pressurant gas: helium or nitrogen. Sets the gas properties of the sizing. Liquid: filled with the pressurant type of the run when it is helium or nitrogen. Hybrid: not filled (the N2O tank is self-pressurised, the run has no pressurant gas). Options: Helium; Nitrogen.

Propellant Volume	`feed_pressurant_propellant_volume`	m^3		(blank)

Effect on the calculation: Propellant volume to expel, m^3 (> 0). Volume the pressurant must displace. Liquid: filled with the propellant volume of the run's oxidizer tank (L $\rightarrow$ m^3, without the ullage) unless you edit it. Hybrid: not filled; the N2O tank is self-pressurised and the run publishes no stored-gas system. *If blank:* The analysis rejects the request naming this field (required).

Gas Temperature	`feed_pressurant_initial_temperature`	K		(blank)

Effect on the calculation: Pressurant gas temperature, K (> 0; at the end of the expansion the gas must stay above 4.222 K for helium and 63.151 K for nitrogen). Sets the gas density and the required mass. Not filled from the motor result. *If blank:* The endpoint uses 293.15 K (20 °C, an HRMA choice) and lists it below the result as a defaulted field.

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Input	Identifier	Unit	Form limit	Form default
Tank Pressure	`feed_pressurant_tank_pressure`	bar		(blank)

Effect on the calculation: Propellant tank pressure, bar (> 0; the storage pressure must exceed (1 + regulator margin) × this value). The regulated outlet pressure the pressurant must hold. Liquid: filled with the tank pressure of the run (for a pump-fed engine this is the pump-NPSH tank pressure, not the chamber pressure). Hybrid: not filled. Used only in regulated mode. *If blank:* The analysis rejects the request naming this field (required in regulated mode).

Storage Pressure	`feed_pressurant_storage_pressure`	bar		(blank)

Effect on the calculation: Pressurant storage bottle pressure, bar (must exceed (1 + regulator margin) × tank pressure). Sets the bottle volume and the gas mass. Not filled from the motor result. Used only in regulated mode. *If blank:* The endpoint uses 200 bar (common industrial bottle) and lists it below the result as a defaulted field.

Regulator Margin [reg]	`feed_pressurant_regulator_margin`			(blank)

Effect on the calculation: Regulator pressure margin, fraction (> 0). The bottle stops feeding below (1 + margin) × tank pressure; the storage pressure must exceed it. Not filled from the motor result. Used only in regulated mode. *If blank:* The endpoint uses 0.10 (an HRMA choice) and lists it below the result as a defaulted field.

Collapse Factor [reg]	`feed_pressurant_collapse_factor`			(blank)

Effect on the calculation: Pressurant collapse factor, × (accepted band 1.0–1.6). Multiplies the gas mass for heat loss to the cold propellant (Sutton Ch. 6); outside the band nothing is sized. Not filled from the motor result. Used only in regulated mode. *If blank:* The endpoint uses 1.0 (no collapse, the optimistic lower bound) and lists it below the result as a defaulted field.

Initial Ullage Vol	`feed_pressurant_initial_ullage_volume`	m^3		(blank)

Effect on the calculation: Initial ullage volume, m^3 (> 0). Gas volume at the start of blowdown. Liquid: filled with the ullage volume of the run's oxidizer tank (L → m^3). Hybrid: not filled. Used only in blowdown mode. *If blank:* The analysis rejects the request naming this field (required in blowdown mode).

Initial Pressure	`feed_pressurant_initial_pressure`	bar		(blank)

Effect on the calculation: Initial tank pressure, bar (> 0). Starting pressure of the blowdown expansion. Not filled from the motor result: a blowdown start pressure is a design choice, and the liquid run's tank pressure is a regulated value. Used only in blowdown mode. *If blank:* The endpoint rejects the request naming this field (required in blowdown mode).

Polytropic n [bd]	`feed_pressurant_polytropic_n`			(blank)

Effect on the calculation: Polytropic exponent n of the ullage gas (accepted band 1.0 up to the γ of the gas: helium 5/3, nitrogen 7/5). Sets how fast the pressure falls during blowdown (1 = isothermal, γ = adiabatic). Not filled from the motor result. Used only in blowdown mode. *If blank:* The endpoint uses 1.2 (an HRMA choice) and lists it below the result as a defaulted field.

Line Fluid	`feed_water_hammer_fluid`			Water

Effect on the calculation: Fluid in the feed line. The analysis takes its density and bulk modulus (wave speed) from it. Liquid: filled with the fluid of the run's oxidizer line. Hybrid: filled with the oxidizer of the run (N2O). Solid: not filled (a solid motor has no liquid line); the selection stays as you leave it. Options: Water; Nitrous oxide (N2O); RP-1 (kerosene); Liquid oxygen.

Line Length	`feed_water_hammer_line_length_m`	m		(blank)

Effect on the calculation: Feed line length, m (> 0). Sets the pressure-wave round-trip time 2L/a (sudden vs slow closure). Liquid: filled with the length of the run's oxidizer line. Hybrid: filled only if you gave a feed line length on the hybrid page (chug loop); the hybrid solver sizes no line. Solid: not filled. *If blank:* The endpoint rejects the request naming this field (required).

continued on next page

Input	Identifier	Unit	Form limit	Form default
Line Inner Diameter	`feed_water_hammer_line_id_mm`	mm		(blank)
Effect on the calculation: Line inner diameter, mm (> 0; the wall must be thinner than the inner radius). Sets the flow velocity and the wave speed. Liquid: filled with the inner diameter of the run's oxidizer line (m → mm). Hybrid: filled only if you gave a feed line diameter on the hybrid page. Solid: not filled. *If blank:* The endpoint rejects the request naming this field (required).				
Wall Thickness	`feed_water_hammer_wall_thickness_mm`	mm		(blank)
Effect on the calculation: Line wall thickness, mm (> 0 and below the inner radius; at or above it the thin-wall pressure rating breaks down and no verdict is issued). Sets the pipe elasticity (wave speed) and the wall stress check. Not filled from the motor result: no motor computes a feed-line wall thickness. *If blank:* The endpoint rejects the request naming this field (required).				
Working Pressure	`feed_water_hammer_working_pressure_bar`	bar		(blank)
Effect on the calculation: Line working pressure, bar. The surge is added to it for the peak-pressure verdict. Liquid: filled with the inlet pressure of the run's oxidizer line (pump discharge; the pressure the motor's own water-hammer block uses). Hybrid: filled with the injector feed pressure P1 of the run (N2O saturation pressure). Solid: not filled. *If blank:* The endpoint rejects the request naming this field (required).				
Working Pressure Reference	`feed_water_hammer_working_pressure_reference`			Not specified (both planes checked)
Effect on the calculation: Reference plane of the working pressure: absolute or gauge. Not specified: both readings are checked; if the verdicts differ the endpoint refuses (400) and you must pick one here. Not filled from the motor result. Options: Not specified (both planes checked); Absolute; Gauge.				
Mass Flow	`feed_water_hammer_mdot_kg_s`	kg/s		(blank)
Effect on the calculation: Line mass flow, kg/s (> 0). Sets the flow velocity that the valve stops. Liquid: filled with the mass flow of the run's oxidizer line; hybrid: with the oxidizer mass flow of the run. Solid: not filled. *If blank:* The endpoint rejects the request (a mass flow or a flow velocity is required).				
Valve Closure Time	`feed_water_hammer_valve_closure_time_ms`	ms		(blank)
Effect on the calculation: Valve closure time, ms (> 0). Compared with the wave round-trip time 2L/a: closure faster than it gives the full Joukowsky surge. Not filled from the motor result (no motor models a closure law). *If blank:* Instantaneous closure is assumed (full surge, the conservative bound) and listed below the result as a defaulted field.				
Pipe Material	`feed_water_hammer_pipe_material`			Stainless 304
Effect on the calculation: Pipe material. Supplies the elastic modulus (wave speed) and the strength for the wall check. Liquid: filled with the structural alloy of the run's oxidizer line. Hybrid and solid: not filled; the selection stays as you leave it. Options: Stainless 304; Stainless 316; Steel AISI 4130; Aluminum 6061; Titanium 6Al-4V.				

Table 13.13: Dock: Comparative Analysis panel inputs

Input	Identifier	Unit	Form limit	Form default
Configuration name	`cmp_name`			(blank)
Effect on the calculation: Name for the snapshot of the current motor result (max 40 characters). Only a label: it does not enter any calculation. *If blank:* An automatic name config-N is used; an existing name is overwritten.				
Design A	`cmp_ov_a`			
Effect on the calculation: First stored snapshot (Design A) of the overlay table. Only selects which saved results are compared; nothing is recalculated. Needs at least two snapshots.				

continued on next page

Input	Identifier	Unit	Form limit	Form default
Design B	`cmp_ov_b`			
Effect on the calculation: Second stored snapshot (Design B); must differ from Design A. Only selects saved results; nothing is recalculated.				

Table 13.14: Dock: Uncertainty Quantification panel inputs

Input	Identifier	Unit	Form limit	Form default
Analysis Level	`uq_level`			Fast Screening (200 samples)
Effect on the calculation: Number of Monte Carlo samples drawn from the input distributions around the motor inputs on this page: Fast Screening 200, Engineering 1000, Extended 3000. More samples give steadier statistics (percentiles, sensitivity ranking), not a different motor. Run time is the sample count times the time of one motor solution, which depends on the motor type and design (one liquid engine sample takes much longer than a hybrid one); it is measured during the run and shown with the remaining time. A run that exceeds the time budget stops by name and publishes no statistics. Options: Fast Screening (200 samples); Engineering (1000 samples); Extended (3000 samples).				
Seed	`uq_seed`			42
Effect on the calculation: Random seed of the sampler (whole number ≥ 0; 0 is valid). The same seed reproduces the same samples. *If blank:* Or non-numeric: 42 is sent; a negative seed is refused.				

Table 13.15: Dock: User Data Validation panel inputs

Input	Identifier	Unit	Form limit	Form default
Static-Fire CSV – paste text or choose a file (columns: time [s], thrust [N])	`vp_csv_text`			(blank)
Effect on the calculation: Static-fire thrust curve as CSV text (columns: time [s], thrust [N]) or loaded from a file. It is compared with the motor's predicted curve; it does not change the motor calculation. *If blank:* Nothing is sent and you are asked for data.				

13.6 Panels in the result section

Some analyses run in their own panel in the result section rather than in the dock.

Table 13.16: Panels in the result section

Panel	Page	Function
Transient Analysis — $P_c(t)$ / $F(t)$	Hybrid	Time-stepped internal ballistics by feed mode; export to Excel. In the N_2O blowdown mode it gives the same impulse and burn time as the Tank Blowdown panel and the delivered numbers of the main result; the tank temperature is filled with the motor's oxidizer temperature. After a new calculation the previous run is cleared.
Tank Blowdown	Hybrid	Pressure, flow and thrust decay of a self-pressurised tank; true total impulse and burn time.
Structural FEA	Hybrid, liquid	Stress field, mesh quality and convergence history on an axisymmetric wall section.
Thermal FEA	Hybrid	Wall temperature field by transient conduction; peak wall temperature at the end of a finite burn. Answers a different question from the equilibrium temperature of the main chain.
Grain Section FEA	Solid	Plane-strain stress and bore strain; the acceptance criterion is the bore-surface strain. For a case-bonded grain the thermal load of cure and cool-down shrinkage is applied as well; the case expands under pressure. The margin badge is green at 1 and above (when the acceptance criterion has converged) and red below 1. If the strain leaves the small-strain range no margin is printed and the exceeded capability is stated.
Flight Dynamics — 6-DOF	All three	Rigid-body flight, Barrowman $C_{N\alpha}$/CP, static margin, rail and wind. Linear small-α aerodynamics ($\alpha < 15°$). Thrust is corrected for altitude: $F(h) = F_{ref} + (p_{ref} - p_a)A_e$. The stability badge is one verdict; if the flight fails it names the end reason. The thrust badge gives the name of the selected thrust source and the flown impulse. The default body is never narrower than the motor envelope.
Injector Design	Hybrid, liquid	Standalone "what if" calculator; after every main calculation it is filled with that design's circuit and does not change the main design.
Trajectory Analysis	All three	Point-mass ballistic flight; not 6-DOF. The load factor is the sensed acceleration (gravity excluded); the kinematic acceleration is given separately. For the hybrid the motor's flight curve is flown. For a vehicle that cannot lift off no chart is drawn and NO LIFTOFF is shown. The assumptions (planar model, launch angle and its source) are in the provenance line.



Figure 13.12: Transient Analysis: time axis, burn-time dashed line, separate $F(t)$ and $P_c(t)$ charts.

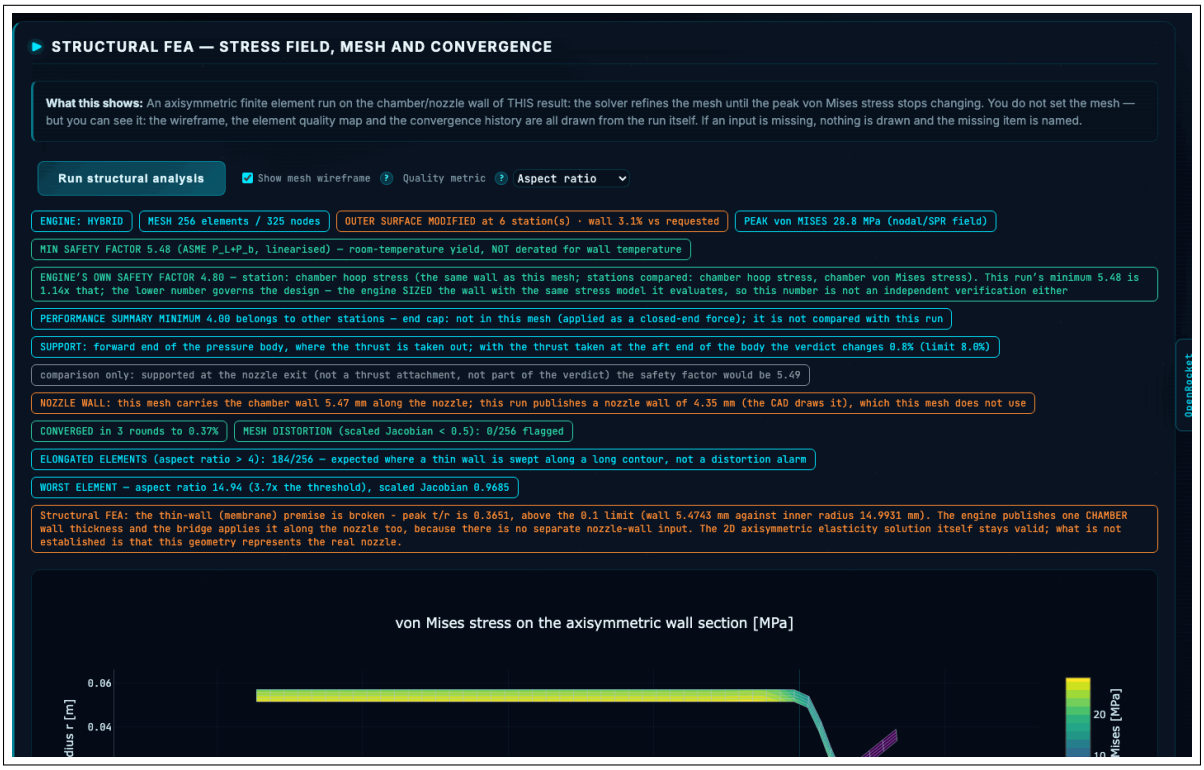


Figure 13.13: Structural FEA: stress field, mesh and convergence.



Figure 13.14: Grain Section FEA: plane-strain stress and bore strain.

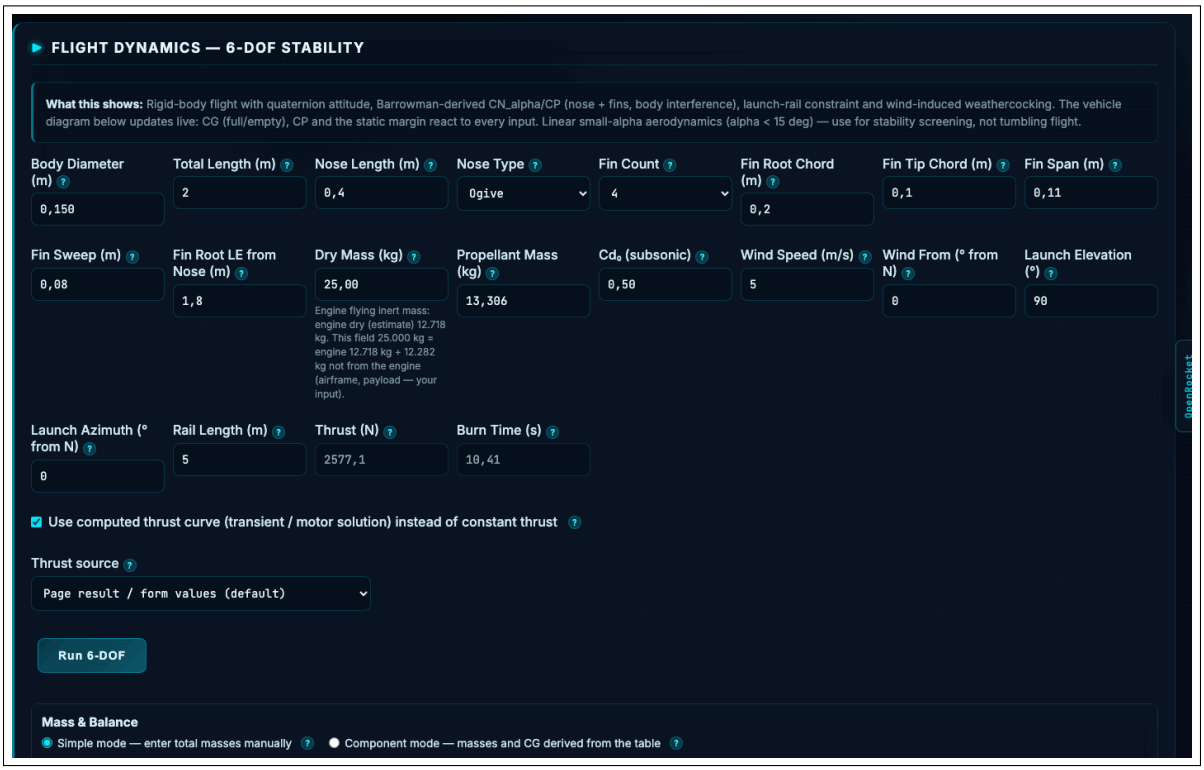


Figure 13.15: Flight Dynamics (6-DOF) panel: vehicle layout, CG, CP and static margin.



Figure 13.16: Standalone Injector Design panel in the hybrid: filled with the main design's circuit.

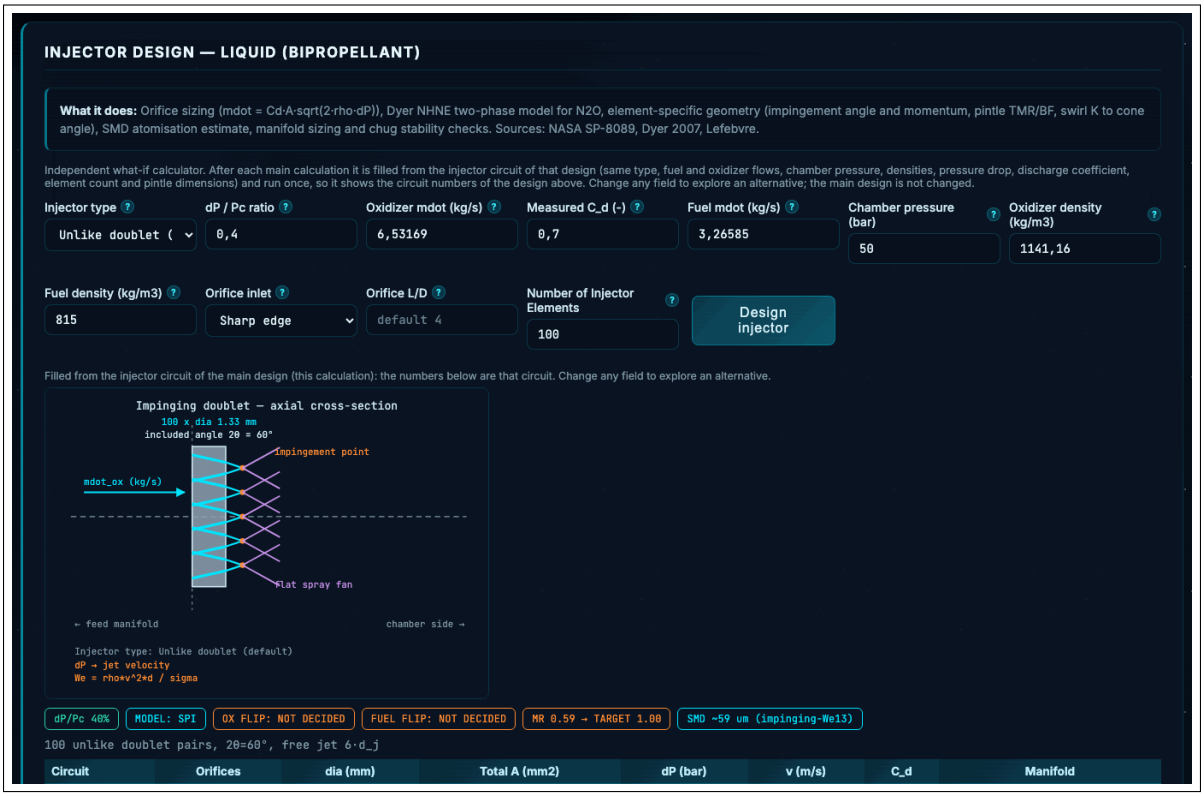


Figure 13.17: Injector Design panel in the liquid (bipropellant).

Table 13.17: Hybrid: Transient Analysis panel inputs

Input	Identifier	Unit	Form limit	Form default
Feed Mode	`tp_feed_mode`			N2O Blowdown (self-pressurizing)
Effect on the calculation: How the oxidizer reaches the injector in this time march. Regulated: the oxidizer mass flow is held at its design value for the whole burn; no tank is modelled, so the fill fraction changes nothing and the tank temperature only enters as the N2O inlet temperature. N2O blowdown: a self-pressurising N2O tank holding the design oxidizer mass drives the flow; as it empties, tank pressure, thrust and Pc fall and the run can end before the design burn time (the end event is marked on the chart). Blowdown is disabled when the main design uses a user-set tank pressure. Choices: Regulated (default) or N2O blowdown. Options: Regulated (constant mass flow); N2O Blowdown (self-pressurizing).				
Tank Temperature	`tp_tank_temp`	K	240 – 309	293.15
Effect on the calculation: Initial N2O tank temperature, in K. In N2O blowdown mode it sets the starting vapour pressure and liquid density from the N2O saturation table, so it drives the flow. In Regulated mode the oxidizer flow is prescribed, so it does not drive the feed; it is still passed to the engine as the N2O inlet temperature, which shifts the curves only slightly. After each engine calculation it is filled with the engine Oxidizer Temperature (the tank temperature the engine was designed with); a value you type is kept and, if it differs from the engine value, the difference is shown below the field. Accepted (blowdown): 240–309 K (saturation table band); a temperature outside the N2O liquid range is rejected in both modes with the reason. *If blank:* Empty: the engine oxidizer temperature is sent, or 293.15 K (the solver's default tank temperature) if the engine has none.				
Liquid Fill Fraction	`tp_fill`		0.05 – 0.98	0.85
Effect on the calculation: Fraction of the tank volume filled with liquid N2O at the start (dimensionless, 0–1); the rest is vapour ullage. Read only in N2O blowdown mode: with the design oxidizer mass fixed it sets the tank volume, and so how fast the tank pressure falls. In Regulated mode it changes no number. Accepted: 0.05–0.98; outside it the blowdown run is rejected with the reason. *If blank:* Empty: 0.85 is sent (the solver's default).				

Table 13.18: Injector Design panel inputs (hybrid)

Input	Identifier	Unit	Form limit	Form default
Injector type	`inj_type`			Showerhead (default)
Effect on the calculation: Injector element family. Drives the geometry model, the discharge-coefficient basis and the atomisation correlation. Options: Showerhead (default); Pressure-swirl / vortex; Pintle (throttleable); Like-impinging (self); Coaxial (single-fluid: jet + annulus).				
dP / Pc ratio	`inj_dp`			0.2
Effect on the calculation: Injector pressure drop as a fraction of chamber pressure. Typical 0.15-0.30; below 0.15 chug instability is likely (Sutton & Biblarz 9th ed. Ch. 8; NASA SP-194 Ch. 5-6).				
Oxidizer mdot	`inj_mdot_ox`	kg/s		2
Effect on the calculation: Oxidizer mass flow rate through the injector, kg/s; must be > 0. After a calculation it is filled from the motor result.				
Inlet pressure P1	`inj_pfeed`	bar		(blank)
Effect on the calculation: Injector inlet pressure P_1 (bar, absolute). Filled after a calculation only when the main design uses a user-supplied tank pressure. *If blank:* Empty: self-pressurised N2O takes P_sat(T_tank), other oxidizers P_c·(1 + dP/P_c).				
Measured C_d	`inj_cd`	-		(blank)
Effect on the calculation: Measured discharge coefficient of the orifices. Not applied to swirl elements, whose C_d follows from their geometry. *If blank:* Empty: C_d comes from the circuit table (inlet + L/D).				
Chamber pressure	`inj_pc`	bar		20
Effect on the calculation: Chamber pressure at the design point, bar. After a calculation it is filled from the motor result.				

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Input	Identifier	Unit	Form limit	Form default
Oxidizer density	`inj_rho_ox`	kg/m3		786
Effect on the calculation: Liquid oxidizer density at injection temperature. N2O at 293 K is about 745-790 kg/m3; LOX about 1141 kg/m3.				
Orifice inlet	`inj_inlet`			Sharp edge
Effect on the calculation: Orifice inlet geometry. A radiused inlet (r/d >= 0.15) raises Cd to about 0.90-0.92; a sharp edge gives 0.63-0.84 and can hydraulically flip. Options: Sharp edge; Radiused.				
Orifice L/D	`inj_ld`			(blank)
Effect on the calculation: Orifice length-to-diameter ratio. It selects the discharge coefficient from the circuit C_d table (inlet type and L/D) and the hydraulic-flip verdict. *If blank:* Hybrid: empty means the circuit default L/D = 4 is used and no flip verdict is issued.				
inj_n2o	`inj_n2o`			checked
Effect on the calculation: Self-pressurised nitrous oxide flashes across the orifice. With this enabled the Dyer NHNE two-phase model sets the flow rate and the pressure drop follows from tank temperature.				
N2O tank temperature	`inj_tox`	K		293
Effect on the calculation: Nitrous oxide tank temperature. Sets saturation pressure (about 50 bar at 293 K) and therefore the available dP.				
Radial flow fraction (0-1)	`inj_radial_fraction`			0.5
Effect on the calculation: Share of the oxidizer flow sent through the radial tip holes (the rest goes through the annulus). This is the real control for a single-fluid hybrid pintle: TMR = f/(1-f) follows from it, and the spray cone angle follows from TMR. Usual band 0.3-0.7.				
Pintle BF target	`inj_bf`			0.58
Effect on the calculation: Pintle blockage factor target BF = n·d/(π·D_p): it sets the tip-hole count when neither the hole count nor the hole diameter is given. 0.58 is an HRMA choice (no literature value); the source of the 0.3-0.74 band is not verified.				
Swirl cone half angle target	`inj_theta`	deg		(blank)
Effect on the calculation: Target spray cone HALF-angle of the swirl elements, met through the Giffen-Muraszew inverse (K -> X -> θ). The solvable envelope is 3.7-71.2 deg: a hybrid design outside it is rejected; the liquid panel clamps it to the envelope with a warning.				
Secondary Holes	`inj_pin_holes`			Radial Holes
Effect on the calculation: Holes drilled through the pintle tip wall, in one row around the post. Radial: hole axes along the radius (classic pintle). Tangential: drilled at an angle to the radius (geometry only; the swirl is not modelled). None: no tip holes - the element becomes a plain annular injector. Options: None; Radial Holes; Tangential Holes.				
Pintle Tip Diameter D_p	`inj_pin_dp`	mm		(blank)
Effect on the calculation: Outer diameter of the pintle post at its tip (D_p) - the surface the tip holes are drilled into. It sets the annular gap, the hole count through the blockage factor and the default skip distance. *If blank:* Empty: sized from the oxidizer flow with the annulus rule t = 0.05·D_p (HRMA choice, no literature value).				
Annular Gap t	`inj_pin_t`	mm		(blank)
Effect on the calculation: Radial clearance between the pintle post (D_p) and the body bore (t) - the annulus the axial oxidizer sheet flows through. If given, it fixes the annulus area and the tip holes take the rest of the flow area. *If blank:* Empty: solved from the annulus flow area.				
Tip Hole Count n	`inj_pin_n`			(blank)
Effect on the calculation: Number of holes in the single circumferential row on the pintle tip (n). If given with the hole diameter, the two fix the tip-hole flow area. *If blank:* Empty: from the blockage factor BF = n·d/(π·D_p) = 0.58 (HRMA choice).				
Tip Hole Diameter d	`inj_pin_d`	mm		(blank)
Effect on the calculation: Diameter of each tip hole (d). Inputs that over-determine the flow (D_p, t, n and d all given and not matching) are rejected with the flow mismatch. *If blank:* Empty: solved so the holes carry their share of the oxidizer flow area.				

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Input	Identifier	Unit	Form limit	Form default
Skip Distance L_s	`inj_pin_ls`	mm		(blank)
Effect on the calculation: Axial distance from the annulus exit (body face) to the axis of the tip-hole row (L_s; Dressler & Bauer 2000). Geometry and drawing only - no model uses it. *If blank:* Empty: L_s = D_p (HRMA choice, no literature value).				
Tangential Hole Angle	`inj_pin_ang`	deg		(blank)
Effect on the calculation: Angle between each hole axis and the local radius (0° = radial, 90° = tangent). Geometry/CAD only: the tangential momentum and the swirl it induces are NOT modelled; the reported spray angle is the radial value (a lower bound).				
Injector Body Outer Diameter	`inj_pin_od`	mm		(blank)
Effect on the calculation: Outer diameter of the injector body (housing) that surrounds the annulus bore. If given, the body wall (OD - bore)/2 is checked and drawn; it must fit inside the chamber. *If blank:* Empty: HRMA does not size the body (it needs housing material, allowable stress and the pressure across the body wall) and draws only the bore D_p + 2t.				
Number of Doublet Pairs	`inj_imp_pairs`			(blank)
Effect on the calculation: Number of like-on-like hole pairs (2 holes each). Given alone, the hole diameter is solved from the oxidizer flow; given together with the hole diameter, the two must deliver the target flow within 5%. *If blank:* Empty: the circuit plans the fewest holes whose diameter falls inside the 0.5-2.5 mm design band, rounded up to an even count.				
Hole Diameter d	`inj_imp_d`	mm		(blank)
Effect on the calculation: Diameter of each doublet hole. If given, the pair count is the nearest whole number for this diameter and the published diameter is the one that delivers the target flow exactly with that count; a plan off by more than 5% is rejected with the nearest valid plans. *If blank:* Empty: the injector circuit sizes it from the oxidizer flow.				
Impingement Angle 2θ	`inj_imp_angle`	deg		(blank)
Effect on the calculation: FULL angle 2θ between the two jets of each doublet. It sets the jet direction (axial velocity component) and the drawn impingement geometry; the hole plan does not depend on it. Must lie strictly between 0° and 180°; outside 40-90° a band note is shown (HRMA heuristic threshold; NASA SP-8089 gives 60° as the design value). *If blank:* Empty: 60° (declared).				
Swirl Chamber Diameter D_s	`inj_sw_ds`	mm		(blank)
Effect on the calculation: Diameter of the swirl (spin) chamber of each element. With the exit orifice diameter d_o it fixes the inlet area that gives the atomizer constant K. Below $1.25 \cdot d_o$ the Giffen-Muraszew geometry assumption breaks down and a warning is shown. *If blank:* Empty: $D_s = 2.5 \cdot d_o$ (simplex atomizer proportion, Lefebvre and McDonell; declared).				
Tangential Slots per Element	`inj_sw_nslots`			(blank)
Effect on the calculation: Number of tangential inlet slots in each swirl element; together they carry the element inlet area $A_p = K \cdot D_s \cdot d_o$. The element count and the exit orifice come from the injector circuit. *If blank:* Empty: 6 (server default, declared).				
Slot Width w	`inj_sw_w`	mm		(blank)
Effect on the calculation: Width of each rectangular tangential slot. Give the width and the height together, or neither. If given, the slots fix the atomizer constant K and with it the spray angle. *If blank:* Empty: the slots are sized from the target spray angle (w = 2h, HRMA proportion).				
Slot Height h	`inj_sw_h`	mm		(blank)
Effect on the calculation: Height of each rectangular tangential slot. *If blank:* Give it together with the slot width, or leave both empty.				
Number of holes	`inj_sh_n`			(blank)
Effect on the calculation: Number of showerhead orifices. *If blank:* Empty: the circuit plans the fewest holes inside the diameter band.				

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Input	Identifier	Unit	Form limit	Form default
Min Hole Diameter	`inj_sh_dmin`	mm		(blank)
Effect on the calculation: Lower edge of the hole-diameter band the injector circuit plans in. The circuit uses the fewest holes whose diameter falls inside [min, max]; a hole count fixed in Injector Design overrides that search. *If blank:* Empty: 0.5 mm (design band 0.5-2.5 mm, Sutton & Biblarz 9th ed. p. 337; declared).				
Max Hole Diameter	`inj_sh_dmax`	mm		(blank)
Effect on the calculation: Upper edge of the hole-diameter band the injector circuit plans in; it must be larger than the minimum. *If blank:* Empty: 2.5 mm (design band 0.5-2.5 mm, Sutton & Biblarz 9th ed. p. 337; declared).				
Inner Jet Flow Share (0.1-0.9)	`inj_coax_f`			(blank)
Effect on the calculation: Share of the oxidizer flow through the inner jet; the rest flows through the outer annulus. In a hybrid BOTH passages carry oxidizer (the fuel is the solid grain). The inner-jet and annulus diameters, the annulus gap and the recess are computed from this share and the oxidizer flow. Values outside 0.1-0.9 are rejected, not clamped. *If blank:* Empty: 0.6 (HRMA choice, no literature value; declared).				

Table 13.19: Injector Design panel inputs (liquid)

Input	Identifier	Unit	Form limit	Form default
Injector type	`inj_type`			Unlike doublet (default)
Effect on the calculation: Injector element family. Drives the geometry model, the discharge-coefficient basis and the atomisation correlation. Options: Unlike doublet (default); Unlike triplet (2:1); Like-impinging (self); Showerhead; Pintle (throttleable); Coaxial swirl; Pressure-swirl.				
dP / Pc ratio	`inj_dp`			0.2
Effect on the calculation: Injector pressure drop as a fraction of chamber pressure. Typical 0.15-0.30; below 0.15 chug instability is likely (Sutton & Biblarz 9th ed. Ch. 8; NASA SP-194 Ch. 5-6).				
Oxidizer mdot	`inj_mdot_ox`	kg/s		2
Effect on the calculation: Oxidizer mass flow rate through the injector, kg/s; must be > 0. After a calculation it is filled from the motor result.				
Measured C_d	`inj_cd`	-		(blank)
Effect on the calculation: Measured discharge coefficient of the orifices, the same quantity as the Discharge Coefficient Cd of the main form. Not applied to swirl elements, whose Cd follows from their geometry. *If blank:* Empty: Cd from the circuit table (inlet + L/D).				
Fuel mdot	`inj_mdot_fuel`	kg/s		0.8
Effect on the calculation: Fuel mass flow rate (liquid engines only). Set by the mixture ratio: $\dot{m}$_fuel = $\dot{m}$_ox / (O/F).				
Chamber pressure	`inj_pc`	bar		100
Effect on the calculation: Chamber pressure at the design point, bar. After a calculation it is filled from the motor result.				
Oxidizer density	`inj_rho_ox`	kg/m3		1141
Effect on the calculation: Liquid oxidizer density at injection temperature. N2O at 293 K is about 745-790 kg/m3; LOX about 1141 kg/m3.				
Fuel density	`inj_rho_fuel`	kg/m3		810
Effect on the calculation: Liquid fuel density. RP-1 about 810 kg/m3, ethanol about 789 kg/m3.				
Orifice inlet	`inj_inlet`			Sharp edge
Effect on the calculation: Orifice inlet geometry. A radiused inlet (r/d >= 0.15) raises Cd to about 0.90-0.92; a sharp edge gives 0.63-0.84 and can hydraulically flip. Options: Sharp edge; Radiused.				

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Input	Identifier	Unit	Form limit	Form default
Orifice L/D	`inj_ld`			(blank)
Effect on the calculation: Orifice length-to-diameter ratio. It selects the discharge coefficient from the circuit C_d table (inlet type and L/D) and the hydraulic-flip verdict. *If blank:* Hybrid: empty means the circuit default L/D = 4 is used and no flip verdict is issued.				
Pintle TMR target	`inj_tmr`			1
Effect on the calculation: Pintle total momentum ratio target. TMR = 1 gives a 60 deg spray half angle (Cheng 2017); usual band 0.5-2.				
Pintle BF target	`inj_bf`			0.58
Effect on the calculation: Pintle blockage factor target BF = n·d/(π·D_p): it sets the tip-hole count when neither the hole count nor the hole diameter is given. 0.58 is an HRMA choice (no literature value); the source of the 0.3-0.74 band is not verified.				
Swirl cone half angle target	`inj_theta`	deg		45
Effect on the calculation: Target spray cone HALF-angle of the swirl elements, met through the Giffen-Muraszew inverse (K -> X -> θ). The solvable envelope is 3.7-71.2 deg: a hybrid design outside it is rejected; the liquid panel clamps it to the envelope with a warning.				
Number of Injector Elements	`inj_n_el`			(blank)
Effect on the calculation: Number of injection elements (count). It fixes the hole count of the injector circuits; the hole diameters then follow from the flow areas. Unlike doublet, showerhead, coaxial: holes per circuit (one fuel + one oxidizer hole per element). Like-on-like doublet: same-fluid PAIRS per circuit (two holes each), 1-60 because the circuit planner allows at most 120 holes per circuit; more is rejected, not clamped. Not used for the pintle (a single element; the field is hidden). Accepted band 1-20000. *If blank:* Empty: each circuit plans the fewest holes inside its 0.5-2.5 mm diameter band and the answer says so.				
Pintle Post Diameter D_p	`inj_pin_dp`			(blank)
Effect on the calculation: Outer diameter of the pintle post at its tip (D_p, mm); the fuel tip holes are drilled into it and the oxidizer flows through the annulus around it. With the annulus area fixed by the oxidizer circuit, D_p sets the annular gap t, and through the blockage factor the tip-hole count. Must be a finite positive length; the annulus bore D_p + 2t must fit inside the chamber. *If blank:* Empty: sized from the oxidizer flow with the annulus rule t = 0.05·D_p (HRMA choice, no literature value).				
Oxidizer Annular Gap t	`inj_pin_t`			(blank)
Effect on the calculation: Radial gap of the oxidizer annulus around the pintle post (t, mm). The annulus area pi·(D_p + t)·t must equal the oxidizer circuit flow area: given alone, D_p is solved from it; given with D_p, the two must agree within 5% (HRMA tolerance) or the design is rejected with the mismatch. *If blank:* Empty: solved from the oxidizer flow area.				
Fuel Tip Hole Count n	`inj_pin_n`			(blank)
Effect on the calculation: Number of radial fuel holes in the single row on the pintle tip (whole number ≥ 1). The holes carry the fuel circuit flow area n·pi·d^2/4, so the hole diameter follows from n. *If blank:* Empty: from the blockage factor n·d/(pi·D_p) = 0.58 (HRMA choice, no literature value).				
Fuel Tip Hole Diameter d	`inj_pin_d`			(blank)
Effect on the calculation: Diameter of each radial fuel tip hole (d, mm). Given alone, the hole count is the nearest whole number for this diameter and the published diameter is the one that delivers the fuel circuit area exactly with that count; a plan more than 5% off (also with a given count) is rejected with the nearest valid plans. *If blank:* Empty: solved from the fuel flow area.				
Skip Distance L_s	`inj_pin_ls`			(blank)
Effect on the calculation: Axial distance from the annulus exit (injector face) to the axis of the tip-hole row (L_s, mm; definition Dressler & Bauer, AIAA 2000-3871). Geometry and CAD only: no performance model uses it. Must be shorter than the chamber. *If blank:* Empty: L_s = D_p (HRMA choice, no literature value).				

Table 13.20: Flight Dynamics (6-DOF) panel inputs

Input	Identifier	Unit	Form limit	Form default
Body Diameter	`sd_body_d`	m		0.150
Effect on the calculation: Outer body diameter (m). Sets the reference area $\pi d^2/4$ and scales the fin normal force (span/diameter)^2 in the Barrowman CP. Must be > 0. Filled from the page vehicle definition when the page publishes one. Otherwise, after each engine calculation a body narrower than the engine envelope (case/chamber outer diameter or nozzle exit, whichever is larger) is raised to the envelope, rounded up to 1 mm (HRMA choice, not a design rule); a wider body is kept. A value you type is kept; a body narrower than the engine envelope is refused by name when you run. *If blank:* Empty or invalid: the field turns red and Run is blocked.				
Total Length	`sd_body_l`	m		2
Effect on the calculation: Total vehicle length from nose tip (m). Must be > 0 and longer than the nose; fin root LE + root chord must fit inside it. Used for the CP/CG diagram and the default fin position. *If blank:* Empty or invalid: the field turns red and Run is blocked.				
Nose Length	`sd_nose_l`	m		0.4
Effect on the calculation: Nose cone length (m). The nose CP sits at (nose-type factor $\times$ this length) from the tip. Must be > 0 and shorter than the total length. *If blank:* Empty or invalid: the field turns red and Run is blocked.				
Nose Type	`sd_nose_type`			Ogive
Effect on the calculation: Nose shape. Only changes the nose CP factor: ogive 0.466, conical 2/3, parabolic 0.5 ($\times$ nose length); the nose normal-force slope stays CN_α = 2. Options: Ogive; Conical; Parabolic.				
Fin Count	`sd_fin_count`			4
Effect on the calculation: Number of fins, 3 or 4. Multiplies the fin normal-force term of the Barrowman CN_α, so it moves the CP and the static margin. Options: 3; 4.				
Fin Root Chord	`sd_fin_root`	m		0.2
Effect on the calculation: Fin root chord (m), the chord along the body. Enters the fin CN_α and fin CP. Must be > 0; root LE + root chord must not exceed the total length. *If blank:* Empty or invalid: the field turns red and Run is blocked.				
Fin Tip Chord	`sd_fin_tip`	m		0.1
Effect on the calculation: Fin tip chord (m). Enters the fin CN_α and fin CP; 0 means a pointed (triangular) fin. Must be ≥ 0. *If blank:* Empty or invalid: the field turns red and Run is blocked.				
Fin Span	`sd_fin_span`	m		0.11
Effect on the calculation: Fin span (m), from the body surface to the fin tip. The fin normal force grows with (span/diameter)^2. Must be > 0. *If blank:* Empty or invalid: the field turns red and Run is blocked.				
Fin Sweep	`sd_fin_sweep`	m		0.08
Effect on the calculation: Fin leading-edge sweep length (m): axial distance from the root LE to the tip LE. Moves the fin CP aft. Must be ≥ 0. *If blank:* Empty or invalid: the field turns red and Run is blocked.				
Fin Root LE from Nose	`sd_fin_pos`	m		1.8
Effect on the calculation: Distance from the nose tip to the fin root leading edge (m). Places the fin CP. Must be $\geq$ 0 and (this + root chord) $\leq$ total length. *If blank:* Empty or invalid: the field turns red and Run is blocked.				
Dry Mass	`sd_dry_m`	kg		25.00
Effect on the calculation: Vehicle mass without propellant (kg). Must be > 0. Read-only in component mode (sum of non-propellant rows). Filled from the page vehicle definition when the page publishes one. Avionics, wiring and the recovery system belong in this mass: the engine dry (flying inert) mass does not contain them. *If blank:* Empty or invalid: the field turns red and Run is blocked.				
Propellant Mass	`sd_prop_m`	kg		4
Effect on the calculation: Propellant mass burned during the flight (kg). Must be > 0. When the page calculation supplies the motor it is overwritten with that motor's propellant mass; read-only in component mode (sum of propellant rows). *If blank:* Empty or invalid: the field turns red and Run is blocked.				
Cd_0 (subsonic)	`sd_cd0`			0.50
Effect on the calculation: Zero-lift (subsonic) drag coefficient, dimensionless, on the reference area $\pi d^2/4$. Must be $>$ 0 here; the endpoint accepts at most 5. Filled from the page vehicle definition when the page publishes one. *If blank:* Empty or invalid: the field turns red and Run is blocked.				

continued on next page

Input	Identifier	Unit	Form limit	Form default
Wind Speed	`sd_wind`	m/s		5
Effect on the calculation: Horizontal wind speed (m/s), added to the airflow seen by the vehicle (drives weathercocking). Must be ≥ 0; the endpoint rejects speeds above the ISA sea-level speed of sound. *If blank:* Empty or invalid: the field turns red and Run is blocked.				
Wind From	`sd_wind_dir`	° from N		0
Effect on the calculation: Compass bearing the wind blows FROM (degrees from north). The endpoint accepts 0–360. *If blank:* Empty or invalid: the field turns red and Run is blocked.				
Launch Elevation	`sd_elev`	°		90
Effect on the calculation: Launch rail elevation above the horizon (degrees); 90 is straight up. Must be > 0; the endpoint accepts at most 90. *If blank:* Empty or invalid: the field turns red and Run is blocked.				
Launch Azimuth	`sd_azimuth`	° from N		0
Effect on the calculation: Launch azimuth (degrees from north): the horizontal direction the rail points. Any finite number. *If blank:* Empty or invalid: the field turns red and Run is blocked.				
Rail Length	`sd_rail`	m		5
Effect on the calculation: Launch rail length (m). Attitude is locked while the vehicle is on the rail; the result badge shows the rail length the solver used. Must be > 0. *If blank:* Empty or invalid: the field turns red and Run is blocked.				
Thrust	`sd_thrust`	N		1200
Effect on the calculation: Constant thrust (N), used only when no thrust curve is used (no imported motor file and the computed-curve box is off or unavailable). When the page calculation supplies the motor it is overwritten with that value. Must be > 0 when used.				
Burn Time	`sd_burn`	s		6
Effect on the calculation: Burn time (s) of the constant-thrust case, used together with Thrust only when no thrust curve is used. When the page calculation supplies the motor it is overwritten with that value. Must be > 0 when used.				
sd_use_curve	`sd_use_curve`			checked
Effect on the calculation: On (default): the run flies the thrust curve (or thrust and burn time) of the page calculation, and the Thrust/Burn Time/Propellant Mass fields are filled from it. Off: the form's constant Thrust and Burn Time are used. An imported motor file overrides both.				
Thrust source	`sd_thrust_src`			Page result / form values (default)
Effect on the calculation: Where thrust comes from. Default: the page calculation (or the form values). Imported: a RASP .eng / RockSim .rse file is parsed and its thrust curve is flown; if the file holds several motors, the list next to it picks one. Its propellant mass, if given, is written to Propellant Mass for review. Options: Page result / form values (default); Time-Resolved panel curve (this calculation); Imported motor file (.eng/.rse).				
Thrust source	`sd_motor_select`			
Effect on the calculation: Where thrust comes from. Default: the page calculation (or the form values). Imported: a RASP .eng / RockSim .rse file is parsed and its thrust curve is flown; if the file holds several motors, the list next to it picks one. Its propellant mass, if given, is written to Propellant Mass for review.				

Table 13.21: Structural FEA panel inputs

Input	Identifier	Unit	Form limit	Form default
fea_show_mesh	`fea_show_mesh`			checked
Effect on the calculation: Display only: draws the element edges (wireframe) over the result maps. The analysis is not re-run and no number changes.				
fea_metric	`fea_metric`			Aspect ratio
Effect on the calculation: Which element-quality measure the mesh-quality map colours: aspect ratio or scaled Jacobian. Display only – the analysis is not re-run; elements outside the quality threshold are marked in red in both cases. Options: Aspect ratio; Scaled Jacobian.				

Table 13.22: Thermal FEA panel inputs (hybrid)

Input	Identifier	Unit	Form limit	Form default
fea_t_ambient	`fea_t_ambient`			293.15
Effect on the calculation: Uniform initial wall temperature and ambient temperature of the transient thermal FEA (K). Must be > 0. *If blank:* Pre-filled with the panel default 293.15 K, which the result labels as the panel's value until you edit it; empty: the analysis names the missing input and no field is drawn.				
fea_t_show_mesh	`fea_t_show_mesh`			checked
Effect on the calculation: Display only: draws the element edges (wireframe) over the temperature maps. The analysis is not re-run and no number changes.				

13.7 Analysis Center

The Analysis Center arranges the same analyses by component: the component tree (component × analysis) on the left, the run card in the middle, the result viewer on the right, and the session's run history at the bottom. Components: chamber and nozzle wall, nozzle internal flow, chamber acoustics, solid grain section, tank / bolted joint / line. Reload from current result refills the run card's inputs from the last calculation. A result opened from the history is a stored result and says so in its title.

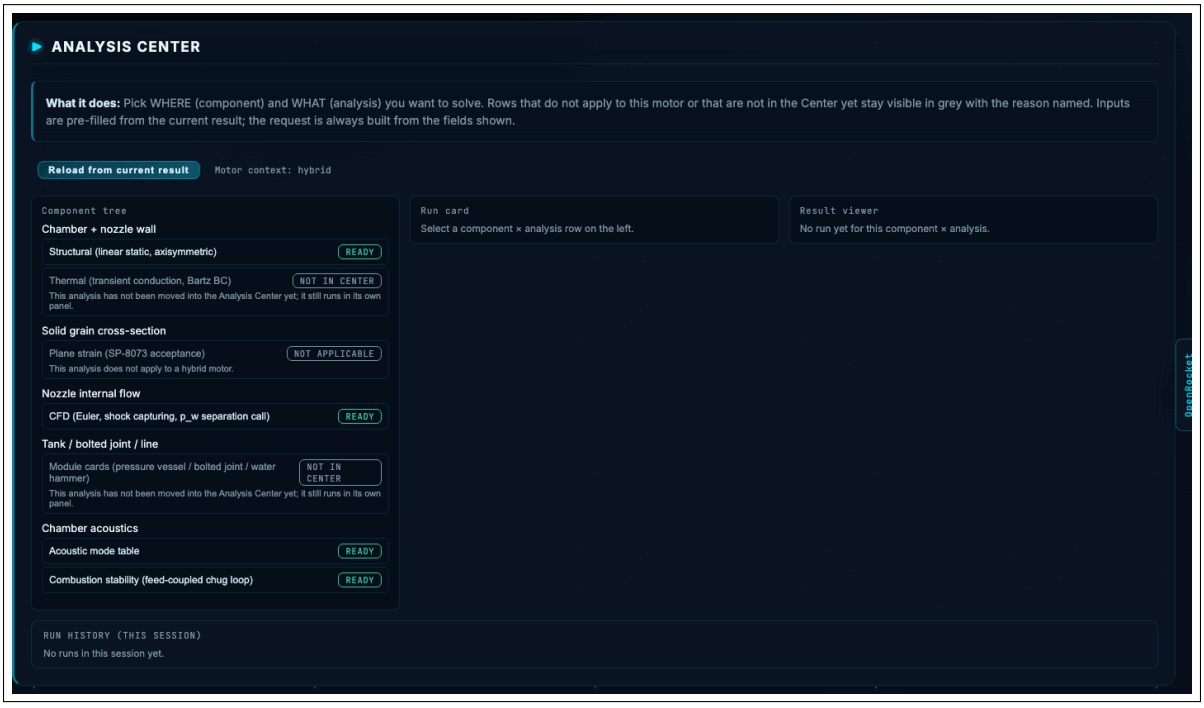


Figure 13.18: Analysis Center: component tree, run card and result viewer.

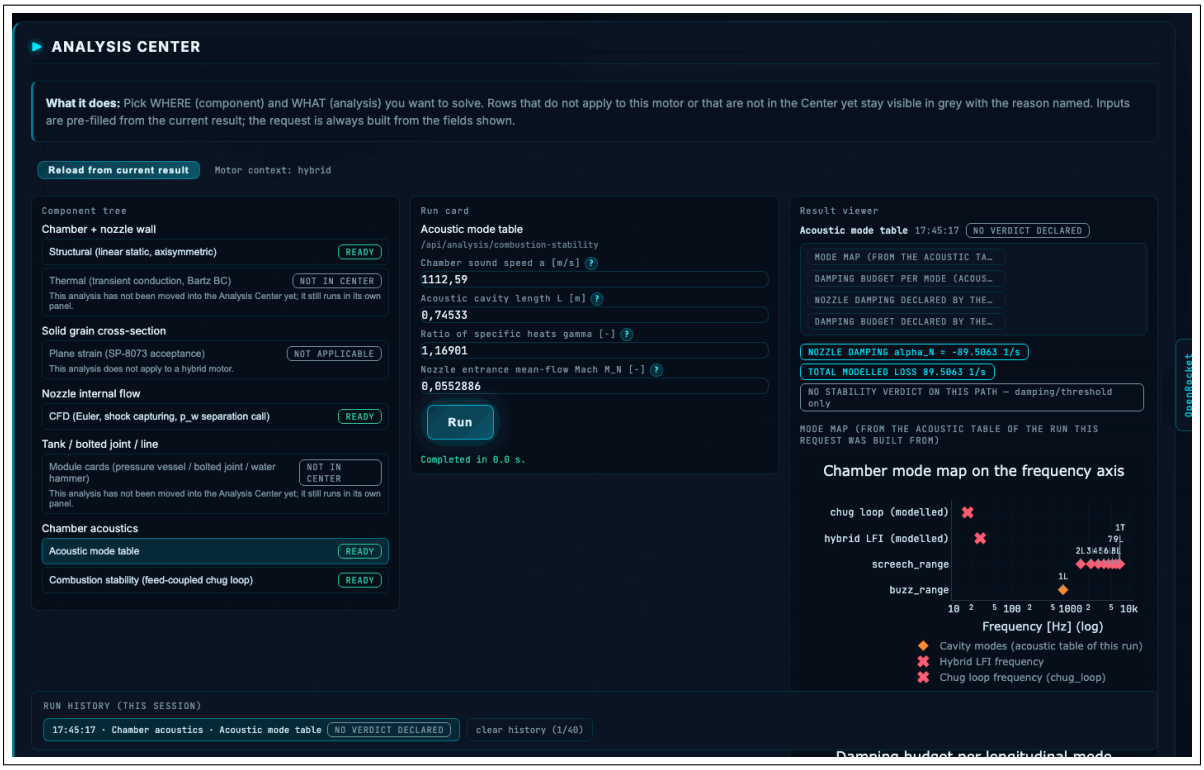


Figure 13.19: Analysis Center: an acoustic-mode table run.

Table 13.23: Analysis Center row states

State	Meaning
READY	The analysis can be run for this motor.
AWAITING RESULT	No calculation on the page yet; calculate the motor first.
NOT APPLICABLE	The analysis does not apply to this motor type; the reason is on the row (example: combustion stability in a solid motor, which has no injector and no feed system). The row is not hidden.
NOT IN CENTER	The analysis runs in its own panel.
NO VERDICT DECLARED	The analysis publishes no acceptance verdict; read the numbers and the list of items not modelled.

14. Launch Site

14.1 Purpose

Selects a launch point on Earth and flies a motor's vehicle from that site, showing the solved flight path (6-DOF) on a 3D globe.

Table 13.24: Verdicts of the analyses in the Center

Analysis	Verdict and reading
Structural (linear static, axisymmetric)	Minimum of the safety-factor field; read with its scope, the governing mechanism (`binding`) and mesh convergence. On an unconverged mesh the number is a trend.
CFD (Euler, shock capturing, p_w separation verdict)	The convergence verdict has three states: CONVERGED, NOT CONVERGED, verdict not published. Only a firm safe verdict is green. The INVISCID (Euler) badge says there is no wall shear and no wall heat flux. Mass and energy imbalance numbers have no threshold. Compare the wall pressure with the distribution of the Nozzle Flow panel.
Acoustic mode table	Chamber mode map and damping budget per mode. This path issues no stability verdict; terms not modelled in the budget are listed separately.
Combustion stability (feed-coupled chug loop)	Neutral curve, operating point, growth rate. Without a feed-line length the loop runs without inertance and the badge says so. Measured example (25 kN LOX/RP-1, 60 bar, O/F 2.2): without a line length “unstable”, 77.4 Hz; with a 2.5 m line “stable”, 33.6 Hz. A half-filled line group is not sent.
Plane strain (SP-8073 acceptance)	Solid only; the acceptance criterion is the bore-surface strain, margin = capacity / peak bore strain. A chamber pressure outside the solid band is rejected.

14.2 Screen



Figure 14.1: Launch Site: globe, SELECTED SITE, VEHICLE TO FLY and FLIGHT PATH panels.

14.3 Procedure

1. Pick the site by clicking the globe or by typing LATITUDE and LONGITUDE and pressing Apply.
2. Choose the SOURCE: Last motor calculated, Saved project (.hrma) or Example vehicle. With a

saved project the vehicle is built with the same calculation as the project's page. The airframe diameter and length come from the selected source: from the project's airframe definition if it has one, otherwise derived from the motor's outer envelope and propellant tanks (HRMA choice); the example vehicle has its own airframe. The airframe dry mass is not derived: if the field is empty, Fly this site starts no flight and asks for the airframe's dry mass (without the motor, kg). An airframe field typed by the user is kept when the vehicle changes and is marked for review. If the propellant tanks do not fit in the airframe, the page states the required volume and the volume the airframe can hold.

3. Press Resolve site, then Fly this site.
4. Play the flight with the time bar; LIVE READOUT gives altitude, speed and downrange at the selected time, FLIGHT SUMMARY the apogee and the downrange at the end of the solution.

14.4 Inputs

AIRFRAME DRY excludes the FLYING INERT MASS; the page adds the two and sends the sum as `dry_mass`. A missing vehicle input is rejected by name, or the default used is listed under `defaults_applied`.

Table 14.1: Launch Site: Selected Site

Input	Identifier	Unit	Form limit	Form default
LATITUDE	`ls-lat`			28.6084
Effect on the calculation: Site latitude, decimal degrees (WGS84, north positive). Apply moves the globe there; Resolve site and Fly this site read it (the flight uses it for Earth rotation). Accepted −90 to 90. *If blank:* Empty or not a number: those buttons do nothing.				
LONGITUDE	`ls-lon`			-80.6043
Effect on the calculation: Site longitude, decimal degrees (WGS84, east positive). Apply moves the globe there and Resolve site reads it; the flight path is placed at this point. Accepted −180 to 180. *If blank:* Empty or not a number: those buttons do nothing.				

Table 14.2: Launch Site: Vehicle to Fly

Input	Identifier	Unit	Form limit	Form default
SOURCE	`ls-vehicle-source`			Last motor calculated
Effect on the calculation: Which vehicle is flown: the last motor calculated on a motor page (handed over through this browser's storage), a saved .hrma project (recalculated on the server, since a project stores only inputs) or the built-in example vehicle. Changing it disables the previous flight until you fly again. Options: Last motor calculated; Saved project (.hrma); Example vehicle.				
Project	`ls-project`			
Effect on the calculation: Saved project whose motor is flown; it is recalculated on the server from its stored inputs. Shown only when Source is a saved project.				
THRUST	`ls-v-thrust`	N		(blank)
Effect on the calculation: Thrust of the flown motor (N), read-only: it comes from the selected vehicle. If the vehicle does not provide it (and carries no thrust curve), the flight is refused by name – no number is substituted.				
BURN TIME	`ls-v-burntime`	s		(blank)
Effect on the calculation: Burn time of the flown motor (s), read-only, from the selected vehicle. If the vehicle does not provide it (and carries no thrust curve), the flight is refused by name – no number is substituted. A liquid engine gives a burn time only when one was entered on the liquid page.				

continued on next page

Input	Identifier	Unit	Form limit	Form default
PROPELLANT	`ls-v-propmass`	kg		(blank)

Effect on the calculation: Loaded propellant of the flown motor (kg), read-only, from the selected vehicle. The part the motor thrust curve burns is consumed during the flight; the rest (the unburnt fuel grain of a hybrid, the oxidizer left in a self-pressurised N_2O tank) flies to burnout and is added to the dry mass, so lift-off mass = Airframe dry mass + Flying inert mass + this value. If the vehicle does not provide it, the flight is refused by name – no number is substituted.

FLYING INERT MASS	`ls-v-inertmass`	kg		(blank)

Effect on the calculation: Inert mass the motor adds to the vehicle in flight (kg), read-only, from the selected vehicle: the motor dry mass (case or chamber, nozzle, injector); for a liquid engine also the propellant tank dry structure and the loaded-minus-nominal (reserve) propellant when the result publishes them. The breakdown is shown under the vehicle name. It is added to Airframe dry mass and the sum is flown as the dry mass. If the vehicle does not provide it, the flight is refused by name.

BODY OD	`ls-a-bodydia`	m		(blank)

Effect on the calculation: Body outer diameter (m); it sets the reference area and the fin scaling of the 6-DOF flight. It is filled from the selected source: the example vehicle's own definition, the saved project's airframe, or – when the source has none – derived from the engine (HRMA choice: the larger of the engine envelope and the widest propellant tank). A value you type is kept; when the vehicle changes it is marked as left from the previous vehicle. Must be greater than 0; a body narrower than the engine is refused by name. *If blank:* Empty: the solver applies its own declared default and names it under the result.

BODY LENGTH	`ls-a-bodylen`	m		(blank)

Effect on the calculation: Body length including the nose (m); it sets the Barrowman centre of pressure and the room for the propellant tanks. Filled from the selected source like the diameter; derived from the engine as nose + engine length + propellant tank lengths in line (HRMA choice). A value you type is kept and marked when the vehicle changes. Must be greater than 0; a body shorter than the engine is refused by name, and tanks that do not fit are reported after the flight. Nose length 0.50 m (ogive) is fixed on this page. *If blank:* Empty: the solver applies its own declared default and names it under the result.

AIRFRAME DRY	`ls-a-drymass`	kg		(blank)

Effect on the calculation: Airframe dry mass (kg), EXCLUDING the flying inert mass shown above; the page adds the two as the vehicle dry mass of the flight. Avionics, wiring and the recovery system belong in this mass: the engine dry (flying inert) mass does not contain them. A value you type is kept and marked when the vehicle changes. Must be 0 or greater. *If blank:* The example vehicle and a saved project with an airframe fill it; for an engine without an airframe it is left empty (no airframe mass can be derived from the engine) and must be entered. Empty or not a number: the flight is refused by name (no mass is assumed).

CD0 (SUBSONIC)	`ls-a-cd0`			0.45

Effect on the calculation: Zero-lift drag coefficient (dimensionless); accepted 0–5. *If blank:* Empty or not a number: not sent – the solver applies its own declared default and names it under the result.

FINS	`ls-a-fincount`			4

Effect on the calculation: Number of fins; must be a whole number ≥ 0 (0 = no fins), otherwise the flight is refused. Fin root chord 0.20 m, tip chord 0.10 m and sweep 0.08 m are fixed on this page. *If blank:* Empty: not sent – the solver applies its own declared default and names it under the result.

FIN SPAN	`ls-a-finspan`	m		0.11

Effect on the calculation: Fin span (m); must be ≥ 0. *If blank:* Empty or not a number: not sent – the solver applies its own declared default and names it under the result.

WIND SPEED	`ls-a-windspeed`	m/s	≥ 0	0

Effect on the calculation: Wind speed (m/s); must be ≥ 0 and below the ISA sea-level speed of sound. Default 0 (no wind). *If blank:* Empty or not a number: not sent – the solver applies its own declared default and names it under the result.

WIND DIRECTION	`ls-a-winddir`	deg	0 – 360	270

Effect on the calculation: Direction the wind blows FROM (degrees from north); accepted 0–360. *If blank:* Empty or not a number: not sent – the solver applies its own declared default and names it under the result.

continued on next page

Input	Identifier	Unit	Form limit	Form default
LAUNCH ELEVATION	`ls-a-elev`	deg		84
Effect on the calculation: Launch elevation above the horizon (degrees); accepted 0–90 (90 = vertical). The launch rail is fixed at 5 m on this page. *If blank:* Empty or not a number: not sent – the solver applies its own declared default and names it under the result.				
AZIMUTH	`ls-a-azim`	deg		90
Effect on the calculation: Launch azimuth (degrees from north); any finite number. *If blank:* Empty or not a number: not sent – the solver applies its own declared default and names it under the result.				
SHORTCUTS (NOT A LIMIT – ANY POINT IS SELECTABLE)	`ls-presets`			–
Effect on the calculation: Shortcut list of known sites: choosing one writes its latitude and longitude into the fields and moves the globe there. Not a limit – any point can be picked on the globe. Options: –; Kennedy Space Center LC-39A; Cape Canaveral SLC-40; Vandenberg SLC-4E; Wallops Flight Facility Pad 0A; Guiana Space Centre ELA-3 (Kourou); Baikonur Site 1/5 (Gagarin Start); Esrange Space Center (Kiruna); Andoya Space; Rocket Lab LC-1 (Mahia); Tanegashima Yoshinobu LP-1; Satish Dhawan SC (Sriharikota); Jiuquan LC-43; Sinop (TR) - area reference, not a launch facility; Konya (TR) - area reference, not a launch facility; Mount Everest (high terrain) - terrain reference, not a launch site.				
ls-online	`ls-online`			unchecked
Effect on the calculation: Off (default): Resolve site uses only packaged offline data. On: it first tries Open-Meteo over the internet for elevation and current surface temperature/pressure, which sends the coordinates to that service; any failure falls back to the offline data.				

Table 14.3: Launch Site: Flight Path

Input	Identifier	Unit	Form limit	Form default
ls-time	`ls-time`		0 – 1000	0
Effect on the calculation: Flight playback timeline: scrubs the solved flight from lift-off (left) to the end (right). Enabled only after Fly this site; display only – nothing is recalculated.				
ls-exagg	`ls-exagg`			unchecked
Effect on the calculation: Display only: stretches the vertical scale of the drawn flight path for readability; off means true scale. The flight is not recalculated.				
ls-follow	`ls-follow`			unchecked
Effect on the calculation: Display only: the camera follows the rocket during playback. The flight is not recalculated.				

Table 14.4: Launch Site: Legend

Input	Identifier	Unit	Form limit	Form default
ls-tile-network	`ls-tile-network`			unchecked
Effect on the calculation: Off (default): no map request leaves this computer; areas outside the map cache show the packaged offline texture. On: satellite tiles are fetched from NASA GIBS and each request tells the provider which area you are viewing. The choice is stored in this browser.				



Figure 14.2: The saved example hybrid project is selected: vehicle identification and the airframe diameter and length derived from the motor. AIRFRAME DRY (KG) is empty, so no flight starts; the line in the FLIGHT PATH panel asks for the field by name.

14.5 Limits

- Earth rotation and the Coriolis effect are not modelled; the ground track is a flat-Earth projection.
- The path is a suborbital, ballistic flight; the vehicle does not enter orbit.
- Exaggerate altitude is for readability only and is off by default (the scale is true).
- The 3D globe requires WebGL; without it the screen says The 3D globe could not be started (WebGL required).
- Online satellite tiles (NASA GIBS) and weather data (Open-Meteo) are off by default; when enabled, only this page goes online. Clear map cache deletes the downloaded tiles.

15. Formulas Page

The Formulas page gives the relations used by the solvers and their sources in fourteen sections: basic performance parameters, nozzle design, combustion analysis, solid motors, liquid engines, hybrid motors, heat transfer, structural analysis, trajectory analysis, safety analysis, manufacturing analysis, cost analysis, advanced analysis, and the relations behind the advanced-analysis figures (Bartz 1957, NASA SP-8124, NASA SP-125, NACA Report 1135 and others). Equation numbers follow the source book (example: Sections 1.1 and 1.2, Sutton and Biblarz). If the maths typesetting layer fails to load, raw markup appears; reload the page.

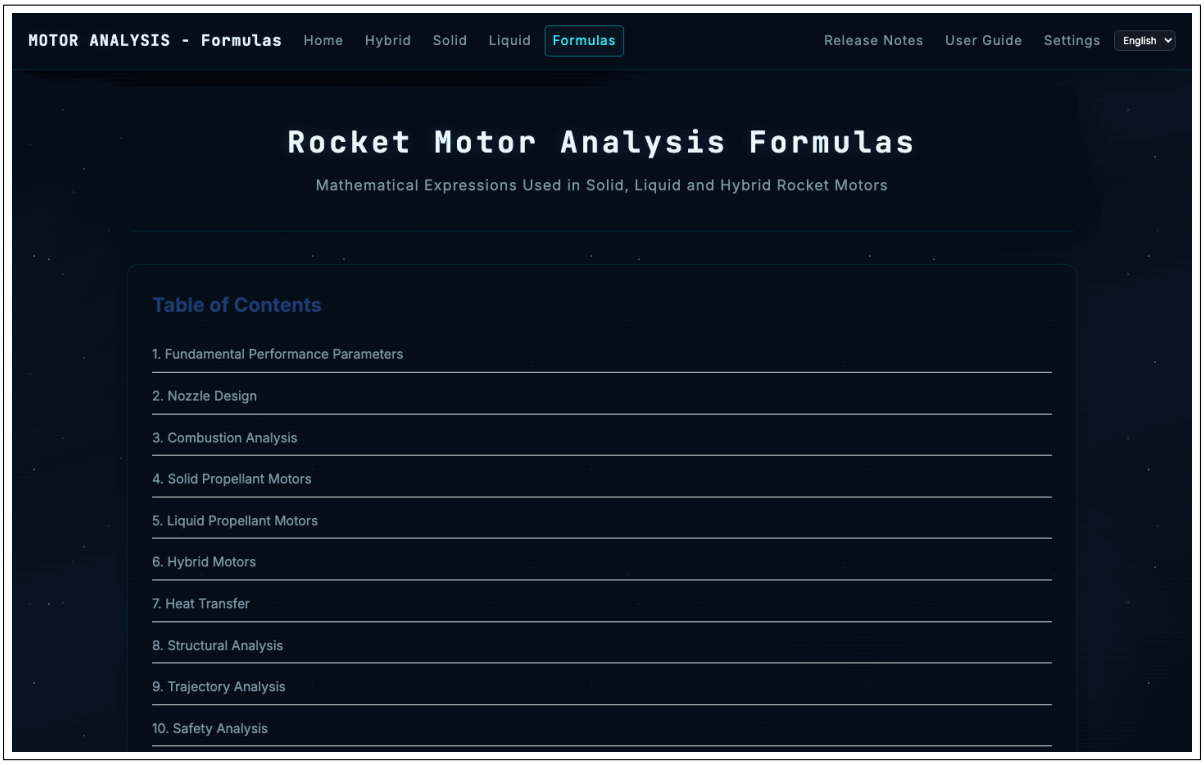


Figure 15.1: Formulas page: section list.



Figure 15.2: Formulas, Section 1.1: the specific impulse relation and the definition of c^*.

16. Exports

16.1 Buttons and files

Table 16.1: Export buttons

Button	Page	File produced
Export Results	Hybrid	Summary PDF (same file as Summary PDF).
Summary PDF, Technical PDF	Hybrid	PDF report (short, detailed).
PDF Summary Report, PDF Technical Report, PDF Complete Report	Solid, liquid	PDF report (three levels).
Excel Workbook	All three	XLSX: performance, geometry, injector, feed-system sheets. At most 100 000 rows per sheet and 20 sheets; a request beyond the limit is rejected.
Configure Design	Hybrid	Wall, safety-factor and injector choices before CAD generation (Table 9.11).
Generate 3D Model (STEP); Technical Drawings (PDF); DXF Drawing; Export STL Files	Hybrid	The 3D view on the page and a parametric solid model from the solver geometry (STEP package: chamber, nozzle contour, injector plate, assembly); dimensioned drawing set with title block; 2D manufacturing profile; STL. There is no pop-up window; each button downloads its own file.
STEP Package, DXF Drawing	Solid, liquid	Parametric solid models and assembly (compressed STEP); 2D manufacturing profile (DXF).
Technical Drawings PDF; Technical Drawings	Liquid; solid	Dimensioned drawing set with title block. On the solid page the button is in the CAD Design tab of the Motor Specifications card; the same tab holds Download STEP/STL Files.
Export CAD Design	Liquid	Next to the calculate button; CAD export of the liquid engine.
Download Motor File (.eng); Download OpenRocket .eng; OpenRocket .eng	Hybrid; solid; liquid	Thrust curve and masses for OpenRocket and similar flight simulators (RASP format). The OpenRocket tab at the page edge also provides the computed motor's `.eng` file with Download .eng.
Export STL Files	Hybrid	Triangle-mesh models.
Generate All Files, Complete Package (ZIP)	All three	STL, DXF, drawing PDF, STEP, `.eng` and geometry JSON files in one archive.
Analyze Regression Rate, Export Excel, Export CSV	Hybrid	Regression rate and port-growth table.

16.2 Export gates

The export section of the hybrid page is in Figure 3.4.

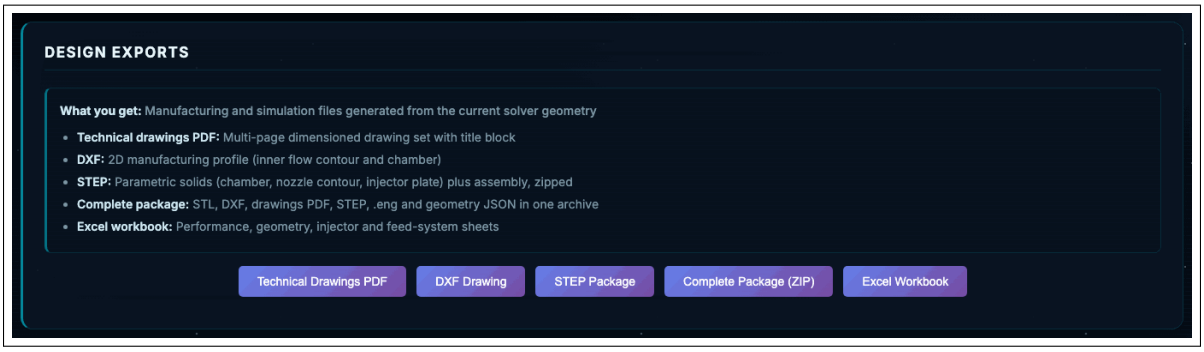


Figure 16.1: The Design Exports section of the liquid page.

- If the geometry is incomplete or a gate is closed no file is produced; the rejection names the missing field, the reason and the value sent. No partial model is drawn.
- If the injector plate thickness has not been earned, the STEP and STL packages do not contain the injector part; the package content matches the file list in the archive.
- The two-layer liquid wall is drawn as two bodies of two materials; the cooling channels are not drawn and the drawing note says so.
- Star and finocyl grains are not exported to CAD. The tangential pintle mode is rejected in CAD.
- The assembly STL is not a closed solid; use STEP for CAD work.
- The `.eng` header lines carry the status and source of the two-phase correction; if the solver published no status, “not published by the solver” is written.

16.3 Safety section of the PDF report

The technical PDF carries a safety assessment: explosive, toxic and fire hazards, protective equipment, emergency response and operational safety items. A provenance line states from which field each input was resolved. Without a usable chamber pressure, propellant mass, chamber diameter or wall thickness no safety verdict is produced. Regulatory compliance (NFPA, OSHA, DOT and local requirements) is not assessed.

17. Verdicts, Badges and Warning Classes

Table 17.1: Warning classes

Class	Page	Meaning	Action
CRITICAL	All three	A limit is exceeded (material allowable or melting temperature, flow, geometry). The design is not valid as it stands.	Apply the change the line proposes and recalculate.
WARNING	Hybrid, liquid	Outside a correlation range or beyond a design-practice threshold.	Assess the assumption for the design.
CHECK	Solid	Non-critical check and declaration line.	Read the declaration.
INPUT	Liquid	Input consistency or a missing-input declaration.	Complete or correct the input.
INFO	Hybrid	Declaration of a model assumption.	

Table 17.2: Verdict codes and badges

Code / badge	Meaning	Action
SAFE	The margin meets the target for the mechanisms checked. The scope of the verdict is stated on the line; a mechanism not on the list has not been called “safe”.	Read the scope.
MARGINAL	The margin is borderline.	Increase the design margin.
UNSAFE	The margin is below the target or below 1.	Change the design.
NOT EVALUATED, NOT_EVALUATED	No verdict issued; the reason is stated next to it. The results section prints it as words, the dock panels as a code.	Supply the input named in the reason.
NOT_MODELLED, NOT MODELLED	The quantity is not computed in this model.	
VERDICT WITHHELD	A condition of the verdict is not met (example: HRMA sized the wall; the safety factor is the target read back).	Enter the wall thickness to be manufactured.
WALL: SIZED BY HRMA	No wall thickness entered; the result is a design proposal.	
LINER REQUIRED, PROTECTED DURING BURN	Hybrid wall-zone verdicts.	Table 4.3.
MARGINAL (chug)	The liquid pintle chug verdict differs between the two atomisation scales; in 6-DOF a stability band (UNSTABLE / MARGINAL).	Increase the stability margin.
rule met, rule marginal, rule not met; STABLE, MARGINAL, UNSTABLE	The two chug columns of the throttle map: ratio rule ($\Delta P/P_c$) and loop model. A point counts as stable only if the loop model says STABLE.	
THERMAL RISK LOW / MEDIUM / HIGH	Risk level of the thermal panel.	
MOTOR DESIGN RISK, FACILITY / SITING HAZARD: LOW / MEDIUM / HIGH / CRITICAL or NOT EVALUATED	The two separate verdicts of the Comprehensive Safety panel: the motor’s own risk and the facility hazard of the propellant on site. If an axis cannot be evaluated and the evaluated axes are low or medium, the motor verdict is NOT EVALUATED; no acceptability verdict is given.	Complete the input of the missing axis; use the facility hazard for siting and distance decisions.

Code / badge	Meaning	Action
Nozzle regime: PERFECT EXPANSION, UNDEREXPANDED, OVEREXPANDED, SEPARATED, NORMAL SHOCK IN NOZZLE, UNCHOKED	Flow regime of the Nozzle Flow panel.	Underexpanded: thrust can be gained with a longer nozzle; overexpanded or separated: reduce the expansion ratio.
CONVERGED, NOT CONVERGED, verdict not published	CFD convergence verdict (Table 13.24).	

18. Input Rules and Error Messages

18.1 Rules

1. **A blank field is not sent.** An optional field left blank is not put in the request; the server applies its canonical default and declares it.
2. **An out-of-band value is not clipped.** On the solid and liquid pages every band is a gate: the request is rejected with `input_not_accepted` and the response gives the field, the value, the accepted range and the unit. On the hybrid page, fields with a form gate are rejected; for fields without a gate the request runs, the value is ignored and declared in the warning list with the field name, the value and the band.
3. **A computed field is not an input.** For example the hybrid oxidizer mass flow is a result of the calculation; putting it in the request gets the request rejected.
4. **A number-like string is not a number.** A quoted value such as `"20.0"`, a non-finite value and a boolean are rejected, each with its own reason code.
5. **Conflicting inputs are not resolved.** Without a rule that picks the winner, the request is rejected and the conflict is named.

18.2 Messages of the hybrid form

18.3 Server rejections

A calculation rejected by the server (HTTP 422) is shown in the interface language; the field is named by its screen label and the machine code stays in the tooltip. Common rejections:

- **The endpoint does not read this field.** If the endpoint never reads the field sent, the request is rejected; silently dropping the field would publish the result under a wrong label.
- **No sweep range.** In a parametric sweep with an empty start or end, no point is computed.
- **Unknown propellant.** Without a record for the requested propellant the request is rejected and the valid names are listed.
- **Missing flow input.** The required fields of the Nozzle Flow analysis are not assumed; the rejection code is `incomplete_flow_input`.

Table 18.1: Browser check messages of the hybrid form

Message	Meaning
Thrust must be positive; Burn time must be positive; O/F ratio must be positive; Chamber pressure must be positive	Zero or negative value.
Burn time too long (max 300 seconds)	Upper limit of the form.
O/F ratio too high (max 20)	Upper limit of the form.
Chamber pressure too high (max 200 bar)	Upper limit of the form.
Chamber diameter too small (min 10mm); Chamber diameter too large (max 1000mm)	Limits of the form (the server accepts 10–5000 mm; a chamber wider than 1000 mm is sized by leaving the diameter blank or through the contraction ratio).
Tank pressure (… bar) must be higher than chamber pressure (… bar). Minimum required: … bar	Lower limit in user tank-pressure mode.
For finite area combustion, fill only ONE parameter	Contraction ratio or chamber mass flux, not both.
… is below (above) the minimum (maximum) this form declares on screen … The request was not sent.	The limit written in the field's label or note is applied.

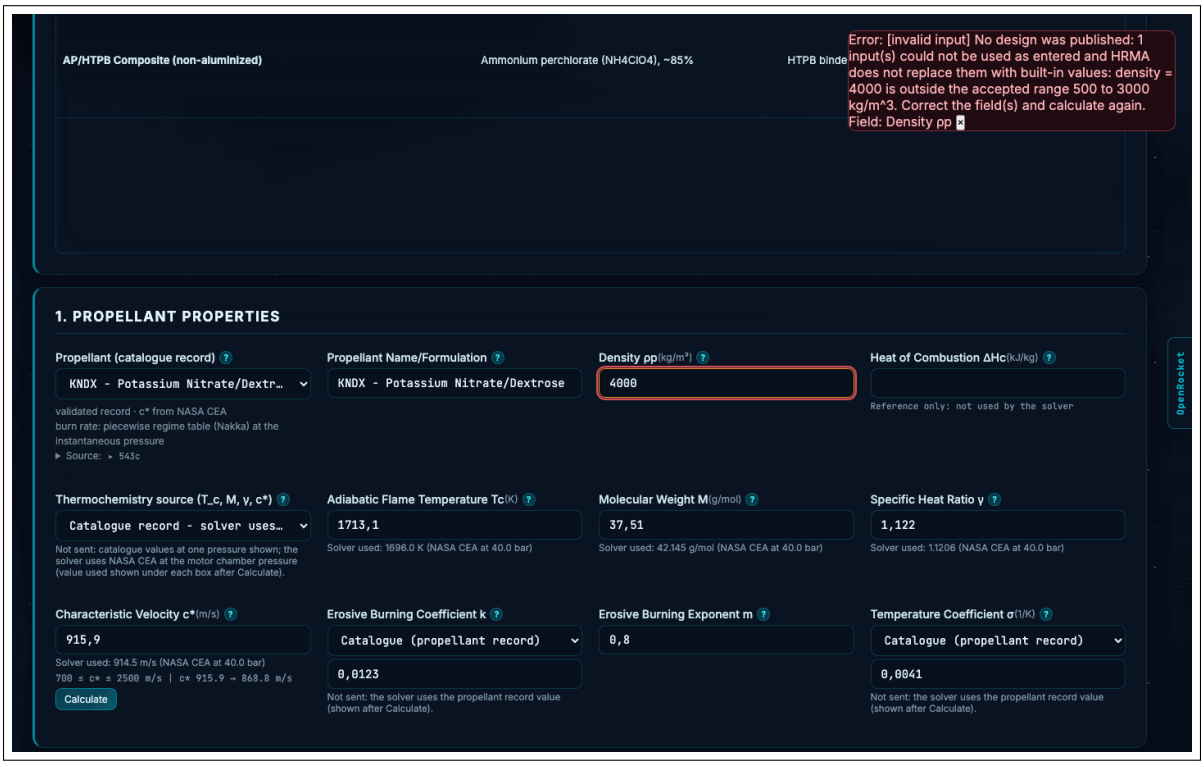


Figure 18.1: A named rejection (top right) and the marked field: density 4000 kg/m^3, accepted range 500–3000 kg/m^3; no design is published.

On a JavaScript error or a network interruption the application shows a box titled ERROR DETECTED: source (network, form validation, data parsing, JavaScript), time, context and the original message. If a server response is not valid JSON nothing is drawn and the raw body is written to the browser console.

19. Scope and Limits

19.1 Model level

The models are closed-form and one-dimensional relations. Turbopump maps, cavitation, rotordynamics, bearing life, start-up and shut-down transients and two- and three-dimensional conjugate heat transfer are not included. The nozzle CFD panel is two-dimensional: it solves the Euler equations with shock capturing and has an optional viscous mode; it is not a validated viscous design tool.

19.2 Measured deviations

Measured relative errors against the shipped experimental database (current numbers in `docs/VALIDATION_STATUS.md`):

Table 19.1: Median relative error measured against the experimental database

Motor	Quantity	N	Median error	Note
Liquid	Vacuum Isp	14	0.9 %	Published engine data
Liquid	Vacuum thrust	4	0.2 %	4/4 weak evidence
Liquid	Sea-level Isp	4	2.0 %	Small sample
Solid	Burn rate	27	0.5 %	27/27 own fit data
Hybrid	Characteristic velocity	18	3.3 %	
Hybrid	Specific impulse	18	5.0 %	
Hybrid	Final port diameter	18	10.1 %	
Hybrid	Chamber pressure	35	10.9 %	
Hybrid	Regression rate	35	36.7 %	17/35 own fit data

The 0.9 % median error on the liquid side shows that Isp is reproduced when the true operating point and the published geometry are given; it does not show that the flight performance of an unknown motor is predicted with this error. Rows marked "own fit data" are scored against the data the coefficients were derived from; they are not independent validation. The scatter of the hybrid regression rate is a known property of hybrid motors.

19.3 Verdicts not published

- The CFD convergence verdict for runs outside the scope of the self-consistency gate.
- A stability verdict on the acoustic-mode path.
- A verified two-phase correction in the solid motor.
- Buckling without a declared external-pressure load case.
- Tungsten nozzle erosion (no published coefficient band).
- Jacket flow and pressure drop of dump cooling.
- Ablative liner thickness without net heating, and when the recession rate exceeds the validity ceiling of the Level-1 Q* model (0.35 mm/s).

19.4 Known limitations (3.0.6)

- The liquid delivered Isp is a preliminary-design estimate; the losses of a real injector and chamber are determined by test.
- For solid motors below about 75 mm the delivered Isp can be optimistic.
- Heat transfer is a Bartz-level estimate; a burn-through margin needs thermocouple data.

- Combustion instability, ignition transients and hard starts are outside the model; the delivered c^* efficiency is determined by static firing.
- Two-layer regenerative wall: low-cycle fatigue, creep, the thermal mismatch between liner and shell, bond strength and buckling are not modelled; the throat axial membrane check is not repeated on the closeout; the staged-cycle preburner pressure ladder is not in the jacket pressure budget.
- RP-1 carbon deposits are modelled only inside the NASA TP-2862 envelope.
- Hybrid with user tank pressure: regulator lock-up and thermal drift are not modelled, the tank structural mass is not computed.
- CEA comparison: solid propellants of uncertain composition show no number; the black-powder card has low confidence; the sea-level Isp of a separated liquid nozzle is RocketCEA's estimate.
- The coaxial hybrid element publishes no atomisation number.
- Hybrid with a gaseous N_2O feed: combustion is solved with a gas inlet, but the compressible orifice circuit and the gas supply are not modelled; injector hole plan, tank volume, sloshing, tank vessel and blowdown history are withheld in the gas phase with a named reason.
- The regenerative jacket model is a constant-section milled channel. Tube-wall jackets and a nozzle extension cooled by turbine exhaust are not modelled; for engines for which this model cannot build a consistent circuit (F-1 class, RL10A-3-3A) the jacket carries the model limit label and cooling-dependent numbers are not published.
- With film, dump, ablative and radiative cooling the hot-wall temperature is not solved; the value used is declared by name and the verdict is given with a validity label.
- In a solid motor the stress concentration of sharp-cornered ports (star, finocyl, slotted, wagon wheel) is not solved; a crack verdict is given only if the circular-port strain alone exceeds the capability, otherwise Not determined is shown.
- The liquid page designs at sea level. A fixed upper-stage nozzle that separates at sea level (e.g. ε 61, RL10 class) is refused by name; the refusal states the largest expansion ratio that stays attached. A vacuum or altitude design ambient is planned.
- Number boxes follow the operating system's decimal separator; the value is read correctly.
- The station march is not extended to thick walls; for a thick wall the declared lumped estimate is used.
- In a back-pressured 2D CFD run, shocked overexpanded cases may show backflow at the exit; these cases return with "not converged / plateau" and a `suspect` separation verdict. Treat the exit-region wall pressure and separation output as qualitative.
- In the API and the PDF report the hybrid TNT equivalent is applied to the total propellant mass (conservatively high); the safety panel does not evaluate the TNT equivalent for hybrids.

19.5 Not in this version

A C++ solver core; 3D CFD, 3D FEA, reacting CFD, modal FEA; detailed CAD (injector head and dome, flanges, bolt circle, O-ring grooves, regenerative channels, closed STL); turbopump, preburner, feed-line, valve and gimbal models and their CAD; exhaust plume visuals. The turbopump power balance on the cycle card is a cycle calculation, not a pump design.

19.6 Flight worthiness

HRMA is a preliminary design and analysis tool. The final design, safety decision and qualification of a motor that will fly require fatigue and creep analysis, physical testing (static firing) and a formal verification process. Every critical result is confirmed by physical testing.

20. Troubleshooting

Table 20.1: Troubleshooting

Symptom	Cause and remedy
Could not reach the update server.	No network access or it is blocked. The application works normally.
GitHub is rate limiting this network	Happens on shared networks. Try again after a few minutes or open the releases page.
Download slow or stuck	Download the installer manually with the browser link in the window.
The installer could not be started	Run the installer from the Downloads folder manually.
Could not reach the HRMA server	The local calculation engine has stopped. Close and reopen the application.
Raw markup on the Formulas page	The maths typesetting layer did not load. Reload the page.
The 3D globe could not be started (WebGL required).	The graphics driver does not support WebGL.
Windows SmartScreen warning	The package is unsigned: More info, then Run anyway.
macOS “developer cannot be verified”	Right-click the application and choose Open; if the dialog has no Open, use System Settings → Privacy & Security → Open Anyway (first launch only, Section 2).
The calculation says the request was not sent	A required field is empty or a limit declared by the form is exceeded; the message names the field (Section 18).

To report a problem use the Report a bug section of Settings.

21. Changes in Version 3.0.6

3.0.6 is the correction release after 3.0.5; it contains no new features. The table gives the changes visible to the user: the panel or field, what was published before and what is published now, and the reason. Example numbers are from the examples shipped with the application and are the same as in the 3.0.6 section of `CHANGELOG.md`. The full list is in the Release Notes window of the application and in `CHANGELOG.md`.

Table 21.1: 3.0.6 changes (user view)

Panel / field	Old → new	Reason
Hybrid impulse	For a self-pressurised N_2O motor the delivered impulse, burn time and average Isp come from the tank blowdown solution (example 29 948 → 26 839 N · s, 10.00 → 10.41 s, 209.8 → 205.7 s); the constant-flow budget is shown alongside by name.	Two impulses were published for the same motor; the curve flown is the blowdown curve (Section 4.4).
Hybrid consumed propellant	The curve integral and the delivery budget were two "consumed" masses → one number (the delivery budget, example 13.306 kg); the flown Isp uses the same definition.	One quantity had two numbers.
Hybrid thermal exposure time	Liner, wall heat load and erosion use the solved burn time, the same as the curve and the flight (this change alone moves the example liner 7.404 → 7.582 mm).	The thermal chain used a different duration from the curve.
Hybrid example	Regression pair Carmicino et al. 2013 (a 6.399e-5, n 0.554; flux 250); fuel grain 1125.0 → 662.7 mm, L/D 13.37 → 6.42, chamber diameter 84.1 → 103.2 mm; the citation names the pair used.	The example produced its own single-port slenderness warning (Section 4.7).
Hybrid mass budget	The loaded oxidizer includes the ullage vapour (example 12.50 → 12.95 kg, of which 0.444 kg vapour); the dry mass includes the injector plate; a Total Propellant Mass (loaded) row.	The vapour in the tank and the plate that flies with the motor were missing.
Hybrid graphite nozzle	Not sized as a pressure shell; an insert in a structural housing (example wall 21.59 → 1.58 + 2.77 mm).	A graphite insert is not a pressure-bearing body.
Hybrid liner and vacuum Isp	Liner from recession and conduction allowances, also at low heat flux (example 1.28 → 7.58 mm); altitude performance with one exit area anchored at the design ambient (example vacuum Isp 237.48 → 234.92 s).	At low flux no liner was sized; the loss model used a different exit area.
Hybrid c^* efficiency, Find Optimum	c^* efficiency: for every pair the median of the records with a valid measurement window (declared default); optimum O/F one definition (button and solver), N_2O sweep range 4–12, form limits equal the server's.	The default policy differed between pairs; one question had two answers.
Hybrid warning list	The 13 unsourced verdict bands of the validator were removed; the C_F upper limit uses the run's own γ; the HTPB bias reason only for a run with total-flux closure; the false "burn time is not read from the form" notice on every N_2O hybrid is gone.	Unsourced thresholds and a false notice produced lines on measured motors.
Hybrid 3D view	O/F and Isp over time follow the delivered blowdown curve (example: over the burn O/F falls 6.0 → 5.36 and Isp 209.7 → 199.5 s; 3.0.5 showed O/F rising to 6.17).	The view drew the constant-flow curve, which is not flown.

Panel / field	Old → new	Reason
Liquid engine Isp	Per total propellant consumption, next to the main thrust chamber Isp (example sea level / vacuum: engine 254.8 / 299.3 s, thrust chamber 257.1 / 301.1 s; 3.0.5 showed 257.1 / 301.1 s as the engine Isp); total flow 9.915 → 10.006 kg/s.	In an open cycle the propellant consumed by the gas generator was missing from Isp and flow.
Liquid engine mass	The fixed 15 kg "Control / Avionics" item was removed (a vehicle item); injector mass only when the plate is sized (example 78.6 → 63.0 kg, T/W 32.4 → 40.4).	An unsourced fixed mass and an unsized part were in the engine mass.
Liquid jacket pressure drop	One value from the station march; the feed budget and the march no longer publish two (example feed budget 5.29 → 6.13 bar); the cooling deck uses the motor's march inputs (example deck 3.96 → 6.13 bar); in staged and expander cycles the jacket inlet is the cycle's fuel pump discharge (one coupled solution).	Budget, march and deck published different jacket pressures.
Liquid unsolved cooling circuit	If the jacket has no valid solution (an F-1 class engine at the jacket model limit) the feed, cycle, pump, turbine, engine thrust, engine Isp and mass numbers that depend on it are withheld with their reason; the thrust-chamber Isp is published with its definition; a flight with such an engine is refused with the reason.	Engine numbers were produced from an unsolved circuit.
Liquid chamber standard	AIAA S-080 → NASA-STD-5012C (acceptance 1.20 × MDP, qualification and burst 1.50 × MDP; MDP from the closeout layer; example MDP 77.2 bar, acceptance 92.7 bar, qualification and burst 115.8 bar; 3.0.5 based these on the 50 bar chamber pressure); S-080 remains for tanks and lines.	Chamber layers are engine hardware.
Test pressures	Each vessel has one test pressure ladder (AIAA S-080A: MEOP → proof 1.5× → burst 2.0×); the solid Safety tab and the deck show the same design pressure; the safety factor is stated with the pressure it was computed at.	Two ladders and two design pressures were published for the same vessel.
Liquid throttle map	Ratio rule and chug cycle model in separate columns (example 40–70 % UNSTABLE, as in the Analysis Center).	The table and the Analysis Center gave different verdicts for the same point.
Liquid nozzle and default form	The unsourced expansion-ratio pre-fill ($\varepsilon = 50$) was removed; the solver matches the nozzle to the ambient; a fixed nozzle that separates at sea level is refused by name with the largest ε that stays attached (this change alone: ε 50 at 30 bar had been sized to Isp 22 s and a 597 kg engine).	A design sized on a separated nozzle was published.

Panel / field	Old → new	Reason
Thermal Safety panel	Heat power, heat energy and hot-gas area on the motor's own nozzle contour (this change alone: liquid example area 0.153 → 0.280 m^2, heat power 2042.6 → 2723.5 kW; hybrid heat power 1125.7 → 1100.9 kW; risk verdicts unchanged).	The panel used a generic nozzle.
Solid nozzle	The convergent section starts at the case inner diameter in the response, the 3D view, STEP, CAD and FEA (example convergent length 28.30 → 33.30 mm); nozzle mass from the drawn contour (example nozzle 379 → 65 g, motor dry mass 3.734 → 3.420 kg); the sea-level Isp is the same nozzle at 101.325 kPa.	The contour differed between outputs; the nozzle mass was a fixed fraction of the case mass.
Solid grain structural	The case strain verdict (NASA SP-8073) replaces the crack-risk classes; at sharp port corners (star, finocyl, slotted) a strain margin that does not converge with the mesh is not published, the reason is shown.	Class boundaries absent from the source and a non-converging number were used.
Solid form, projects, VALIDATION	"Star fillet radius" was read by no calculation and was removed (old projects open with a named warning); number of slots 2–16; solver-filled component masses no longer load as user edits (an opened 3.0.5 example showed 3.733 kg against the solver's 3.420 kg); the VALIDATION panel uses the motor's own thrust curve; TOTAL IMPULSE card 3.0 kN · s → 2957 N · s.	An unread field, stale masses and a comparison curve the motor did not publish.
Comprehensive Safety	An incomplete motor assessment is NOT EVALUATED, with the number of axes evaluated, instead of a green LOW.	A missing evaluation was shown as a low risk.
Structural FEA, solid case	The two-layer liquid wall is analysed layer by layer (this change alone: example lowest SF 1.93 → 2.59, the liner governs); support at the forward body; no burst pressure for a composite case; no safety factor printed under a withdrawn verdict.	The liner was not solved; unknown or withdrawn numbers were published.
Hybrid flight mass, .eng	The unburned grain strip and the propellant left in the tank fly with the vehicle; the lift-off mass equals the motor's total mass; the `.eng` propellant mass is the mass burned during the firing (example 14.556 → 13.306 kg; 1.694 kg stays in the loaded mass).	The flight chain did not carry the residuals; the `.eng` file wrote the loaded mass as propellant.

Panel / field	Old → new	Reason
Trajectory, 6-DOF, sweep	Load factor is the sensed acceleration (about +1 g on the pad); 6-DOF thrust is corrected for altitude; the 6-DOF default body is never narrower than the motor envelope (solid example Ø100 → 116 mm); a parametric sweep point gives the same apogee as Calculate for the same motor and vehicle; trajectory captions describe the plotted cells.	The load factor was the kinematic acceleration; the default body did not carry the motor; sweep and Calculate disagreed.
Projects, package, Excel	A saved project keeps its flight setup and an opened project carries no state from the previous one; when the CAD generator refuses a geometry the package manifest says FAILED with the reason, the STL refusal shows its reason and range; the liquid Excel "Feed system" sheet lists each line with its items and total.	Saving lost the setup and the previous project's settings reached the new calculation; refusals gave no reason.
Interface language	No HTTP codes, raw keys or endpoint names in user texts; help notes quote no API field names or unsourced "typical values"; charts do not overflow their panels; remaining English sentences on Turkish screens translated, one Turkish term per English term; the Turkish PDF in the interface language.	Language and readability findings of the 3.0.5 user test.
Update (macOS), licence	The old version is not opened while an update is installing; a real progress window; notifications in the interface language. HRMA is distributed under GPL-3.0-or-later; the licence text and third-party notices are in the package (Section 1.5).	A half-installed version could open during an update; the package includes the GPLv3-licensed RocketCEA.

22. Changes in Version 3.0.5

3.0.5 is the correction release after 3.0.0; the work prepared under the name "3.0.1" ships with this version. The full list, with every changed published number, is in the Release Notes window and in `CHANGELOG.md`.

Table 22.1: 3.0.5 changes (user view)

Area	Change
Inputs	A "?" note next to every field: what it is and its unit, what it changes in the calculation (or that it is informational), what happens if blank, the accepted range.
Rejections	Server rejections in the interface language and with the field's screen label; out-of-range input rejected on the solid and liquid pages (`input_not_accepted`); blank fields are not sent. The liquid page used to send a prefilled number for 45 blank fields, the solid page had 55 hidden substitutions, the hybrid had hidden 1000 / 10 / 10000 values; removed.

Area	Change
Data source	The grey "NASA CEA / NIST WebBook" badges are gone; each panel states the source the calculation uses (CoolProp/REFPROP, HRMA catalogue, Cantera, NASA CEA at the motor's pressure). A NASA CEA comparison block was added to the three pages (Section 12).
Hybrid	c^* efficiency became an input (sourced 0.89 for N_2O/HTPB); three-zone chamber wall; a single canonical pintle geometry; tank-pressure source selection; the end-of-burn port reaches the chamber wall (the old "chamber = 1.5 × final port" rule is gone).; when the injector plate is not earned, plate thickness and L/D are not published
Solid	Propellant selector; default KNDX, grain 100 / 48 / 400 mm; thermochemistry from CEA at the motor's chamber pressure; case and grain gates; catalogue / user choice for erosive and temperature coefficients; machined and effective throat diameters shown separately.; Grain Support input (case-bonded or cartridge-loaded; changes only the grain structural results)
Liquid	Wall temperature target and solved value separate; fin effect and Sieder–Tate correction in regenerative cooling; RP-1 carbon deposit; two-layer wall; RP-1 coking in two levels (561 K warning, 728 K critical); no jacket with film cooling; pintle chug MARGINAL; band and cycle gates.; Fuel Density is optional: if blank the server uses its own table value and declares it
Charts	Transient analysis on a real time axis; thrust and chamber pressure on separate charts; minimum font sizes; counting animation removed.
Design with AI	Context text, suggestion check, before/after table (Section 8.6).
Flight and projects	A saved project flies the page's motor; old projects open; unmodified example copies are refreshed; Turkish letters allowed in project names; no field is lost when a project is saved and reopened; FLYING INERT MASS on the Launch Site page; the range axis of the trajectory profile.
Safety	Motor design risk and facility / siting hazard are separate verdicts; for liquid pairs the TNT equivalent comes from 14 CFR Part 420, Appendix E; the risk matrix is translated (Table 13.2).
Export	The propellant and dry masses in the `.eng` file are the result's own numbers; the `.eng` in the liquid package was corrected.
Examples	The three examples were regenerated with the real page payload: hybrid O/F 6.0; liquid 1.0 mm CuCrZr liner and closeout shell, O/F 2.0, 50 bar, 30 s; solid with all its fields.
CAD	Pintle post, tip and radial holes; the nozzle from a single wall profile; the hybrid nozzle wall by one rule with the nozzle's own material; the two-layer liquid wall as two bodies.

A. Menu and Button Reference

A.1 Window menu

In the desktop application the window has its own menu; the labels are fixed English strings in the launcher and do not follow the interface language.

Table A.1: Window menu

Item	macOS	Windows
Settings…	Application menu	HRMA menu
Check for Updates…	Application menu	HRMA menu
Open Output Folder	Application menu	HRMA menu
User Guide…	Help menu	Help menu
Release Notes…	Help menu	Help menu
HRMA on GitHub	Help menu	Help menu
Releases and Downloads	Help menu	Help menu

A.2 Navigation

Links at the bottom of the home page: FORMULA REFERENCE, LAUNCH SITE (3D EARTH), RELEASE NOTES, SETTINGS, USER GUIDE. The navigation bar of the designer and Formulas pages: Home, Hybrid, Solid, Liquid, Formulas. The only navigation link on the Launch Site page is Back to app. Unsaved changes are shown by an asterisk (∗) in the project bar. Save As asks for confirmation before overwriting a project of the same name.

A.3 Designer-page action buttons

Table A.2: Action buttons

Button	Page	Function
Calculate; Calculate Motor Performance; Calculate Engine Performance	Hybrid; solid; liquid	Sends the form to the solver.
Find Optimum	Hybrid	Sweeps O/F for the selected fuel and oxidizer and finds the ratio that maximises Isp. Does not run, with an explanatory message, when a custom mixture or custom fuel composition is selected.
Preview Profile	Hybrid	In the Altitude Profile tab plots ambient pressure against altitude (ISA 1976); performs no motor calculation.
Single Point Analysis, Parametric Analysis, Trajectory Analysis	Hybrid, liquid	Analysis-type tabs. The parametric analysis sweeps one parameter over a given range; the trajectory analysis solves a point-mass ballistic flight.
Compute Trajectory	Solid, liquid	Ballistic flight with the trajectory inputs.
Reset to catalogue values, Keep my edits	Solid	When a field loaded from the catalogue is edited by hand, a banner in 0. PROPELLANT CATALOGUE states the number of changed fields. The first button returns the fields to the catalogue value; the second closes the banner and keeps the user's values. A hand-edited field is not overwritten when a catalogue row is selected.

A.4 3D viewer toolbar

Table A.3: 3D viewer buttons

Button	Function
Cutaway	Cuts the model lengthwise.
Dimensions	Shows dimension labels.
Exhaust	Draws the exhaust cone at the nozzle exit.
Exploded	Separates the parts along the axis.
Port	Cycles the port section (motors with a port).
Heat Map	Temperature layer on the model; without data the layer does not open.
Design Mode	Live sliders: in the hybrid chamber diameter, L* and expansion ratio; in the solid chamber diameter, expansion ratio and grain length; in the liquid chamber diameter, expansion ratio and chamber length. A preview: performance, thermal and structural results are not updated.
Orbit, Cam, HQ, 1×, Reset View	Automatic rotation, preset camera angles, render quality, timeline speed, initial view.

B. Glossary

Ablative. Protective material that carries heat away by controlled recession of its surface.

Blowdown. Emptying of a tank under its own pressure; the feed pressure falls during the burn.

Characteristic length, L*. Ratio of chamber volume to throat area.

Characteristic velocity, c^*. Measure of combustion-chamber performance; independent of the nozzle: $c^* = P_c\, A_t / \dot{m}$.

Chug. Low-frequency pressure oscillation between the feed system and the chamber.

Coking. Carbon left on a hot wall by a hydrocarbon coolant.

Contraction ratio. Ratio of chamber cross-section to throat area.

Divergence loss. Thrust loss from the non-axial component of the exit velocity.

Expansion ratio. Ratio of nozzle exit area to throat area.

Fail-closed. The behaviour of showing a reason instead of a value when a result cannot be computed.

Fuel grain. The shaped body of a solid or hybrid fuel.

Kn. Ratio of burning surface to throat area (solid motor).

Nozzle. Converging-diverging duct that accelerates the combustion products.

O/F. Ratio of oxidizer mass flow to fuel mass flow.

Port. The flow channel inside a fuel grain.

Provenance line. A line stating the source of a published number.

Regenerative cooling. Cooling by passing a propellant through channels in the chamber wall.

Regression rate. Recession rate of the fuel surface in a hybrid motor; mm/s.

Regulated design point. Hybrid design point with the oxidizer flow taken as constant through the burn.

Safety factor. Ratio of strength to applied stress.

Self-pressurised. Feeding of an oxidizer (N_2O) by its own vapour pressure.

Slosh. Free-surface oscillation of the liquid in a tank.

Spallation. Detachment of the carbon deposit on a hot wall; the wall underneath sees the deposit-free heat flux.

Specific impulse, Isp. Impulse per unit weight of propellant; s. The basis of a published value (design point, sea level, vacuum) is stated on its label.

Thrust coefficient, C_F. Dimensionless measure of the nozzle's contribution; $F = C_F\, P_c\, A_t$.

Throat. Smallest cross-section of the nozzle; the station where the flow reaches the speed of sound.

Total impulse. Integral of thrust over the burn; N · s.

Water hammer. Pressure surge in a line caused by a sudden change of flow.

Web. Thickness between the burning surface and the outer boundary of a grain.

C. Symbols and Units

Table C.1: Symbols

Symbol	Quantity	Unit (on screen)
a, n	Regression or burn-rate law coefficient and exponent ($r = a\,G^n$, $r = a\,P^n$)	SI
A_b, A_t, A_e	Burning surface, throat and exit area	m^2, cm^2
c^*	Characteristic velocity	m/s
C_d	Discharge coefficient	–
C_F	Thrust coefficient	–
D_c, D_t, D_e	Chamber, throat and exit diameter	mm
ε	Expansion ratio A_e/A_t	–
ε_c	Contraction ratio	–
η_{c^*}	c^* (combustion) efficiency	% or ratio
F	Thrust	N
g_0	Standard gravity, 9.80665	m/s^2
G	Mass flux	$kg/(m^2 \cdot s)$
γ	Ratio of specific heats	–
I_{sp}	Specific impulse	s
I_t	Total impulse	N · s
K_n	Burning surface / throat area	–
L^*	Characteristic length	m
$\mathcal{M}$	Molar mass	g/mol, kg/kmol
$\dot{m}$	Mass flow	kg/s
O/F	Oxidizer-to-fuel mass ratio	–
P_a, P_c, P_e	Ambient, chamber and exit pressure	bar
r	Regression or burn rate	mm/s
ρ	Density	kg/m^3

Symbol	Quantity	Unit (on screen)
σ_y	Yield strength	MPa
T_c	Chamber (flame) temperature	K
t_b	Burn time	s

D. Parameter Index

Index from a field identifier (the name used in the project file and in server messages) to its input table. The same identifier can appear on several pages or panels.

Identifier	Table	Table subject
`ad_f_cooling_chamber_pressure`	13.5	Dock: Regenerative Cooling panel inputs (liquid)
`ad_f_cooling_chamber_temperature`	13.5	Dock: Regenerative Cooling panel inputs (liquid)
`ad_f_cooling_channel_height`	13.5	Dock: Regenerative Cooling panel inputs (liquid)
`ad_f_cooling_channel_width`	13.5	Dock: Regenerative Cooling panel inputs (liquid)
`ad_f_cooling_coolant`	13.5	Dock: Regenerative Cooling panel inputs (liquid)
`ad_f_cooling_coolant_inlet_pressure`	13.5	Dock: Regenerative Cooling panel inputs (liquid)
`ad_f_cooling_coolant_inlet_temp`	13.5	Dock: Regenerative Cooling panel inputs (liquid)
`ad_f_cooling_coolant_mdot`	13.5	Dock: Regenerative Cooling panel inputs (liquid)
`ad_f_cooling_expansion_ratio`	13.5	Dock: Regenerative Cooling panel inputs (liquid)
`ad_f_cooling_flow_direction`	13.5	Dock: Regenerative Cooling panel inputs (liquid)
`ad_f_cooling_gamma`	13.5	Dock: Regenerative Cooling panel inputs (liquid)
`ad_f_cooling_molecular_weight`	13.5	Dock: Regenerative Cooling panel inputs (liquid)
`ad_f_cooling_n_channels`	13.5	Dock: Regenerative Cooling panel inputs (liquid)
`ad_f_cooling_throat_diameter`	13.5	Dock: Regenerative Cooling panel inputs (liquid)
`ad_f_cooling_wall_material`	13.5	Dock: Regenerative Cooling panel inputs (liquid)
`ad_f_cooling_wall_thickness`	13.5	Dock: Regenerative Cooling panel inputs (liquid)
`ad_f_flow_ambient_pressure`	13.11	Dock: Nozzle Flow panel inputs
`ad_f_flow_chamber_pressure`	13.11	Dock: Nozzle Flow panel inputs
`ad_f_flow_chamber_temperature`	13.11	Dock: Nozzle Flow panel inputs
`ad_f_flow_exit_diameter`	13.11	Dock: Nozzle Flow panel inputs
`ad_f_flow_fidelity`	13.11	Dock: Nozzle Flow panel inputs
`ad_f_flow_fuel_type`	13.11	Dock: Nozzle Flow panel inputs
`ad_f_flow_gamma`	13.11	Dock: Nozzle Flow panel inputs
`ad_f_flow_kinetic_fidelity`	13.11	Dock: Nozzle Flow panel inputs
`ad_f_flow_molecular_weight`	13.11	Dock: Nozzle Flow panel inputs
`ad_f_flow_n_stations`	13.11	Dock: Nozzle Flow panel inputs

Identifier	Table	Table subject
ad_f_flow_of_ratio	13.11	Dock: Nozzle Flow panel inputs
ad_f_flow_oxidizer_type	13.11	Dock: Nozzle Flow panel inputs
ad_f_flow_throat_diameter	13.11	Dock: Nozzle Flow panel inputs
ad_f_joint_bolt_count	13.7	Dock: Bolted Joint panel inputs
ad_f_joint_external_axial_load_n	13.7	Dock: Bolted Joint panel inputs
ad_f_joint_grip_length_mm	13.7	Dock: Bolted Joint panel inputs
ad_f_joint_lubricated	13.7	Dock: Bolted Joint panel inputs
ad_f_joint_member_material	13.7	Dock: Bolted Joint panel inputs
ad_f_joint_pressure_bar	13.7	Dock: Bolted Joint panel inputs
ad_f_joint_property_class	13.7	Dock: Bolted Joint panel inputs
ad_f_joint_reusable	13.7	Dock: Bolted Joint panel inputs
ad_f_joint_seal_diameter_mm	13.7	Dock: Bolted Joint panel inputs
ad_f_joint_size	13.7	Dock: Bolted Joint panel inputs
ad_f_performance_base_heat_flux	13.9	Dock: Advanced Performance panel inputs
ad_f_performance_base_isp	13.9	Dock: Advanced Performance panel inputs
ad_f_performance_burn_time	13.9	Dock: Advanced Performance panel inputs
ad_f_performance_chamber_diameter	13.9	Dock: Advanced Performance panel inputs
ad_f_performance_chamber_length	13.9	Dock: Advanced Performance panel inputs
ad_f_performance_chamber_pressure	13.9	Dock: Advanced Performance panel inputs
ad_f_performance_chamber_temperature	13.9	Dock: Advanced Performance panel inputs
ad_f_performance_critical_heat_flux	13.9	Dock: Advanced Performance panel inputs
ad_f_performance_expansion_ratio	13.9	Dock: Advanced Performance panel inputs
ad_f_performance_gamma	13.9	Dock: Advanced Performance panel inputs
ad_f_performance_mdot_total	13.9	Dock: Advanced Performance panel inputs
ad_f_performance_nozzle_length	13.9	Dock: Advanced Performance panel inputs
ad_f_performance_optimal_of_ratio	13.9	Dock: Advanced Performance panel inputs
ad_f_performance_throat_area	13.9	Dock: Advanced Performance panel inputs
ad_f_protection_burn_time_s	13.4	Dock: Thermal Protection panel inputs
ad_f_protection_design_margin	13.4	Dock: Thermal Protection panel inputs
ad_f_protection_emissivity	13.4	Dock: Thermal Protection panel inputs
ad_f_protection_h_gas_W_m2K	13.4	Dock: Thermal Protection panel inputs
ad_f_protection_material	13.4	Dock: Thermal Protection panel inputs
ad_f_protection_mode	13.4	Dock: Thermal Protection panel inputs
ad_f_protection_q_net_W_m2	13.4	Dock: Thermal Protection panel inputs
ad_f_protection_q_star_MJ_kg	13.4	Dock: Thermal Protection panel inputs
ad_f_protection_radiation_material	13.4	Dock: Thermal Protection panel inputs
ad_f_protection_T_initial_K	13.4	Dock: Thermal Protection panel inputs
ad_f_protection_T_recovery_K	13.4	Dock: Thermal Protection panel inputs
ad_f_protection_wall_material	13.4	Dock: Thermal Protection panel inputs
ad_f_protection_wall_thickness_m	13.4	Dock: Thermal Protection panel inputs
ad_f_safety_burn_time	13.8	Dock: Comprehensive Safety panel inputs
ad_f_safety_chamber_diameter	13.8	Dock: Comprehensive Safety panel inputs

Identifier	Table	Table subject
ad_f_safety_chamber_length	13.8	Dock: Comprehensive Safety panel inputs
ad_f_safety_chamber_pressure	13.8	Dock: Comprehensive Safety panel inputs
ad_f_safety_chamber_temperature	13.8	Dock: Comprehensive Safety panel inputs
ad_f_safety_facility_type	13.8	Dock: Comprehensive Safety panel inputs
ad_f_safety_material	13.8	Dock: Comprehensive Safety panel inputs
ad_f_safety_motor_type	13.8	Dock: Comprehensive Safety panel inputs
ad_f_safety_propellant_mass	13.8	Dock: Comprehensive Safety panel inputs
ad_f_safety_propellant_type	13.8	Dock: Comprehensive Safety panel inputs
ad_f_safety_throat_diameter	13.8	Dock: Comprehensive Safety panel inputs
ad_f_safety_thrust	13.8	Dock: Comprehensive Safety panel inputs
ad_f_safety_wall_temperature_cold	13.8	Dock: Comprehensive Safety panel inputs
ad_f_safety_wall_temperature_hot	13.8	Dock: Comprehensive Safety panel inputs
ad_f_safety_wall_thickness	13.8	Dock: Comprehensive Safety panel inputs
ad_f_structural_burn_time	13.6	Dock: Structural Safety panel inputs
ad_f_structural_chamber_diameter	13.6	Dock: Structural Safety panel inputs
ad_f_structural_chamber_length	13.6	Dock: Structural Safety panel inputs
ad_f_structural_chamber_pressure	13.6	Dock: Structural Safety panel inputs
ad_f_structural_chamber_temperature	13.6	Dock: Structural Safety panel inputs
ad_f_structural_material	13.6	Dock: Structural Safety panel inputs
ad_f_structural_safety_factor	13.6	Dock: Structural Safety panel inputs
ad_f_structural_throat_diameter	13.6	Dock: Structural Safety panel inputs
ad_f_structural_thrust	13.6	Dock: Structural Safety panel inputs
ad_f_structural_wall_temperature_cold	13.6	Dock: Structural Safety panel inputs
ad_f_structural_wall_temperature_hot	13.6	Dock: Structural Safety panel inputs
ad_f_structural_wall_thickness	13.6	Dock: Structural Safety panel inputs
ad_f_thermal_burn_time	13.3	Dock: Thermal Safety panel inputs
ad_f_thermal_c_star	13.3	Dock: Thermal Safety panel inputs
ad_f_thermal_chamber_diameter	13.3	Dock: Thermal Safety panel inputs
ad_f_thermal_chamber_length	13.3	Dock: Thermal Safety panel inputs
ad_f_thermal_chamber_pressure	13.3	Dock: Thermal Safety panel inputs
ad_f_thermal_chamber_temperature	13.3	Dock: Thermal Safety panel inputs
ad_f_thermal_cooling_type	13.3	Dock: Thermal Safety panel inputs
ad_f_thermal_gamma	13.3	Dock: Thermal Safety panel inputs
ad_f_thermal_material	13.3	Dock: Thermal Safety panel inputs
ad_f_thermal_mdot_total	13.3	Dock: Thermal Safety panel inputs
ad_f_thermal_molecular_weight	13.3	Dock: Thermal Safety panel inputs
ad_f_thermal_throat_diameter	13.3	Dock: Thermal Safety panel inputs
ad_f_thermal_wall_protection	13.3	Dock: Thermal Safety panel inputs
ad_f_thermal_wall_temperature_cold	13.3	Dock: Thermal Safety panel inputs

Identifier	Table	Table subject
ad_f_thermal_wall_temperature_hot	13.3	Dock: Thermal Safety panel inputs
ad_f_thermal_wall_thickness	13.3	Dock: Thermal Safety panel inputs
ad_f_thermal_x_CO2	13.3	Dock: Thermal Safety panel inputs
ad_f_thermal_x_H2O	13.3	Dock: Thermal Safety panel inputs
ad_f_vessel_code_mode	13.10	Dock: Pressure Vessel panel inputs
ad_f_vessel_head_type	13.10	Dock: Pressure Vessel panel inputs
ad_f_vessel_inner_diameter_mm	13.10	Dock: Pressure Vessel panel inputs
ad_f_vessel_material	13.10	Dock: Pressure Vessel panel inputs
ad_f_vessel_meop_bar	13.10	Dock: Pressure Vessel panel inputs
ad_f_vessel_temperature_K	13.10	Dock: Pressure Vessel panel inputs
ad_f_vessel_wall_thickness_mm	13.10	Dock: Pressure Vessel panel inputs
ad_f_vessel_weld_efficiency	13.10	Dock: Pressure Vessel panel inputs
alt_end	9.2	Hybrid page: Flight Environment
alt_points	9.2	Hybrid page: Flight Environment
alt_start	9.2	Hybrid page: Flight Environment
ambient_temp	9.11	Hybrid page: Design Configuration window
ambient_temp	10.9	Solid page: 6. Initial and Environmental Conditions
annular_gap	11.4	Liquid page: 4. Injector System Parameters
atm_pressure	10.9	Solid page: 6. Initial and Environmental Conditions
avionics_mass	10.10	Solid page: 7. Mass and Weight Breakdown
burn_rate_a	10.8	Solid page: 5. Efficiency and Empirical Coefficients
burn_rate_n	10.8	Solid page: 5. Efficiency and Empirical Coefficients
burn_rate_preset	10.8	Solid page: 5. Efficiency and Empirical Coefficients
burn_time	9.3	Hybrid page: Basic Parameters
burn_time_impulse	9.3	Hybrid page: Basic Parameters
calculate_trajectory	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
calibration_factor	10.11	Solid page: 8. Test Stand Planning (not used in the motor calculation)
case_mass	10.10	Solid page: 7. Mass and Weight Breakdown
case_material	10.6	Solid page: 3. Combustion Chamber and Casing
case_thickness	10.6	Solid page: 3. Combustion Chamber and Casing
cf_efficiency	10.8	Solid page: 5. Efficiency and Empirical Coefficients
chamber_diameter	10.6	Solid page: 3. Combustion Chamber and Casing
chamber_diameter	11.5	Liquid page: 5. Combustion Chamber Design
chamber_diameter_input	9.4	Hybrid page: Advanced Parameters
chamber_length_override	9.11	Hybrid page: Design Configuration window
chamber_material	9.9	Hybrid page: Materials
chamber_material	11.5	Liquid page: 5. Combustion Chamber Design
chamber_pressure	9.3	Hybrid page: Basic Parameters
chamber_pressure	10.7	Solid page: 4. Nozzle
chamber_pressure	11.3	Liquid page: 3. Feed System / Engine Cycle
chamber_roughness	11.5	Liquid page: 5. Combustion Chamber Design

Identifier	Table	Table subject
chamber_volume	10.6	Solid page: 3. Combustion Chamber and Casing
chamber_wall_thickness	11.5	Liquid page: 5. Combustion Chamber Design
char_velocity	10.2	Solid page: 1. Propellant Properties
characteristic_length	11.5	Liquid page: 5. Combustion Chamber Design
closeout_material	11.6	Liquid page: 6. Nozzle and Cooling System
closeout_thickness	11.6	Liquid page: 6. Nozzle and Cooling System
closure_bolt_class	10.6	Solid page: 3. Combustion Chamber and Casing
closure_bolt_class	11.8	Liquid page: 8. Mass and Structural Parameters
closure_bolt_count	10.6	Solid page: 3. Combustion Chamber and Casing
closure_bolt_count	11.8	Liquid page: 8. Mass and Structural Parameters
closure_bolt_size	10.6	Solid page: 3. Combustion Chamber and Casing
closure_bolt_size	11.8	Liquid page: 8. Mass and Structural Parameters
closure_mass	10.10	Solid page: 7. Mass and Weight Breakdown
cmp_name	13.13	Dock: Comparative Analysis panel inputs
cmp_ov_a	13.13	Dock: Comparative Analysis panel inputs
cmp_ov_b	13.13	Dock: Comparative Analysis panel inputs
combustion_efficiency	9.4	Hybrid page: Advanced Parameters
combustion_efficiency	10.8	Solid page: 5. Efficiency and Empirical Coefficients
combustion_efficiency	11.2	Liquid page: 2. O/F Optimization Data
combustion_type	9.4	Hybrid page: Advanced Parameters
contraction_ratio	9.4	Hybrid page: Advanced Parameters
contraction_ratio	11.5	Liquid page: 5. Combustion Chamber Design
convergent_angle	10.7	Solid page: 4. Nozzle
coolant_flow_percent	11.6	Liquid page: 6. Nozzle and Cooling System
coolant_inlet_temp	11.6	Liquid page: 6. Nozzle and Cooling System
cooling_channels	9.11	Hybrid page: Design Configuration window
cooling_channels	11.6	Liquid page: 6. Nozzle and Cooling System
cooling_type	11.6	Liquid page: 6. Nozzle and Cooling System
core_diameter	10.3	Solid page: 2. Grain Geometry
data_collection_time	10.11	Solid page: 8. Test Stand Planning (not used in the motor calculation)
density	10.2	Solid page: 1. Propellant Properties
discharge_coeff	10.8	Solid page: 5. Efficiency and Empirical Coefficients
discharge_coefficient	9.8	Hybrid page: Injector Configuration
discharge_coefficient	11.4	Liquid page: 4. Injector System Parameters
divergence_loss	10.8	Solid page: 5. Efficiency and Empirical Coefficients
divergent_angle	10.7	Solid page: 4. Nozzle
drag_coefficient	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
dry_mass	10.10	Solid page: 7. Mass and Weight Breakdown
engine_cycle	11.3	Liquid page: 3. Feed System / Engine Cycle
engine_life_cycles	11.8	Liquid page: 8. Mass and Structural Parameters
engine_mount	11.8	Liquid page: 8. Mass and Structural Parameters
erosion_factor	10.7	Solid page: 4. Nozzle

Identifier	Table	Table subject
erosive_k	10.2	Solid page: 1. Propellant Properties
erosive_k_source	10.2	Solid page: 1. Propellant Properties
erosive_m	10.2	Solid page: 1. Propellant Properties
exit_diameter	10.7	Solid page: 4. Nozzle
expansion_ratio	9.4	Hybrid page: Advanced Parameters
expansion_ratio	10.7	Solid page: 4. Nozzle
fea_metric	13.21	Structural FEA panel inputs
fea_show_mesh	13.21	Structural FEA panel inputs
fea_t_ambient	13.22	Thermal FEA panel inputs (hybrid)
fea_t_show_mesh	13.22	Thermal FEA panel inputs (hybrid)
feed_line_diameter_mm	11.3	Liquid page: 3. Feed System / Engine Cycle
feed_line_inner_diameter_mm	9.6	Hybrid page: Oxidizer Configuration
feed_line_length_m	9.6	Hybrid page: Oxidizer Configuration
feed_line_length_m	11.3	Liquid page: 3. Feed System / Engine Cycle
feed_line_material	11.3	Liquid page: 3. Feed System / Engine Cycle
feed_line_wall_thickness	11.3	Liquid page: 3. Feed System / Engine Cycle
feed_pressurant_collapse_factor	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_pressurant_gas	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_pressurant_initial_pressure	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_pressurant_initial_temperature	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_pressurant_initial_ullage_volume	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_pressurant_mode	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_pressurant_polytropic_n	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_pressurant_propellant_volume	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_pressurant_regulator_margin	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_pressurant_storage_pressure	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_pressurant_tank_pressure	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_pressure	11.3	Liquid page: 3. Feed System / Engine Cycle
feed_slosh_baffle_depth_ratio	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_slosh_baffle_width_ratio	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_slosh_fill_height	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_slosh_fluid_density	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_slosh_g_eff	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_slosh_radius	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_water_hammer_fluid	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_water_hammer_line_id_mm	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)

Identifier	Table	Table subject
feed_water_hammer_line_length_m	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_water_hammer_mdot_kg_s	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_water_hammer_pipe_material	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_water_hammer_valve_closure_time_ms	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_water_hammer_wall_thickness_mm	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_water_hammer_working_pressure_bar	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
feed_water_hammer_working_pressure_reference	13.12	Dock: Feed System panel inputs (slosh, pressurant, water hammer)
film_cooling_percent	11.6	Liquid page: 6. Nozzle and Cooling System
filter_cutoff	10.11	Solid page: 8. Test Stand Planning (not used in the motor calculation)
fin_count	10.4	Solid page: grain-type specific fields (Star, Finocyl, Wagon Wheel, Slotted)
fin_length	10.4	Solid page: grain-type specific fields (Star, Finocyl, Wagon Wheel, Slotted)
fin_width	10.4	Solid page: grain-type specific fields (Star, Finocyl, Wagon Wheel, Slotted)
final_mass	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
finned_length_fraction	10.4	Solid page: grain-type specific fields (Star, Finocyl, Wagon Wheel, Slotted)
flame_temp	10.2	Solid page: 1. Propellant Properties
fuel_boiling_point	11.1	Liquid page: 1. Propellant Data
fuel_density	9.5	Hybrid page: Fuel Configuration
fuel_density	11.1	Liquid page: 1. Propellant Data
fuel_heat_capacity	11.1	Liquid page: 1. Propellant Data
fuel_injection_velocity	11.4	Liquid page: 4. Injector System Parameters
fuel_orifice_diameter	11.4	Liquid page: 4. Injector System Parameters
fuel_type	9.5	Hybrid page: Fuel Configuration
fuel_type	11.1	Liquid page: 1. Propellant Data
fuel_viscosity	11.1	Liquid page: 1. Propellant Data
gamma	10.2	Solid page: 1. Propellant Properties
generator_gas_temp	11.3	Liquid page: 3. Feed System / Engine Cycle
gimbal_actuator_arm	11.8	Liquid page: 8. Mass and Structural Parameters
gimbal_bolt_circle	11.8	Liquid page: 8. Mass and Structural Parameters
gimbal_bolt_count	11.8	Liquid page: 8. Mass and Structural Parameters
gimbal_range	11.8	Liquid page: 8. Mass and Structural Parameters
gimbal_ring_offset	11.8	Liquid page: 8. Mass and Structural Parameters
grain_count	10.3	Solid page: 2. Grain Geometry
grain_gap	10.5	Solid page: Inhibitor Coating
grain_length	10.3	Solid page: 2. Grain Geometry
grain_support	10.3	Solid page: 2. Grain Geometry
grain_type	10.3	Solid page: 2. Grain Geometry
heat_value	10.2	Solid page: 1. Propellant Properties
hole_diameter_max	9.8	Hybrid page: Injector Configuration

Identifier	Table	Table subject
hole_diameter_min	9.8	Hybrid page: Injector Configuration
hole_pattern	9.11	Hybrid page: Design Configuration window
humidity	10.9	Solid page: 6. Initial and Environmental Conditions
igniter_pressure_fraction	10.9	Solid page: 6. Initial and Environmental Conditions
ignition_delay	10.9	Solid page: 6. Initial and Environmental Conditions
inhibit_front	10.5	Solid page: Inhibitor Coating
inhibit_outer	10.5	Solid page: Inhibitor Coating
inhibit_rear	10.5	Solid page: Inhibitor Coating
initial_mass	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
initial_port_diameter	9.4	Hybrid page: Advanced Parameters
initial_temp	10.6	Solid page: 3. Combustion Chamber and Casing
inj_bf	13.18	Injector Design panel inputs (hybrid)
inj_bf	13.19	Injector Design panel inputs (liquid)
inj_cd	13.18	Injector Design panel inputs (hybrid)
inj_cd	13.19	Injector Design panel inputs (liquid)
inj_coax_f	13.18	Injector Design panel inputs (hybrid)
inj_dp	13.18	Injector Design panel inputs (hybrid)
inj_dp	13.19	Injector Design panel inputs (liquid)
inj_imp_angle	13.18	Injector Design panel inputs (hybrid)
inj_imp_d	13.18	Injector Design panel inputs (hybrid)
inj_imp_pairs	13.18	Injector Design panel inputs (hybrid)
inj_inlet	13.18	Injector Design panel inputs (hybrid)
inj_inlet	13.19	Injector Design panel inputs (liquid)
inj_ld	13.18	Injector Design panel inputs (hybrid)
inj_ld	13.19	Injector Design panel inputs (liquid)
inj_mdot_fuel	13.19	Injector Design panel inputs (liquid)
inj_mdot_ox	13.18	Injector Design panel inputs (hybrid)
inj_mdot_ox	13.19	Injector Design panel inputs (liquid)
inj_n2o	13.18	Injector Design panel inputs (hybrid)
inj_n_el	13.19	Injector Design panel inputs (liquid)
inj_pc	13.18	Injector Design panel inputs (hybrid)
inj_pc	13.19	Injector Design panel inputs (liquid)
inj_pfeed	13.18	Injector Design panel inputs (hybrid)
inj_pin_ang	13.18	Injector Design panel inputs (hybrid)
inj_pin_d	13.18	Injector Design panel inputs (hybrid)
inj_pin_d	13.19	Injector Design panel inputs (liquid)
inj_pin_dp	13.18	Injector Design panel inputs (hybrid)
inj_pin_dp	13.19	Injector Design panel inputs (liquid)
inj_pin_holes	13.18	Injector Design panel inputs (hybrid)
inj_pin_ls	13.18	Injector Design panel inputs (hybrid)
inj_pin_ls	13.19	Injector Design panel inputs (liquid)
inj_pin_n	13.18	Injector Design panel inputs (hybrid)
inj_pin_n	13.19	Injector Design panel inputs (liquid)
inj_pin_od	13.18	Injector Design panel inputs (hybrid)
inj_pin_t	13.18	Injector Design panel inputs (hybrid)
inj_pin_t	13.19	Injector Design panel inputs (liquid)
inj_radial_fraction	13.18	Injector Design panel inputs (hybrid)
inj_rho_fuel	13.19	Injector Design panel inputs (liquid)
inj_rho_ox	13.18	Injector Design panel inputs (hybrid)
inj_rho_ox	13.19	Injector Design panel inputs (liquid)
inj_sh_dmax	13.18	Injector Design panel inputs (hybrid)

Identifier	Table	Table subject
inj_sh_dmin	13.18	Injector Design panel inputs (hybrid)
inj_sh_n	13.18	Injector Design panel inputs (hybrid)
inj_sw_ds	13.18	Injector Design panel inputs (hybrid)
inj_sw_h	13.18	Injector Design panel inputs (hybrid)
inj_sw_nslots	13.18	Injector Design panel inputs (hybrid)
inj_sw_w	13.18	Injector Design panel inputs (hybrid)
inj_theta	13.18	Injector Design panel inputs (hybrid)
inj_theta	13.19	Injector Design panel inputs (liquid)
inj_tmr	13.19	Injector Design panel inputs (liquid)
inj_tox	13.18	Injector Design panel inputs (hybrid)
inj_type	13.18	Injector Design panel inputs (hybrid)
inj_type	13.19	Injector Design panel inputs (liquid)
injector_elements	11.4	Liquid page: 4. Injector System Parameters
injector_material	9.9	Hybrid page: Materials
injector_material	11.4	Liquid page: 4. Injector System Parameters
injector_pressure_drop	11.4	Liquid page: 4. Injector System Parameters
injector_type	9.8	Hybrid page: Injector Configuration
injector_type	11.4	Liquid page: 4. Injector System Parameters
insulation_mass	10.10	Solid page: 7. Mass and Weight Breakdown
insulation_thickness	10.5	Solid page: Inhibitor Coating
kinetic_efficiency	10.8	Solid page: 5. Efficiency and Empirical Coefficients
l_star	9.4	Hybrid page: Advanced Parameters
launch_altitude	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
launch_angle	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
liner_density	10.6	Solid page: 3. Combustion Chamber and Casing
liner_mass	10.10	Solid page: 7. Mass and Weight Breakdown
liner_thickness	10.6	Solid page: 3. Combustion Chamber and Casing
load_cell_capacity	10.11	Solid page: 8. Test Stand Planning (not used in the motor calculation)
ls-a-azim	14.2	Launch Site: Vehicle to Fly
ls-a-bodydia	14.2	Launch Site: Vehicle to Fly
ls-a-bodylen	14.2	Launch Site: Vehicle to Fly
ls-a-cd0	14.2	Launch Site: Vehicle to Fly
ls-a-drymass	14.2	Launch Site: Vehicle to Fly
ls-a-elev	14.2	Launch Site: Vehicle to Fly
ls-a-fincount	14.2	Launch Site: Vehicle to Fly
ls-a-finspan	14.2	Launch Site: Vehicle to Fly
ls-a-winddir	14.2	Launch Site: Vehicle to Fly
ls-a-windspeed	14.2	Launch Site: Vehicle to Fly
ls-exagg	14.3	Launch Site: Flight Path
ls-follow	14.3	Launch Site: Flight Path
ls-lat	14.1	Launch Site: Selected Site
ls-lon	14.1	Launch Site: Selected Site
ls-online	14.2	Launch Site: Vehicle to Fly
ls-presets	14.2	Launch Site: Vehicle to Fly
ls-project	14.2	Launch Site: Vehicle to Fly
ls-tile-network	14.4	Launch Site: Legend
ls-time	14.3	Launch Site: Flight Path
ls-v-burntime	14.2	Launch Site: Vehicle to Fly
ls-v-inertmass	14.2	Launch Site: Vehicle to Fly
ls-v-propmass	14.2	Launch Site: Vehicle to Fly

Identifier	Table	Table subject
ls-v-thrust	14.2	Launch Site: Vehicle to Fly
ls-vehicle-source	14.2	Launch Site: Vehicle to Fly
mass_flux_chamber	9.4	Hybrid page: Advanced Parameters
material_catalog_target	10.6	Solid page: 3. Combustion Chamber and Casing
max_burn_duration	11.7	Liquid page: 7. Throttling and Burn Duration
max_wall_temp	11.6	Liquid page: 6. Nozzle and Cooling System
min_throttle	11.7	Liquid page: 7. Throttling and Burn Duration
mixture_ratio	11.2	Liquid page: 2. O/F Optimization Data
molecular_weight	10.2	Solid page: 1. Propellant Properties
motor_description	9.1	Hybrid page: Motor Information
motor_name	9.1	Hybrid page: Motor Information
n_holes_override	9.11	Hybrid page: Design Configuration window
nozzle_efficiency	10.7	Solid page: 4. Nozzle
nozzle_expansion_ratio	11.6	Liquid page: 6. Nozzle and Cooling System
nozzle_mass	10.10	Solid page: 7. Mass and Weight Breakdown
nozzle_material	9.9	Hybrid page: Materials
nozzle_material	10.7	Solid page: 4. Nozzle
nozzle_type	9.4	Hybrid page: Advanced Parameters
nozzle_type	11.6	Liquid page: 6. Nozzle and Cooling System
of_max	11.2	Liquid page: 2. O/F Optimization Data
of_min	11.2	Liquid page: 2. O/F Optimization Data
of_ratio	9.7	Hybrid page: Mixture Ratio
outer_diameter	10.3	Solid page: 2. Grain Geometry
overall_efficiency	10.8	Solid page: 5. Efficiency and Empirical Coefficients
oxidizer_density	9.6	Hybrid page: Oxidizer Configuration
oxidizer_density	11.1	Liquid page: 1. Propellant Data
oxidizer_injection_velocity	11.4	Liquid page: 4. Injector System Parameters
oxidizer_orifice_diameter	11.4	Liquid page: 4. Injector System Parameters
oxidizer_phase	9.6	Hybrid page: Oxidizer Configuration
oxidizer_temp	9.6	Hybrid page: Oxidizer Configuration
oxidizer_type	9.6	Hybrid page: Oxidizer Configuration
oxidizer_type	11.1	Liquid page: 1. Propellant Data
oxidizer_viscosity	9.6	Hybrid page: Oxidizer Configuration
oxidizer_viscosity	11.1	Liquid page: 1. Propellant Data
parachute_area	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
parachute_area	10.12	Solid page: Trajectory Analysis inputs
parachute_cd	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
parachute_cd	10.12	Solid page: Trajectory Analysis inputs
parachute_deploy_delay	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
parachute_deploy_delay	10.12	Solid page: Trajectory Analysis inputs
param_end	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
param_start	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
param_steps	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
param_type	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
pintle_diameter	11.4	Liquid page: 4. Injector System Parameters
pintle_radial_hole_count	11.4	Liquid page: 4. Injector System Parameters
pintle_radial_hole_diameter	11.4	Liquid page: 4. Injector System Parameters

Identifier	Table	Table subject
plate_thickness	9.8	Hybrid page: Injector Configuration
port_count	9.4	Hybrid page: Advanced Parameters
preburner_type	11.3	Liquid page: 3. Feed System / Engine Cycle
pressure_drop_percent	9.11	Hybrid page: Design Configuration window
pressure_sensor_range	10.11	Solid page: 8. Test Stand Planning (not used in the motor calculation)
pressurization_type	11.3	Liquid page: 3. Feed System / Engine Cycle
propellant_family_filter	10.1	Solid page: 0. Propellant Catalogue
propellant_mass	10.10	Solid page: 7. Mass and Weight Breakdown
propellant_name	10.2	Solid page: 1. Propellant Properties
propellant_type	10.2	Solid page: 1. Propellant Properties
reference_area	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
regression_a	9.5	Hybrid page: Fuel Configuration
regression_n	9.5	Hybrid page: Fuel Configuration
residual_web_thickness	9.4	Hybrid page: Advanced Parameters
safety_factor	9.11	Hybrid page: Design Configuration window
safety_factor	10.6	Solid page: 3. Combustion Chamber and Casing
safety_factor	11.8	Liquid page: 8. Mass and Structural Parameters
sampling_freq	10.11	Solid page: 8. Test Stand Planning (not used in the motor calculation)
sd_azimuth	13.20	Flight Dynamics (6-DOF) panel inputs
sd_body_d	13.20	Flight Dynamics (6-DOF) panel inputs
sd_body_l	13.20	Flight Dynamics (6-DOF) panel inputs
sd_burn	13.20	Flight Dynamics (6-DOF) panel inputs
sd_cd0	13.20	Flight Dynamics (6-DOF) panel inputs
sd_dry_m	13.20	Flight Dynamics (6-DOF) panel inputs
sd_elev	13.20	Flight Dynamics (6-DOF) panel inputs
sd_fin_count	13.20	Flight Dynamics (6-DOF) panel inputs
sd_fin_pos	13.20	Flight Dynamics (6-DOF) panel inputs
sd_fin_root	13.20	Flight Dynamics (6-DOF) panel inputs
sd_fin_span	13.20	Flight Dynamics (6-DOF) panel inputs
sd_fin_sweep	13.20	Flight Dynamics (6-DOF) panel inputs
sd_fin_tip	13.20	Flight Dynamics (6-DOF) panel inputs
sd_motor_select	13.20	Flight Dynamics (6-DOF) panel inputs
sd_nose_l	13.20	Flight Dynamics (6-DOF) panel inputs
sd_nose_type	13.20	Flight Dynamics (6-DOF) panel inputs
sd_prop_m	13.20	Flight Dynamics (6-DOF) panel inputs
sd_rail	13.20	Flight Dynamics (6-DOF) panel inputs
sd_thrust	13.20	Flight Dynamics (6-DOF) panel inputs
sd_thrust_src	13.20	Flight Dynamics (6-DOF) panel inputs
sd_use_curve	13.20	Flight Dynamics (6-DOF) panel inputs
sd_wind	13.20	Flight Dynamics (6-DOF) panel inputs
sd_wind_dir	13.20	Flight Dynamics (6-DOF) panel inputs
single_altitude	9.2	Hybrid page: Flight Environment
single_pressure	9.2	Hybrid page: Flight Environment
skip_distance	11.4	Liquid page: 4. Injector System Parameters
slot_count	10.4	Solid page: grain-type specific fields (Star, Finocyl, Wagon Wheel, Slotted)
slot_depth	10.4	Solid page: grain-type specific fields (Star, Finocyl, Wagon Wheel, Slotted)
slot_width	10.4	Solid page: grain-type specific fields (Star, Finocyl, Wagon Wheel, Slotted)

Identifier	Table	Table subject
star_points	10.4	Solid page: grain-type specific fields (Star, Finocyl, Wagon Wheel, Slotted)
star_radius	10.4	Solid page: grain-type specific fields (Star, Finocyl, Wagon Wheel, Slotted)
tank_pressure	9.3	Hybrid page: Basic Parameters
tank_pressure_source	9.3	Hybrid page: Basic Parameters
target_thrust_to_weight	11.8	Liquid page: 8. Mass and Structural Parameters
target_velocity	9.8	Hybrid page: Injector Configuration
temp_coeff	10.2	Solid page: 1. Propellant Properties
temp_coeff_source	10.2	Solid page: 1. Propellant Properties
test_altitude	10.9	Solid page: 6. Initial and Environmental Conditions
thermo_source	10.2	Solid page: 1. Propellant Properties
throat_diameter	10.7	Solid page: 4. Nozzle
throat_diameter	11.6	Liquid page: 6. Nozzle and Cooling System
thrust	9.3	Hybrid page: Basic Parameters
thrust	11.3	Liquid page: 3. Feed System / Engine Cycle
thrust_impulse	9.3	Hybrid page: Basic Parameters
total_impulse	9.3	Hybrid page: Basic Parameters
tp_feed_mode	13.17	Hybrid: Transient Analysis panel inputs
tp_fill	13.17	Hybrid: Transient Analysis panel inputs
tp_tank_temp	13.17	Hybrid: Transient Analysis panel inputs
traj_alt_end	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
traj_alt_start	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
traj_burn_time	10.12	Solid page: Trajectory Analysis inputs
traj_burn_time	11.9	Liquid page: Trajectory Inputs
traj_drag_coeff	10.12	Solid page: Trajectory Analysis inputs
traj_drag_coeff	11.9	Liquid page: Trajectory Inputs
traj_final_mass	10.12	Solid page: Trajectory Analysis inputs
traj_final_mass	11.9	Liquid page: Trajectory Inputs
traj_initial_mass	10.12	Solid page: Trajectory Analysis inputs
traj_initial_mass	11.9	Liquid page: Trajectory Inputs
traj_points	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
traj_ref_area	10.12	Solid page: Trajectory Analysis inputs
traj_ref_area	11.9	Liquid page: Trajectory Inputs
turbine_expansion_ratio	11.3	Liquid page: 3. Feed System / Engine Cycle
turbine_inlet_pressure	11.3	Liquid page: 3. Feed System / Engine Cycle
turbopump_efficiency	11.3	Liquid page: 3. Feed System / Engine Cycle
two_phase_loss	10.8	Solid page: 5. Efficiency and Empirical Coefficients
uncertainty_level	10.11	Solid page: 8. Test Stand Planning (not used in the motor calculation)
uq_level	13.14	Dock: Uncertainty Quantification panel inputs
uq_seed	13.14	Dock: Uncertainty Quantification panel inputs
valve_closure_time_ms	11.3	Liquid page: 3. Feed System / Engine Cycle
vp_csv_text	13.15	Dock: User Data Validation panel inputs
wagon_hole_count	10.4	Solid page: grain-type specific fields (Star, Finocyl, Wagon Wheel, Slotted)
wagon_hole_diameter	10.4	Solid page: grain-type specific fields (Star, Finocyl, Wagon Wheel, Slotted)
wagon_pitch_diameter	10.4	Solid page: grain-type specific fields (Star, Finocyl, Wagon Wheel, Slotted)

Identifier	Table	Table subject
`wall_thickness`	9.11	Hybrid page: Design Configuration window
`web_thickness`	10.3	Solid page: 2. Grain Geometry
`wet_mass`	10.10	Solid page: 7. Mass and Weight Breakdown
`wind_direction`	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
`wind_speed`	9.10	Hybrid page: Analysis Type (parametric sweep and trajectory)
`wind_speed`	10.9	Solid page: 6. Initial and Environmental Conditions
`yield_strength`	10.6	Solid page: 3. Combustion Chamber and Casing
`yield_strength_source`	10.6	Solid page: 3. Combustion Chamber and Casing

E. Canonical Bands and Hybrid Design Checks

E.1 Canonical bands

The canonical envelope kept by the endpoint layer in SI units (`CANONICAL_LIMITS`). The form layer's bands in user units (mm, bar) are derived from it.

Table E.1: Canonical bands

Quantity	Canonical band	Unit
Thrust	10 – 1 000 000	N
Chamber pressure	1 – 500	bar
Tank pressure	5 – 700	bar
Burn time	0.1 – 300	s
Chamber diameter	0.01 – 5.0	m
Chamber length	0.05 – 20.0	m
Throat diameter	0.001 – 2.0	m
Exit diameter	0.002 – 5.0	m
Port diameter	0.005 – 1.0	m
Fuel-grain length	0.05 – 10.0	m
Web thickness	0.001 – 0.5	m
O/F and mixture ratio	0.5 – 20.0	–
Chamber temperature	200 – 5000	K
Ambient temperature	180 – 340	K

No band is defined for fields without a product-wide measured envelope (oxidizer temperature, stoichiometric O/F, initial and final mass, drag coefficient, reference area); the value must be finite and positive.

E.2 Hybrid design checks

The following checks do not reject the calculation; when the condition holds, a named line is added to the warning list. The checks come from `validation_system.py`. No unsourced "expected range" verdict is given: the bands for Isp, γ, chamber pressure, initial and final port diameter, chamber length, L*, regression rate, injector Reynolds number and injector exit velocity were removed in 3.0.6.

Table E.2: Hybrid design checks

Criterion	**Condition of the line and its class**	**Basis**
Delivered c^*	Outside the run's theoretical c^* times the pair's measured η_{c^*} band: WARNING.	The band comes from the same validation records as the sourced η_{c^*} default; without a theoretical c^* or η_{c^*} no verdict is given.
O/F	Outside the band in which the optimum was searched: WARNING. For N_2O pairs the sourced usable band is 4.0–12.0.	The optimum search and the warning use the same band; the program's own optimum gets no warning.
Injector $\Delta P/P_c$	Below 0.15: CRITICAL. Above 0.35: WARNING depending on the feed mode; INFO for a self-pressurised tank.	Lower bound chug stability (Sutton and Biblarz, Ch. 8; NASA SP-194), upper bound feed system cost.
Final port envelope	The envelope of the final ports exceeds the chamber inner diameter: CRITICAL.	Geometric condition, not a threshold.
Expansion ratio	$\varepsilon \leq 1$: WARNING. Acceptability is read from the run's own separation screen (Summerfield criterion).	No fixed band.
Thrust coefficient	$(F/A_t)/P_c$ exceeds the infinite-expansion vacuum C_F for the run's γ: CRITICAL.	Sutton and Biblarz, Eq. 3-30; no verdict if γ cannot be read.